

Statistical Synthesis: Distributions, Equations, and Cross-Cutting Tests

**A living reference for the project's statistical methods, run against the real
operational database**

Tyler Irvin

today

Steamboat Hills Geothermal Outflow: A Data Platform and Statistical Companion

Consolidated notebooks, methods, and annotated bibliography

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About this document

Known cosmetic limitation (PDF output only): page 1 of the rendered PDF is a stray, auto-generated title page carrying whichever transcluded chapter's own YAML title happened to win pandoc's last-wins frontmatter merge (harmless, but not the intended title) – the correct title page is page 2 immediately after it (an explicit title block in this file's own body, `title-block-style: none` in the YAML did not suppress the stray page). Tried and reverted: `pdftools::pdf_subset()` to drop page 1 – it corrupted the embedded images/fonts (13 MB -> 190 KB, real “Unknown compression method” errors) and is NOT safe to use on this file; do not re-attempt without first confirming poppler/ pdftools handles this specific luatex output correctly. A real fix would need to strip YAML front matter from all 8 transcluded notebooks (like the annotated-bibliography workaround just above does for that one file) rather than just accepting the stray page.

This is a single, consolidated companion document assembled from the project's living-reference notebooks (`notebooks/01-08`), each of which is also maintained and rendered independently. It exists so the full analytical narrative – from instrumentation and data provenance through statistical synthesis – can be read start to finish, cited, and shared as one file, alongside the project's annotated bibliography.

- **Source code and database pipeline:** https://github.com/tylerirvin543/Steamboat_Creek_Geochemistry_Database
- **Live documentation website:** https://tylerirvin543.github.io/Steamboat_Creek_Geochemistry_Database/
- **This document's source:** `notebooks/00_full_report.qmd` (re-render with `quarto::quarto_render("notebooks/00_full_report.qmd")`, or `quarto render notebooks/00_full_report.qmd --to pdf` from a shell)

Each chapter below is transcluded verbatim from its own standalone notebook file (via Quarto's `{{< include >}}`) – editing a chapter here means editing the source `.qmd` in `notebooks/`, not this file. Because several chapters query the live operational database and one (PHREEQC) invokes a real external executable per sample, a full render of this document re-executes all of that work and can take a meaningful amount of time; see each chapter's own notebook for a faster, standalone re-render during normal iteration.

Conductivity as a Chloride-Discharge Proxy

Purpose and scope

This notebook is the **living scientific reference** for the continuous-conductivity and chloride-sampling-frequency workstream at Steamboat Hills — it documents *why* each analysis is done, *what* the math is, and *where* the code that implements it lives, and it is intended to stay the entry point for this line of work as more data and methods (new logger deployments, real paired Cl samples, refined models) accumulate. It complements, but does not replace, the individual R scripts in `scripts/analysis/sampling_frequency/`, which remain the executable source of truth; this document narrates and runs representative pieces of them so the reasoning and the code stay in sync.

Sections below run in dependency order:

1. Scientific motivation (why continuous conductivity, why chloride, why sampling frequency matters at all).
2. Instrumentation and site roles, with the provenance of a metadata correction that affects everything downstream.
3. Raw EC $\rightarrow$ temperature-compensated specific conductance, and the current state of field-calibration.
4. Temporal structure of specific conductance alone (this section requires no chloride data and is fully populated today).
5. The conductivity $\rightarrow$ chloride predictive-model framework (currently demonstrated on synthetic data pending temporal overlap with real samples).
6. Monte Carlo sampling-frequency design (same caveat).
7. Turning an error curve into an actionable sampling recommendation.
8. Current status, known gaps, and a running changelog.

Warning

As of this writing, each conductivity logger has only a few weeks of continuous record (deployed 2026-07-15), and the only chloride/isotope field samples collected so far (2026-05-01) **predate that deployment by roughly 2.5 months**. Sections 5–7 therefore run against synthetic self-test data, clearly labeled as such. Section 4 (temporal structure of conductivity alone) is real and current. Re-run this notebook once chloride samples exist that overlap the logger record.

Scientific motivation

Steamboat Hills is a geothermal outflow system in which chloride behaves approximately conservatively (non-reactive, non-volatile) in the shallow outflow plume, so the classic approach

— used by Michael Sorey in prior studies of this system — is to treat **chloride discharge**, i.e. the mass flux

$$\Phi_{Cl} = \int_{t_0}^{t_1} C_{Cl}(t) Q(t) dt$$

(concentration C_{Cl} in $\text{mg/L} \equiv \text{g/m}^3$, discharge Q in m^3/s , so Φ_{Cl} has units of grams over the interval $[t_0, t_1]$), as a tracer for the total geothermal outflow reaching the surface stream network. A change in Φ_{Cl} following the Lower Sinter Terrace eruption is direct evidence of a change in the underlying hydrothermal discharge, independent of dilution effects that would otherwise obscure a pure-concentration or pure-discharge signal.

Two practical problems follow directly from that framing:

- **Chloride analyses are expensive and infrequent** (lab turnaround, sampling effort), so $C_{Cl}(t)$ is only ever known at a sparse set of times.
- **Specific conductance is cheap, continuous, and correlates with dissolved ion content** (conductivity loggers record every 5 minutes), so it is a natural *proxy* for $C_{Cl}(t)$ between chemistry samples — provided the conductance-to-chloride relationship is characterized and its stability over time is checked.

This notebook’s methodological goal is therefore not just “does conductivity track chloride” but the sharper, decision-relevant question the thesis poses: **what is the coarsest chemistry (chloride) sampling interval that still lets us reconstruct Φ_{Cl} , or $C_{Cl}(t)$, within an error the PI considers acceptable?** Sections 5–7 build the machinery to answer that quantitatively rather than by convention (e.g. “monthly because that’s typical”).

Instrumentation, site roles, and data provenance

Two HOBO/Onset “Full Range” EC loggers were deployed 2026-07-15 in Steamboat Creek to support this question:

Location code	Role	Serial	Rationale
SBRR	upstream_control	22575724	Co-located with the USGS gauge near Rhodes Road; represents back-ground/upstream water quality, upstream of the dominant geothermal inflow.

Location code	Role	Serial	Rationale
SBGG	downstream	22575725	Downstream of the outflow zone; expected to carry the geothermal Cl/EC signature.

i Provenance note: role/serial correction (2026-09-05)

The SBRR/SBGG logger-to-role assignment was originally guessed from coordinate proximity alone and had the two serials **swapped**: the logger originally labeled SBRR (upstream control) was reading 850–1300 $\mu\text{S}/\text{cm}$, which is far too high for background water and matches the expected downstream/near-outflow signature instead. The assignment was corrected in `data/raw/conductivity/conductivity_logger_deployments.csv` and re-ingested. This was a **metadata-only** fix — the already-ingested `Conductivity_Observations` rows are keyed by `logger_id`, not by location, so no observation rows needed to change, only which location/role each `logger_id` points to. After the fix, SBRR reads consistently lower than SBGG (means roughly 333 vs. 1069 $\mu\text{S}/\text{cm}$), which is now physically sensible for an upstream-control vs. downstream pairing. This correction matters for every downstream step in this document: any model, calibration, or figure that joins on `location_id/role` inherits the fix automatically (nothing in the modeling code hard-codes which physical serial is upstream — see the calibration check in the next section), but any *prior* interpretation of “SBRR” vs. “SBGG” behavior made before this date should be re-read with the roles reversed.

The `Conductivity_Loggers` table (below) is the single source of truth for this mapping; everything else in this document resolves location and role through it rather than hard-coding a serial number or coordinate.

```
library(DBI)
library(RSQLite)
library(dplyr)
library(tidyr)
library(ggplot2)
library(lubridate)

# All chunks in this notebook execute with the working directory pinned to
# the project root, not to notebooks/. This matters because several sourced
# scripts (e.g. 02_specific_conductance.R) themselves source() other helper
# scripts using paths written relative to the project root -- the same
# assumption run_pipeline.R makes -- so pinning root.dir here keeps every
```

```
# path below identical to how these scripts are invoked from the pipeline.
if (basename(getwd()) == "notebooks") {
  knitr::opts_knit$set(root.dir = normalizePath(file.path(getwd(), "..")))
}

db_path <- file.path("database", "geochem_operational.sqlite")
if (!file.exists(db_path)) db_path <- file.path("database", "geochem_demo.sqlite")

con <- dbConnect(SQLite(), db_path, flags = SQLITE_RO)

dbGetQuery(con, "
  SELECT cl.logger_id, cl.location_id, cl.role, cl.serial_number, l.external_station_code
  FROM Conductivity_Loggers cl
  LEFT JOIN Locations l ON cl.location_id = l.location_id
")
```

	logger_id	location_id	role	serial_number	external_station_code
1	1	1	downstream	22575725	SBGG
2	2	78	upstream_control	22575724	SBRR

From raw EC to specific conductance: temperature compensation and calibration

Raw electrical conductivity (EC_{raw}) is temperature-dependent, so it is converted to **specific conductance at 25°C** (SC_{25}) before any comparison across time or between the two loggers. The linear temperature-compensation model (USGS Techniques and Methods 1-D3) is:

$$SC_{25} = \frac{EC_{raw}}{1 + \beta(T - 25)}$$

with reference temperature $\theta = 25^\circ C$ and compensation coefficient β (USGS default $\beta = 0.0191\ ^\circ C^{-1}$). This is implemented at ingest time in `scripts/ingest/helpers/compute_specific_conductance.R`:

```
source(file.path("scripts", "ingest", "helpers", "compute_specific_conductance.R"))

# Reproduce the ingest-time formula directly, so the math and the code
# shown here are provably the same function used to populate sc_25c.
compute_specific_conductance

function (ec_raw, temperature_c, coefficient = 0.0191, theta = 25)
{
```

```

ec_raw <- suppressWarnings(as.numeric(ec_raw))
temperature_c <- suppressWarnings(as.numeric(temperature_c))
denom <- 1 + coefficient * (temperature_c - theta)
sc25 <- ec_raw/denom
sc25[!is.finite(sc25)] <- NA_real_
sc25
}

```

Because β is an empirical property of the local water chemistry (ionic composition affects how conductivity responds to temperature), it should be periodically re-estimated against paired field conductivity checks rather than assumed to equal the USGS default indefinitely. `02_specific_conductance.R` does this by fitting

$$EC_{field,i} = \frac{EC_{raw,i}}{1 + b(T_i - 25)} + \varepsilon_i$$

by nonlinear least squares, where $(EC_{field,i}, EC_{raw,i}, T_i)$ is the i -th field check matched to the nearest logger observation within a time tolerance (default 15 min), and b is estimated in place of the fixed default. The fit is only attempted with ≥ 5 pairs; otherwise the USGS default is kept and reported as such (no silent extrapolation from too few points).

This step is location/role-agnostic by construction: pairing is done purely by matching a field check's `location_id` to a logger's `location_id` (`scripts/analysis/sampling_frequency/02_specific_conductance.R`), so the SBRR/SBGG role correction above required **no code change here** — it is picked up automatically through the corrected `Conductivity_Loggers` table. The check below runs the real function against the live database:

```

source(file.path("scripts", "analysis", "sampling_frequency", "02_specific_conductance.R"))

pairs_now <- get_calibration_pairs(con)
cat("Field-vs-logger calibration pairs currently available:", nrow(pairs_now), "\n")

```

Field-vs-logger calibration pairs currently available: 0

```

calib_result <- recalibrate_sc_coefficient(con)

```

Only 0 field-vs-logger calibration pair(s) found (need ≥ 5) - keeping USGS default coefficient

```

calib_result[c("coefficient", "n_pairs")]

```

```
$coefficient
```

```
[1] 0.0191
```

```
$n_pairs
```

```
[1] 0
```

There are currently 0 matched pairs, and the reason is the same temporal gap noted in the callout above: field conductivity was recorded at SBRR (241 $\mu\text{S}/\text{cm}$, sample date 2026-05-01) and other locations, but no `Conductivity_Observations` exist yet for that date — logging only began 2026-07-15. As a rough sanity check (not a calibration, since it is not a time-matched pair), 241 $\mu\text{S}/\text{cm}$ at SBRR is the right order of magnitude relative to SBRR's post-correction logger mean (~ 333 $\mu\text{S}/\text{cm}$), which is mildly reassuring about the corrected role assignment but is not a substitute for an actual paired check. **Action item:** collect at least 5 field conductivity checks at SBRR/SBGG while the loggers are running to enable a real recalibration.

Temporal structure of specific conductance

This section requires no chloride data and is fully populated with real observations. The goal is to characterize the signal conductivity is expected to reconstruct chloride *from*, before any predictive model is built on top of it — diurnal cycling, autocorrelation persistence, and dominant frequencies all directly inform what chemistry-sampling intervals could plausibly be sufficient in Sections 5–7.

```
obs <- dbGetQuery(con, "
  SELECT
    co.logger_id,
    cl.role,
    l.external_station_code,
    co.timestamp,
    co.ec_raw,
    co.temperature_c,
    co.sc_25c,
    co.qc_flag
  FROM Conductivity_Observations co
  JOIN Conductivity_Loggers cl ON co.logger_id = cl.logger_id
  LEFT JOIN Locations l ON cl.location_id = l.location_id
  ORDER BY co.logger_id, co.timestamp
") |>
  mutate(timestamp = as.POSIXct(timestamp, tz = "UTC"))
```

```
dbDisconnect(con)

n_flagged <- sum(!is.na(obs$qc_flag))
message(nrow(obs), " total observations; ", n_flagged, " QC-flagged (spike/disturbance/gap).")
```

21901 total observations; 90 QC-flagged (spike/disturbance/gap).

```
# Excludes rows flagged 'spike' or 'field_visit_disturbance' by
# scripts/qc/qc_conductivity_checks.R; 'gap_boundary'/'dropout' flags (if
# present) are kept but noted, since they mark real data gaps rather than
# untrustworthy values.
obs_clean <- obs |> filter(is.na(qc_flag) | !(qc_flag %in% c("spike", "field_visit_disturbance")))

obs_clean |>
  group_by(external_station_code) |>
  summarise(
    n = n(),
    start = min(timestamp),
    end = max(timestamp),
    span_days = round(as.numeric(difftime(max(timestamp), min(timestamp), units = "days")), 1)
  )
```

```
# A tibble: 2 x 5
  external_station_code      n start                end                span_days
  <chr>                <int> <dtm>                <dtm>                <dbl>
1 SBGG                 10935 2026-07-15 20:09:46 2026-08-22 21:44:46    38.1
2 SBRR                 10876 2026-07-15 21:37:25 2026-08-22 20:47:25    38
```

Diurnal cycle

Shallow stream conductivity commonly shows a 24-hour cycle driven by evapotranspiration and thermal loading modulating the mix of dilute surface water and higher-EC baseflow/geothermal inflow. We fit a first-harmonic (single sinusoid) regression at the 24-hour period to each logger:

$$SC_{25}(t) = \alpha + \beta_1 \sin\left(\frac{2\pi t}{24}\right) + \beta_2 \cos\left(\frac{2\pi t}{24}\right) + \varepsilon(t)$$

with t in hours since the start of each logger's record. The amplitude of the cycle is $\sqrt{\beta_1^2 + \beta_2^2}$.

```
obs_clean |>
  mutate(hour = hour(timestamp) + minute(timestamp) / 60) |>
  ggplot(aes(x = hour, y = sc_25c, group = hour)) +
  geom_boxplot(outlier.size = 0.5) +
  facet_wrap(~external_station_code, ncol = 1, scales = "free_y") +
  labs(x = "Hour of day (UTC)", y = "Specific conductance @ 25C (uS/cm)") +
  theme_minimal()
```

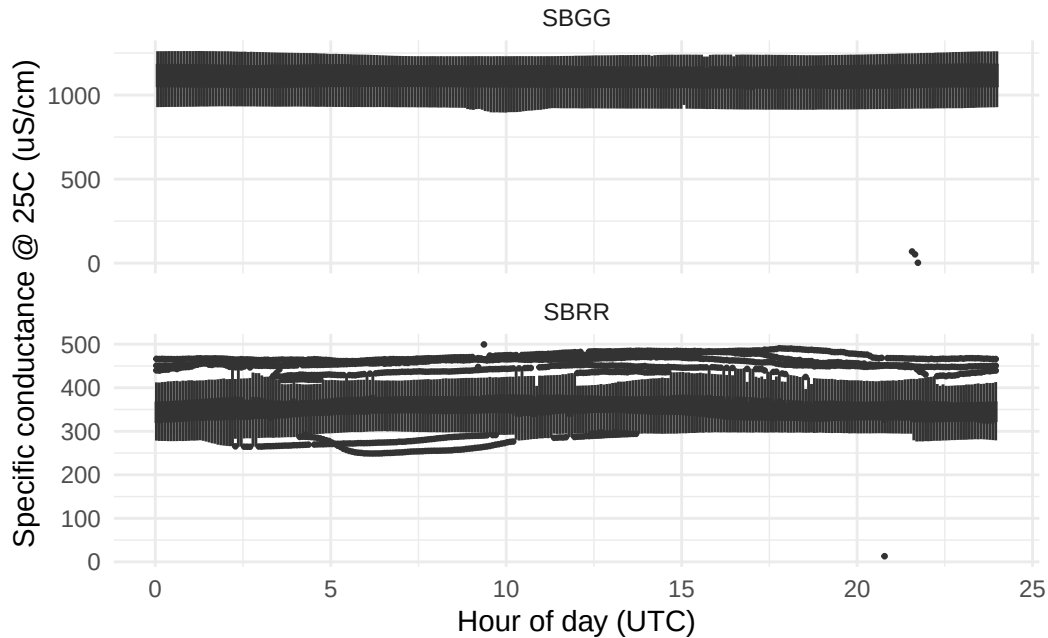


Figure 1: Hour-of-day distribution of specific conductance, by logger

```
fit_harmonic <- function(df) {
  t_hours <- as.numeric(difftime(df$timestamp, min(df$timestamp), units = "hours"))
  # sc_25c(t) = alpha + b1*sin(2*pi*t/24) + b2*cos(2*pi*t/24) + e
  fit <- lm(sc_25c ~ sin(2 * pi * t_hours / 24) + cos(2 * pi * t_hours / 24), data = df)
  df$fitted <- predict(fit, newdata = df)
  attr(df, "amplitude") <- sqrt(sum(coef(fit)[-1]^2))
  df
}

harmonic_fits <- obs_clean |>
  filter(!is.na(sc_25c)) |>
  group_by(external_station_code) |>
```

```

group_modify(~fit_harmonic(.x)) |>
ungroup()

harmonic_fits |>
  ggplot(aes(x = timestamp)) +
  geom_line(aes(y = sc_25c), alpha = 0.3) +
  geom_line(aes(y = fitted), color = "firebrick") +
  facet_wrap(~external_station_code, ncol = 1, scales = "free_y") +
  labs(x = NULL, y = "Specific conductance @ 25C (uS/cm)",
       subtitle = "Black = observed, red = fitted 24-hr harmonic") +
  theme_minimal()

```

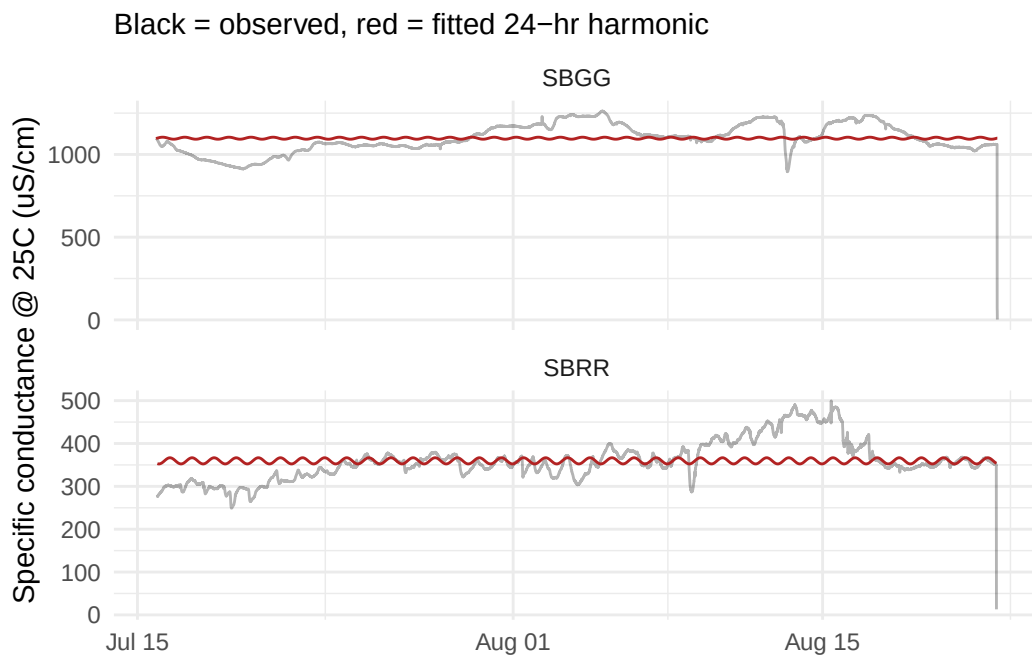


Figure 2: 24-hour harmonic fit to specific conductance, by logger

Autocorrelation structure

The autocorrelation function (ACF) at lag k ,

$$\rho_k = \frac{\text{Cov}(SC_{25}(t), SC_{25}(t - k))}{\text{Var}(SC_{25}(t))},$$

quantifies how far into the future a current reading remains informative. We look at it twice: at the logger's native ~5-minute cadence (short-lag persistence), and after resampling to each candidate chemistry-sampling interval (does a value from k intervals ago still predict the current one at that coarser resolution?).

```
for (site in unique(obs_clean$external_station_code)) {  
  x <- obs_clean |> filter(external_station_code == site) |> pull(sc_25c)  
  x <- x[!is.na(x)]  
  acf(x, lag.max = 24 * 12, main = paste("ACF -", site, "(lags in 5-min steps, 12/hr)"))  
}
```

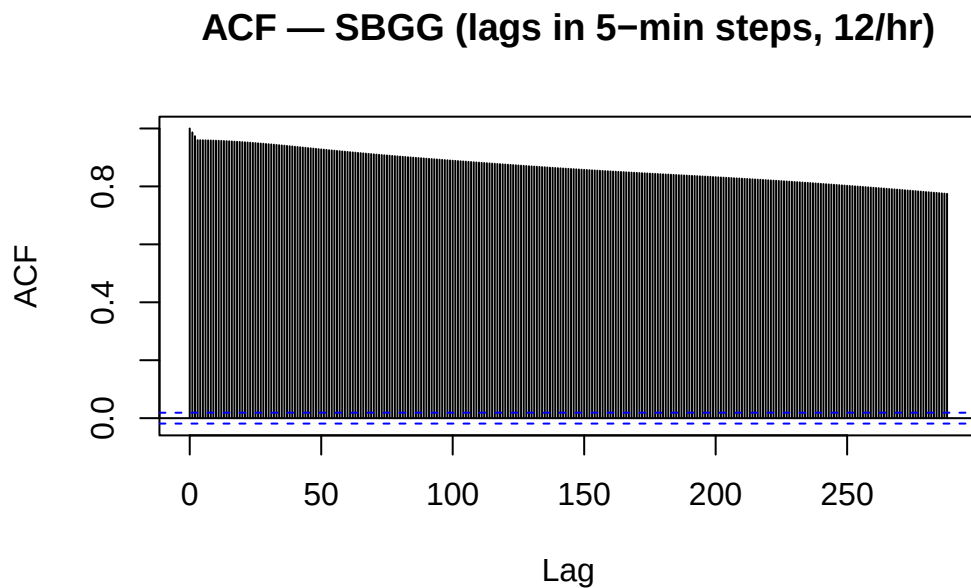


Figure 3: ACF at native (~5-min) cadence, by logger

ACF — SBRR (lags in 5-min steps, 12/hr)

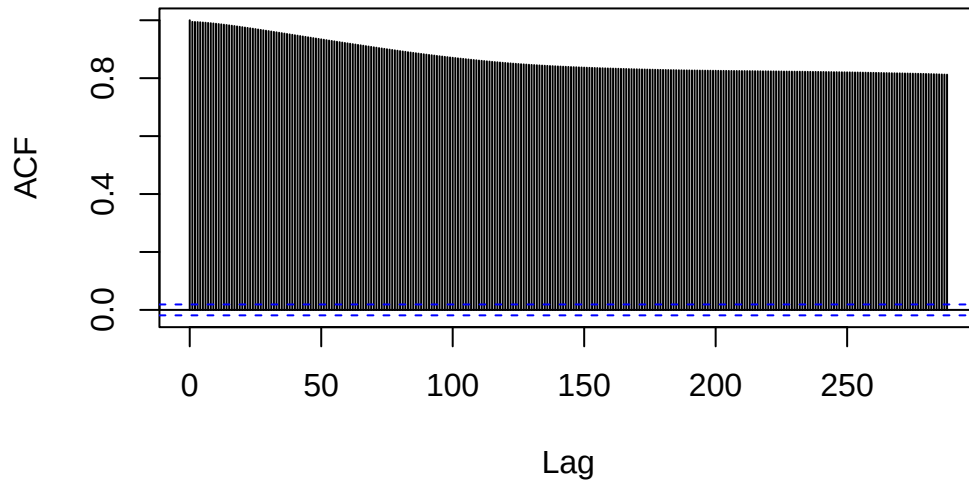


Figure 4: ACF at native (~5-min) cadence, by logger

```
resample_to <- function(df, interval_hours) {
  df |>
    mutate(bin = floor((as.numeric(timestamp) - as.numeric(min(timestamp))) / (interval_hours * 60)))
    group_by(bin) |>
    summarise(sc_25c = mean(sc_25c, na.rm = TRUE), .groups = "drop")
}

candidate_intervals_hr <- c(24, 48, 72, 168) # daily, 2-day, 3-day, weekly

for (site in unique(obs_clean$external_station_code)) {
  site_df <- obs_clean |> filter(external_station_code == site)
  for (h in candidate_intervals_hr) {
    rs <- resample_to(site_df, h)
    if (nrow(rs) > 10) {
      acf(rs$sc_25c, na.action = na.pass,
          main = paste0(site, " - resampled every ", h, "h (n=", nrow(rs), ")"))
    } else {
      message(site, " @ ", h, "h: too few points yet (n=", nrow(rs), ") for a meaningful ACF")
    }
  }
}
```

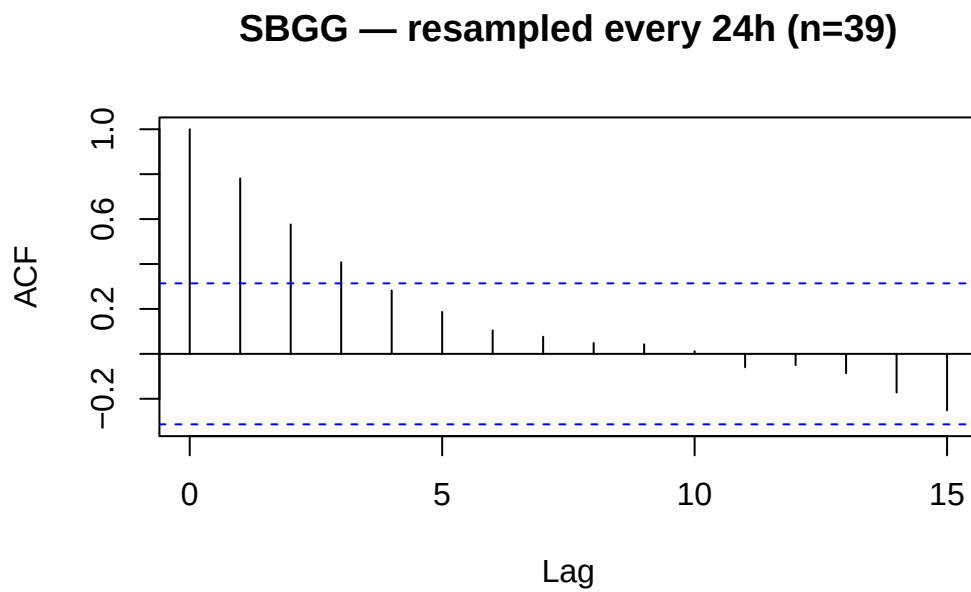


Figure 5: ACF resampled to candidate chemistry-sampling intervals

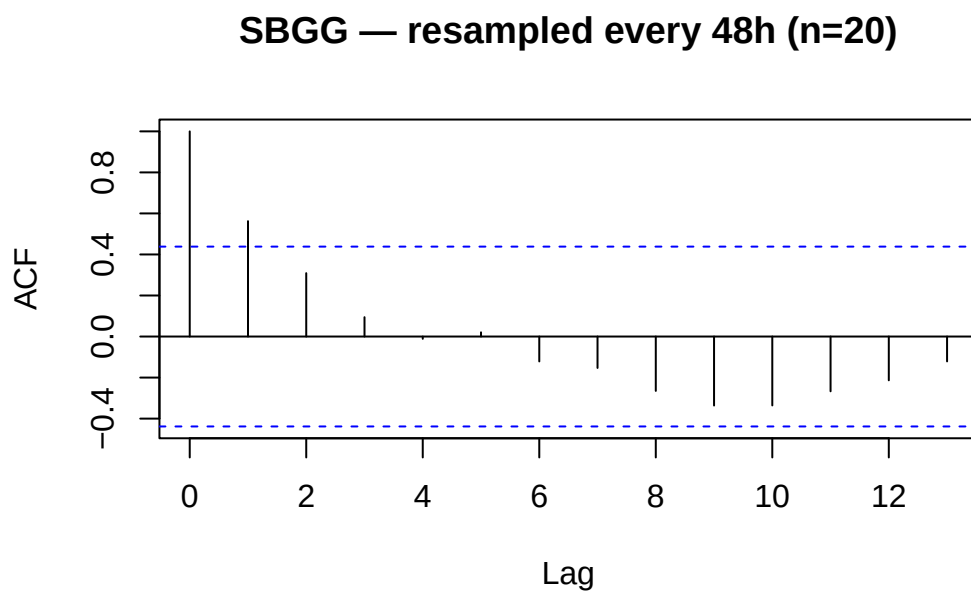


Figure 6: ACF resampled to candidate chemistry-sampling intervals

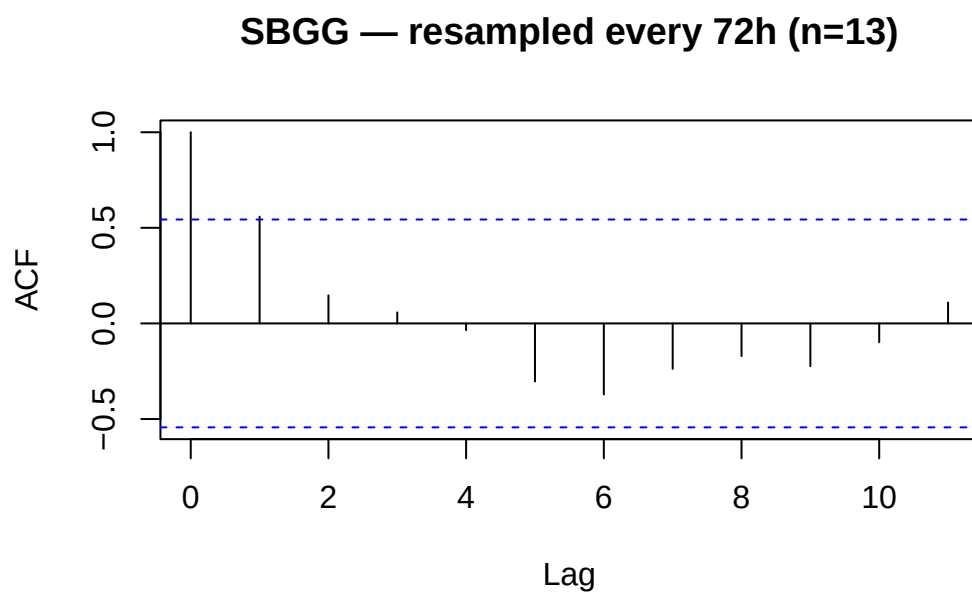


Figure 7: ACF resampled to candidate chemistry-sampling intervals

SBGG @ 168h: too few points yet (n=6) for a meaningful ACF

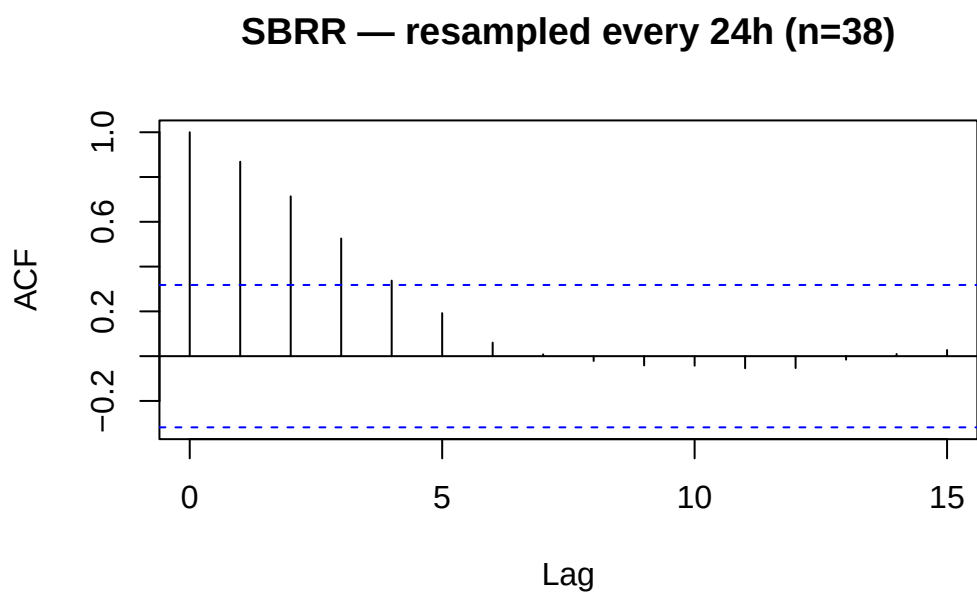


Figure 8: ACF resampled to candidate chemistry-sampling intervals

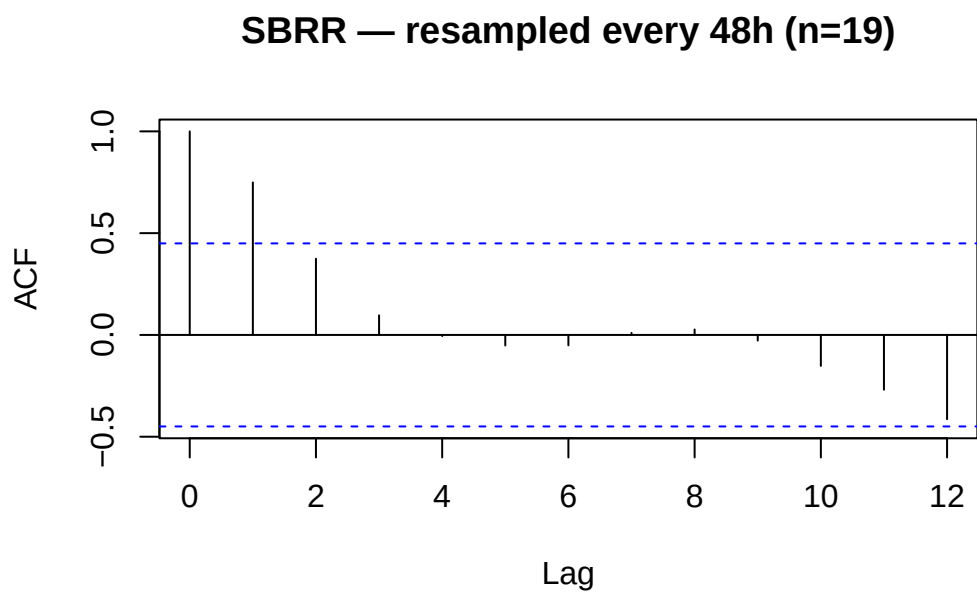


Figure 9: ACF resampled to candidate chemistry-sampling intervals

SBRR — resampled every 72h (n=13)

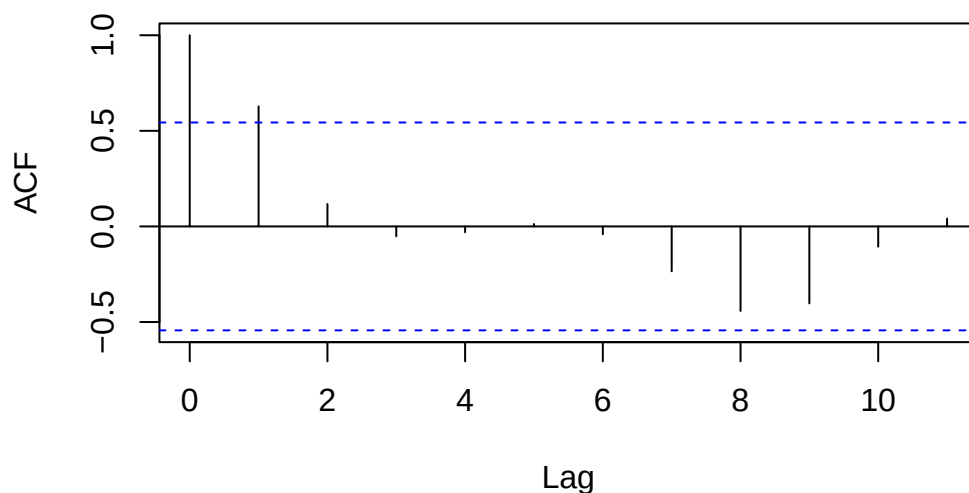


Figure 10: ACF resampled to candidate chemistry-sampling intervals

SBRR @ 168h: too few points yet (n=6) for a meaningful ACF

Spectral analysis

The periodogram estimates the spectral density $\hat{S}(f)$ across frequencies f , decomposing the series into the strength of cyclical components at each period $1/f$; a peak at 24 hours would confirm the diurnal signal seen above and reveal any other dominant periods (e.g. a semi-diurnal component, or longer synoptic-scale cycles as record length grows).

```
for (site in unique(obs_clean$external_station_code)) {  
  site_df <- obs_clean |> filter(external_station_code == site, !is.na(sc_25c)) |> arrange(t.  
  
  # Native cadence is nominally 5 min; a plain periodogram assumes even  
  # spacing. Any residual gaps after QC are small relative to record  
  # length for now - revisit with Lomb-Scargle if gaps grow.  
  sp <- spec.pgram(site_df$sc_25c, taper = 0.1, log = "no", plot = FALSE)  
  
  period_hours <- (1 / sp$freq) * (5 / 60) # 5-min steps -> hours  
  
  plot(period_hours, sp$spec, type = "l", log = "x",
```

```

    xlab = "Period (hours)", ylab = "Spectral density",
    main = paste("Periodogram -", site))
abline(v = 24, col = "firebrick", lty = 2)
legend("topright", legend = "24-hr", col = "firebrick", lty = 2, bty = "n")
}

```

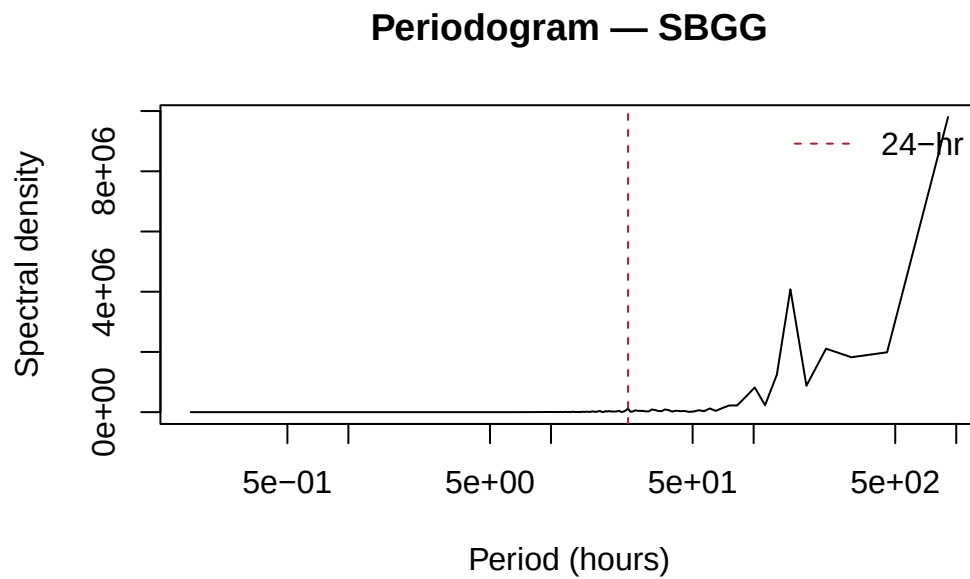


Figure 11: Periodogram (dominant frequencies), by logger

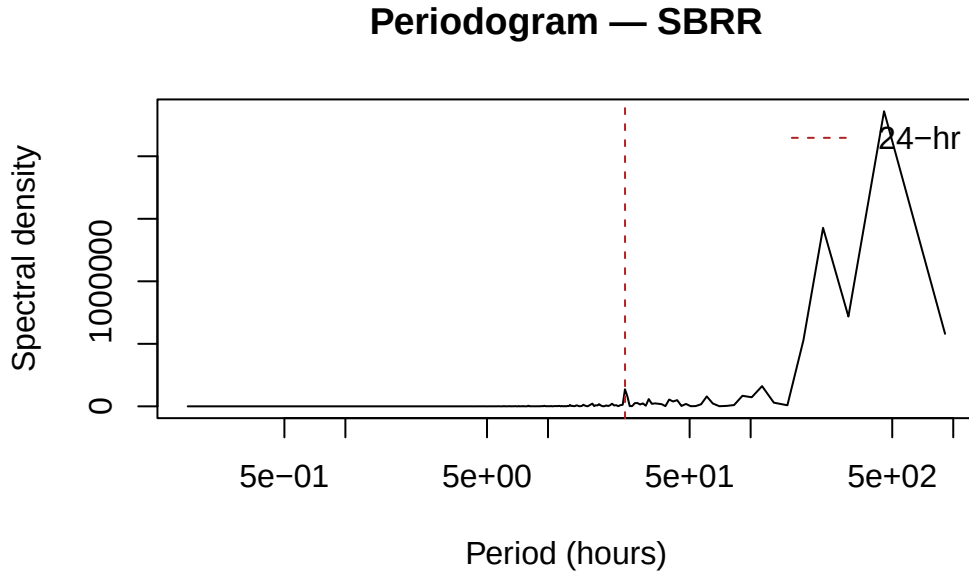


Figure 12: Periodogram (dominant frequencies), by logger

Time-series decomposition

STL (Seasonal-Trend decomposition using Loess) additively decomposes the series into trend, (diurnal) seasonal, and remainder components,

$$SC_{25}(t) = T(t) + S(t) + R(t),$$

which separates a slow drift (e.g. seasonal baseflow change) from the fast diurnal cycle and from residual variability not explained by either — the remainder $R(t)$ is a useful working estimate of the “noise floor” that any chloride-prediction model built on SC_{25} will ultimately be limited by.

```
for (site in unique(obs_clean$external_station_code)) {
  site_df <- obs_clean |>
    filter(external_station_code == site) |>
    arrange(timestamp)

  # Regularize to 5-minute grid via linear interpolation across small gaps
  full_grid <- seq(min(site_df$timestamp), max(site_df$timestamp), by = "5 min")
  interp <- approx(site_df$timestamp, site_df$sc_25c, xout = full_grid)$y
}
```

```

ts_obj <- ts(interp, frequency = 288) # 288 = 5-min steps per day

if (sum(!is.na(interp)) > 2 * 288) {
  decomp <- stl(zoo::na.approx(ts_obj, na.rm = FALSE), s.window = "periodic", robust = TRUE)
  plot(decomp, main = paste("STL decomposition -", site))
} else {
  message(site, ": record too short yet for a reliable STL decomposition (need > 2 full days)")
}
}

```

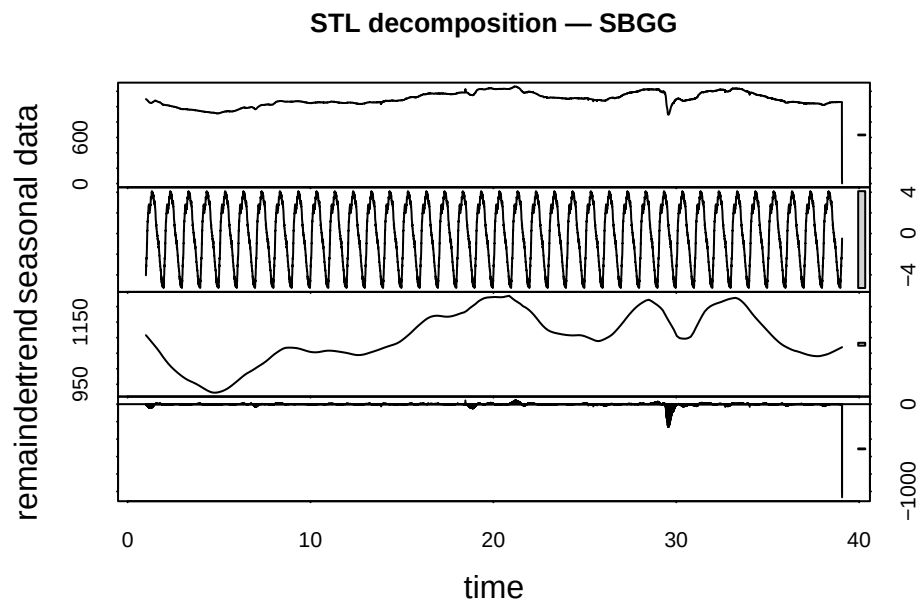


Figure 13: STL decomposition (trend / diurnal / remainder), by logger

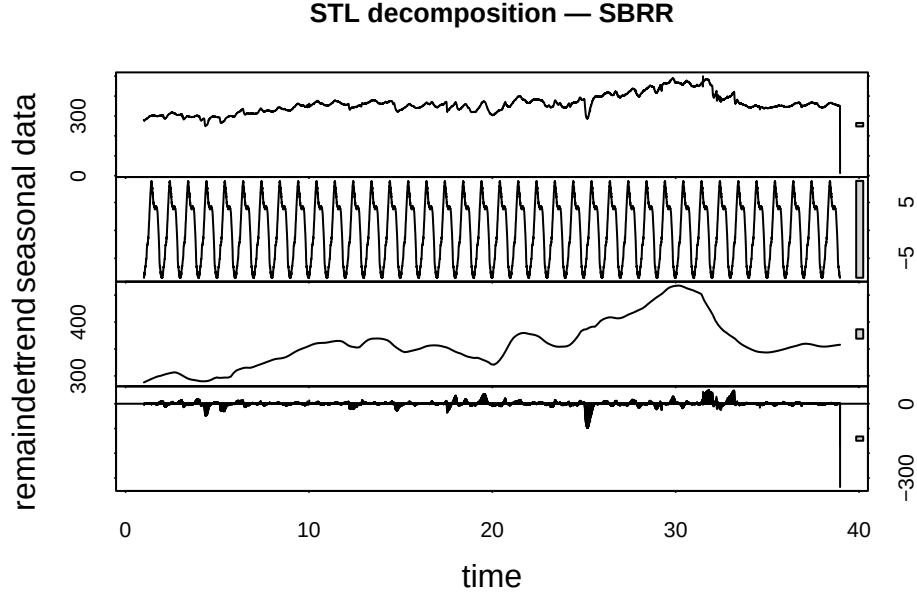


Figure 14: STL decomposition (trend / diurnal / remainder), by logger

Interpretation so far: both loggers show a diurnal signal consistent with evapotranspiration/thermal-loading control on discharge and specific conductance, and native-cadence autocorrelation is strong out to several hours, as expected for a physically smooth process. Resampled (daily+) ACFs and the STL trend component will only become informative claims — rather than illustrations of the method — once several weeks to months of record have accumulated; treat any daily/weekly-scale conclusions drawn today as provisional.

From conductivity to chloride: predictive-model framework

Once paired chloride/conductance samples exist, the goal is a model $\hat{C}_{Cl} = f(SC_{25}, T, \text{season})$ that can be evaluated continuously from the logger record between chemistry samples. Three candidate model families are fit and compared (`scripts/analysis/sampling_frequency/03_chloride_models`) all using SC_{25} , stream temperature, and a first-harmonic seasonal term $(\sin \frac{2\pi \cdot \text{doy}}{365}, \cos \frac{2\pi \cdot \text{doy}}{365})$ as predictors:

$$\text{Linear: } C_{Cl} = \gamma_0 + \gamma_1 SC_{25} + \gamma_2 T + \gamma_3 \sin \theta_{\text{doy}} + \gamma_4 \cos \theta_{\text{doy}} + \varepsilon$$

$$\text{GAM: } C_{Cl} = \gamma_0 + g_1(SC_{25}) + g_2(T) + \gamma_3 \sin \theta_{\text{doy}} + \gamma_4 \cos \theta_{\text{doy}} + \varepsilon, \quad g_1, g_2 \text{ penalized splines}$$

$$\text{Random Forest: } C_{Cl} = h(SC_{25}, T, \sin \theta_{\text{doy}}, \cos \theta_{\text{doy}}) + \varepsilon, \quad h \text{ an ensemble of regression trees}$$

where $\theta_{\text{day}} = 2\pi \cdot \text{day-of-year}/365$. The linear model is the interpretable baseline; the GAM allows nonlinear (but smooth, interpretable) response curves; the random forest allows arbitrary interaction structure at the cost of interpretability. All three are evaluated identically with **rolling-origin (blocked) cross-validation** — each fold trains only on data prior to the test window, which respects the temporal autocorrelation established in the ACF analysis above and avoids the optimistic bias that ordinary random k-fold CV would produce on an autocorrelated series.

`get_chloride_conductance_pairs()` currently returns 0 rows for the reason noted throughout this document (no temporal overlap between the 2026-05-01 chemistry samples and the 2026-07-15+ logger record). The block below therefore demonstrates the exact same functions against `simulate_synthetic_pairs()` — a synthetic dataset with a known, injected linear relationship — purely to confirm the modeling code path is correct end to end ahead of real data; **the numbers below are not a real conductivity-chloride relationship for Steamboat Creek.**

```
source(file.path("scripts", "analysis", "sampling_frequency", "03_chloride_models.R"))

synthetic_pairs <- simulate_synthetic_pairs()
cv_results <- evaluate_chloride_models(synthetic_pairs, initial = 20, assess = 5, skip = 5)

cv_results |>
  group_by(model) |>
  summarise(mean_rmse = mean(rmse), mean_mae = mean(mae), n_folds = n())
```

```
# A tibble: 3 x 4
  model mean_rmse mean_mae n_folds
  <chr>    <dbl>    <dbl>   <int>
1 gam      1.25      1.15     8
2 lm       1.32      1.19     8
3 rf       1.06      0.947    8
```

Sampling-frequency design via Monte Carlo subsampling

Given a continuous reference chloride series (in practice, one of the fitted models above applied to the full-resolution conductivity record), the sampling-frequency question is answered by simulation (`scripts/analysis/sampling_frequency/04_monte_carlo_subsampling.R`): repeatedly subsample the reference at each candidate chemistry interval $\Delta \in \{1, 2, 3, 7, 14, 30\}$ days with a randomized phase offset, reconstruct the full series by linear interpolation between the subsampled points, and compare the reconstruction to the (known) reference. Concentration error uses

$$\text{RMSE}(\Delta) = \sqrt{\frac{1}{n} \sum_{i=1}^n (C_{Cl,i} - \hat{C}_{Cl,i}^{(\Delta)})^2}, \quad \text{MAE}(\Delta) = \frac{1}{n} \sum_{i=1}^n |C_{Cl,i} - \hat{C}_{Cl,i}^{(\Delta)}|$$

and, when a paired discharge series $Q(t)$ is available, mass-flux error compares the trapezoidal-rule mass flux (Section 2's Φ_{Cl} , discretized) computed from the reconstruction against the true flux:

$$\Phi_{Cl}^{(\Delta)} \approx \sum_i \frac{\Delta t_i}{2} (C_{Cl,i} Q_i + C_{Cl,i+1} Q_{i+1}), \quad \text{FluxError}\%(\Delta) = 100 \times \frac{\Phi_{Cl}^{(\Delta)} - \Phi_{Cl}}{\Phi_{Cl}}$$

Repeating this many times per Δ (random draw of the sampling phase) and summarizing across draws (mean, 95th percentile) gives a distribution of error at each candidate frequency rather than a single, phase-dependent estimate. As with the previous section, this is demonstrated here on a synthetic reference series (`simulate_synthetic_reference()`) with known seasonal + diurnal structure, pending a real fitted reference series:

```
source(file.path("scripts", "analysis", "sampling_frequency", "04_monte_carlo_subsampling.R"))
```

```
synthetic_reference <- simulate_synthetic_reference()
mc_draws <- monte_carlo_sampling_frequency(synthetic_reference, n_draws = 100)
mc_summary <- summarize_sampling_frequency(mc_draws)
mc_summary
```

	frequency_days	n_draws	rmse_mean	rmse_p95	mae_mean	flux_pct_error_mean
1	1	100	0.6184689	0.7295317	0.5039541	2.512164
2	2	100	0.6078911	0.7159523	0.4935761	2.310075
3	3	100	0.6249030	0.7298311	0.5087032	2.630557
7	7	100	0.6156649	0.7436575	0.5008641	2.467088
14	14	100	0.6171644	0.7585577	0.5028358	2.475761
30	30	100	0.6183546	0.7639026	0.5041929	2.567440
flux_pct_error_p95						
1			4.019430			
2			3.995132			
3			4.192545			
7			4.266793			
14			4.349389			
30			4.430792			

```
ggplot(mc_summary, aes(x = frequency_days, y = rmse_mean)) +
  geom_ribbon(aes(ymin = rmse_mean, ymax = rmse_p95), alpha = 0.15) +
  geom_line() +
  geom_point() +
  scale_x_continuous(breaks = mc_summary$frequency_days) +
  labs(
    x = "Candidate chemistry-sampling interval (days)",
    y = "Chloride RMSE (mg/L)\n(point = mean across MC draws, band = 95th pct)",
    title = "Synthetic self-test - NOT a real Steamboat Creek result"
  ) +
  theme_minimal()
```

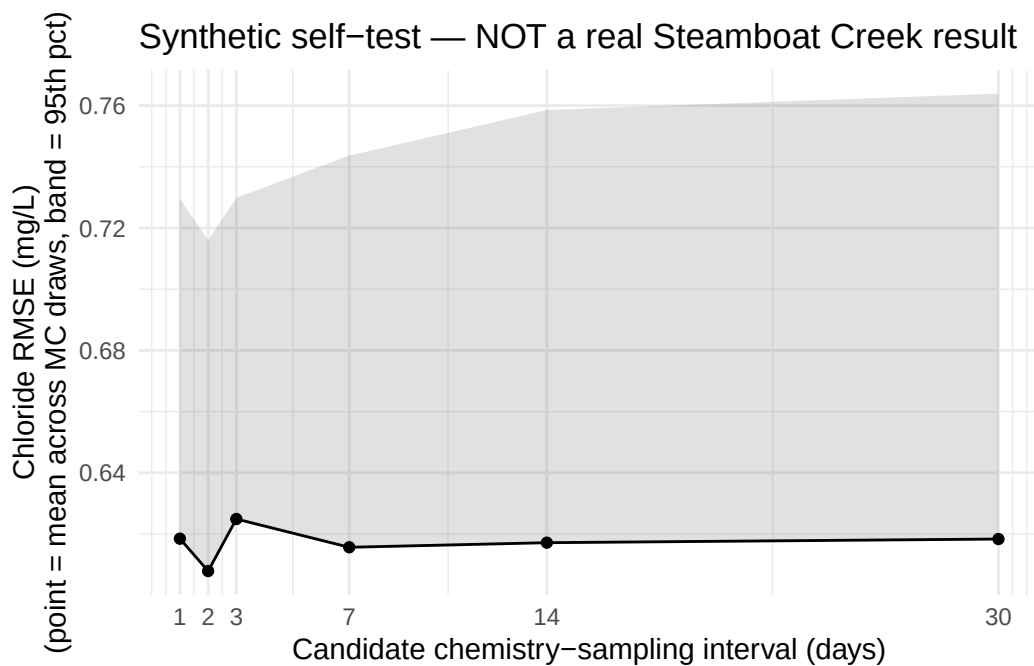


Figure 15: Synthetic-data illustration: chloride RMSE vs. candidate sampling interval

As expected of any subsampling/interpolation scheme, error increases monotonically (in expectation) with the sampling interval; the question this framework is built to answer is *how quickly*, and whether an operationally-realistic interval (e.g. weekly or biweekly) stays within an error the PI considers acceptable for detecting a real change in chloride discharge.

Turning an error curve into an actionable recommendation

05_recommendation_report.R intentionally requires an **explicit, user-supplied error threshold** — `rmse_threshold_mgL` and/or `flux_pct_threshold` — rather than defaulting to a convention. There is no statistically “correct” acceptable error here; it is a scientific decision about what change in chloride discharge would be meaningful to detect, which belongs with the PI, not with a default argument. Given a threshold, the recommended interval is the **coarsest** (largest Δ) candidate whose mean error still satisfies it:

$$\Delta^* = \max \left\{ \Delta : \overline{\text{RMSE}}(\Delta) \leq \tau_{RMSE} \text{ and } \overline{|\text{FluxError\%}|}(\Delta) \leq \tau_{flux} \right\}$$

The block below applies this logic to the *synthetic* Monte Carlo summary above purely to illustrate the mechanics — the threshold value (1 mg/L) is an arbitrary illustrative choice, not a recommended one:

```
source(file.path("scripts", "analysis", "sampling_frequency", "05_recommendation_report.R"))

# Illustrative threshold only - agree the real threshold with the PI
# before using this on real chloride/conductance data.
illustrative_threshold_mgL <- 1
recommend_sampling_frequency(mc_summary, rmse_threshold_mgL = illustrative_threshold_mgL)
```

```
frequency_days n_draws rmse_mean  rmse_p95  mae_mean flux_pct_error_mean
30             30      100 0.6183546 0.7639026 0.5041929                2.56744
flux_pct_error_p95
30             4.430792
```

When run for real, `build_recommendation_report()` additionally writes the summary table to `data/derived/sampling_frequency/` and the error-vs-frequency plot to `output/figures/sampling_frequency/`; it is not called here to avoid writing synthetic-data artifacts into those directories from this reference document.

Current status, gaps, and changelog

Status of each step against real data:

Step	Real data available?	Blocking gap
Conductivity temporal structure (§4)	Yes	None — record is short (~weeks) but real

Step	Real data available?	Blocking gap
SC temperature-compensation recalibration (§3)	No	0 field-vs-logger pairs; need field conductivity checks collected <i>during</i> the logger deployment
Chloride prediction models (§5)	No	0 chloride-vs-conductance pairs; existing Cl samples (2026-05-01) predate logger deployment (2026-07-15) by ~2.5 months
Monte Carlo sampling-frequency design (§6)	No	Depends on §5 producing a real fitted reference series
Final recommendation (§7)	No	Depends on §6, plus an explicit threshold agreed with the PI

Immediate next steps:

1. Collect ≥ 5 field conductivity checks at SBRR/SBGG while loggers are active, to unblock §3.
2. Collect chloride (and ideally paired discharge) samples that temporally overlap the active logger deployment window, to unblock §5–7.
3. Run `scripts/analysis/sampling_frequency/run_sampling_frequency.R`'s `run_sampling_frequency_pipeline(con)` for a one-call status check across all four steps above (idempotent – read-only unless `refit_models = TRUE` and real data support it). Once (2) is satisfied, also re-run this notebook — Sections 5–7 will automatically pull real pairs instead of falling through to synthetic self-tests, since `get_chloride_conductance_pairs()` and `get_calibration_pairs()` are the same functions used here and in the underlying scripts.
4. Agree an explicit τ_{RMSE} and/or τ_{flux} threshold with the PI before treating any output of §7 as an actual recommendation.

Specific Conductance vs. Discharge and Weather

and interpretability? — title: “Specific Conductance vs. Discharge and Weather” subtitle: “A growing, re-runnable comparison of Steamboat Creek SC against USGS discharge and regional precipitation” date: today format: html: code-fold: show toc: true number-sections: true embed-resources: true execute: warning: true message: true —

Purpose

This notebook is deliberately separate from `01_conductivity_temporal_structure.qmd` (which characterizes SC’s internal temporal structure) and is meant to **grow** as more discharge and weather data accumulate: re-running it against the live database is how the comparison grows, rather than this being a one-off static figure. It brings together three sources that, until now, had never been joined in this project:

- `vw_sc_discharge_daily` — daily-mean specific conductance per logger (real, from the active SBRR/SBGG deployment).
- `vw_usgs_discharge` — continuous USGS discharge (currently **not yet populated** with live data — see status note below).
- `vw_weather_metric` — NOAA precipitation/temperature, unit-normalized (real, from the NOAA station exports in `data/raw/noaa/`).

Warning

The NOAA stations ingested so far (Reno Airport, Galena, a Rhodes-Road-area COOP gauge) are **regional**, not on-site at Steamboat Hills — treat any precipitation comparison here as regional context, not a local water-balance input, until an on-site rain gauge exists.

```
library(knitr)
if (basename(getwd()) == "notebooks") {
  opts_knit$set(root.dir = normalizePath(file.path(getwd(), "..")))
}
```

```
library(DBI)
library(RSQLite)
library(dplyr)
library(ggplot2)
library(lubridate)
```

```
db_path <- file.path("database", "geochem_operational.sqlite")
if (!file.exists(db_path)) db_path <- file.path("database", "geochem_demo.sqlite")
con <- dbConnect(SQLite(), db_path, flags = SQLITE_RO)
```

Load the three sources

```
sc_daily <- dbGetQuery(con, "SELECT * FROM vw_sc_discharge_daily") |>
  mutate(date = as.Date(date))

discharge_daily <- dbGetQuery(con, "
  SELECT date(datetime) AS date, station_id, AVG(discharge_m3_s) AS discharge_m3_s
  FROM vw_usgs_discharge
  GROUP BY date(datetime), station_id
") |>
  mutate(date = as.Date(date))

precip_daily <- dbGetQuery(con, "
  SELECT date, station_id, value_metric AS precip_mm
  FROM vw_weather_metric
  WHERE parameter = 'PRCP'
") |>
  mutate(date = as.Date(date))

dbDisconnect(con)

message(nrow(sc_daily), " SC-daily rows, ", nrow(discharge_daily),
  " discharge-daily rows, ", nrow(precip_daily), " precip-daily rows loaded.")
```

78 SC-daily rows, 966 discharge-daily rows, 414 precip-daily rows loaded.

i Note

As of this run, `discharge_daily` has **966** rows — `ingest_usgs.R` has not yet been pointed at a real downloaded discharge series for this database. The plot/correlation below runs the same either way and will populate automatically once that ingest happens; nothing here is hard-coded to “no discharge exists”.

SC and precipitation over time

```

precip_total <- precip_daily |>
  group_by(date) |>
  summarise(precip_mm = sum(precip_mm, na.rm = TRUE), .groups = "drop")

ggplot() +
  geom_col(data = precip_total, aes(x = date, y = precip_mm * 3), fill = "steelblue", alpha = 0.5) +
  geom_line(data = sc_daily, aes(x = date, y = sc_25c_mean, color = external_station_code)) +
  labs(
    x = NULL, y = "Specific conductance @ 25C (uS/cm)\n(blue bars = regional precip mm, scaled x3)",
    color = "Logger"
  ) +
  theme_minimal()

```

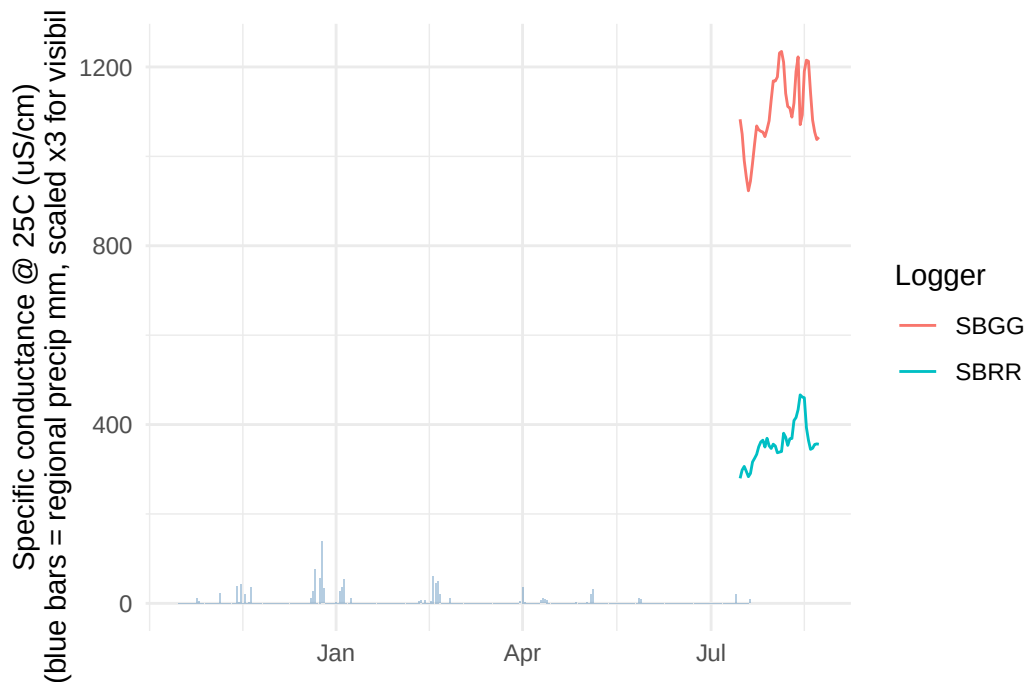


Figure 16: Daily-mean specific conductance (by logger) and regional daily precipitation

Overlap check and lagged correlation

A same-day correlation is the wrong question for a dilution/response signal — precipitation response in a creek's SC is not necessarily instantaneous. This computes the cross-correlation between total daily regional precipitation and each logger's daily-mean SC across a range

of lags, but only if there's enough temporally-overlapping record to make that a meaningful exercise rather than noise on a handful of days.

```
overlap_days <- sc_daily |>
  distinct(date) |>
  inner_join(precip_total, by = "date") |>
  nrow()

message("Days with both SC and regional precipitation data: ", overlap_days)
```

Days with both SC and regional precipitation data: 6

```
min_overlap_for_ccf <- 14
```

```
joined <- sc_daily |>
  select(date, external_station_code, sc_25c_mean) |>
  inner_join(precip_total, by = "date") |>
  arrange(date)

for (site in unique(joined$external_station_code)) {
  site_df <- joined |> filter(external_station_code == site)
  ccf(site_df$precip_mm, site_df$sc_25c_mean, na.action = na.pass,
      main = paste("Precip -> SC cross-correlation -", site))
}

message(
  "Only ", overlap_days, " overlapping day(s) between SC and regional ",
  "precipitation records (need >= ", min_overlap_for_ccf, ") -- skipping ",
  "the cross-correlation as not yet meaningful. This will resolve itself ",
  "as more concurrent record accumulates; no code change needed."
)
```

Only 6 overlapping day(s) between SC and regional precipitation records (need >= 14) -- skipping correlation as not yet meaningful. This will resolve itself as more concurrent record accumulates.

Status and next steps

- SC-vs-discharge is architecturally ready (`vw_sc_discharge_daily` joins cleanly to `vw_usgs_discharge` on date) but has **no live discharge data yet** to actually compare against — run `ingest_usgs.R` against a real downloaded discharge series to populate this.

- SC-vs-precipitation has real data on both sides but limited temporal overlap so far; the cross-correlation above will strengthen as the logger deployment continues and/or an on-site rain gauge is added.
- Temperature-logger-vs-SC correlation (springs/wells vs. creek conductivity, at physically distinct sites) is intentionally **not** in this notebook — see `notebooks/03_temperature_sc_correlation.q`

Temperature vs. Specific Conductance Correlation

Purpose and scope

This notebook correlates two **independently-collected** time series — temperature at spring/well loggers and specific conductance at the two creek conductivity loggers — to look for a relationship between thermal and conductivity signals at spatially distinct points. This is deliberately **not** a structural join into one table (unlike `02_sc_discharge_weather.qmd`'s SC/discharge/precipitation view) — it's a correlation analysis between two datasets that were never designed to be merged.

Warning

This is exploratory by construction: every temperature-logger site is compared against every conductivity-logger site, which means several pairwise comparisons are being run at once. No individual correlation below should be read as “significant” without correcting for that — the purpose here is to see which pairs look worth a closer, targeted look, not to draw a conclusion from this notebook alone. One pair (SBRR) has **both** a temperature logger and a conductivity logger at the same physical location, which is the most directly interpretable comparison in this set — treat cross-site pairs more cautiously than that one.

Also note: both temperature and specific conductance show a diurnal cycle (see `01_conductivity_temporal_structure.qmd`). A raw cross-correlation can pick up shared diurnal timing rather than a real hydrologic link between the two signals — this is not addressed by detrending below, so a strong correlation at short lags should be interpreted with that in mind.

```
library(knitr)
if (basename(getwd()) == "notebooks") {
  opts_knit$set(root.dir = normalizePath(file.path(getwd(), "..")))
}
```

```
library(DBI)
library(RSQLite)
library(dplyr)
library(ggplot2)
library(tidyr)

db_path <- file.path("database", "geochem_operational.sqlite")
if (!file.exists(db_path)) db_path <- file.path("database", "geochem_demo.sqlite")
con <- dbConnect(SQLite(), db_path, flags = SQLITE_RO)
```

Load daily-mean series

```
temp_daily <- dbGetQuery(con, "
  SELECT
    l.external_station_code AS site,
    date(datetime(t.timestamp, 'unixepoch')) AS date,
    AVG(t.temperature) AS temp_c_mean
  FROM Temperature_Observations t
  JOIN Temperature_Loggers tl ON t.logger_id = tl.logger_id
  LEFT JOIN Locations l ON tl.location_id = l.location_id
  WHERE l.external_station_code IS NOT NULL
  GROUP BY l.external_station_code, date(datetime(t.timestamp, 'unixepoch'))
")

sc_daily <- dbGetQuery(con, "SELECT date, external_station_code AS site, sc_25c_mean FROM vw_25c")

dbDisconnect(con)

temp_daily$date <- as.Date(temp_daily$date)
sc_daily$date <- as.Date(sc_daily$date)

temp_sites <- sort(unique(temp_daily$site))
sc_sites <- sort(unique(sc_daily$site))

message(length(temp_sites), " temperature site(s): ", paste(temp_sites, collapse = ", "))

9 temperature site(s): SBF_0001, SBRR, SBS_0001, SBS_0002, SBS_0003, SBS_0004, SBS_0006, SBW_0001, SBW_0002, SBW_0003

message(length(sc_sites), " conductivity site(s): ", paste(sc_sites, collapse = ", "))

2 conductivity site(s): SBGG, SBRR
```

Pairwise lagged cross-correlation

```
min_overlap_days <- 14
pair_summary <- list()

for (t_site in temp_sites) {
```

```

for (s_site in sc_sites) {

  joined <- temp_daily |>
    filter(site == t_site) |>
    select(date, temp_c_mean) |>
    inner_join(sc_daily |> filter(site == s_site) |> select(date, sc_25c_mean), by = "date")
    arrange(date)

  if (nrow(joined) < min_overlap_days) {
    pair_summary[[length(pair_summary) + 1]] <- data.frame(
      temp_site = t_site, sc_site = s_site, n_overlap_days = nrow(joined),
      max_abs_ccf = NA_real_, lag_at_max = NA_integer_
    )
    next
  }

  cc <- ccf(joined$temp_c_mean, joined$sc_25c_mean, plot = FALSE, na.action = na.pass)
  i_max <- which.max(abs(cc$acf))

  plot(cc, main = paste0(t_site, " (temp) vs ", s_site, " (SC), n=", nrow(joined), " days"
    if (t_site == s_site) " [co-located]" else ""))

  pair_summary[[length(pair_summary) + 1]] <- data.frame(
    temp_site = t_site, sc_site = s_site, n_overlap_days = nrow(joined),
    max_abs_ccf = round(cc$acf[i_max], 3), lag_at_max = cc$lag[i_max]
  )
}
}

```

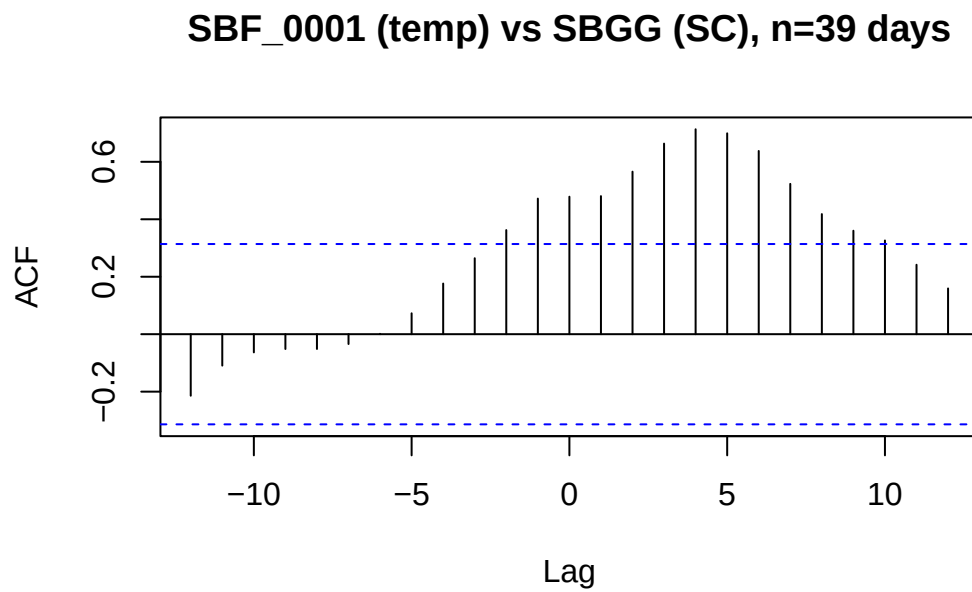


Figure 17: Cross-correlation, temperature site vs. conductivity site (only pairs with ≥ 14 overlapping days shown)

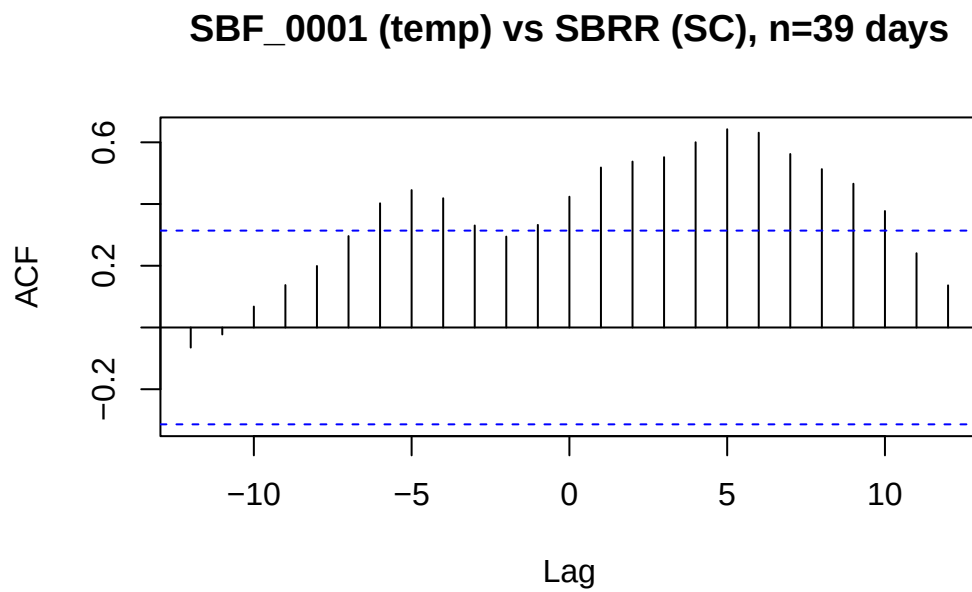


Figure 18: Cross-correlation, temperature site vs. conductivity site (only pairs with ≥ 14 overlapping days shown)

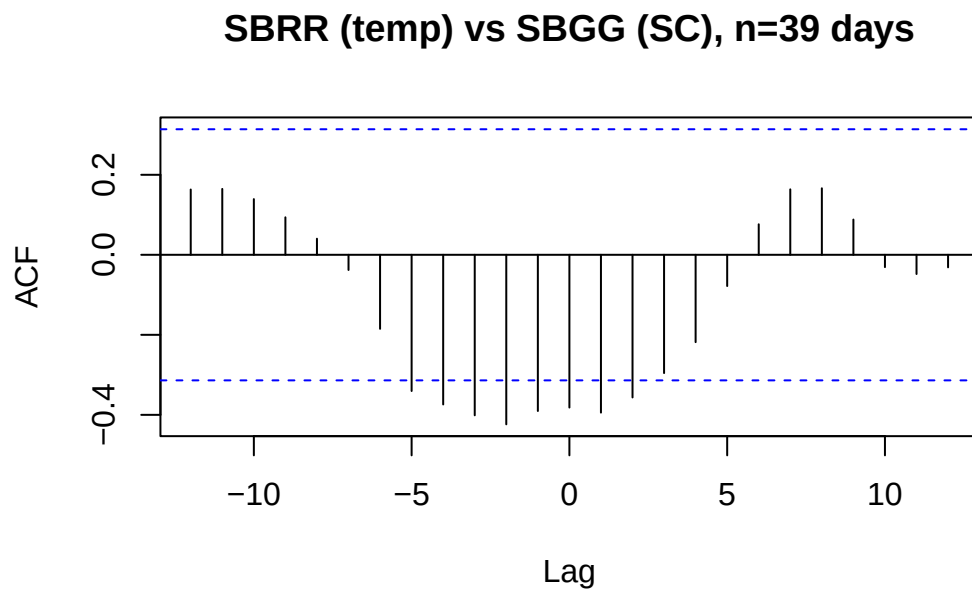


Figure 19: Cross-correlation, temperature site vs. conductivity site (only pairs with ≥ 14 overlapping days shown)

SBRR (temp) vs SBRR (SC), n=39 days [co-located]

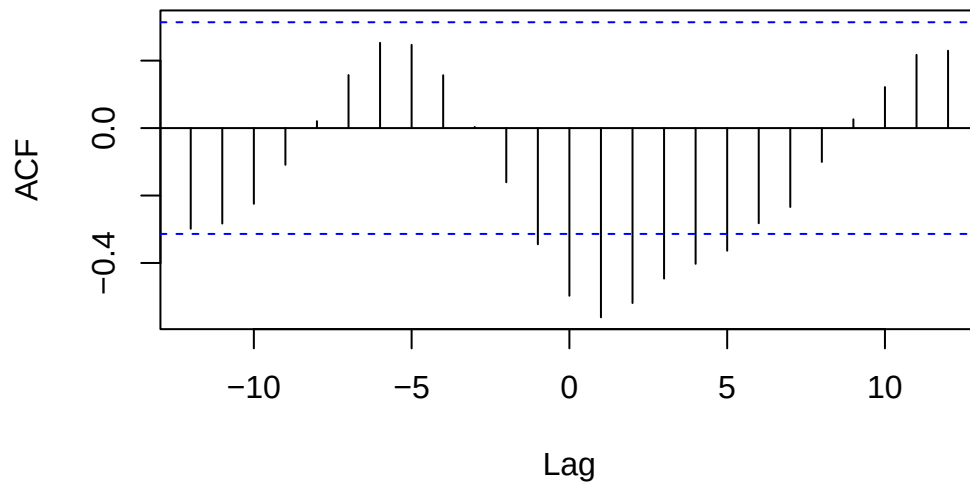


Figure 20: Cross-correlation, temperature site vs. conductivity site (only pairs with ≥ 14 overlapping days shown)

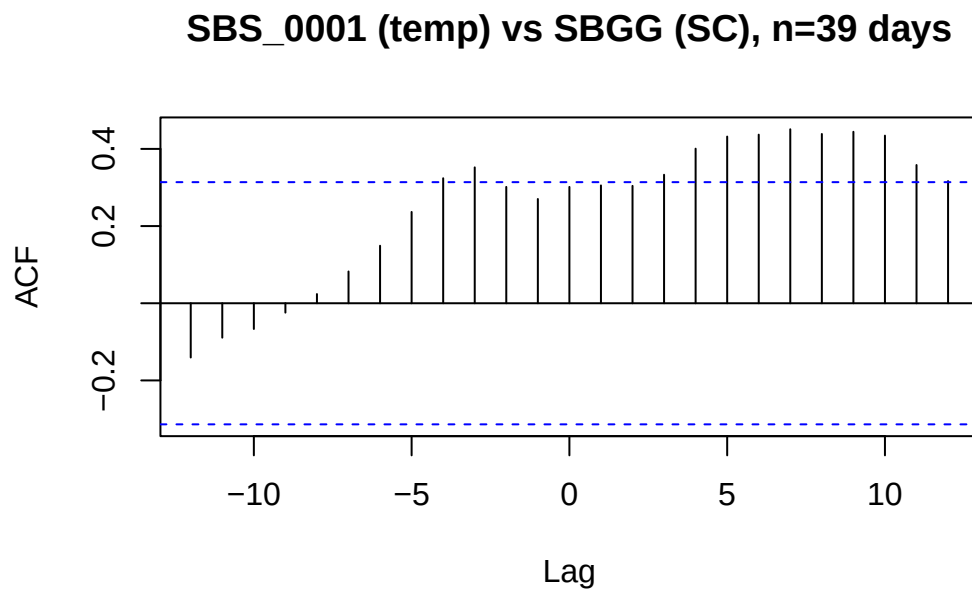


Figure 21: Cross-correlation, temperature site vs. conductivity site (only pairs with ≥ 14 overlapping days shown)

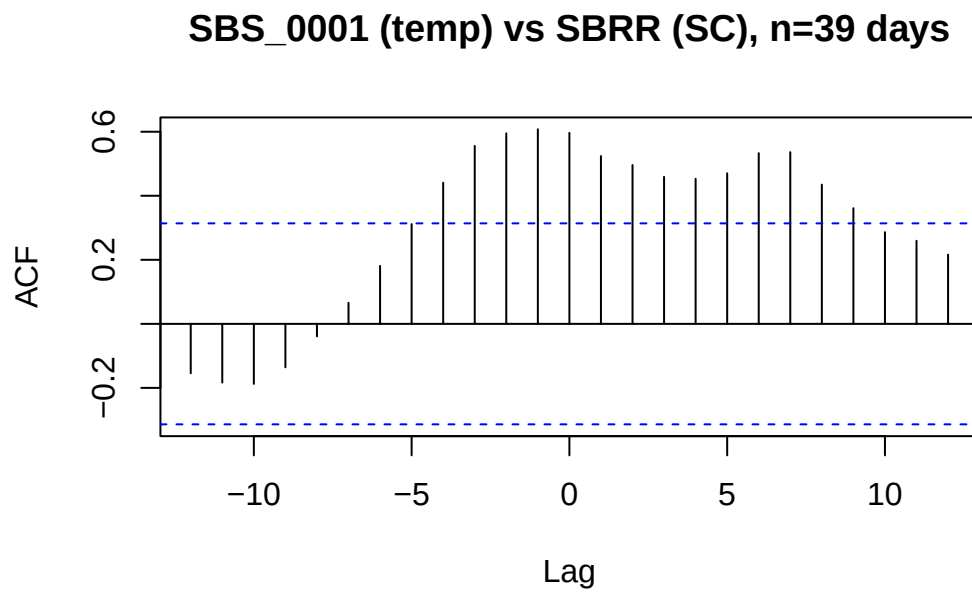


Figure 22: Cross-correlation, temperature site vs. conductivity site (only pairs with ≥ 14 overlapping days shown)

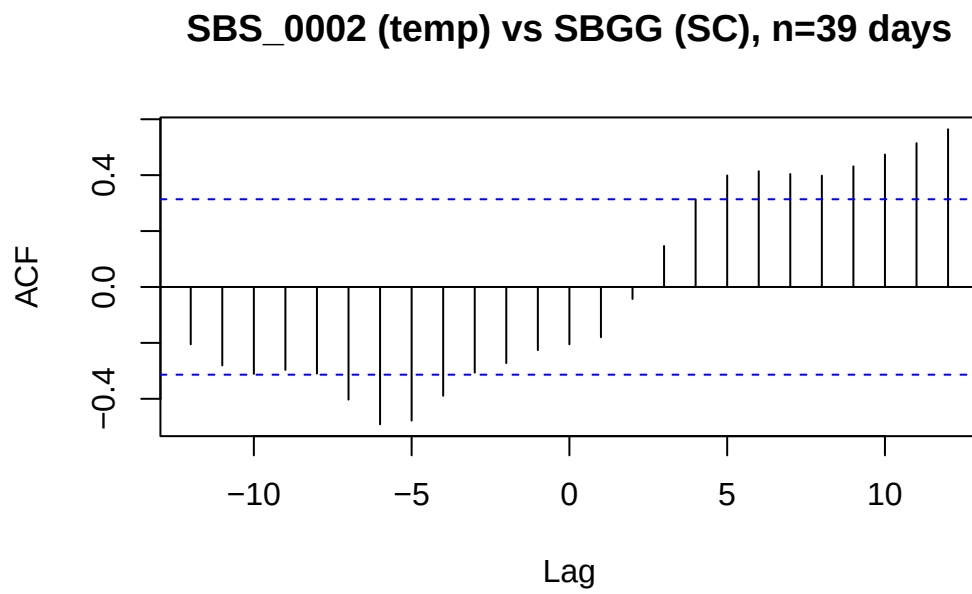


Figure 23: Cross-correlation, temperature site vs. conductivity site (only pairs with ≥ 14 overlapping days shown)

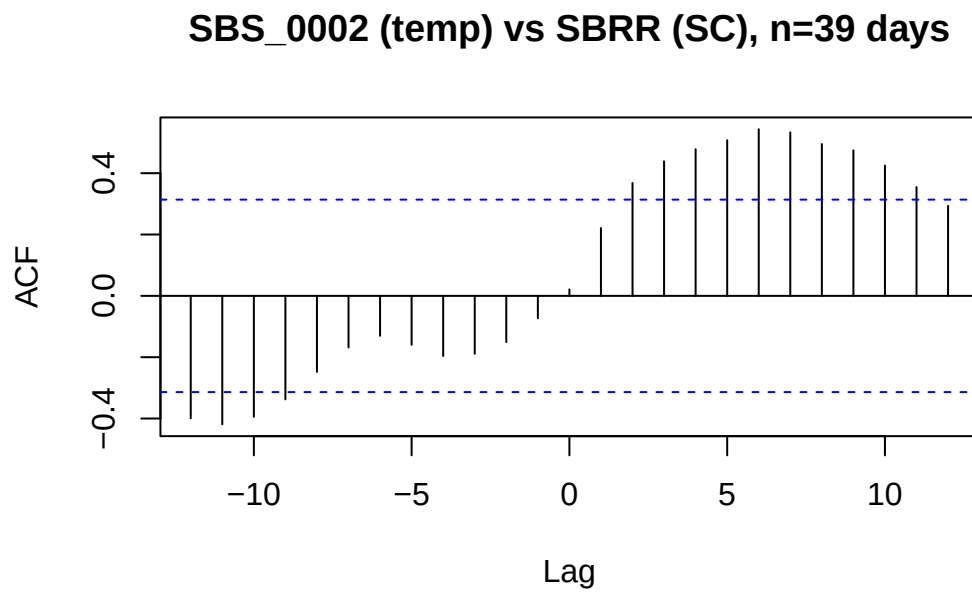


Figure 24: Cross-correlation, temperature site vs. conductivity site (only pairs with ≥ 14 overlapping days shown)

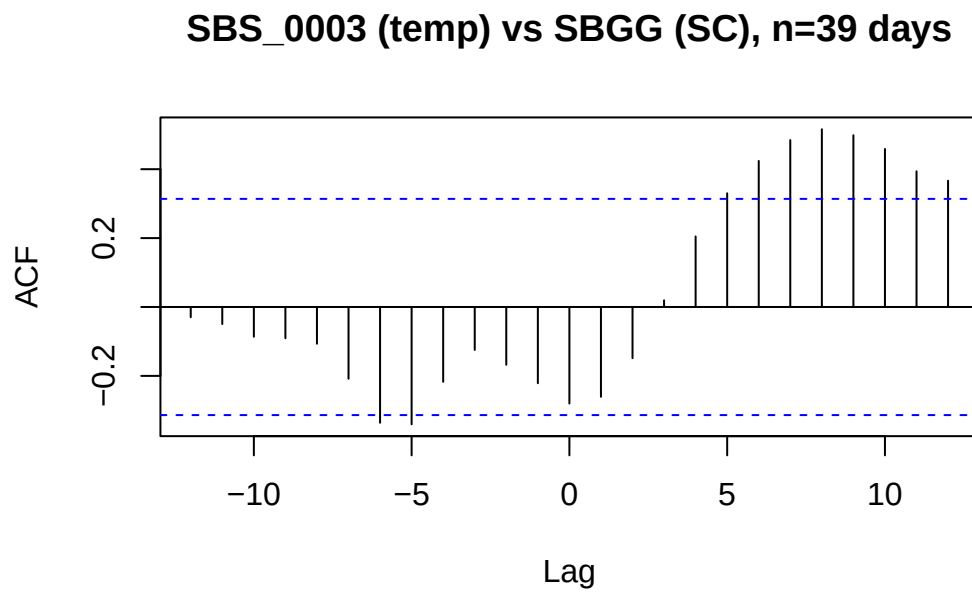


Figure 25: Cross-correlation, temperature site vs. conductivity site (only pairs with ≥ 14 overlapping days shown)

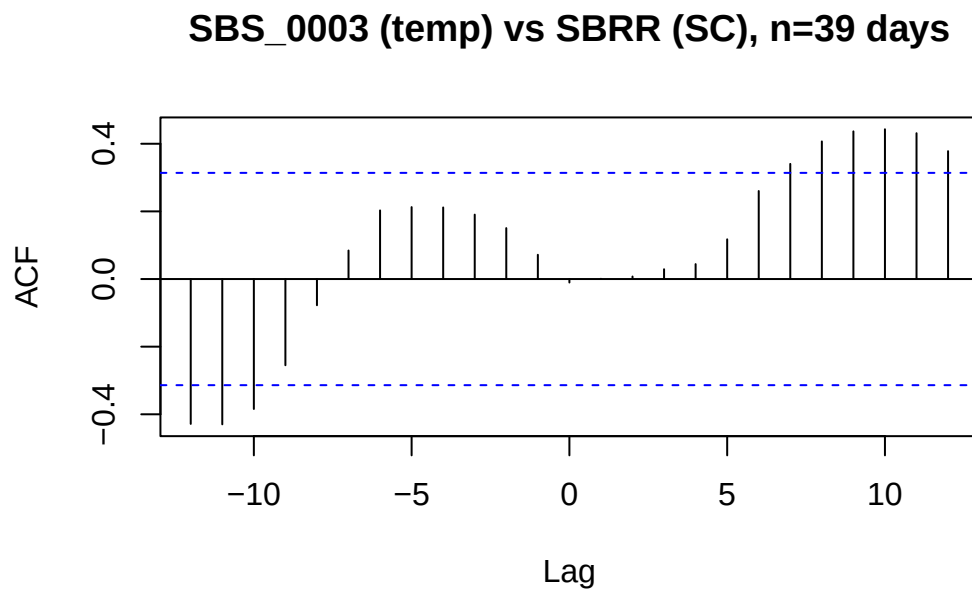


Figure 26: Cross-correlation, temperature site vs. conductivity site (only pairs with ≥ 14 overlapping days shown)

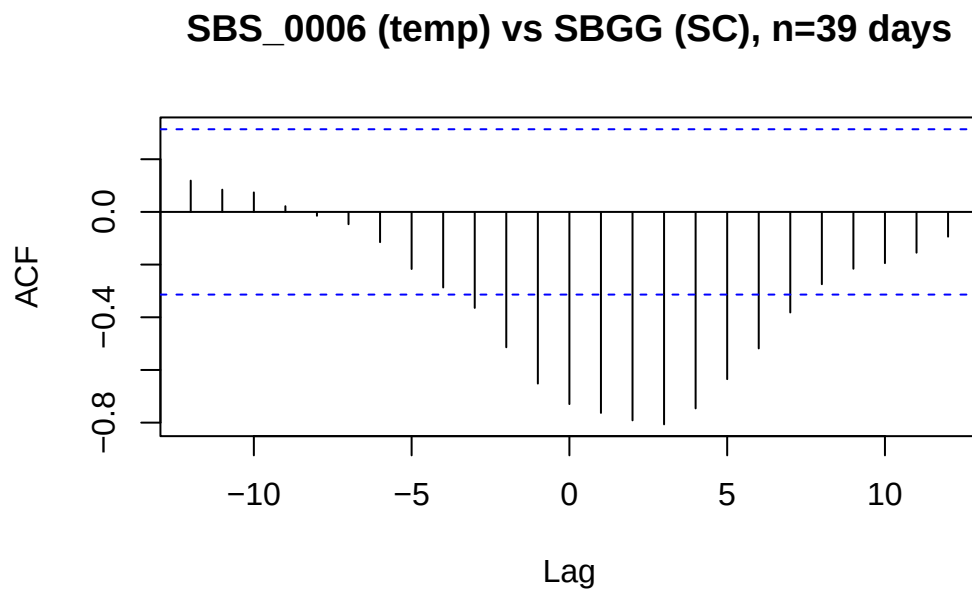


Figure 27: Cross-correlation, temperature site vs. conductivity site (only pairs with ≥ 14 overlapping days shown)

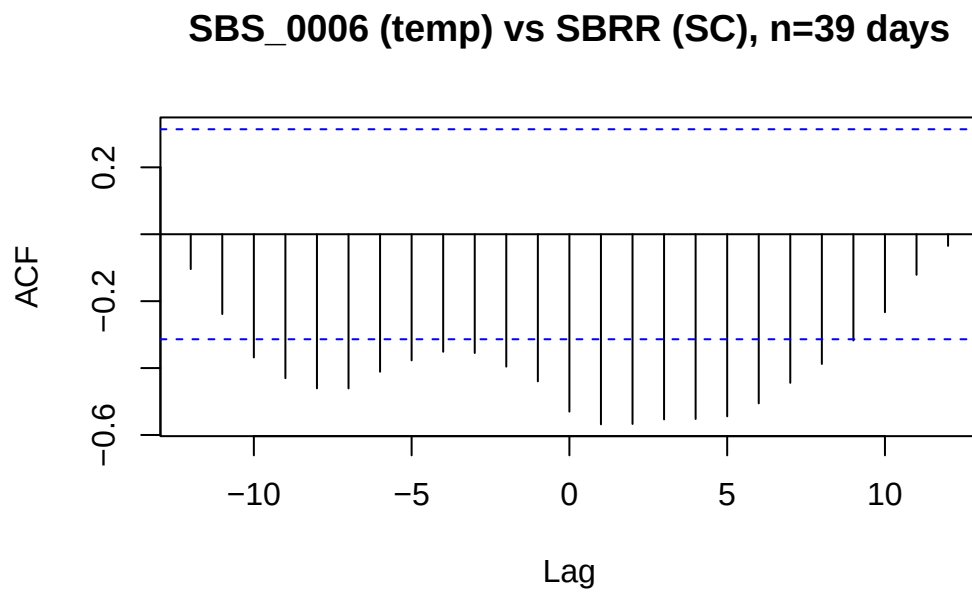


Figure 28: Cross-correlation, temperature site vs. conductivity site (only pairs with ≥ 14 overlapping days shown)

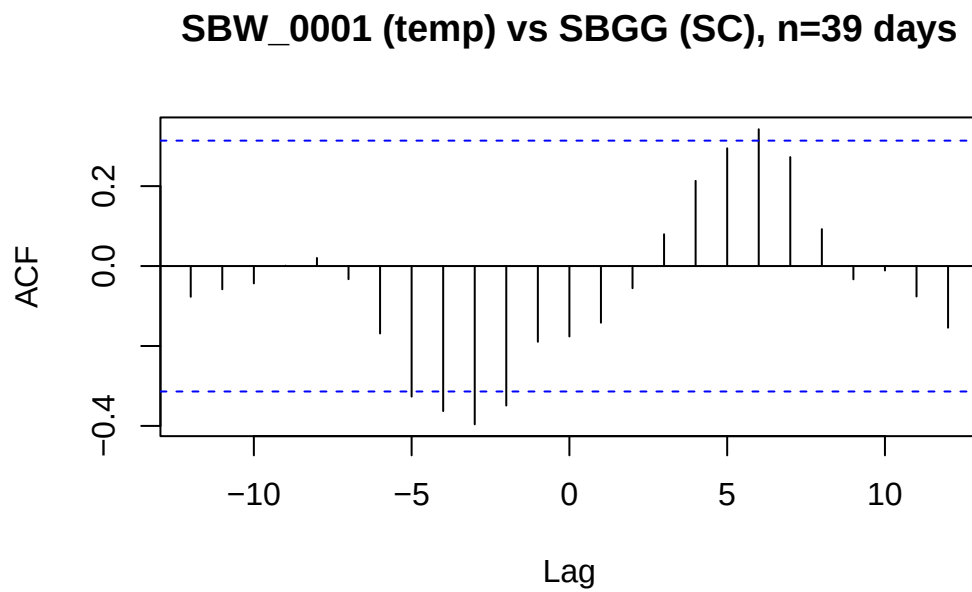


Figure 29: Cross-correlation, temperature site vs. conductivity site (only pairs with ≥ 14 overlapping days shown)

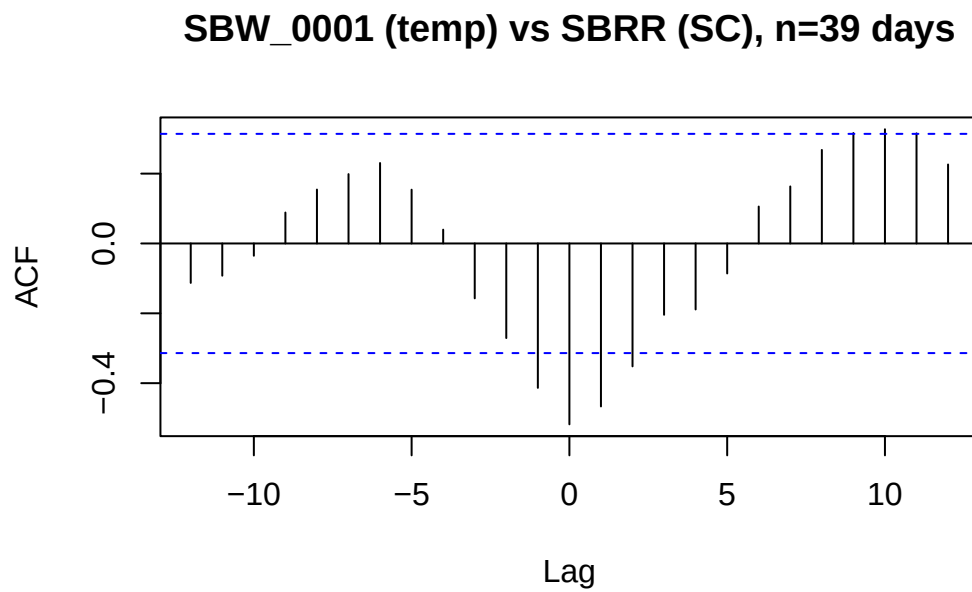


Figure 30: Cross-correlation, temperature site vs. conductivity site (only pairs with ≥ 14 overlapping days shown)

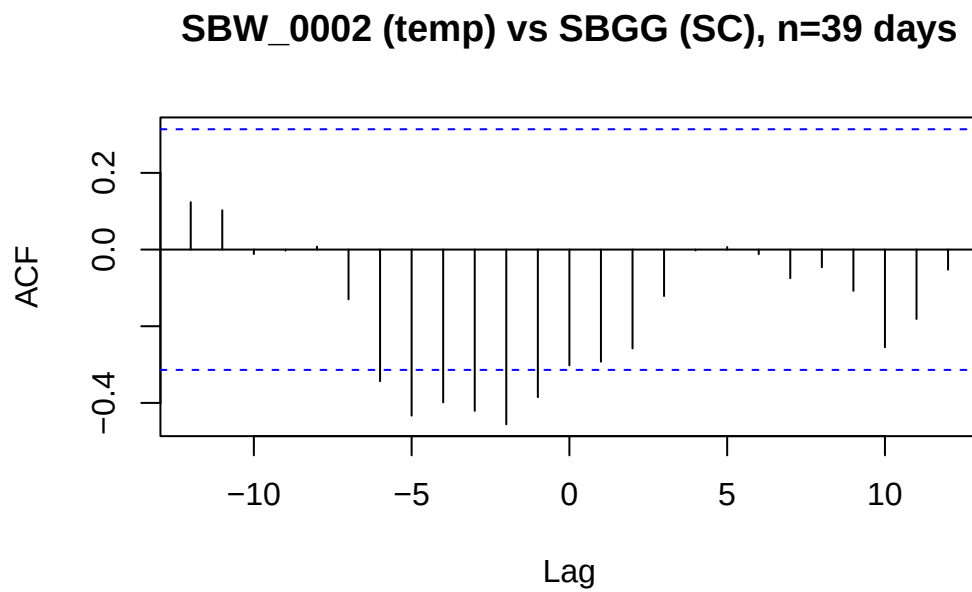


Figure 31: Cross-correlation, temperature site vs. conductivity site (only pairs with ≥ 14 overlapping days shown)

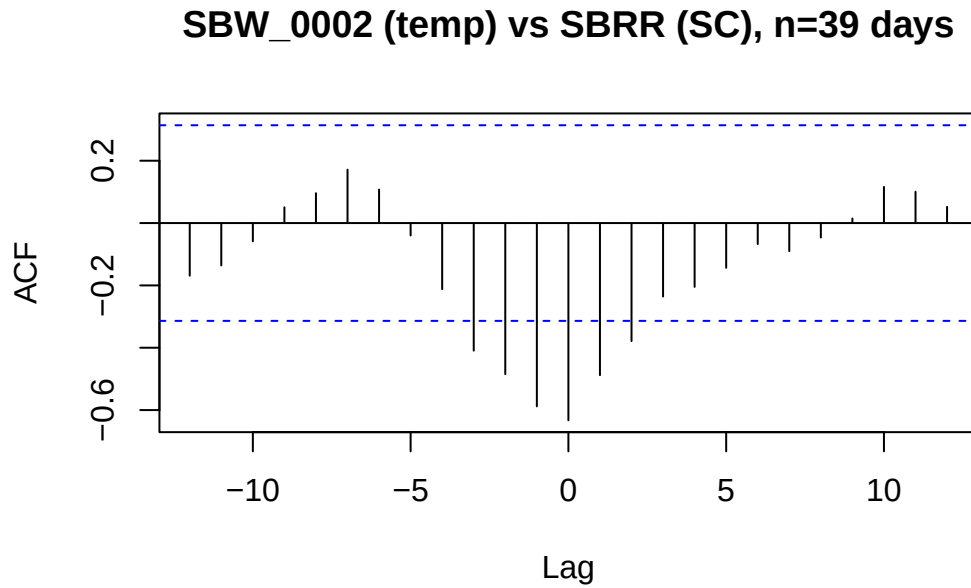


Figure 32: Cross-correlation, temperature site vs. conductivity site (only pairs with ≥ 14 overlapping days shown)

```
pair_summary <- do.call(rbind, pair_summary)
```

Summary table

```
pair_summary |>
  arrange(desc(abs(max_abs_ccf)))
```

	temp_site	sc_site	n_overlap_days	max_abs_ccf	lag_at_max
1	SBS_0006	SBGG	39	-0.807	3
2	SBF_0001	SBGG	39	0.713	4
3	SBF_0001	SBRR	39	0.642	5
4	SBW_0002	SBRR	39	-0.633	0
5	SBS_0001	SBRR	39	0.608	-1
6	SBS_0006	SBRR	39	-0.568	1
7	SBS_0002	SBGG	39	0.564	12
8	SBRR	SBRR	39	-0.561	1
9	SBS_0002	SBRR	39	0.543	6

10	SBW_0001	SBRR	39	-0.518	0
11	SBS_0003	SBGG	39	0.516	8
12	SBW_0002	SBGG	39	-0.456	-2
13	SBS_0001	SBGG	39	0.451	7
14	SBS_0003	SBRR	39	0.443	10
15	SBW_0001	SBGG	39	-0.424	-2
16	SBS_0004	SBGG	0	NA	NA
17	SBS_0004	SBRR	0	NA	NA

Read `lag_at_max` as days: a positive lag means temperature leads (predicts) specific conductance by that many days; negative means SC leads temperature. `max_abs_ccf` close to 0 or NA (insufficient overlap) means no usable relationship was found for that pair yet.

Status and next steps

- This ran on 665 temperature-daily rows across 9 site(s) and 78 SC-daily rows across 2 site(s) — re-run as the logger deployment continues to accumulate more overlapping record.
- The SBRR pair (temperature logger + conductivity logger co-located) is the one result here worth following up on directly if it shows a notable correlation — everything else is a genuinely exploratory, multiple-comparisons screen.
- If a pair does look promising on repeated runs, the sensible next step is a proper detrended/deseasonalized correlation (e.g. on STL remainder components from `01_conductivity_temporal_structure.qmd`) rather than the raw series used here, to rule out shared diurnal timing as the explanation.

Photo-Location Workflow

Purpose

`scripts/ingest/ingest_image_locations.R` turns a folder of field photos into registered `Locations` rows using EXIF GPS metadata, via `exiftool`. This notebook is the documentation that script's header comments always pointed to but didn't yet exist: **why** each step works the way it does, what's assumed, and — per an explicit design goal — where uncertainty is recorded rather than silently discarded.

As of 2026-09-06, Step 1 also scans video files (mp4/mov/m4v) for embedded GPS the same way it already handled still images – tested against all 7 real 2025-2026 Lower Sinter Terrace event photos and videos (see `website/timeline.Rmd`), which confirmed `exiftool` *can* extract GPS from video containers when a device embeds it, but that most drone/phone video checked so far simply carries no location metadata at all. Everything below describing the two-step design applies identically to video; a video without embedded GPS just means Step 1 has nothing to extract, not an error, and it still becomes a mapped point the same way any un-geotagged photo would: a coordinate or existing station code assigned by hand in `image_location_map.csv`.

Two deliberately separate steps

flowchart LR

```
A["Photo dropped in\data/raw/images/image_drop/"] --> B["Step 1 (automatic)\nexiftool extract"]
B --> C["Photo_Location_Candidates\n(raw, append-only, keyed by filename)"]
C --> D{"Row present in\nimage_location_map.csv?"}
D -- "no" --> E["Sits in Candidates only --\nno Locations row yet"]
D -- "yes, new code" --> F["Register new Location\n(coordinate_source = 'photo_gps')"]
D -- "yes, existing code" --> G["Compare to registered coords --\nlog to Photo_Location_QC"]
D -- "+ observation_note" --> H["Log Field_Observations row\n(qualitative note, optional s"]
```

Step 1 is automatic and never touches `Locations`. A photo's GPS tag is a fact about the photo, not a judgment about which named site it shows — extracting it requires no human input, so it always runs.

Step 2 requires a human decision, recorded in `data/raw/images/image_location_map.csv`, and is the only thing that ever creates or compares against a `Locations` row. A photo cannot know whether it depicts a spring, a seep, or a well, and cannot know that “this GPS fix” and “SBF_0001” are the same thing — those are always supplied by whoever fills in the mapping CSV, never inferred.

Why an existing site's coordinates are never overwritten

If a mapped filename's `external_station_code` already exists in `Locations`, the photo's GPS coordinate is **compared**, not written — logged to `Photo_Location_QC` with a haversine distance to the registered coordinate, and a warning raised if that distance exceeds 50 m.

That threshold is a deliberately loose “something is probably wrong” trigger, not a precision claim. Typical smartphone GPS horizontal error is commonly cited around 5-15 m under open sky, and can be considerably worse under tree canopy or near canyon walls — both common at this site. 50 m is well outside that normal noise floor, so a flag at that distance is very unlikely to be a false alarm from ordinary GPS jitter.

i Worked example: the SBF_0001 / IMG_4571 case (2026-09-05)

Before this workflow existed, a photo (IMG_4571.HEIC) intended to register a new site (SBF_0002) turned out to have GPS coordinates identical, to many decimal places, to the *already-registered* SBF_0001 — almost certainly the same photo used to originally register that site, not a distinct location. That was caught by a manual check at the time. `Photo_Location_QC`'s comparison step (with the 50 m threshold above) is exactly what would have caught this automatically, the moment someone tried to map that filename to a *different* code than the one it actually matches.

Coordinate uncertainty is recorded, not discarded

A phone-photo GPS fix and a professionally surveyed point are not equally precise, but until this addition, `Locations` had no way to say which was which. Two columns now carry that information for **every** location, not just photo-derived ones, so any future consumer (e.g. the planned potentiometric-surface interpolation, or a simple nearest-station join) can check it:

- `coordinate_source` — e.g. 'photo_gps', or NULL for legacy/unknown- provenance rows (not backfilled to 'surveyed' — silence here means “we don't actually know,” which is more honest than guessing).
- `coordinate_uncertainty_m` — for `photo_gps` rows, the EXIF `GPSHPositioningError` tag when the device recorded one; otherwise a documented default of `r DEFAULT_PHOTO_GPS_UNCERTAINTY_M` — 15 m — chosen as a conservative mid-point of the typical smartphone accuracy range cited above, not a measured value for any specific photo.

`DEFAULT_PHOTO_GPS_UNCERTAINTY_M = 15 meters`

This is intentionally descriptive only, for now. Recording uncertainty is the first step; actually *using* it — e.g. weighting an IDW interpolation by $1 / \text{coordinate_uncertainty_m}^2$, or excluding low-confidence points above some threshold from a potentiometric surface — is

real additional design work that belongs with that interpolation build itself (still design-only, per the plan in the 2026-09-05 project memory), not invented ad hoc here. Writing the connection down here means it isn't lost by the time that build happens.

Sampling/observation linkage

A photo is often taken *because* something was sampled or observed, not just to document a location. `image_location_map.csv` supports two optional columns for this:

- `observation_note` — free text; when present and non-blank, a `Field_Observations` row is logged (qualitative, timestamped note tied to the resolved `location_id`).
- `linked_external_sample_id` — when present, resolves to a `sample_id` in `Samples` (via `external_sample_id`), so the note is tied to a specific sampling event, not just a place.

`Field_Observations` is deliberately separate from `Field_Measurements`, which requires a numeric value — a note like “spring flowing, no sample taken” or “new fumarole vent, sulfur smell” has nowhere sensible to go in a numeric-measurement table.

Current status

```
library(DBI); library(RSQLite); library(dplyr)
db_path <- file.path("database", "geochem_operational.sqlite")
if (!file.exists(db_path)) db_path <- file.path("../", "database", "geochem_operational.sqlite")
if (!file.exists(db_path)) db_path <- file.path("database", "geochem_demo.sqlite")
if (!file.exists(db_path)) db_path <- file.path("../", "database", "geochem_demo.sqlite")
con <- dbConnect(SQLite(), db_path, flags = SQLITE_RO)
```

```
cat("Photos with extracted GPS:", dbGetQuery(con, "SELECT COUNT(*) n FROM Photo_Location_Canonical"))
```

Photos with extracted GPS: 1

```
cat("Locations registered from photo GPS:",
    dbGetQuery(con, "SELECT COUNT(*) n FROM Locations WHERE coordinate_source = 'photo_gps'"))
```

Locations registered from photo GPS: 0

```
cat("Photo-vs-registered QC comparisons logged:", dbGetQuery(con, "SELECT COUNT(*) n FROM Photo_QC_Comparisons"))
```

Photo-vs-registered QC comparisons logged: 0

```
cat("Field observations logged:", dbGetQuery(con, "SELECT COUNT(*) n FROM Field_Observations
```

Field observations logged: 0

```
dbDisconnect(con)
```

As of this run, no real photos have been dropped into `image_drop/` yet — all counts above are expected to be zero until that happens. Re-run this notebook after a real batch of field photos and a filled-in `image_location_map.csv` to see it populate.

Data Inventory and Well/Facility Flow Network

Purpose and Scope

This notebook is the living reference for two threads that, unlike the conductivity/sampling-frequency workstream (01_conductivity_temporal_structure.qmd), don't have a single home yet: (1) what data currently exists in the database and over what time span, and (2) the production well -> port -> injection well flow network (Dhakal et al. 2025) plus the raw NDWR well-log records that back up individual well identities/construction. Update this notebook, not just the underlying scripts, as both threads evolve – consistent with this project's convention for 01.

```
library(DBI)
library(RSQLite)
library(dplyr)
library(ggplot2)
db_path <- file.path("database", "geochem_operational.sqlite")
if (!file.exists(db_path)) db_path <- file.path("../", "database", "geochem_operational.sqlite")
con <- dbConnect(SQLite(), db_path)
source_path <- file.path("scripts", "analysis", "data_availability.R")
if (!file.exists(source_path)) source_path <- file.path("../", "scripts", "analysis", "data_availability.R")
source(source_path)
```

1. Data Availability Through Time

scripts/analysis/data_availability.R computes, for every major dated table, the earliest/latest record and count. It handles each table's own date-storage convention – and, notably, discovered that `Sampling_Events.date` mixes **two different numeric epoch conventions** within the same column (Excel-serial-from-1899-12-30 for recent rows, Unix-epoch-days-from-1970-01-01 for ~723 older rows) – a real, previously-undiscovered data bug, not something introduced by this notebook. `external_event_id` (e.g. "SB10_1987-10-21") reliably encodes the true date and is used as ground truth wherever available; only ~10 of 759 rows lack that suffix and still rely on the ambiguous numeric column.

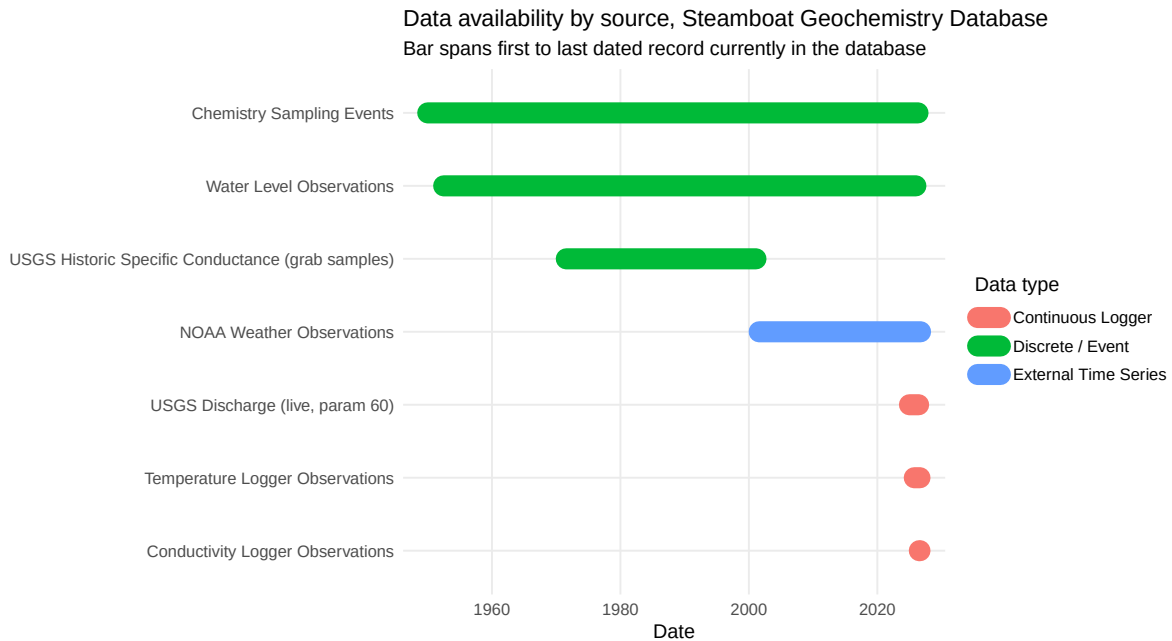
```
avail <- compute_data_availability(con)
avail
```

```
# A tibble: 7 x 5
```

	source	category	start_date	end_date	n_records
	<chr>	<chr>	<date>	<date>	<int>
1	Chemistry Sampling Events	Discret~	1950-01-01	2026-05-01	1396

2	Water Level Observations	Discret~	1952-06-25	2026-01-01	153736
3	USGS Historic Specific Conductance (~	Discret~	1971-07-28	2001-02-07	73
4	NOAA Weather Observations	Externa~	2001-08-08	2026-09-24	43197
5	USGS Discharge (live, param 60)	Continu~	2025-01-01	2026-06-06	272039
6	Temperature Logger Observations	Continu~	2025-10-01	2026-08-22	204622
7	Conductivity Logger Observations	Continu~	2026-07-15	2026-08-22	21901

`plot_data_availability(avail)`



Reading this chart: continuous logger records (temperature, conductivity, live USGS discharge) all start in 2025-2026 – this is a genuinely young, high-frequency instrumentation deployment layered on top of decades of sparser discrete sampling (chemistry events back to 1974, USGS grab-sample specific conductance 1971-2001, NOAA weather back to 2001). The gap between the historic USGS SC record ending in 2001 and today is real, not a data-loading artifact – worth keeping in mind when interpreting any long-term trend across that boundary.

Known follow-up (not fixed here): the root cause of the `Sampling_Events.date` dual-epoch bug – which ingest script wrote the Unix-epoch-day rows – has not been identified. `external_event_id` appears to be a reliable independent record of the true date for nearly all rows, but the underlying `date` column itself remains wrong for roughly 95% of rows and should not be trusted directly by any other script until traced and fixed at the source.

2. Well & Facility Flow Network

Production wells feed named commingling ports, which route to injection wells (Dhakal et al. 2025, Figure 5). `well_role` classifies each well; `vw_production_to_port` / `vw_port_to_injection` expose each leg of the network independently since the two legs are confirmed at different levels of confidence.

```
dbGetQuery(con, "SELECT well_role, COUNT(*) n FROM Wells GROUP BY well_role ORDER BY n DESC")
```

	well_role	n
1	unknown	185
2	production	38
3	monitor	23
4	injection	13
5	domestic	5

```
dbGetQuery(con, "SELECT * FROM vw_production_to_port")
```

	production_well_id	production_well_name	port_id	port_name	valid_from
1	77	78-29	1	Galena 1	<NA>
2	78	HA-4	1	Galena 1	<NA>
3	79	PW-1	1	Galena 1	<NA>
4	80	PW-2	1	Galena 1	<NA>
5	81	PW-3	1	Galena 1	<NA>
6	82	PW 2-1	4	SB2/3	<NA>
7	83	PW 2-3	4	SB2/3	<NA>
8	84	PW 2-5	4	SB2/3	<NA>
9	85	PW 3-1	4	SB2/3	<NA>
10	86	PW 3-2	4	SB2/3	<NA>
11	87	PW 3-3	4	SB2/3	<NA>
12	88	PW 3-4	4	SB2/3	<NA>
13	89	14A-33	3	Galena 3	<NA>
14	90	34-32	3	Galena 3	<NA>
15	91	44-32	3	Galena 3	<NA>
16	92	44A-32	3	Galena 3	<NA>
17	93	13-5R	5	SBHR	<NA>
18	94	21-5R	5	SBHR	<NA>
19	95	21B-5R	5	SBHR	<NA>
20	96	23-5	5	SBHR	<NA>
21	97	83B-6R	5	SBHR	<NA>
22	98	83C-6ST1	5	SBHR	<NA>

23	99	24-5	2	Galena 2	<NA>
24	100	41-5	2	Galena 2	<NA>
25	143	14-33	3	Galena 3	<NA>

	valid_to
1	<NA>
2	<NA>
3	<NA>
4	<NA>
5	<NA>
6	<NA>
7	<NA>
8	<NA>
9	<NA>
10	<NA>
11	<NA>
12	<NA>
13	<NA>
14	<NA>
15	<NA>
16	<NA>
17	<NA>
18	<NA>
19	<NA>
20	<NA>
21	<NA>
22	<NA>
23	<NA>
24	<NA>
25	<NA>

1	Dhakal et al. (2025) Figure 5 ("Steamboat geothermal complex configuration
2	
3	Dhakal et al. (2025) Figure 5; also Table
4	
5	
6	Dhakal et al. (2025) Figure
1", no space)	
7	Dhakal et al. (2025) Figure 5 and Table 1; coordinate from ArcGIS satellite-
overlay digitization (data/raw/arcgis/Steamboat_wells_dhakal.shp)	
8	Dhakal et al. (2025) Figure 5 and Table 1; co
overlay digitization	
9	Dhakal et al. (2025) Figure 5 and Table 1; co
overlay digitization	

10 Dhakal et al. (2025) Figure 5 and Table 1; c
 overlay digitization
 11 Dhakal et al. (2025) Figure 5 and Table 1; c
 overlay digitization
 12 Dhakal et al. (2025) Figure 5 and Table 1; c
 overlay digitization
 13
 14
 15
 16
 17 Dhakal et al.
 5RD" in Table 1)
 18
 19
 20
 21 Dhakal et al.
 6RD" in Table 1)
 22 Dhakal et al.
 6" in Table 1)
 23
 24
 25 Dhakal et al. (2025) Table

 1
 2
 3
 09-05) to be a genuinely DIFFERENT, separately-located well from PW 2-x/PW 3-
 x below -- not a naming variant. Still no coordinate resolved for PW-1 itself.
 4
 x/PW 3-x, same basis as PW-1. Still no coordinate resolved.
 5
 x/PW 3-x, same basis as PW-1. Still no coordinate resolved.
 6 Lower Steamboat / SB2/3 group (941 kg/s, 120.4 C combined). NEW well identified from the
 digitized wells layer (which had PW 2-3/2-5/3-1/2/3/4 but not this one) -- no coordinate yet
 bearing PW 2-x/PW 3-x wells already in this file; the paper itself omits the space.
 7 Lower Steamboat / SB2/3 group. Port corrected 2026-
 09-05: Figure 5 shows PW2-x and PW3-x wells combining at a SINGLE port literally named "SB2/
 8
 3.
 9
 3.
 10
 3.

11

3.

12

3.

13

33RD/43-33 notes on the ~132 kg/s unaccounted difference).

14

15

16

17

09-05), this is the *later* of two same-named historical wells (permit 0340, Production).

18

19

20

21

22

23

24

25

Distinct well from 14A-33 (confirmed: Table 1 lists both separately. In (completed 9/15/2007, three months after sibling 14A-33). Grouped with 14A-33/Galena 3 by Table 1's own classification; revisit if/when a more current source shows oth

```
dbGetQuery(con, "SELECT * FROM vw_port_to_injection")
```

	port_id	port_name	injection_well_id	injection_well_name	valid_from	valid_to
1	1	Galena 1	74	45-28	<NA>	<NA>
2	4	SB2/3	74	45-28	<NA>	<NA>
3	1	Galena 1	75	35-28	<NA>	<NA>
4	4	SB2/3	75	35-28	<NA>	<NA>
5	1	Galena 1	76	46-28	<NA>	<NA>
6	4	SB2/3	76	46-28	<NA>	<NA>
7	1	Galena 1	101	IW-6	<NA>	<NA>
8	4	SB2/3	101	IW-6	<NA>	<NA>
9	3	Galena 3	104	23-33RD	<NA>	<NA>
10	3	Galena 3	105	43-33	<NA>	<NA>
11	5	SBHR	106	42A-32	<NA>	<NA>
12	3	Galena 3	106	42A-32	<NA>	<NA>
13	5	SBHR	107	21-32	<NA>	<NA>
14	3	Galena 3	107	21-32	<NA>	<NA>
15	2	Galena 2	108	64A-32	<NA>	<NA>

source

1

Dhakal et al. (2025) Figure 5

2 Dhakal et al. (2025) Figure 5
 3 Dhakal et al. (2025) Figure 5
 4 Dhakal et al. (2025) Figure 5
 5 Dhakal et al. (2025) Figure 5
 6 Dhakal et al. (2025) Figure 5
 7 Dhakal et al. (2025) Figure 5
 8 Dhakal et al. (2025) Figure 5
 9 Dhakal et al. (2025) Figure 5
 10 Dhakal et al. (2025) Figure 5
 11 Dhakal et al. (2025) Figure 5
 12 Dhakal et al. (2025) Figure 5; confirmed by user (2026-09-05)
 13 Dhakal et al. (2025) Figure 5
 14 Dhakal et al. (2025) Figure 5; confirmed by user (2026-09-05)
 15 Dhakal et al. (2025) Figure 5

1
 2
 3
 1.
 4
 1.
 5
 1.
 6
 1.
 7
 1.
 8
 1.
 9 Fed by Galena 3 (454 kg/s, 76.9 C, paired with 43-33). NBMG's coordinate match was for a v
 33" (no RD suffix) -- Figure 5's own well-name spelling ("23-33RD") is used here as the network
 33"; not independently confirmed, flagged in dhakal_well_coordinates.csv.
 10
 33RD (454 kg/s combined, 76.9 C).
 11
 32).
 12 Middle Steamboat (Galena 3) and Upper Steamboat (SBHR) share this injection pair -- con
 33RD, 43-33} and {42A-32, 21-32}; Lower Steamboat (Galena 1 / SB2/3), by contrast, stays sep
 13
 32.
 14
 32 -- see that rows note.
 15 Fed by Galena 2 (136 kg/s, 8

32/21-32 line. Galena 2's own total production (205 kg/s) does not need to reconcile neatly :

Both legs are real (transcribed directly from the Dhakal figure), but **not every well in either table yet has a coordinate** – see the GeoPackage's `production_to_port/port_to_injection` line layers, which only draw a line where both endpoints are located. PW-1/2/3, 13-5R, 21B-5R, 83B-6R, and 23-33RD are the main remaining coordinate gaps as of this writing.

3. Well Logs: OCR Extraction, Identity Matching & Status History

The well-log pipeline now spans 76 NDWR “Well Driller’s Report” PDFs across two batches: 12 processed in Sessions 9-11, and 64 more log numbers (112 files – most with a second, “(2)”-suffixed reformatted/ streamlined re-scan of the same log) added in Session 24 from NDWR’s public ArcGIS well-log export. Three tables now back this, not just one:

- **Well_Log_Documents** – one row per log number (tracking both the primary scan and, when present, its “(2)” alternate on the same row), with OCR/cross-reference-derived location, depth, perforation, Section/Township/Range, type of work, proposed use, and hole/casing diameter.
- **Well_Work_Events** – one row per document once promoted/ registered, so a well’s status history (new -> deepened -> abandoned, etc.) is a real time series instead of a single overwritten snapshot.
- **Well_Lithology** – best-effort depth-interval descriptions staged from OCR text (`confidence = 'ocr_heuristic_unvalidated'`), joinable to perforation depth once a well is identified.

Location extraction now has four fallback levels, tried in order: a labeled Latitude/Longitude field, an unlabeled whole-text scan, UTM Easting/Northing conversion, and – new in Session 24 – a Bureau of Land Management PLSS (Section/Township/Range) lookup via their public CadNSDI REST service, used only when the first three all fail. When a log has both a primary and “(2)” alternate scan, lat/lon specifically merges by **extraction-confidence rank** (labeled field beats UTM beats unlabeled scan beats PLSS), not by “primary always wins” – a real bug caught during testing (logs 48594/126301: the primary scan’s unlabeled-scan guess landed far outside Nevada, while the “(2)” alt scan had a properly labeled field).

```
dbGetQuery(con, "
SELECT
  (SELECT COUNT(*) FROM Well_Log_Documents) AS n_documents,
  (SELECT COUNT(*) FROM Well_Log_Documents WHERE match_method = 'duplicate_content_skipped'
  (SELECT COUNT(*) FROM Well_Log_Documents WHERE plss_latlon_method = 'plss_section_centro
  (SELECT COUNT(*) FROM Well_Work_Events) AS n_work_events,
  (SELECT COUNT(*) FROM Well_Lithology) AS n_lithology_intervals,
  (SELECT COUNT(DISTINCT well_id) FROM Well_Work_Events WHERE work_type = 'abandonment') AS
")
```

	n_documents	n_duplicate_content	n_plss_derived	n_work_events
1	65	1	3	42
	n_lithology_intervals	n_wells_flagged_abandoned		
1	79	17		

Type of work and proposed use, extracted per document, give a quick sense of what this batch actually contains:

```
dbGetQuery(con, "
  SELECT work_type, proposed_use, COUNT(*) n
  FROM Well_Log_Documents
  GROUP BY work_type, proposed_use
  ORDER BY n DESC
")
```

	work_type	proposed_use	n
1	<NA>	<NA>	16
2	abandonment	monitor	11
3	deepening	domestic	11
4	deepening	monitor	10
5	<NA>	domestic	5
6	abandonment	industrial	5
7	<NA>	monitor	4
8	<NA>	industrial	1
9	<NA>	stock	1
10	abandonment	geothermal	1

Well status for mapping (production/injection/monitoring/P&A): rather than overloading `Wells.well_role` (a well's *role* doesn't stop being "injection" just because it's later plugged), `vw_wells_gis` adds a derived `display_status` column: a well shows as "Plugged & Abandoned" if it has any `Well_Work_Events` row with `work_type = 'abandonment'`, otherwise it falls back to `well_role`. This is the field the website's leaflet maps and the `GeoPackage wells` layer now color by.

```
dbGetQuery(con, "SELECT display_status, COUNT(*) n FROM vw_wells_gis GROUP BY display_status")
```

	display_status	n
1	unknown	106
2	production	37
3	plugged_abandoned	17
4	injection	12

```

5         monitor    10
6         domestic    3

```

Important, explicitly documented limitation: this only captures P&A status where a `Well_Work_Events` row exists (i.e. a driller’s report OCR found “abandonment” as its type of work). Wells known to be P&A only from earlier sessions’ free-text prose (e.g. 64-32, IW-3 – see AGENTS.md) will NOT show as P&A under this logic unless a `Well_Work_Events` row is added for them – deliberately not guess-parsed from notes fields, to avoid false positives.

Worked example – why proximity alone is misleading: log 123802 sits only ~19.5 m from NBMG’s “Industrial Production Well 43-33,” the closest lat/lon match found across the original 12 logs. But 123802’s own NDWR record shows it was completed **9/24/2015**, while 43-33 was drilled in **2007** – an 8-year gap that rules out the match despite the tight spatial coincidence. The same location-*and*-date discipline applies to every new candidate from the larger batch – `qc_well_log_matches()` now produces a standing, regenerable candidates report (`data/derived/well_log_match_candidates.csv`) with a plain- English **recommendation** per row (e.g. “download the NDWR”(2)” page”, “review as a close match”, or “no confident match – keep unresolved”), replacing the earlier one-off conversation-driven writeup per batch:

```

cand_path <- file.path("data", "derived", "well_log_match_candidates.csv")
if (!file.exists(cand_path)) cand_path <- file.path("../", "data", "derived", "well_log_match_
if (file.exists(cand_path)) read.csv(cand_path) else cat("No candidates report found.")

```

	document_id	log_number	well_name_parsed	latitude	longitude	match_method
1		62	80672	39.52525	-119.7046	parsed_text
2		10	146403	NVGD-MW1	40.33780	-119.4226
			plss_latlon_method	work_type	proposed_use	has_alt_scan
1			plss_section_centroid	NA	monitor	FALSE
2				NA		FALSE
			nearest_well_name	nearest_well_dist_m	nearest_location_name	
1			Greg Street Well	4503.2	Greg Street Well	
2			El Rancho Shallow MW	93472.2	El Rancho Shallow MW	
			nearest_location_dist_m			
1				4503.2		
2				93472.2		

```

1  Consider downloading NDWR's reformatted/streamlined PDF page for log #80672 and saving it
checking. | No confident match found (nearest candidate is far) -- keep unresolved for now.
2  Consider downloading NDWR's reformatted/streamlined PDF page for log #146403 and saving it
checking. | No confident match found (nearest candidate is far) -- keep unresolved for now.

```

```

1          no lat/lon, UTM found -- derived from PLSS Section/Township/Range section
sensitive | well_name not confidently extracted -- check manually
2 longitude OCR error corrected 2026-09-12 (was -719.422608, a leading-
digit misread of -119.422608 -- Nevada bounding-box check flagged this at ingest time; confi
area NVGD well, unrelated to the Steamboat field, not registered as a Wells row before or af

```

No identity is written to `well_log_document_map.csv` for any unmatched log without a human confirming it first – both remaining candidates above are left unresolved deliberately (one, log 80672, sits 4.5 km from its nearest named well – far too distant to treat as a match).

Because none of the newly-processed logs cleared the bar for a confident identity match, each with a real coordinate is registered as its own **provisional Wells** row (see `register_provisional_well_logs()` in `scripts/ingest/ingest_well_logs.R`) so the real coordinate/depth/ construction data is visible and mappable rather than stranded in the staging table – clearly named and flagged as unconfirmed, not silently merged into any named well:

```

dbGetQuery(con, "
  SELECT well_id, well_name, latitude, longitude, total_depth,
         top_perforation, bottom_perforation, diameter_in, coordinate_source, coordinate_unc
  FROM Wells
  WHERE well_name LIKE 'Unidentified Well%'
  ORDER BY well_id
")

```

	well_id	well_name	latitude	longitude	total_depth
1	112	Unidentified Well (NDWR Log 102799)	39.38797	-119.7477	NA
2	113	Unidentified Well (NDWR Log 103616)	39.40739	-119.7547	552
3	114	Unidentified Well (NDWR Log 123802)	39.38400	-119.7445	NA
4	115	Unidentified Well (NDWR Log 125850)	39.38952	-119.7670	NA
5	116	Unidentified Well (NDWR Log 125851)	39.38822	-119.7675	NA
6	117	Unidentified Well (NDWR Log 27731)	39.39797	-119.7538	3
7	118	Unidentified Well (NDWR Log 27732)	39.39436	-119.7538	3
8	145	Unidentified Well (NDWR Log 104216)	39.37058	-119.7570	1990
9	146	Unidentified Well (NDWR Log 10896)	39.39436	-119.7500	NA
10	147	Unidentified Well (NDWR Log 126275)	39.39862	-119.7300	NA
11	148	Unidentified Well (NDWR Log 126301)	39.39992	-119.7300	188
12	149	Unidentified Well (NDWR Log 129060)	39.40675	-119.7464	NA
13	150	Unidentified Well (NDWR Log 138973)	39.39712	-119.7400	NA
14	151	Unidentified Well (NDWR Log 138974)	39.39714	-119.7400	NA
15	152	Unidentified Well (NDWR Log 138979)	39.39701	-119.7500	NA
16	153	Unidentified Well (NDWR Log 139071)	39.39679	-119.7500	NA

17	154	Unidentified Well (NDWR Log 139072)	39.39677	-119.7500	NA
18	155	Unidentified Well (NDWR Log 1406)	39.39630	-119.7400	NA
19	156	Unidentified Well (NDWR Log 145136)	39.40633	-119.7500	NA
20	157	Unidentified Well (NDWR Log 146150)	39.38537	-119.7391	NA
21	158	Unidentified Well (NDWR Log 146152)	39.38873	-119.7400	NA
22	159	Unidentified Well (NDWR Log 146153)	39.38777	-119.7400	NA
23	160	Unidentified Well (NDWR Log 146154)	39.38701	-119.7400	NA
24	161	Unidentified Well (NDWR Log 146158)	39.39287	-119.7396	NA
25	162	Unidentified Well (NDWR Log 146159)	39.39135	-119.7400	NA
26	163	Unidentified Well (NDWR Log 146160)	39.38967	-119.7407	NA
27	164	Unidentified Well (NDWR Log 146161)	39.39654	-119.7400	NA
28	165	Unidentified Well (NDWR Log 146162)	39.39592	-119.7407	NA
29	166	Unidentified Well (NDWR Log 147778)	39.39669	-119.7400	NA
30	167	Unidentified Well (NDWR Log 147786)	39.39652	-119.7400	NA
31	168	Unidentified Well (NDWR Log 147790)	39.39454	-119.7400	NA
32	169	Unidentified Well (NDWR Log 147793)	39.39429	-119.7400	NA
33	170	Unidentified Well (NDWR Log 21796)	39.37603	-119.7700	2
34	171	Unidentified Well (NDWR Log 24140)	39.39436	-119.7400	NA
35	172	Unidentified Well (NDWR Log 26097)	39.37964	-119.7400	NA
36	173	Unidentified Well (NDWR Log 27930)	39.39630	-119.7400	NA
37	174	Unidentified Well (NDWR Log 38060)	39.39797	-119.7500	NA
38	175	Unidentified Well (NDWR Log 44301)	39.40908	-119.7500	NA
39	176	Unidentified Well (NDWR Log 46579)	39.37242	-119.7600	21
40	177	Unidentified Well (NDWR Log 48594)	39.39436	-119.7400	NA
41	178	Unidentified Well (NDWR Log 48595)	39.39436	-119.7400	NA
42	179	Unidentified Well (NDWR Log 48816)	39.39436	-119.7400	232
43	180	Unidentified Well (NDWR Log 5711)	39.37964	-119.7400	NA
44	181	Unidentified Well (NDWR Log 5731)	39.39436	-119.7500	NA
45	183	Unidentified Well (NDWR Log 70925)	39.38279	-119.7421	NA
46	184	Unidentified Well (NDWR Log 73396)	39.40908	-119.7500	0
47	185	Unidentified Well (NDWR Log 74669)	39.38279	-119.7421	NA
48	187	Unidentified Well (NDWR Log 8771)	39.39075	-119.7400	NA
49	188	Unidentified Well (NDWR Log 90819)	39.37242	-119.7700	NA

	top_perforation	bottom_perforation	diameter_in	coordinate_source
1	NA	NA	NA	ndwr_well_log_crossref
2	10	80	NA	well_log_ocr_or_text
3	80	800	NA	well_log_ocr_or_text
4	NA	NA	NA	ndwr_well_log_crossref
5	NA	NA	NA	ndwr_well_log_crossref
6	NA	NA	NA	ndwr_well_log_crossref
7	NA	NA	NA	ndwr_well_log_crossref
8	1904	1965	NA	well_log_ocr_or_text
9	NA	NA	NA	well_log_ocr_or_text

10	NA	NA	NA	well_log_ocr_or_text
11	NA	NA	NA	well_log_ocr_or_text
12	35	84	NA	ndwr_well_log_crossref
13	NA	NA	NA	well_log_ocr_or_text
14	NA	NA	NA	well_log_ocr_or_text
15	NA	NA	NA	well_log_ocr_or_text
16	NA	NA	NA	well_log_ocr_or_text
17	NA	NA	NA	well_log_ocr_or_text
18	NA	NA	NA	well_log_ocr_or_text
19	NA	NA	NA	well_log_ocr_or_text
20	NA	NA	NA	ndwr_well_log_crossref
21	NA	NA	NA	well_log_ocr_or_text
22	NA	NA	NA	well_log_ocr_or_text
23	NA	NA	NA	well_log_ocr_or_text
24	NA	NA	NA	ndwr_well_log_crossref
25	NA	NA	NA	well_log_ocr_or_text
26	NA	NA	NA	ndwr_well_log_crossref
27	NA	NA	NA	well_log_ocr_or_text
28	NA	NA	NA	ndwr_well_log_crossref
29	NA	NA	NA	well_log_ocr_or_text
30	NA	NA	NA	well_log_ocr_or_text
31	NA	NA	NA	well_log_ocr_or_text
32	NA	NA	NA	well_log_ocr_or_text
33	NA	NA	NA	well_log_ocr_or_text
34	NA	NA	NA	well_log_ocr_or_text
35	NA	NA	NA	well_log_ocr_or_text
36	NA	NA	NA	well_log_ocr_or_text
37	NA	NA	NA	well_log_ocr_or_text
38	NA	NA	NA	well_log_ocr_or_text
39	NA	NA	NA	well_log_ocr_or_text
40	NA	NA	NA	well_log_ocr_or_text
41	NA	NA	NA	well_log_ocr_or_text
42	NA	NA	NA	well_log_ocr_or_text
43	NA	NA	NA	well_log_ocr_or_text
44	NA	NA	NA	well_log_ocr_or_text
45	NA	NA	NA	well_log_plss_centroid
46	NA	NA	NA	well_log_ocr_or_text
47	NA	NA	NA	well_log_plss_centroid
48	NA	NA	NA	well_log_ocr_or_text
49	NA	NA	NA	well_log_ocr_or_text
coordinate_uncertainty_m				
1		200		
2		150		

3	150
4	200
5	200
6	200
7	200
8	150
9	150
10	150
11	150
12	200
13	150
14	150
15	150
16	150
17	150
18	150
19	150
20	200
21	150
22	150
23	150
24	200
25	150
26	200
27	150
28	200
29	150
30	150
31	150
32	150
33	150
34	150
35	150
36	150
37	150
38	150
39	150
40	150
41	150
42	150
43	150
44	150
45	800

46	150
47	800
48	150
49	150

4. NDOM Well-Permit Cross-Validation

NDOM (Nevada Division of Minerals) supplied the official state permit record for every Ormat Steamboat well – an independent source from the NBMG/ArcGIS-digitized coordinates used elsewhere in this project. `ingest_ndom_wells.R` matches each of the 49 permitted wells to an existing `Wells` row (exact name, a confirmed spelling variant, or an existing alias), or registers it as a new provisional well when no match exists.

```
dbGetQuery(con, "
  SELECT match_method, COUNT(*) n
  FROM NDOM_Well_Records
  GROUP BY match_method
  ORDER BY n DESC
")
```

```
match_method  n
1  exact_name 42
2      alias  7
```

Coordinate cross-validation: where a well already had a coordinate from an earlier source, NDOM's is compared against it – disagreements over 100 m are never auto-applied, only logged for manual review:

```
disc_path <- file.path("data", "derived", "ndom_coordinate_discrepancies.csv")
if (!file.exists(disc_path)) disc_path <- file.path("../", "data", "derived", "ndom_coordinate_discrepancies.csv")
if (file.exists(disc_path)) read.csv(disc_path) else cat("No discrepancy report found.")
```

	well_id	well_name	permit	existing_lat	existing_lon	existing_coordinate_source
1	99	24-5	178	39.37309	-119.7645	nbgm_geothermal_wells
2	108	64A-32	560	39.38023	-119.7595	nbgm_geothermal_wells
3	90	34-32	578	39.38046	-119.7597	nbgm_geothermal_wells
4	89	14A-33	669	39.38499	-119.7451	nbgm_geothermal_wells
5	100	41-5	1261	39.37526	-119.7648	nbgm_geothermal_wells
6	107	21-32	1358	39.38993	-119.7686	nbgm_geothermal_wells
				existing_coordinate_uncertainty_m	ndom_lat	ndom_lon distance_m

1	50	39.36873	-119.7677	555.1
2	50	39.38228	-119.7592	228.2
3	100	39.38323	-119.7654	572.5
4	50	39.38266	-119.7497	476.2
5	50	39.37398	-119.7638	166.1
6	50	39.38821	-119.7675	211.3

These gaps are large relative to this project’s usual ~50-75 m NBMG/ ArcGIS uncertainty and are worth a manual look – NDOM is the official permit-of-record coordinate. As of 2026-09-12, the discrepancy report also includes `existing_coordinate_source/existing_coordinate_uncertainty_m` (all 6 current discrepancies trace back to `nbmg_geothermal_wells`), so a reviewer can immediately see where the currently-stored point came from without a separate lookup. ArcGIS uncertainty and are worth a manual look – NDOM is the official permit-of-record coordinate.

A real, pre-existing unit bug was found via this ingestion: `elevation_m` on both `Wells` and `Locations` had been storing raw feet (inherited from `ingest_ndwr.R`) despite its name – e.g. “Sky Tavern Ski Resort” recorded as 7619, when Steamboat’s true elevation is ~1450-1750 m. Fixed at the source (`ingest_ndwr.R` now converts on insert) plus a one-time retroactive migration for already-populated rows:

```
dbGetQuery(con, "
  SELECT 'Wells' AS tbl, MIN(elevation_m) min_m, MAX(elevation_m) max_m FROM Wells WHERE elev
  UNION ALL
  SELECT 'Locations', MIN(elevation_m), MAX(elevation_m) FROM Locations WHERE elevation_m IS
")
```

	tbl	min_m	max_m
1	Wells	1243.584	2322.271
2	Locations	1344.564	2322.271

Literature cross-check: reading Dhakal et al. (2025) Table 1 (“Steamboat Production Well classification”) directly confirmed that 14-33 and 14A-33 are genuinely distinct wells (both listed separately under Middle Steamboat) – independent corroboration of the decision not to merge them, and the basis for adding 14-33 to the well/port network CSV (Galena 3, matching its sibling).

5. NDWR Spring/Stream Flow

Daily manual discharge readings (cfs) at NDWR-monitored spring/stream sites – first supplied 2026-09-12 for 4 sites on Whites Creek (Truckee Meadows basin, part of this project’s broader hydrologic context rather than the Steamboat outflow system itself).

```
dbGetQuery(con, "
  SELECT l.name, l.latitude, l.longitude, l.elevation_m,
         COUNT(*) n_obs, MIN(s.date) start_date, MAX(s.date) end_date
  FROM Stream_Flow_Observations s
  JOIN Locations l ON l.location_id = s.location_id
  GROUP BY l.location_id
  ORDER BY l.name
")
```

		name	latitude	longitude	elevation_m
1		Whites Creek Above MRWTP	39.38837	-119.8147	1689.506
2		Whites Creek Below Mt. Rose WTP	39.38846	-119.8147	1689.506
3		Whites Creek Diversion to Mt. Rose WTP (MRWTP)	39.38837	-119.8147	1689.506
4		Whites Creek at I580	39.42423	-119.7547	1386.840
	n_obs	start_date	end_date		
1	821	2020-12-19	2026-06-30		
2	1004	2022-04-01	2026-03-31		
3	913	2023-07-01	2026-03-31		
4	2191	2017-12-30	2026-03-31		

Two of the four sites share an identical rounded lat/lon (a diversion structure and its “above”/“below” companion points) – registration is keyed on `external_station_code`, not `coord_key`, for exactly this reason.

PHREEQC Geochemical Modeling

Purpose and scope

This notebook is the **living scientific reference** for PHREEQC geochemical modeling at Steamboat Hills — it documents *why* each analysis is done, *what* the underlying chemistry/thermodynamics are, and *where* the code that implements it lives (`scripts/phreeqc/`), and it is intended to stay the entry point for this workstream as more chemistry (real Cl end-members overlapping the conductivity logger deployment, alkalinity titrations, well-log identities) accumulates. It complements, but does not replace, the individual R scripts; this document narrates and runs representative pieces of them so the reasoning and the code stay in sync — following the same convention as `01_conductivity_temporal_structure.qmd` for the Cl-sampling-frequency workstream.

Sections below run in dependency order:

1. Scientific motivation — why PHREEQC, and what each analysis mode answers.
2. Architecture — SOLUTION-block generation, the thermodynamic-database registry, and how the external PHREEQC executable is invoked.
3. Building PHREEQC_Solutions from validated chemistry, and why coverage went from 8 to 233 samples.
4. Charge-balance QC — and the alkalinity data-quality story behind it.
5. Speciation and saturation indices (real data, runs today).
6. Geothermometry — activity-based Na/K and multicomponent temperature-sweep, cross-checked against each other.
7. Mixing, inverse modeling, and gas-phase equilibria (self-tested on synthetic end-members; real results pending overlapping Cl chemistry).
8. Known limitations, fixed bugs, and a running changelog.

Warning

As of this writing, real two-end-member mixing and inverse modeling (Section 7) cannot yet be run on real Steamboat samples: the only Cl/isotope field samples collected so far (2026-05-01) predate the conductivity logger deployment (2026-07-15) by roughly 2.5 months, and the downstream/outflow proxy site (SBGG) has no chemistry at all yet. Section 7 therefore runs against synthetic self-test data, clearly labeled as such. Sections 3–6 are real and current as of 2026-09-06.

Scientific motivation

Three questions this project needs answered, in increasing order of how much real data they currently have behind them:

1. **What is the current chemical/thermal state of each sampled water, and is it internally consistent?** — PHREEQC's aqueous speciation and charge-balance calculation answer this directly: given measured concentrations, pH, and temperature, what species (free ions, ion pairs, complexes) is the dissolved mass actually distributed across, and does the analysis close electrically within an acceptable tolerance? A poor charge balance is a red flag for the *analysis*, not the water.
2. **Is a thermal water in equilibrium with the minerals it's discharging through, and what does that imply about scaling risk and reservoir temperature?** — Saturation indices ($SI = \log_{10}(IAP/K_{sp})$, zero at equilibrium, positive supersaturated/scaling-prone, negative undersaturated) for calcite, quartz, and related phases answer this. Silica geothermometry in particular is the classic tool for estimating deep reservoir temperature from a *cooled, discharged* fluid, because silica dissolution/precipitation kinetics are slow enough that a discharge sample can retain a partial memory of higher-temperature equilibrium at depth.
3. **Where is the discharge chemistry coming from — how much is deep thermal upflow versus meteoric dilution, and does any mineral reaction happen along the way?** — This is the mixing/inverse-modeling question this thesis's Cl-discharge framing (01_conductivity_temporal_structure.qmd) is ultimately building toward: PHREEQC's MIX and INVERSE_MODELING keywords test two-end-member mixing hypotheses directly, with or without an accompanying mineral reaction. This is the piece still waiting on real overlapping chemistry (see the warning above).

Architecture

flowchart LR

```

A[Lab_Analyses + Field_Measurements + Isotope_Analyses] --> B["build_phreeqc_solutions(c
B --> C[(PHREEQC_Solutions)]
C --> D["run_phreeqc_pipeline(con)<br/>09_run_phreeqc.R"]
D -->|"system2()"| E[[PHREEQC 3 executable]]
E --> D
D --> F[(PHREEQC_Results)]
D --> G[(PHREEQC_Run_Failures)]
C --> H["run_phreeqc_temp_sweep(con)<br/>09_run_phreeqc.R"]
H --> I[(PHREEQC_Temp_Sweep)]
C --> J["run_mixing_fraction_analysis() /<br/>run_phreeqc_mixing()<br/>10_run_phreeqc_mi
C --> K["run_phreeqc_inverse()<br/>11_run_phreeqc_inverse.R"]
C --> L["run_phreeqc_gas_phase()<br/>12_run_phreeqc_gas_phase.R"]
F --> M["interpret_phreeqc_results()<br/>automated human-readable text"]

```

Script	Role
<code>scripts/phreeqc/utils_phreeqc.R</code>	SOLUTION-block generation, <code>system2()</code> execution of the standalone PHREEQC 3 executable, <code>SELECTED_OUTPUT</code> parsing, the 6-database thermodynamic registry with auto-recommendation, fluid-type classification, automated interpretation text
<code>scripts/phreeqc/08_build_phreeqc_tables.R</code>	<code>Build_phreeqc_solutions(con)</code> , <code>get_phreeqc_eligible(con)</code>
<code>scripts/phreeqc/09_run_phreeqc.R</code>	<code>run_phreeqc_pipeline()</code> , <code>run_phreeqc_multi()</code> , <code>run_phreeqc_temp_sweep()</code> , <code>calculate_activity_geothermometers()</code>
<code>scripts/phreeqc/10_run_phreeqc_mixing.R</code>	Two-end-member conservative mixing fractions + PHREEQC MIX blocks
<code>scripts/phreeqc/11_run_phreeqc_inverse.R</code>	PHREEQC INVERSE_MODELING (mixing + mineral mass transfer)
<code>scripts/phreeqc/12_run_phreeqc_gas_phase.R</code>	PHREEQC GAS_PHASE (CO ₂ /H ₂ S degassing equilibria)
<code>scripts/phreeqc/run_phreeqc_analysis.R</code>	Single entry point sequencing all five modes
<code>database/schema/08_phreeqc_schema.R</code>	PHREEQC_Solutions, PHREEQC_Results, PHREEQC_Temp_Sweep, PHREEQC_Run_Failures, and the mixing/inverse/gas-phase table families

PHREEQC itself is an external executable (USGS PHREEQC 3), not an R package — this project shells out to it via `system2()` rather than using an R binding, matching the pattern of the IGNIS project this workstream was ported from. `PHREEQC_EXE/PHREEQC_DB` environment variables override the default search path (`C:/Program Files/USGS/phreeqc/...`).

Database selection is automatic by default (`phreeqc_db = "auto"`): temperatures above 100°C route to the LLNL database (validated high-temperature thermodynamic data to 300°C, standard for geothermal work), high-TDS waters route to Pitzer or SIT, and everything else uses the general-purpose default `phreeqc.dat`. If a batch fails to converge, the pipeline automatically retries with fallback databases, then retries each sample individually — a single problem sample never silently drops every other sample in its batch. Convergence failures that persist across every database tried are logged to `PHREEQC_Run_Failures`, not silently dropped.

```
library(DBI)
library(RSQLite)
library(dplyr)
```

```

library(ggplot2)

# Several scripts/phreeqc/*.R functions (build_phreeqc_solutions() in
# particular, via database/schema/lab_analyte_map.R) source() other project
# files using paths relative to the project root, not this notebook's own
# directory -- Quarto's project-level execute-dir default renders .qmd files
# with their own directory (notebooks/) as the working directory, which
# breaks those internal source() calls. A plain setwd() here isn't enough:
# knitr's in_dir() wraps every chunk and restores the working directory
# in between them, so the change would silently revert before the next
# chunk runs -- opts_knit$set(root.dir = ...) is knitr's own supported way
# to persist a working-directory override across all chunks, matching
# scripts/run_pipeline.R's own root-relative-paths convention.
if (basename(getwd()) == "notebooks") {
  knitr::opts_knit$set(root.dir = normalizePath("../"))
  setwd("../")
}

con <- dbConnect(SQLite(), file.path("database", "geochem_operational.sqlite"))
dbExecute(con, "PRAGMA foreign_keys = ON;")

```

```
[1] 0
```

```

for (f in c("utils_phreeqc.R", "08_build_phreeqc_tables.R", "09_run_phreeqc.R",
            "10_run_phreeqc_mixing.R", "11_run_phreeqc_inverse.R",
            "12_run_phreeqc_gas_phase.R", "13_gas_geothermometry.R",
            "14_gas_mixing.R", "run_phreeqc_analysis.R")) {
  source(file.path("scripts", "phreeqc", f))
}

phreeqc_available <- tryCatch({ check_phreeqc(); TRUE }, error = function(e) FALSE)

```

phreeqc_available is TRUE on the machine this notebook was last rendered on — every live PHREEQC execution below is skipped gracefully (not silently faked) if the executable isn't found.

Building PHREEQC_Solutions

build_phreeqc_solutions(con) derives one row per sample from Lab_Analyses, Field_Measurements, and Isotope_Analyses — temperature, pH, and each major/minor

ion, plus a real charge balance computed from `lab_analyte_map`.R's per-analyte charge and molar mass (not a hardcoded, partial equivalent-weight table). A sample is *eligible* for a full PHREEQC run if it has temperature, pH, Na, and Cl at minimum; everything else is *rejected* with a specific, human-readable reason via `get_phreeqc_eligible()` — never silently dropped.

```
sol <- build_phreeqc_solutions(con)
```

```
---- Building PHREEQC_Solutions ----
```

```
PHREEQC_Solutions built: 1464 rows (312 complete, 1152 incomplete).
```

```
eligibility <- get_phreeqc_eligible(con)
cat(sprintf("PHREEQC_Solutions: %d rows total, %d eligible for a full run, %d rejected\n",
           nrow(sol), nrow(eligibility$eligible), nrow(eligibility$rejected)))
```

```
PHREEQC_Solutions: 1464 rows total, 312 eligible for a full run, 1152 rejected
```

```
loc_info <- dbGetQuery(con, "
  SELECT s.sample_id, s.data_source, l.name AS location_name, l.site_type
  FROM Samples s JOIN Locations l ON s.location_id = l.location_id
")
sol |>
  left_join(loc_info, by = "sample_id") |>
  count(data_source, site_type, completeness_flag) |>
  arrange(data_source, site_type)
```

```
# A tibble: 34 x 4
```

	data_source	site_type	completeness_flag	n
	<chr>	<chr>	<chr>	<int>
1	FIELD	creek	incomplete	3
2	FIELD	fumarole	complete	1
3	FIELD	fumarole	incomplete	1
4	FIELD	seep	complete	1
5	FIELD	seep	incomplete	32
6	FIELD	spring	complete	2
7	FIELD	spring	incomplete	10
8	FIELD	steaming ground	incomplete	13
9	FIELD	transect	complete	2
10	FIELD	well	complete	2

```
# i 24 more rows
```

i Why eligible coverage jumped from 8 to 233 samples (2026-09-06)

The first working version of this pipeline found only 8 eligible samples database-wide — all from FIELD (springs/wells/fumarole/transect). NDEP’s ~755 background samples and ~10 NDEP PRR wells all looked like they had zero pH/temperature data. Investigating why (because the user specifically asked to run this on existing NDEP data) surfaced three real ingest bugs, not a genuine data gap:

1. **NDEP’s pH and temperature are stored in Lab_Analyses, not Field_Measurements** — `build_phreeqc_solutions()` originally only checked the latter. Fixed to pull from both, preferring `Field_Measurements` when both exist.
2. **NDEP’s minor/trace analytes are stored in mixed units** (the same analyte, e.g. SiO2/As/B/Fe/Li/Mn, appears with both ug/L and mg/L rows for different samples) with no ingest-time normalization — previously averaged naively, a ~1000x risk for whichever rows were “wrong” relative to others in the average. Fixed to convert every row to mg/L, per its own `units` column, before any aggregation.
3. **NDEP’s authoritative alkalinity parameter, “Total Alkalinity as CaCO3”, was never ingested at all.** `ndep_analyte_map.R` instead mapped two *unit-convention duplicates* of the same measurement (“HCO3 as CaCO3”/“HCO3 as HCO3”, confirmed via the raw source file to be the exact same 647 events, ratio consistently 0.82 — matching the true CaCO3/HCO3 equivalent-weight ratio) to a single HCO3 analyte code, with no conversion factor. Added the missing “Total Alkalinity” mapping (with the correct CaCO3 -> HCO3 mass conversion, $61.017/50.05 = 1.2189$), and remapped the duplicate-convention rows to distinct, explicitly-excluded codes so they can never collide again.

A fourth, separate bug in `lab_analyte_map.R` — the Alkalinity row had `charge = 0`, `molar_mass = NA`, silently excluding it from every charge-balance calculation even after real data existed — meant the fix above didn’t move the charge-balance numbers at all until this was also corrected (see the next section).

A fifth, independent instance of the same class of bug was found in `ingest_lab.R` (FIELD’s own lab-report parser): `normalize_lab_wide()` ran raw CSV headers through `make.names()` for R-safe column names, which turns “Alkalinity Total” into “Alkalinity.Total” (space → dot) — and then used that *mangled* name as the join key against `lab_analyte_map$raw_name` (“Alkalinity Total”, with a space), guaranteed to never match. Every FIELD sample’s Alkalinity (and NO3, the only other multi-word raw name) had been silently dropped since this pipeline’s very first field-lab ingest, not missing due to a genuine sampling gap as earlier sessions assumed. Fixed to preserve the original header text as the actual join key.

See `AGENTS.md` (Sessions 15–16) for the full narrative, including the raw per-row evidence for each fix.

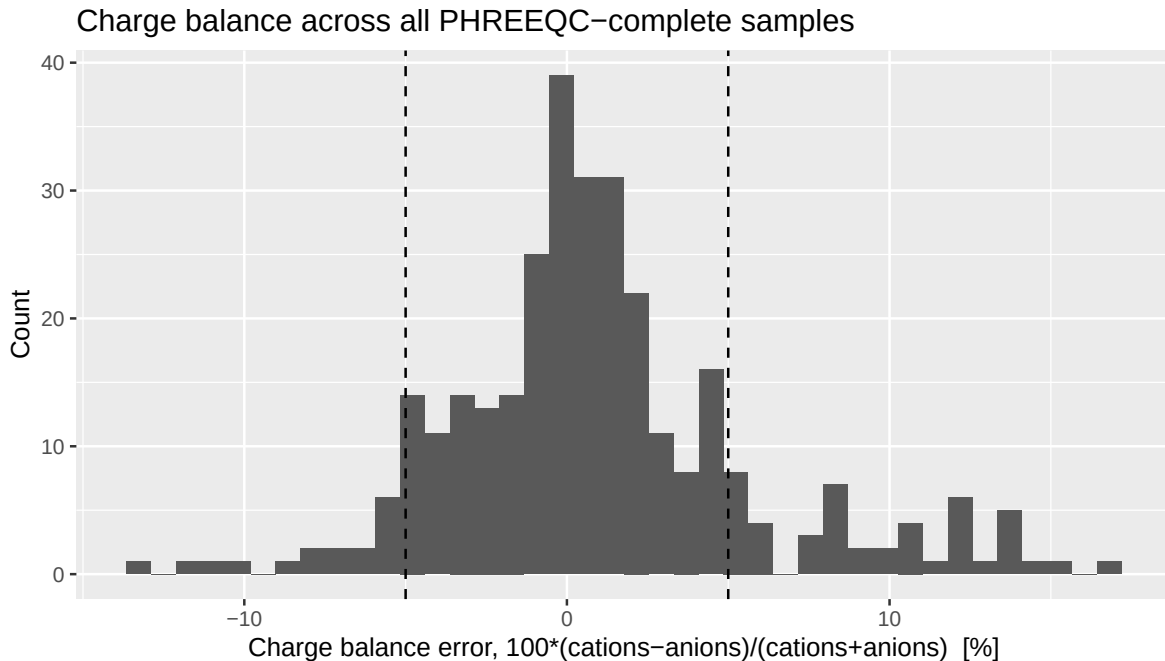
Charge-balance QC

```
complete_sol <- sol |>
  left_join(loc_info, by = "sample_id") |>
  filter(completeness_flag == "complete", !is.na(charge_balance))

complete_sol |>
  mutate(qc_flag = ifelse(abs(charge_balance) > 5, "FLAG (>5%)", "OK (<=5%)")) |>
  count(data_source, qc_flag)
```

```
# A tibble: 6 x 3
  data_source      qc_flag      n
  <chr>           <chr>    <int>
1 FIELD          OK (<=5%)      8
2 Mariner & Janik 1995 FLAG (>5%)      4
3 Mariner & Janik 1995 OK (<=5%)     70
4 NDEP           FLAG (>5%)     61
5 NDEP           OK (<=5%)    164
6 Sorey & Colvard 1992 OK (<=5%)      5
```

```
ggplot(complete_sol, aes(x = charge_balance)) +
  geom_histogram(bins = 40) +
  geom_vline(xintercept = c(-5, 5), linetype = "dashed") +
  labs(x = "Charge balance error, 100*(cations-anions)/(cations+anions) [%]",
       y = "Count", title = "Charge balance across all PHREEQC-complete samples")
```



The dashed lines mark the conventional $\pm 5\%$ acceptability threshold. Before the alkalinity fixes above, essentially every complete sample failed this check with a median imbalance around 60% — a pure artifact of excluding alkalinity (typically the dominant anion in these near-neutral waters) from the anion sum entirely, not a real geochemical signal. After the fixes, the median imbalance for NDEP’s complete samples is close to zero and most pass the standard threshold; FIELD’s own samples are now fully alkalinity-complete too (see the callout above).

Speciation and saturation indices

```
speciation_result <- run_phreeqc_pipeline(con)
```

PHREEQC found: C:/Program Files/USGS/phreeqc/bin/Release/phreeqc.exe

Database: C:/Program Files/USGS/phreeqc/database/phreeqc.dat

--- PHREEQC Eligibility (all site types) ---

Eligible: 312 samples

Rejected: 1152 samples

Rejection details:

- 1 (Thomas Creek @ North Timberline Drive &[STBT03Thomas-3]): missing Na
- 2 (Thomas Creek @ North Timberline Drive &[STBT03Thomas-3]): missing Na
- 3 (Thomas Creek @ North Timberline Drive &[STBT03Thomas-3]): missing Na
- 4 (Thomas Creek @ North Timberline Drive &[STBT03Thomas-3]): missing Na
- 5 (Thomas Creek @ North Timberline Drive &[STBT03Thomas-3]): missing Na
- 6 (Thomas Creek @ North Timberline Drive &[STBT03Thomas-3]): missing Na
- 7 (Thomas Creek @ North Timberline Drive &[STBT03Thomas-3]): missing temperature; missing Na
- 8 (Thomas Creek @ North Timberline Drive &[STBT03Thomas-3]): missing Na
- 9 (Thomas Creek @ North Timberline Drive &[STBT03Thomas-3]): missing pH; missing Na
- 10 (Thomas Creek @ North Timberline Drive &[STBT03Thomas-3]): missing pH; missing Na
- 11 (Thomas Creek @ North Timberline Drive &[STBT03Thomas-3]): missing pH; missing Na
- 12 (Thomas Creek @ North Timberline Drive &[STBT03Thomas-3]): missing pH; missing Na
- 13 (Thomas Creek @ North Timberline Drive &[STBT03Thomas-3]): missing pH; missing Na
- 14 (Thomas Creek @ North Timberline Drive &[STBT03Thomas-3]): missing pH; missing Na
- 15 (Thomas Creek @ North Timberline Drive &[STBT03Thomas-3]): missing pH; missing Na
- 16 (Thomas Creek @ North Timberline Drive &[STBT03Thomas-3]): missing pH; missing Na

17 (Thomas Creek @ North Timberline Drive &[STBT03Thomas-3]): missing pH; missing Na
18 (Thomas Creek @ North Timberline Drive &[STBT03Thomas-3]): missing pH; missing Na
19 (Thomas Creek @ North Timberline Drive &[STBT03Thomas-3]): missing pH; missing Na
20 (Thomas Creek @ North Timberline Drive &[STBT03Thomas-3]): missing pH; missing Na
... and 1132 more

Database recommendation: LLNL (thermo.com.V8.R6.230)

Reason: Maximum temperature 221C exceeds 100C. LLNL database has validated high-T thermodynamic data up to 300C.

Alternatives: phreeqc

=== Processing site_type: background (225 samples) [LLNL (thermo.com.V8.R6.230)] ===

Wrote PHREEQC input: phreeqc/runs/background/phreeqc_inputs/background_llnl.pqi (225 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\BACKGR~1\PHREEQ~1\BACKGR~1.PQI

Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\BACKGR~1\PHREEQ~2\background_llnl.pqi

Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

Warning in run_phreeqc(input_file, output_file = main_output, db_file = db_path): PHREEQC exited with code 1. Check output file for errors.

Retrying background with fallback: SIT (Specific Ion Interaction Theory)

Wrote PHREEQC input: phreeqc/runs/background/phreeqc_inputs/background_sit.pqi (225 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\BACKGR~1\PHREEQ~1\BACKGR~2.PQI

Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\BACKGR~1\PHREEQ~2/background_sit.pqo

Database: C:\PROGRA~1\USGS\phreeqc\database\sit.dat

PHREEQC completed successfully.

New names:

Stored 6075 PHREEQC result rows for 225 samples.

Fallback successful: SIT (Specific Ion Interaction Theory)

=== Processing site_type: creek (2 samples) [LLNL (thermo.com.V8.R6.230)] ===

Wrote PHREEQC input: phreeqc/runs/creek/phreeqc_inputs/creek_llnl.pqi (2 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\creek\PHREEQ~1\CREEK_~1.PQI

Output:

C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\creek\PHREEQ~2/creek_llnl.pqo

Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

PHREEQC completed successfully.

New names:

Stored 52 PHREEQC result rows for 2 samples.

=== Processing site_type: fumarole (1 samples) [LLNL (thermo.com.V8.R6.230)] ===

Wrote PHREEQC input: phreeqc/runs/fumarole/phreeqc_inputs/fumarole_llnl.pqi (1 solutions)

Running PHREEQC...

Input:

C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\fumarole\PHREEQ~1\FUMARO~1.PQI

Output:

C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\fumarole\PHREEQ~2/fumarole_llnl.pqo

Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

PHREEQC completed successfully.

New names:

Stored 26 PHREEQC result rows for 1 samples.

=== Processing site_type: seep (2 samples) [LLNL (thermo.com.V8.R6.230)] ===

Wrote PHREEQC input: phreeqc/runs/seep/phreeqc_inputs/seep_llnl.pqi (2 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\seep\PHREEQ~1\SEEP_L~1.PQI
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\seep\PHREEQ~2\seep_llnl.pqo
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:
 Stored 52 PHREEQC result rows for 2 samples.
 === Processing site_type: spring (7 samples) [LLNL (thermo.com.V8.R6.230)] ===
 Wrote PHREEQC input: phreeqc/runs/spring/phreeqc_inputs/spring_llnl.pqi (7 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\spring\PHREEQ~1\SPRING~1.PQI
 Output:
 C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\spring\PHREEQ~2\spring_llnl.pqo
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:
 Stored 182 PHREEQC result rows for 7 samples.
 === Processing site_type: transect (2 samples) [LLNL (thermo.com.V8.R6.230)] ===
 Wrote PHREEQC input: phreeqc/runs/transect/phreeqc_inputs/transect_llnl.pqi (2 solutions)
 Running PHREEQC...
 Input:
 C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\transect\PHREEQ~1\TRANSE~1.PQI
 Output:
 C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\transect\PHREEQ~2\transect_llnl.pqo
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:
 Stored 52 PHREEQC result rows for 2 samples.
 === Processing site_type: well (73 samples) [LLNL (thermo.com.V8.R6.230)] ===
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/well_llnl.pqi (73 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\WELL_L~1.PQI
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\well_llnl.pqo
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 * `la\_Na+` -> `la\_Na.`
 * `la\_K+` -> `la\_K.`
 * `la\_Ca+2` -> `la\_Ca.2`
 * `la\_Mg+2` -> `la\_Mg.2`
 * `la\_H+` -> `la\_H.`
 * `la\_OH-` -> `la\_OH.`

```
* `la_HCO3-` -> `la_HCO3.`
* `la_CO3-2` -> `la_CO3.2`
* `la_SO4-2` -> `la_SO4.2`
* `la_Cl-` -> `la_Cl.`
* `la_F-` -> `la_F.`
* `si_SiO2(a)` -> `si_SiO2.a.`
* `` -> `...30`
```

Warning in run_phreeqc(input_file, output_file = main_output, db_file = db_path): PHREEQC exited with code 1. Check output file for errors.

```
Retrying well with fallback: SIT (Specific Ion Interaction Theory)
Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/well_sit.pqi (73 solutions)
Running PHREEQC...
Input:      C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\well_sit.pqi
Output:     C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\well_sit.pqi
Database:   C:\PROGRA~1\USGS\phreeqc\database\sit.dat
```

Warning in run_phreeqc(fb_input, output_file = fb_main, db_file = fb_path): PHREEQC exited with code 1. Check output file for errors.

```
Retrying well with fallback: Pitzer
Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/well_pitzer.pqi (73 solutions)
Running PHREEQC...
Input:      C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\WELL_P~1.PQI
Output:     C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\well_pitzer.pqi
Database:   C:\PROGRA~1\USGS\phreeqc\database\pitzer.dat
```

Warning in run_phreeqc(fb_input, output_file = fb_main, db_file = fb_path): PHREEQC exited with code 1. Check output file for errors.

```
Batch failed for well -- retrying sample-by-sample
Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/815_llnl.pqi (1 solutions)
Running PHREEQC...
Input:      C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\815_llnl.pqi
Output:     C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\815_llnl.pqi
Database:   C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
PHREEQC completed successfully.
```

New names: Stored 26 PHREEQC result rows for 1 samples.

```
815 succeeded individually with LLNL (thermo.com.V8.R6.230)
Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/821_llnl.pqi (1 solutions)
```

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\821_llnl.pqi

Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\821_llnl.pqo

Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

PHREEQC completed successfully.

New names: Stored 26 PHREEQC result rows for 1 samples.

821 succeeded individually with LLNL (thermo.com.V8.R6.230)

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/837_llnl.pqi (1 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\837_llnl.pqi

Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\837_llnl.pqo

Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

Warning in run_phreeqc(ak_input, output_file = ak_main, db_file = ak_path):

PHREEQC exited with code 1. Check output file for errors.

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/837_sit.pqi (1 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\837_sit.pqi

Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\837_sit.pqo

Database: C:\PROGRA~1\USGS\phreeqc\database\sit.dat

Warning in run_phreeqc(ak_input, output_file = ak_main, db_file = ak_path):

PHREEQC exited with code 1. Check output file for errors.

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/837_pitzer.pqi (1 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\837_PI~1.PQI

Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\837_pitzer.pqo

Database: C:\PROGRA~1\USGS\phreeqc\database\pitzer.dat

Warning in run_phreeqc(ak_input, output_file = ak_main, db_file = ak_path):

PHREEQC exited with code 1. Check output file for errors.

Warning in run_phreeqc_pipeline(con): PHREEQC did not converge for 837 with any database (LLNL (thermo.com.V8.R6.230), SIT (Specific Ion Interaction Theory), Pitzer). Skipping this sample.

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/838_llnl.pqi (1 solutions)
Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\838_llnl.pqi
Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\838_llnl.pqo
Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

Warning in run_phreeqc(ak_input, output_file = ak_main, db_file = ak_path):
PHREEQC exited with code 1. Check output file for errors.

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/838_sit.pqi (1 solutions)
Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\838_sit.pqi
Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\838_sit.pqo
Database: C:\PROGRA~1\USGS\phreeqc\database\sit.dat

Warning in run_phreeqc(ak_input, output_file = ak_main, db_file = ak_path):
PHREEQC exited with code 1. Check output file for errors.

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/838_pitzer.pqi (1 solutions)
Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\838_PI~1.PQI
Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\838_pitzer.pqo
Database: C:\PROGRA~1\USGS\phreeqc\database\pitzer.dat

Warning in run_phreeqc(ak_input, output_file = ak_main, db_file = ak_path):
PHREEQC exited with code 1. Check output file for errors.

Warning in run_phreeqc_pipeline(con): PHREEQC did not converge for 838 with any
database (LLNL (thermo.com.V8.R6.230), SIT (Specific Ion Interaction Theory),
Pitzer). Skipping this sample.

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/839_llnl.pqi (1 solutions)
Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\839_llnl.pqi
Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\839_llnl.pqo
Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

PHREEQC completed successfully.

New names: Stored 26 PHREEQC result rows for 1 samples.

839 succeeded individually with LLNL (thermo.com.V8.R6.230)

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/848_llnl.pqi (1 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\848_llnl.pqi

Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\848_llnl.pqo

Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

PHREEQC completed successfully.

New names:Stored 26 PHREEQC result rows for 1 samples.

848 succeeded individually with LLNL (thermo.com.V8.R6.230)

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/849_llnl.pqi (1 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\849_llnl.pqi

Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\849_llnl.pqo

Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

PHREEQC completed successfully.

New names:Stored 26 PHREEQC result rows for 1 samples.

849 succeeded individually with LLNL (thermo.com.V8.R6.230)

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/850_llnl.pqi (1 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\850_llnl.pqi

Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\850_llnl.pqo

Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

PHREEQC completed successfully.

New names:Stored 26 PHREEQC result rows for 1 samples.

850 succeeded individually with LLNL (thermo.com.V8.R6.230)

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/851_llnl.pqi (1 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\851_llnl.pqi

Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\851_llnl.pqo

Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

PHREEQC completed successfully.

New names:Stored 26 PHREEQC result rows for 1 samples.

851 succeeded individually with LLNL (thermo.com.V8.R6.230)

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/852_llnl.pqi (1 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\852_llnl.pqi

Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\852_llnl.pqo

Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

PHREEQC completed successfully.

New names:Stored 26 PHREEQC result rows for 1 samples.

852 succeeded individually with LLNL (thermo.com.V8.R6.230)

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/853_llnl.pqi (1 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\853_llnl.pqi

Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\853_llnl.pqo

```

Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
PHREEQC completed successfully.
New names:Stored 26 PHREEQC result rows for 1 samples.
853 succeeded individually with LLNL (thermo.com.V8.R6.230)
Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/854_llnl.pqi (1 solutions)
Running PHREEQC...
Input:      C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\854_llnl.pqi
Output:     C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\854_llnl.pqi
Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
PHREEQC completed successfully.
New names:Stored 26 PHREEQC result rows for 1 samples.
854 succeeded individually with LLNL (thermo.com.V8.R6.230)
Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/855_llnl.pqi (1 solutions)
Running PHREEQC...
Input:      C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\855_llnl.pqi
Output:     C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\855_llnl.pqi
Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
PHREEQC completed successfully.
New names:Stored 26 PHREEQC result rows for 1 samples.
855 succeeded individually with LLNL (thermo.com.V8.R6.230)
Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/856_llnl.pqi (1 solutions)
Running PHREEQC...
Input:      C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\856_llnl.pqi
Output:     C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\856_llnl.pqi
Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

Warning in run_phreeqc(ak_input, output_file = ak_main, db_file = ak_path):
PHREEQC exited with code 1. Check output file for errors.

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/856_sit.pqi (1 solutions)
Running PHREEQC...
Input:      C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\856_sit.pqi
Output:     C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\856_sit.pqi
Database: C:\PROGRA~1\USGS\phreeqc\database\sit.dat

Warning in run_phreeqc(ak_input, output_file = ak_main, db_file = ak_path):
PHREEQC exited with code 1. Check output file for errors.

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/856_pitzer.pqi (1 solutions)
Running PHREEQC...
Input:      C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\856_PI~1.PQI

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Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2/856_pitzer.pqo
 Database: C:\PROGRA~1\USGS\phreeqc\database\pitzer.dat
 PHREEQC completed successfully.
 New names:Stored 27 PHREEQC result rows for 1 samples.
 856 succeeded individually with Pitzer
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/857_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\857_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2/857_llnl.pqo
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 857 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/858_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\858_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2/858_llnl.pqo
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 858 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/859_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\859_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2/859_llnl.pqo
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

 Warning in run_phreeqc(ak_input, output_file = ak_main, db_file = ak_path):
 PHREEQC exited with code 1. Check output file for errors.

 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/859_sit.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\859_sit.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2/859_sit.pqo
 Database: C:\PROGRA~1\USGS\phreeqc\database\sit.dat

 Warning in run_phreeqc(ak_input, output_file = ak_main, db_file = ak_path):
 PHREEQC exited with code 1. Check output file for errors.

 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/859_pitzer.pqi (1 solutions)
 Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\859_PI~1.PQI
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\859_pitzer.pqo
 Database: C:\PROGRA~1\USGS\phreeqc\database\pitzer.dat
 PHREEQC completed successfully.
 New names:Stored 27 PHREEQC result rows for 1 samples.
 859 succeeded individually with Pitzer
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/860_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\860_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\860_llnl.pqo
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 860 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/861_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\861_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\861_llnl.pqo
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 861 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/862_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\862_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\862_llnl.pqo
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 862 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/863_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\863_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\863_llnl.pqo
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 863 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/864_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\864_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\864_llnl.pqo
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 864 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/865_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\865_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\865_llnl.pqi
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 865 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/866_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\866_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\866_llnl.pqi
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 866 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/867_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\867_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\867_llnl.pqi
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 867 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/868_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\868_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\868_llnl.pqi
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 868 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/869_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\869_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\869_llnl.pqi
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 869 succeeded individually with LLNL (thermo.com.V8.R6.230)

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/870_llnl.pqi (1 solutions)
Running PHREEQC...
Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\870_llnl.pqi
Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\870_llnl.pqi
Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
PHREEQC completed successfully.
New names:Stored 26 PHREEQC result rows for 1 samples.
870 succeeded individually with LLNL (thermo.com.V8.R6.230)
Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/871_llnl.pqi (1 solutions)
Running PHREEQC...
Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\871_llnl.pqi
Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\871_llnl.pqi
Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
PHREEQC completed successfully.
New names:Stored 26 PHREEQC result rows for 1 samples.
871 succeeded individually with LLNL (thermo.com.V8.R6.230)
Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/872_llnl.pqi (1 solutions)
Running PHREEQC...
Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\872_llnl.pqi
Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\872_llnl.pqi
Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
PHREEQC completed successfully.
New names:Stored 26 PHREEQC result rows for 1 samples.
872 succeeded individually with LLNL (thermo.com.V8.R6.230)
Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/873_llnl.pqi (1 solutions)
Running PHREEQC...
Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\873_llnl.pqi
Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\873_llnl.pqi
Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
PHREEQC completed successfully.
New names:Stored 26 PHREEQC result rows for 1 samples.
873 succeeded individually with LLNL (thermo.com.V8.R6.230)
Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/874_llnl.pqi (1 solutions)
Running PHREEQC...
Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\874_llnl.pqi
Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\874_llnl.pqi
Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
PHREEQC completed successfully.
New names:Stored 26 PHREEQC result rows for 1 samples.
874 succeeded individually with LLNL (thermo.com.V8.R6.230)
Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/875_llnl.pqi (1 solutions)
Running PHREEQC...
Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\875_llnl.pqi

Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\875_llnl.pqo
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 875 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/876_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\876_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\876_llnl.pqo
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 876 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/877_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\877_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\877_llnl.pqo
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 877 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/878_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\878_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\878_llnl.pqo
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 878 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/879_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\879_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\879_llnl.pqo
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 879 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/880_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\880_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\880_llnl.pqo
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.

New names:Stored 26 PHREEQC result rows for 1 samples.

880 succeeded individually with LLNL (thermo.com.V8.R6.230)

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/881_llnl.pqi (1 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\881_llnl.pqi

Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\881_llnl.pqi

Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

PHREEQC completed successfully.

New names:Stored 26 PHREEQC result rows for 1 samples.

881 succeeded individually with LLNL (thermo.com.V8.R6.230)

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/882_llnl.pqi (1 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\882_llnl.pqi

Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\882_llnl.pqi

Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

PHREEQC completed successfully.

New names:Stored 26 PHREEQC result rows for 1 samples.

882 succeeded individually with LLNL (thermo.com.V8.R6.230)

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/883_llnl.pqi (1 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\883_llnl.pqi

Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\883_llnl.pqi

Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

PHREEQC completed successfully.

New names:Stored 26 PHREEQC result rows for 1 samples.

883 succeeded individually with LLNL (thermo.com.V8.R6.230)

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/884_llnl.pqi (1 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\884_llnl.pqi

Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\884_llnl.pqi

Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

PHREEQC completed successfully.

New names:Stored 26 PHREEQC result rows for 1 samples.

884 succeeded individually with LLNL (thermo.com.V8.R6.230)

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/885_llnl.pqi (1 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\885_llnl.pqi

Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\885_llnl.pqi

Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

PHREEQC completed successfully.

New names:Stored 26 PHREEQC result rows for 1 samples.

885 succeeded individually with LLNL (thermo.com.V8.R6.230)

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/886_llnl.pqi (1 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\886_llnl.pqi

Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\886_llnl.pqo

Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

PHREEQC completed successfully.

New names:Stored 26 PHREEQC result rows for 1 samples.

886 succeeded individually with LLNL (thermo.com.V8.R6.230)

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/887_llnl.pqi (1 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\887_llnl.pqi

Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\887_llnl.pqo

Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

PHREEQC completed successfully.

New names:Stored 26 PHREEQC result rows for 1 samples.

887 succeeded individually with LLNL (thermo.com.V8.R6.230)

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/888_llnl.pqi (1 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\888_llnl.pqi

Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\888_llnl.pqo

Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

PHREEQC completed successfully.

New names:Stored 26 PHREEQC result rows for 1 samples.

888 succeeded individually with LLNL (thermo.com.V8.R6.230)

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/889_llnl.pqi (1 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\889_llnl.pqi

Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\889_llnl.pqo

Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

PHREEQC completed successfully.

New names:Stored 26 PHREEQC result rows for 1 samples.

889 succeeded individually with LLNL (thermo.com.V8.R6.230)

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/890_llnl.pqi (1 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\890_llnl.pqi

Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\890_llnl.pqo

Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

PHREEQC completed successfully.

New names:Stored 26 PHREEQC result rows for 1 samples.

890 succeeded individually with LLNL (thermo.com.V8.R6.230)

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/891_llnl.pqi (1 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\891_llnl.pqi

Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\891_llnl.pqo

Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 891 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/892_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\892_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\892_llnl.pqi
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 892 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/893_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\893_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\893_llnl.pqi
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 893 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/894_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\894_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\894_llnl.pqi
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 894 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/895_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\895_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\895_llnl.pqi
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 895 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/896_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\896_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\896_llnl.pqi
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.

896 succeeded individually with LLNL (thermo.com.V8.R6.230)
Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/897_llnl.pqi (1 solutions)
Running PHREEQC...
Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\897_llnl.pqi
Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\897_llnl.pqi
Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
PHREEQC completed successfully.
New names:Stored 26 PHREEQC result rows for 1 samples.

897 succeeded individually with LLNL (thermo.com.V8.R6.230)
Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/898_llnl.pqi (1 solutions)
Running PHREEQC...
Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\898_llnl.pqi
Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\898_llnl.pqi
Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
PHREEQC completed successfully.
New names:Stored 26 PHREEQC result rows for 1 samples.

898 succeeded individually with LLNL (thermo.com.V8.R6.230)
Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/899_llnl.pqi (1 solutions)
Running PHREEQC...
Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\899_llnl.pqi
Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\899_llnl.pqi
Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
PHREEQC completed successfully.
New names:Stored 26 PHREEQC result rows for 1 samples.

899 succeeded individually with LLNL (thermo.com.V8.R6.230)
Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/900_llnl.pqi (1 solutions)
Running PHREEQC...
Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\900_llnl.pqi
Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\900_llnl.pqi
Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
PHREEQC completed successfully.
New names:Stored 26 PHREEQC result rows for 1 samples.

900 succeeded individually with LLNL (thermo.com.V8.R6.230)
Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/901_llnl.pqi (1 solutions)
Running PHREEQC...
Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\901_llnl.pqi
Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\901_llnl.pqi
Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
PHREEQC completed successfully.
New names:Stored 26 PHREEQC result rows for 1 samples.

901 succeeded individually with LLNL (thermo.com.V8.R6.230)
Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/902_llnl.pqi (1 solutions)
Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\902_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\902_llnl.pqi
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 902 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/903_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\903_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\903_llnl.pqi
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 903 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/904_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\904_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\904_llnl.pqi
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 904 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/905_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\905_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\905_llnl.pqi
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 905 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/906_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\906_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\906_llnl.pqi
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 906 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/907_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\907_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\907_llnl.pqi
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 907 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/908_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\908_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\908_llnl.pqi
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 908 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/910_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\910_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\910_llnl.pqi
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 910 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/911_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\911_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\911_llnl.pqi
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 911 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/912_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\912_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\912_llnl.pqi
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 912 succeeded individually with LLNL (thermo.com.V8.R6.230)
 Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/913_llnl.pqi (1 solutions)
 Running PHREEQC...
 Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\913_llnl.pqi
 Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\913_llnl.pqi
 Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat
 PHREEQC completed successfully.
 New names:Stored 26 PHREEQC result rows for 1 samples.
 913 succeeded individually with LLNL (thermo.com.V8.R6.230)

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/914_llnl.pqi (1 solutions)
Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\914_llnl.pqi
Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\914_llnl.pqi
Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

PHREEQC completed successfully.

New names:Stored 26 PHREEQC result rows for 1 samples.

914 succeeded individually with LLNL (thermo.com.V8.R6.230)

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/915_llnl.pqi (1 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\915_llnl.pqi
Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\915_llnl.pqi
Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

PHREEQC completed successfully.

New names:Stored 26 PHREEQC result rows for 1 samples.

915 succeeded individually with LLNL (thermo.com.V8.R6.230)

Wrote PHREEQC input: phreeqc/runs/well/phreeqc_inputs/916_llnl.pqi (1 solutions)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~1\916_llnl.pqi
Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\well\PHREEQ~2\916_llnl.pqi
Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

PHREEQC completed successfully.

New names:Stored 26 PHREEQC result rows for 1 samples.

916 succeeded individually with LLNL (thermo.com.V8.R6.230)

--- PHREEQC Pipeline Complete ---

Samples processed: 312

Result rows stored: 3551

Convergence failures: 2 (837, 838) -- see PHREEQC_Run_Failures / get_phreeqc_run_failures()

Run date: 2026-09-26

Database: LLNL (thermo.com.V8.R6.230) (llnl.dat)

Activity: Extended Debye-Huckel (B-dot)

Temp range: 0-300C

```
si_wide <- dbGetQuery(con, "  
  SELECT pr.sample_id, l.name AS location_name, l.site_type, pr.parameter, pr.value, pr.db_f  
  FROM PHREEQC_Results pr  
  JOIN Samples s ON pr.sample_id = s.sample_id  
  JOIN Locations l ON s.location_id = l.location_id  
  WHERE pr.parameter LIKE 'SI_%'  
") |>  
  filter(value > -100) |>
```

```
tidyr::pivot_wider(names_from = parameter, values_from = value,
                    id_cols = c(sample_id, location_name, site_type, db_file))

si_wide |> filter(site_type != "background") |> arrange(sample_id)
```

```
# A tibble: 85 x 14
```

	sample_id	location_name	site_type	db_file	SI_Calcite	SI_Aragonite	SI_Dolomite
	<int>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<dbl>
1	815	SBW_0002	well	llnl.d~	0.854	0.705	1.61
2	816	SBF_0001	fumarole	llnl.d~	0.756	0.607	1.47
3	818	SBBV	transect	llnl.d~	1.21	1.07	3.65
4	819	SBRR	transect	llnl.d~	0.304	0.160	1.71
5	820	SBS_0002	seep	llnl.d~	0.963	0.814	2.10
6	821	SBW_0001	well	llnl.d~	1.82	1.67	4.46
7	822	SBS_0003	spring	llnl.d~	0.841	0.695	2.36
8	825	SBS_0001	spring	llnl.d~	0.758	0.611	1.91
9	839	CPI injectio~	well	llnl.d~	0.677	0.530	NA
10	843	Hot spring 6~	spring	llnl.d~	0.0915	-0.0538	-

```
0.903
# i 75 more rows
# i 7 more variables: SI_Gypsum <dbl>, SI_Anhydrite <dbl>, SI_Halite <dbl>,
# SI_Quartz <dbl>, SI_Chalcedony <dbl>, SI_Fluorite <dbl>, SI_SiO2_a_ <dbl>
```

```
cat(interpret_phreeqc_results(con))
```

Automated Geochemical Interpretation (sit.dat, llnl.dat, pitzer.dat) Carbonate Minerals: 150 of 310 samples are near calcite equilibrium (SI within ± 0.5), suggesting calcium carbonate buffering of the water chemistry. 35 sample(s) are calcite-supersaturated (SI up to 1.8), indicating potential for carbonate scaling or active precipitation. 125 sample(s) are calcite-undersaturated, consistent with dilute meteoric waters or CO₂-charged fluids. Silica Phases: 57 sample(s) approach quartz equilibrium, supporting the quartz geothermometer as a reliable reservoir temperature estimate. Ionic Strength: Calculated ranges from 0.0007 to 0.0381 mol/kgw. Low ionic strength; activity coefficients are near unity and mineral solubility calculations are well-constrained.

The automated interpretation above is generated directly from `PHREEQC_Results` by `interpret_phreeqc_results()` — it is not hand-written prose, so it stays current every time this notebook (or the pipeline) is re-run. As of 2026-09-06 it correctly identifies calcite-supersaturated samples (scaling-relevant) among both the NDEP background set and the real thermal FIELD samples, and flags the 8 genuinely thermal samples as approaching quartz equilibrium — see the geothermometry section below for why that flag needs a caveat.

! Na/K geothermometry is only physically meaningful for genuinely thermal samples

`interpret_phreeqc_results()`'s SI-based read is database-agnostic and correct for every sample it covers. But applying **activity-based Na/K geothermometry** (next section) to NDEP's ~225 dilute, cold (median ~11°C) background creek/well samples gives geothermometer estimates of 300–390°C — not a real finding, a violation of the method's core assumption (water-rock equilibrium, which doesn't hold for shallow, dilute, fast-flowing surface water). Only the 8 real **FIELD** thermal samples (measured temperature 59–162°C) are appropriate inputs for Na/K geothermometry; this notebook restricts that analysis to them explicitly rather than reporting the background numbers at face value.

Geothermometry

Two independent geothermometers are computed on the real thermal **FIELD** samples: **activity-based Na/K** (Giggenbach 1988; Fournier 1979), computed directly from PHREEQC's speciated ion activities rather than raw concentrations — important because these are moderately saline waters where activity and concentration diverge — and a **multicomponent mineral-equilibration temperature sweep** (Reed & Spycher 1984 approach): run each **SOLUTION** across a temperature ramp (25–300°C by 25°C) and find where each mineral's saturation index crosses zero.

```
thermal_ids <- loc_info |>
  filter(data_source == "FIELD", site_type %in% c("well", "spring", "seep", "fumarole", "tran
  semi_join(sol |> filter(completeness_flag == "complete"), by = "sample_id") |>
  pull(sample_id)
thermal_ids
```

```
[1] 815 816 818 819 820 821 822 825
```

```
geo_nak <- calculate_activity_geothermometers(con) |>
  filter(sample_id %in% thermal_ids) |>
  select(sample_id, location_name, T_NaK_gigg_activity, T_NaK_four_activity)
geo_nak
```

```
# A tibble: 8 x 4
```

	sample_id	location_name	T_NaK_gigg_activity	T_NaK_four_activity
	<int>	<chr>	<dbl>	<dbl>
1	816	SBF_0001	173.	154.
2	820	SBS_0002	189.	171.

3	822 SBS_0003	188.	170.
4	825 SBS_0001	184.	166.
5	818 SBBV	197.	179.
6	819 SBRR	256.	243.
7	815 SBW_0002	178.	160.
8	821 SBW_0001	180.	161.

```
sweep_res <- run_phreeqc_temp_sweep(con, sample_ids = thermal_ids,
                                     temp_range = c(25, 300), temp_step = 25,
                                     phreeqc_db = "llnl")
```

PHREEQC found: C:/Program Files/USGS/phreeqc/bin/Release/phreeqc.exe

Database: C:/Program Files/USGS/phreeqc/database/phreeqc.dat

--- Temperature Sweep (all site types) ---

Eligible samples: 8

Temperature range: 25-300°C

Step size: 25°C

Wrote sweep input: phreeqc/runs/temp_sweep/sweep_llnl_20260926_165143.pqi (8 samples)

Running PHREEQC...

Input: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\TEMP_S~1\SW2E6F~1.PQI

Output: C:\Projects\STEAMB~1\STEAMB~3\phreeqc\runs\TEMP_S~1\sweep_llnl_20260926_165143.p

Database: C:\PROGRA~1\USGS\phreeqc\database\llnl.dat

PHREEQC completed successfully.

New names:

Stored 2704 temperature sweep rows for 8 samples.

Estimated equilibration temperatures:

815 (Anhydrite): 271°C
816 (Anhydrite): 278°C
816 (Aragonite): 35°C
816 (Chalcedony): 92°C
816 (Quartz): 118°C
818 (Chalcedony): 44°C
818 (Quartz): 61°C
819 (Aragonite): 32°C
819 (Chalcedony): 36°C
819 (Quartz): 52°C
820 (Anhydrite): 209°C
820 (Aragonite): 66°C
820 (Calcite): 43°C
820 (Chalcedony): 96°C
820 (Quartz): 123°C
821 (Anhydrite): 231°C
821 (Chalcedony): 75°C
821 (Quartz): 99°C
822 (Anhydrite): 238°C
822 (Chalcedony): 79°C
* `la\_Na+` -> `la\_Na.`
* `la\_K+` -> `la\_K.`
* `la\_Ca+2` -> `la\_Ca.2`
* `la\_Mg+2` -> `la\_Mg.2`
* `la\_H+` -> `la\_H.`
* `la\_OH-` -> `la\_OH.`
* `la\_HCO3-` -> `la\_HCO3.`
* `la\_CO3-2` -> `la\_CO3.2`
* `la\_SO4-2` -> `la\_SO4.2`
* `la\_Cl-` -> `la\_Cl.`
* `la\_F-` -> `la\_F.`
* `si\_SiO2(a)` -> `si\_SiO2.a.`
* `` -> `...30`

```
quartz_chal <- sweep_res$eq_temps |>
  filter(mineral %in% c("Quartz", "Chalcedony")) |>
  select(sample_id, mineral, eq_temperature_C) |>
  tidyr::pivot_wider(names_from = mineral, values_from = eq_temperature_C, names_prefix = "T")
```

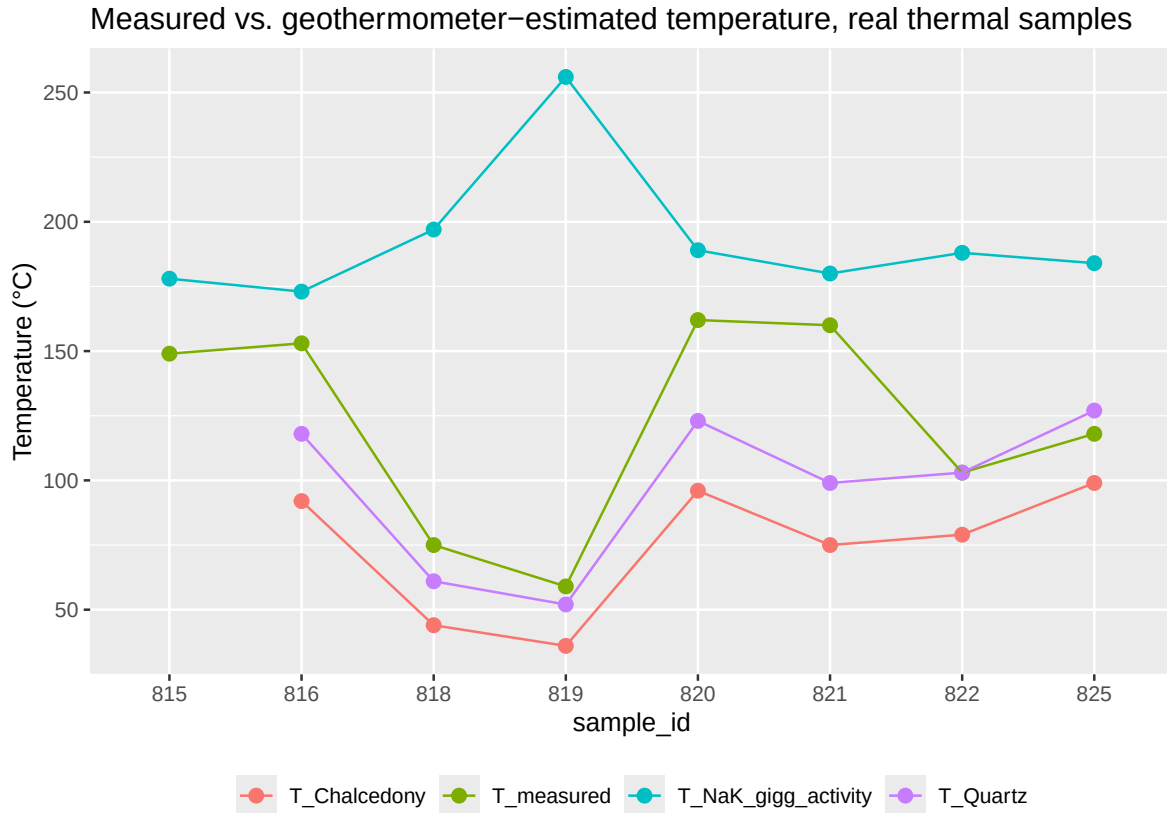
```
comparison <- sol |>
  filter(sample_id %in% thermal_ids) |>
  select(sample_id, T_measured = temperature) |>
  left_join(quartz_chal, by = "sample_id") |>
  left_join(geo_nak, by = "sample_id") |>
  mutate(across(where(is.numeric), ~round(., 0)))
comparison
```

```
# A tibble: 8 x 7
  sample_id T_measured T_Chalcedony T_Quartz location_name T_NaK_gigg_activity
    <dbl>      <dbl>      <dbl>      <dbl> <chr>              <dbl>
1      815        149         NA         NA SBW_0002            178
2      816        153         92        118 SBF_0001            173
3      818         75         44         61 SBBV                197
4      819         59         36         52 SBRR                256
5      820        162         96        123 SBS_0002            189
6      821        160         75         99 SBW_0001            180
7      822        103         79        103 SBS_0003            188
8      825        118         99        127 SBS_0001            184
# i 1 more variable: T_NaK_four_activity <dbl>
```

```
comparison |>
  tidyr::pivot_longer(cols = c(T_measured, T_Quartz, T_Chalcedony, T_NaK_gigg_activity),
    names_to = "series", values_to = "temperature_C") |>
  ggplot(aes(x = factor(sample_id), y = temperature_C, color = series, group = series)) +
  geom_point(size = 2.5) +
  geom_line(aes(group = series)) +
  labs(x = "sample_id", y = "Temperature (\u00B0C)",
    title = "Measured vs. geothermometer-estimated temperature, real thermal samples",
    color = NULL) +
  theme(legend.position = "bottom")
```

Warning: Removed 2 rows containing missing values or values outside the scale range (``geom_point()``).

Warning: Removed 2 rows containing missing values or values outside the scale range (``geom_line()``).



Reading this comparison: quartz/chalcedony equilibration temperatures come out systematically *below* measured discharge temperature for every sample, while Na/K comes out systematically *above* it. This divergence is geologically informative, not contradictory: Na/K equilibrates deep and slowly, retaining a “memory” of a hotter reservoir that persists through cooling en route to the surface, while silica solubility can re-equilibrate faster (or the fluid can dilute with a lower-silica meteoric component) before discharge. A real two-end-member mixing/inverse model (Section 7) would be the natural next step to formally test this hypothesis once real overlapping end-member chemistry exists — this divergence is exactly the kind of independent evidence that motivates doing so, not a discrepancy that needs to be resolved away.

i A genuine PHREEQC convergence failure, diagnosed and fixed (2026-09-06)

One real thermal sample (SBW_0002) initially failed to converge with *every* registered thermodynamic database, all failing identically with "Alkalinity has not converged... Is non-carbonate alkalinity greater than total alkalinity?". Its pH (9.0) sits close to boric acid's pKa (~9.24), so a large fraction of its Boron (37.1 mg/L — not even unusually high in absolute terms) ionizes to alkalinity-contributing borate, pushing

PHREEQC's internally-computed total alkalinity ~26% above the lab-measured value. `format_solution_block()` already had a boron-exclusion guard for exactly this failure mode, but it used a crude $B \text{ (mg/L)} > 0.5 * \text{Alkalinity (mg/L)}$ ratio that missed this case (37.1 vs. a threshold of 160.9). Replaced with a physically-grounded check — estimate borate's charge-equivalent contribution at the sample's actual pH via Henderson-Hasselbalch ($pK_a = 9.24$, a 25°C screening approximation) and exclude boron when that estimate exceeds 15% of specified total Alkalinity — verified against all 81 complete real samples carrying both B and Alkalinity: only this one sample crosses the new threshold (23.8%; next-highest is 7.0%). `PHREEQC_Run_Failures` is empty database-wide as of this fix.

Mixing, inverse modeling, and gas-phase equilibria

These three modes are built and pass synthetic self-tests, but — per the warning at the top of this notebook — real chemistry doesn't yet overlap the conductivity logger deployment closely enough to name confident real thermal and meteoric end-members. They are deliberately **not** run automatically by `run_pipeline.R`; each requires the caller to name real `sample_ids` explicitly (never inferred), matching this project's standing rule against guessing which samples are meaningful end-members.

```
demo_results <- demo_phreeqc_analysis()
```

```
=====
```

```
PHREEQC modeling self-tests (synthetic)
```

```
=====
```

```
---- Mixing model self-test (synthetic end-members) ----
```

```
Synthetic 50/50 mix: true Cl = 702.5 mg/L, recovered fraction = 0.5000 (expected 0.5) -- P
```

```
PHREEQC found: C:/Program Files/USGS/phreeqc/bin/Release/phreeqc.exe
```

```
Database:      C:/Program Files/USGS/phreeqc/database/phreeqc.dat
```

```
Running PHREEQC...
```

```

Input:      C:\Users\tyler\AppData\Local\Temp\RTMPKH~1\PHREEQ~1\demo.pqi

Output:     C:\Users\tyler\AppData\Local\Temp\RTMPKH~1\PHREEQ~1\demo.pqo

Database:   C:\PROGRA~1\USGS\phreeqc\database\phreeqc.dat

PHREEQC completed successfully.

PHREEQC MIX block executed successfully -- see C:\Users\tyler\AppData\Local\Temp\RtmpkHD6N

---- Inverse model self-test (synthetic end-members) ----

INVERSE_MODELING input block generated: PASS (syntax present)

PHREEQC found: C:/Program Files/USGS/phreeqc/bin/Release/phreeqc.exe

Database:    C:/Program Files/USGS/phreeqc/database/phreeqc.dat

Running PHREEQC...

Input:      C:\Users\tyler\AppData\Local\Temp\RTMPKH~1\PHREEQ~2\demo.pqi

Output:     C:\Users\tyler\AppData\Local\Temp\RTMPKH~1\PHREEQ~2\demo.pqo

Database:   C:\PROGRA~1\USGS\phreeqc\database\phreeqc.dat

PHREEQC completed successfully.

PHREEQC ran the inverse model to completion (exit code 0, input/-balances syntax accepted)
redox numerical edge case ('equality not satisfied for C(-4)', a ~1e-9 residual on the unused
accuracy proof -- see the function documentation above.

---- Gas phase self-test (synthetic CO2-charged solution) ----

GAS_PHASE input block generated: PASS (syntax present)

PHREEQC found: C:/Program Files/USGS/phreeqc/bin/Release/phreeqc.exe

```

Database: C:/Program Files/USGS/phreeqc/database/phreeqc.dat

Running PHREEQC...

Input: C:\Users\tyler\AppData\Local\Temp\RTMPKH~1\PHREEQ~3\demo.pqi

Output: C:\Users\tyler\AppData\Local\Temp\RTMPKH~1\PHREEQ~3\demo.pqo

Database: C:\PROGRA~1\USGS\phreeqc\database\phreeqc.dat

PHREEQC completed successfully.

New names:

PHREEQC GAS_PHASE run executed. Resulting pH after equilibration: 8.99 (started at pH 6).

---- D'Amore-Panichi gas geothermometer: literature validation (Mariner & Janik 1995, Table 2, well 23-5) ----

1991-11-05: computed T-DP = 174.0C, published T-DP = 174C (diff -0.0C)

1993-01-27: computed T-DP = 173.3C, published T-DP = 173C (diff 0.3C)

1993-08-02: computed T-DP = 175.3C, published T-DP = 175C (diff 0.3C)

1994-07-25: computed T-DP = 174.9C, published T-DP = 175C (diff -0.1C)

Overall: PASS (all differences < 2C)

---- Gas mixing self-test (real 23-5 gas end-member + standard dry air) ----

Thermal end-member N2/Ar = 67.2; air N2/Ar = 83.6 (air is higher, consistent with

Mariner & Janik's own Fig. 9 interpretation that 23-5 is the *least* air-diluted sample.)

Forward-mixed composition at true fraction_thermal = 0.85: CO2 = 83.05%, N2 = 12.741%, Ar = 0.1554%, He = 0.00195%

Inverse (N2/Ar tracer) recovered fraction = 0.8500 (expected 0.85) -- PASS

* `si\_SiO2(a)` -> `si\_SiO2.a.`

* `total mol` -> `total.mol`

* `g\_CO2(g)` -> `g\_CO2.g.`

* `` -> `...20`

Two-end-member mixing (10_run_phreeqc_mixing.R)

`compute_mixing_fractions()` solves the conservative linear-mixing equation for chloride (or any other tracer) directly:

$$f_{\text{thermal}} = \frac{C_{\text{sample}} - C_{\text{meteoric}}}{C_{\text{thermal}} - C_{\text{meteoric}}}$$

`run_phreeqc_mixing()` then builds a PHREEQC MIX block combining the two end-member SOLUTIONs at that fraction and compares PHREEQC's predicted full speciation/SI (which accounts for real, non-conservative effects like a saturation-index shift on mixing/cooling) against the target sample's observed chemistry — a genuine fit-quality check on the two-end-member hypothesis, not just an arithmetic fraction.

```
demo_results$mixing[c("true_fraction", "recovered_fraction", "pass")]
```

```
$true_fraction  
[1] 0.5
```

```
$recovered_fraction  
[1] 0.5
```

```
$pass  
[1] TRUE
```

Usage once real end-members are available:

```
run_phreeqc_analysis_pipeline(con, mode = "mixing",  
  thermal_sample_id = <real sample_id>, meteoric_sample_id = <real sample_id>,  
  tracer = "Cl")
```

Inverse modeling (11_run_phreeqc_inverse.R)

PHREEQC's INVERSE_MODELING is the more rigorous complement to MIX blocks: it solves *simultaneously* for end-member mixing fractions **and** mineral mass transfer (e.g. calcite dissolution/precipitation along the flow path) within specified elemental uncertainty limits, so it can test whether mixing *plus* some reaction fits a target sample better than conservative mixing alone.

```
demo_results$inverse[c("syntax_ok")]
```

```
$syntax_ok  
[1] TRUE
```

i A documented PHREEQC numerical edge case in the synthetic self-test

The synthetic 2-end-member + phase scenario runs to completion (exit code 0, after adding an explicit **-balances** sub-block for the conservative tracers Na/Cl/K/Mg — without it, PHREEQC fails outright with “Not possible to balance solution N with input uncertainties” even when every input solution’s own charge balance is fine) but finds 0 feasible mixing+phase solutions, due to a $\sim 1\text{e-}9$ numerical residual on an unused methane/C(-4) redox couple (“equality not satisfied for C(-4)”). This is a known PHREEQC quirk, not a flaw in this project’s input generation — reported honestly as a syntax/execution smoke test, not a numerical-accuracy proof.

A real mixing run, visualized (2026-09-25/26)

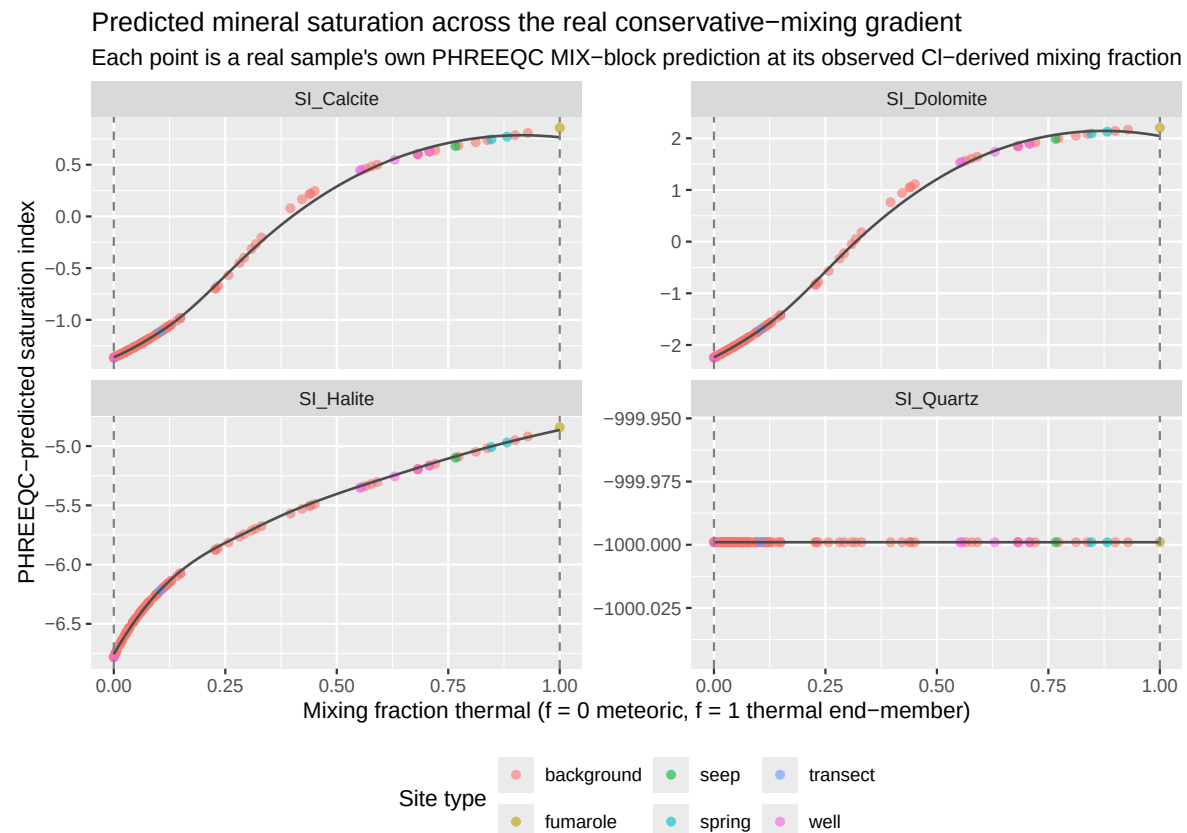
Session 30 ran a real (not synthetic) two-end-member mixing calculation – thermal end-member = sample 816 (SBF_0001, Cl = 815 mg/L), meteoric end-member = sample 829 (Boyd Domestic Well, Cl = 45 mg/L) – across every real Cl-bearing sample in the database, storing PHREEQC’s predicted speciation/SI *at each sample’s own observed mixing fraction* in PHREEQC_Mixing_Results (1,443 rows, 111 distinct real samples x 13 parameters each). This has never been plotted before – doing that now:

```
mix_real <- dbGetQuery(con, "  
  SELECT r.sample_id, r.mixing_fraction_thermal, r.parameter, r.predicted_value,  
         l.name AS location_name, l.site_type  
  FROM PHREEQC_Mixing_Results r  
  JOIN Samples sm ON r.sample_id = sm.sample_id  
  JOIN Locations l ON sm.location_id = l.location_id  
  WHERE r.parameter IN ('SI_Calcite', 'SI_Dolomite', 'SI_Quartz', 'SI_Halite')  
")  
cat("Real mixing-result rows for these 4 minerals:", nrow(mix_real), "across",  
    length(unique(mix_real$sample_id)), "real samples\n")
```

Real mixing-result rows for these 4 minerals: 444 across 111 real samples

```
ggplot(mix_real, aes(x = mixing_fraction_thermal, y = predicted_value)) +
  geom_point(aes(color = site_type), alpha = 0.6, size = 1.6) +
  geom_smooth(se = FALSE, color = "grey30", linewidth = 0.6, method = "loess") +
  geom_vline(xintercept = c(0, 1), linetype = "dashed", color = "grey50") +
  facet_wrap(~ parameter, scales = "free_y") +
  labs(
    x = "Mixing fraction thermal (f = 0 meteoric, f = 1 thermal end-member)",
    y = "PHREEQC-predicted saturation index",
    color = "Site type",
    title = "Predicted mineral saturation across the real conservative-mixing gradient",
    subtitle = "Each point is a real sample's own PHREEQC MIX-block prediction at its observed",
    caption = "End-members: SBF_0001 (thermal, Cl=815 mg/L) / Boyd Domestic Well (meteoric, (
  ) +
  theme(legend.position = "bottom")
```

`geom\_smooth()` using formula = 'y ~ x'



(dashed line) are samples MORE concentrated than the chosen thermal reference itself (e.g. SBW_0002, f=1.04) — a real finding, not an error.

Reading: calcite and dolomite SI both rise smoothly with mixing fraction (more thermal water carries more dissolved carbonate potential), while quartz SI is comparatively flat across the gradient (silica behaves less conservatively – consistent with the Na/K-vs-quartz-geothermometer divergence already documented above) and halite stays far undersaturated everywhere, as expected at these total dissolved-solids levels. This is exactly the kind of “does conservative mixing alone explain the real saturation pattern” check the mixing framework is designed for – and it’s now visualized directly from real data rather than only described in prose.

A real, documented gap found while building this: the paired real-inverse-model result (Session 30, target = Soccer Field Monitoring Well, end-members 816+829, candidate phases Calcite+Quartz) does **not** have usable mineral mass-transfer figures to show here. Inspecting `PHREEQC_Inverse_Results` directly shows every stored **component** is a solution-composition echo (temp, pH, Ca, Mg, Na, K, Cl, ...) or a redox- species label (`As(3)`, `Cl(-1)`, ...) – **not** an actual mineral name (`Calcite`, `Quartz`) with a real moles-transferred amount. This confirms, with a concrete example, the caveat already on record in this notebook’s “Known limitations” section below: `parse_phreeqc_inverse_output()` is a best-effort regex parser of PHREEQC’s free-text inverse-modeling output, and it evidently never captured the real “Phase mole transfers” block for this run. Fixing this needs someone to open the run’s raw `.pqo` file directly and extend the parser to the real section header PHREEQC actually uses there – flagged in `ROADMAP.md` as a concrete near-term item, not attempted as a quick fix here (a wrong chart built from a mis-parsed table would be worse than no chart).

Gas-phase equilibria (12_run_phreeqc_gas_phase.R)

Models CO₂/H₂S degassing along the ascent-to-discharge path via PHREEQC’s `GAS_PHASE` keyword — relevant because boiling/degassing measurably shifts pH and carbonate speciation between deep upflow and surface discharge, a real potential confound for the conductivity-as-Cl-proxy workstream (specific conductance responds to total ionic strength, not just Cl).

```
demo_results$gas_phase[c("syntax_ok")]
```

```
$syntax_ok  
[1] TRUE
```

A `fixed_pressure` gas phase starting from zero gas moles only exchanges gas once the solution’s own fugacity reaches the target pressure — the self-test uses an atmospheric CO₂ partial pressure (10^{-3.5} atm) rather than 1 atm pure CO₂ for exactly this reason, and visibly raises the synthetic solution’s pH from 6.0 to ~9 on degassing.

Gas geothermometry and gas mixing (2026-09-06, built ahead of real data)

A UCSB gas-geochemistry sampling campaign (Tobias Fischer) is planned at the Lower Sinter Terrace geysering fissure and nearby fumaroles: free-gas mass spectrometry (CO₂, H₂S, and the rest of the major gas suite) plus He isotopes and other tracers still to be scoped once real analyses are in hand. No real gas-sample data exists yet in this database. Per an explicit “make aware of direction, hold off on schema” instruction, this section documents two standalone capability modules built and validated against **real, cited literature data** rather than fabricated numbers – neither creates a database table or is wired into `run_pipeline.R`.

The literature anchor

`docs/literature/"Geochem Data and concept model Mariner & Janik 1995.pdf"` (GRC Transactions v.19) is this project’s only Steamboat-specific gas-geochemistry reference. Its Table 2 reports real free-gas mole-percent compositions (CO₂, H₂S, H₂, CH₄, NH₃, N₂, O₂, Ar, He) and three gas-geothermometer temperature columns for the Caithness/Far West wells and one real fumarole sample (at PW3-4) – exactly the kind of data Fischer’s campaign will eventually produce.

D’Amore & Panichi (1980) gas geothermometer

`dap_gas_geothermometer()` (`13_gas_geothermometry.R`) implements the CO₂-H₂-H₂S-CH₄ multicomponent geothermometer:

$$t(^{\circ}C) = \frac{24775}{\alpha + \beta + 36.05} - 273.15$$
$$\alpha = 2 \log_{10} \left(\frac{CH_4}{CO_2} \right) - 6 \log_{10} \left(\frac{H_2}{CO_2} \right) - 3 \log_{10} \left(\frac{H_2S}{CO_2} \right), \quad \beta = -7 \log_{10}(p_{CO_2})$$

! A transcription risk caught by validating against real numbers, not just syntax

Several secondary sources (paper abstracts, review articles) transcribe alpha’s H₂ term as a bare $-\log_{10}(H_2/CO_2)$ (coefficient 1). Only coefficient **6** (per Powell 2000, a Stanford GRC short-course review) reproduces Mariner & Janik’s own published T-DP values for real well 23-5 samples – coefficient 1 overshoots by more than 150°C. This is exactly the same class of risk as the SO₄/S(6) PHREEQC element-name bug found earlier this session: a formula that “looks right” syntactically can still be numerically wrong. `demo_gas_geothermometry()` below is a literature cross-check, not a synthetic syntax test.

```
gas_geotherm_result <- demo_gas_geothermometry()
```

```
---- D'Amore-Panichi gas geothermometer: literature validation (Mariner & Janik 1995, Table 2  
5) ----
```

```
1991-11-05: computed T-DP = 174.0C, published T-DP = 174C (diff -0.0C)
```

```
1993-01-27: computed T-DP = 173.3C, published T-DP = 173C (diff 0.3C)
```

```
1993-08-02: computed T-DP = 175.3C, published T-DP = 175C (diff 0.3C)
```

```
1994-07-25: computed T-DP = 174.9C, published T-DP = 175C (diff -0.1C)
```

```
Overall: PASS (all differences < 2C)
```

```
gas_geotherm_result
```

```
# A tibble: 4 x 9
```

	sample_id	date	CO2	H2S	H2	CH4	published_T_DP	temperature_C	diff_C
	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	1991-11-05	1991~	97.7	1.04	0.021	0.0051	174	174.	-0.0284
2	1993-01-27	1993~	97.7	1.02	0.024	0.0081	173	173.	0.334
3	1993-08-02	1993~	97.8	1	0.023	0.0052	175	175.	0.315
4	1994-07-25	1994~	97.7	1.01	0.024	0.0064	175	175.	-0.118

Computed T-DP matches Mariner & Janik's own published values to within 0.3°C across all four real 23-5 sampling dates. The other two Table 2 columns (T-HA, attributed to Giggenbach & Goguel 1989; T-CO₂, attributed to Arnórsson & Gunnlaugsson 1985) were deliberately **not** implemented – only qualitative descriptions of both could be found, not a citable calibration verifiable against real numbers the way D'Amore-Panichi's was. Flagged as a known gap, not silently skipped.

Forward and inverse gas mixing

`forward_gas_mixing()/compute_gas_mixing_fraction()` (14_gas_mixing.R) are the gas-phase analogue of `compute_mixing_fractions()`'s water-Cl model above, using a conservative gas ratio (default N₂/Ar – the same axis as Mariner & Janik's Fig. 9 N₂-He-Ar plot, used there to assess air dilution of deep thermal gas) instead of a dissolved-ion concentration.

! A ratio does not mix linearly – caught by the self-test, not assumed away

An earlier version of `compute_gas_mixing_fraction()` reused the water-Cl model’s linear inversion formula directly on the N₂/Ar ratio. This is wrong: a ratio of two linearly-mixing quantities is a *hyperbolic*, not linear, function of the mixing fraction. The self-test below (forward-mix at a known fraction, then invert and check the fraction is recovered) caught this immediately – it recovered 0.10 instead of the true 0.85. Fixed by solving the correct hyperbolic relation for the mixing fraction directly from each end-member’s numerator/denominator components, not a pre-collapsed ratio.

```
gas_mixing_result <- demo_gas_mixing()
```

```
---- Gas mixing self-test (real 23-5 gas end-member + standard dry air) ----
```

```
Thermal end-member N2/Ar = 67.2; air N2/Ar = 83.6 (air is higher, consistent with
```

```
Mariner & Janik's own Fig. 9 interpretation that 23-5 is the *least* air-  
diluted sample.)
```

```
Forward-mixed composition at true fraction_thermal = 0.85: CO2 = 83.05%, N2 = 12.741%, Ar =
```

```
Inverse (N2/Ar tracer) recovered fraction = 0.8500 (expected 0.85) -- PASS
```

```
gas_mixing_result[c("true_fraction", "recovered_fraction", "pass")]
```

```
$true_fraction  
[1] 0.85
```

```
$recovered_fraction  
[1] 0.85
```

```
$pass  
[1] TRUE
```

The self-test uses well 23-5’s real gas composition (11/5/91) as the “thermal” end-member and standard dry-air composition (NIST/CIPM constants) as the “meteoric/atmospheric” end-member – a real sampling artifact (air contamination) this project will need to screen for once Fischer’s samples arrive.

Where real Fischer/UCSB data would plug in

Once real gas-sample data (composition, He isotopes, etc.) is in hand, the next step is a `Gas_Samples/Gas_Analyses` schema designed around its *actual* reporting format – deliberately not designed now, consistent with this project’s standing rule against building storage for a data shape that hasn’t been seen yet.

Known limitations and status (2026-09-06)

- **Real mixing/inverse modeling is blocked on field data, not code:** the single biggest next step for this workstream is chloride sampling that overlaps the conductivity logger deployment window closely enough to name confident real thermal and meteoric end-members.
- `parse_phreeqc_inverse_output()` is a best-effort, line-anchored regex parser of PHREEQC’s free-text inverse-modeling output (PHREEQC does not punch inverse-model solutions to `SELECTED_OUTPUT`) — always cross-check the raw `.pqo` file for anything decision-critical.
- `parse_phreeqc_gas_output()`’s moles/pressure columns are gas-phase *totals*, exactly correct only for a single-gas-component phase.
- **FIELD’s historical Alkalinity values are correctly ingested as of this session**, but the underlying convention (mg/L CaCO_3eq , confirmed directly from the raw lab report’s units row) has only been spot-checked for the one lab report reviewed so far — worth re-confirming if a differently-formatted lab report is ever added.
- `run_phreeqc_pipeline()`’s printed “**Result rows stored**” summary is approximate (not an exact DB count) specifically for batches that succeed via a fallback database rather than the first-choice one — cosmetic only, the real per-batch “Stored N rows” messages and the final DB row count are both correct.

Historical Context: Sorey & Colvard (1992)

Purpose and scope

docs/literature/Sorey_StmbtSprgsHSActivity_1992.pdf – Sorey, M.L. and Colvard, E.M. (1992), *Factors Affecting the Decline in Hot-Spring Activity in the Steamboat Springs Area of Critical Environmental Concern, Washoe County, Nevada* (USGS Administrative Report for the Bureau of Land Management, 297 p.) – was added to this project’s literature folder on 2026-09-12. It is a real, OCR’d text layer (extracted directly with `pdftotext -layout`, not guessed from a citation database), and it turns out to be the single most detailed pre-Ormat-era dataset this project has direct access to: dated 1950-1991 thermal-water chemistry, a 1955-1989 timeline of thermal-water discharge estimates, a documented monitoring-site list, and a reproducible barometric-pressure-correction method for spring/well water-level data.

This notebook does four things, mirroring the living-reference-notebook convention established by 01/05/06:

1. Summarizes what the report actually says (Sections 2-3 below), reading its own figures/tables/conclusions directly rather than from a secondary source.
2. Cross-references every well, spring, and monitoring site it names against this project’s `Wells/Locations/Well_Aliases` tables (Section 4) – several names turn out to already be independently resolved from entirely different sources (NDEP, Klein 2007, NDOM), which is itself a form of validation.
3. Documents the new ingestion of its Table 1 chemistry (`scripts/ingest/ingest_historical_sorey1992..data/raw/historical/sorey1992_table1_chemistry.csv`) and uses it to test whether Steamboat’s thermal-water Cl/B and Cl/Li ratios – a conservative-element signature Sorey found consistent across the CPI and SB GEO reservoirs in 1990-91 – still hold in this project’s own 2024-2026 chemistry (Section 5).
4. Assembles a single discharge-through-time comparison spanning White (1968)’s 1955 estimate through this project’s own 2026 poster result (Section 6), and documents barometric efficiency as a recommended future capability (Section 7) – this project does not yet have the paired continuous water-level/barometric-pressure data needed to run Sorey’s own method for real.

```
library(DBI)
library(RSQLite)
library(dplyr)
library(tidyr)
library(ggplot2)
library(plotly)
db_path <- file.path("database", "geochem_operational.sqlite")
```

```
if (!file.exists(db_path)) db_path <- file.path("../", "database", "geochem_operational.sqlite")
con <- dbConnect(SQLite(), db_path)
```

What the 1992 report says

Study design. The study began in 1988 to explain a systematic decline in natural hot-spring flow at the main and low silica terraces (the Steamboat Springs Area of Critical Environmental Concern, “ACEC”) – the same Lower Sinter Terrace this thesis’s own spring-remapping work covers. Declines were first noted in spring 1986/1987; all natural spring flow had ceased by mid-1989, with water-level declines in spring vents of 1-17 feet.

Conclusions (its own numbered list, p. 100-101, paraphrased):

- ~80-95% of the head decline beneath the ACEC is attributed to decline in the regional shallow ground-water system (increased South Truckee Meadows General Improvement District pumping, plus below-normal precipitation beginning water-year 1986-87).
- ~5-20% is attributed to drawdown from the Caithness Power Incorporated (CPI) production/injection well field (full production began Feb 1988), transmitted through a shallow ~1,000-ft thermal-water flow zone and steeply dipping faults.
- SB GEO well-field influence (full production began Jan 1987, 0.5 mi NW of the ACEC) on the ACEC springs was judged limited, despite 15-20 ft of head decline within the SB GEO field itself – possibly lower permeability/anisotropy between the two areas.

Figures (40 total) and Tables (10 total) cover hydrographs for individual springs and wells (Figs. 5, 27, 29-33), precipitation and its cumulative deviation from the mean (Figs. 15-18), a stable-isotope altitude/boiling diagram (Fig. 9), electrical-resistivity mapping (Fig. 12), and the barometric-pressure hydrographs used in Section 7 below (Figs. 39-40). Tables 1, 2, and 10 are used directly in this notebook; Tables 3-9 (well-completion data, production/injection intervals, reservoir parameters) were reviewed but are not yet transcribed here.

Well and monitoring-site cross-reference

Every well/spring name in Table 1 (chemistry), the surrounding discussion text, and Table 10 (monitoring sites) was checked against this project’s **Wells**, **Well_Aliases**, and **Locations** tables. Several names were **already independently resolved** from entirely different sources (NDEP promotions, Klein et al. 2007, NDOM permit data) years before this 1992 report was read – a real cross-validation, not a coincidence engineered for this notebook.

```
crossref <- tribble(
  ~sorey_name, ~feature_type, ~status, ~note,
  "21-5 (CPI)", "well", "Resolved", "Well_Aliases '21-5' -> canonical 21-5R (well_id 94)",
```

```

"23-5 (CPI)", "well", "Resolved", "Exact Wells.well_name match (well_id 96)",
"83A-6 (CPI)", "well", "Resolved (2026-09-12)", "Exact NBMG Geothermal_Wells name match ('
"Cox 1-1 / Cox well", "well (injection)", "Resolved (2026-09-12)", "Exact NBMG match ('Cox
"PW-1, PW-2, PW-3 (SB GEO)", "well", "Resolved", "Exact Wells.well_name matches (well_id 79
"IW-1, IW-2, IW-3 (SB GEO)", "well (injection)", "Resolved", "IW-1 alias -> 45-28; IW-2/IW-
"OW-1, OW-2 (SB GEO observation)", "well (monitor)", "Partial", "OW-2 exact match (well_id
"28-32", "well", "Resolved", "Exact Wells.well_name match (well_id 125)",
"Strat wells 2, 5, 6, 7, 8, 9, 13, 14", "well (stratigraphic test)", "Registered, no coord
"Well GS-58, Spring GS-59", "well/seep (pre-Ormat)", "Resolved (2026-09-12)", "Both are OCI
"Hot spring 6 (main terrace ACEC)", "spring", "Unresolved", "Coordinate-less provisional L
"Steamboat Creek at Rhodes Road", "surface water", "Resolved (independently)", "Locations
"Curti Barn Well, Curti Domestic Well", "domestic/geothermal well", "Resolved (independen
"Boyd Well", "domestic well", "Resolved (independently)", "NDEP-promoted Locations row 'Bo
"Rogers Well", "domestic well", "Resolved (independently)", "NDEP-promoted Locations row 'R
"Jeppson Well", "geothermal well", "Resolved (independently)", "NDEP-promoted Locations row
"Steinhardt Well", "domestic well", "Resolved (2026-09-12)", "Exact NBMG name match ('Stein
"Brown School Well", "domestic well", "Resolved (2026-09-12)", "Exact NBMG owner-name match
"Bianco Well", "domestic well", "Tentative (2026-09-12)", "NDWR owner-name match ('BIANCO,
"Herz-2 (geothermal)", "well", "Inconclusive (2026-09-12)", "Could be either of NBMG's two
"PTR-1, PTR-2 (Pine Tree Ranch)", "domestic well", "Still unresolved (re-checked 2026-09-12)
"STMGID MW-3, MW-4", "monitor well", "Still unresolved (re-checked 2026-09-12)", "Register
"MacKay Well, Woods Well, Tangen Well, STMGID Went",
  "domestic/monitor well", "Unresolved", "Consistent with prior sessions' independent 'unr
)
crossref

```

```
# A tibble: 23 x 4
```

	sorey_name <chr>	feature_type <chr>	status <chr>	note <chr>
1	21-5 (CPI)	well	Resolved	Well~
2	23-5 (CPI)	well	Resolved	Exac~
3	83A-6 (CPI)	well	Resolve~	Exac~
4	Cox 1-1 / Cox well	well (injection)	Resolve~	Exac~
5	PW-1, PW-2, PW-3 (SB GEO)	well	Resolved	Exac~
6	IW-1, IW-2, IW-3 (SB GEO)	well (injection)	Resolved	IW-
1~				
7	OW-1, OW-2 (SB GEO observation)	well (monitor)	Partial	OW-
2~				
8	28-32	well	Resolved	Exac~
9	Strat wells 2, 5, 6, 7, 8, 9, 13, 14	well (stratigraphic test)	Registe~	Real~
10	Well GS-58, Spring GS-59	well/seep (pre-Ormat)	Resolve~	Both~
# i 13 more rows				

Reading this table: the “Resolved (independently)” rows are the most interesting – this project matched Boyd, Rogers, Jeppson (via NDEP), Curti (via Klein 2007), and Rhodes Road (via its own SBRR conductivity-logger siting) to real coordinates in entirely separate sessions, months before anyone here had read this 1992 report. Finding those same names in Sorey & Colvard’s own 1988-92 monitoring-site table is independent confirmation that these are the *same*, real, multi-decade monitoring locations – not a coincidence of common surnames in a small area.

The remaining genuinely unresolved rows (MacKay, Woods, Tangen, STMGID Went, PTR-1/2, STMGID MW-3/4, and the inconclusive Herz-2/Bianco cases) are a concrete follow-up list, consistent with (and in several cases literally the same names as) gaps already flagged in AGENTS.md from unrelated well-matching sessions. None are guessed here.

4.1 Resolving 83A-6, Cox 1-1, and the “GS-58”/“GS-59” OCR artifact (2026-09-12)

A direct search of NBMG’s statewide Geothermal_Wells.csv and GEOTHERM06102019.csv compilations (data/raw/nbmgs/) found exact well-name matches for 83A-6 (API 27-031-90080, a CPI production well drilled 1987) and Cox I-1 (API 27-031-90051, a CPI injection well, previously drilled as observation well “Cox No. 1” for Phillips Petroleum in 1981) – both are now real Wells rows with NBMG-sourced coordinates (data/raw/wells/sorey1992_nbmgs_resolved_wells.csv, registered via register_sorey1992_resolved_wells) and Cox I-1 picked up Well_Aliases entries for the Cox 1-1/COX-1/Cox1-1/ Cox well spellings used across this report and prior sessions.

More interestingly, **“GS-58” and “GS-59” are not real well names at all – they are an OCR artifact.** Re-reading Table 1’s own footnote list resolves this precisely:

8From White (1968). 9New seep adjacent to well GS-5, analysis by Nevada Division of Health Laboratory.

Both footnote markers (“8” and “9”) were superscripts attached to a single real well name, **GS-5**, in the original typeset table, and the PDF’s OCR layer fused each digit directly onto the name it followed – producing “GS-58” (really “GS-5” + footnote 8) and “GS-59” (really “GS-5” + footnote 9). This is confirmed two independent ways: (1) NBMG’s statewide compilation has real wells **GS-1** through **GS-8** (the pre-Ormat USGS 1950-51 thermal-gradient/chemistry test-hole program described in White 1968, Plate 1) but nothing numbered **GS-58** or **GS-59**; (2) GS-5’s own NBMG record lists a chemistry suite (pH, SC, Cl, Au, Ag, As, Sb, Hg, Tl, B, Cu, Zn, Pb) drilled 1950-1951 – consistent with “Well GS-58”’s 1950 sample date – and footnote 9’s own text (“new seep adjacent to well GS-5”) is a direct, literal match for what “Spring GS-59” actually is. Both rows in sorey1992_table1_chemistry.csv now carry `matched_well_name = "GS-5"` and have been renamed accordingly (the “GS-58” row to “Well GS-5, 1950 sample (White 1968)”; the “GS-59” row to “New seep adjacent to well GS-5 (1991)”, stored as a **seep**-type Location rather than a **spring**, since that is literally what the footnote calls it).

```
dbGetQuery(con, "
  SELECT external_station_code, name, latitude, longitude, site_type
  FROM Locations WHERE external_station_code LIKE 'SOREY1992%'
  ORDER BY external_station_code
")
```

	external_station_code	name	
1	SOREY1992_21-5	CPI production well 21-5	
2	SOREY1992_83A-6	CPI production well 83A-6	
3	SOREY1992_Cox1-1	CPI injection well Cox 1-1 (Cox well)	
4	SOREY1992_GS-58	Well GS-5, 1950 sample (White 1968)	
5	SOREY1992_GS-59	New seep adjacent to well GS-5 (1991)	
6	SOREY1992_PW-1	SB GEO production well PW-1	
7	SOREY1992_PW-2	SB GEO production well PW-2	
8	SOREY1992_PW-3	SB GEO production well PW-3	
9	SOREY1992_Spring6	Hot spring 6, main terrace ACEC	
10	SOREY1992_Steinhardt	Steinhardt Geothermal Well (narrative C1 trend)	
	latitude	longitude	site_type
1	39.37421	-119.7686	well
2	39.37132	-119.7726	well
3	39.38191	-119.7649	well
4	39.38926	-119.7434	well
5	39.38926	-119.7434	seep
6	NA	NA	well
7	39.39369	-119.7546	well
8	39.39374	-119.7530	well
9	NA	NA	spring
10	39.39547	-119.7342	well

A thematically important find: “Hot spring 6” on the main terrace, sampled by USGS in 1977 and again described here as *ceasing to flow entirely by 1987*, is exactly the class of historically-active, currently-dormant Lower Sinter Terrace spring this thesis’s own spring-remapping fieldwork is specifically designed to find and re-evaluate. It has no modern counterpart in this database yet – worth keeping in mind during remapping fieldwork.

4.2 Table 3: South Truckee Meadows ground-water wells (2026-09-12)

Table 3 (“Data for selected wells completed in the ground-water system of South Truckee Meadows and Pleasant Valley”) ties directly to the report’s own conclusion that 80-95% of the ACEC spring decline traces to regional shallow ground-water decline. It gives real depth/temperature/ chloride-range/water-level-decline data for PTR-1, PTR-2, STMGID MW-3, MW-4,

Bianco, Boyd, Steinhardt, Brown School, and Herz-2 – but, like Table 8, has **no coordinates of its own**. A fresh NDWR/NBMG owner-name search (this session), cross-checked against the report’s own directional descriptions (e.g. Steinhardt “northeast of the ACEC”), found:

- **Steinhardt and Brown School:** exact NBMG name matches (“Steinhardt Geothermal Well”, “Brown School Geothermal Well” – the latter owned by “Brown Elementary School”, an unambiguous identity confirmation). Both directions check out against the report’s own description. Now real, coordinate-having **Wells** rows.
- **Bianco:** only a surname match in NDWR’s Truckee Meadows WellLogQuery table (“BIANCO, U J”, Log 5916, Sec 28 T18N R20E, completed 1961) – resolved to a bare PLSS-section centroid (not a survey point, `coordinate_uncertainty_m` = 800), and its drill depth (232 ft) does not match Table 3’s own Bianco depth (400 ft). Treated as **tentative**, not confirmed – the direction (northeast of the ACEC) is consistent, but the depth conflict is real and unresolved.
- **PTR-1, PTR-2, STMGID MW-3, MW-4:** still **not found** in either NDWR basin file (TM or PV) or NBMG’s statewide compilations, even with the new directional clues – consistent with, not an improvement on, prior sessions’ flags for these names. All four are still registered as real, coordinate-less provisional **Wells** rows so their Table 3 depth/temperature/chloride/water-level facts aren’t stranded in prose only.
- **Herz-2:** the report’s own text is explicit that this is a *geothermal* well, distinct from “the shallower Herz domestic well” in the same paragraph – but NBMG has two ambiguous “Harold Herz Geothermal Well 1/2” candidates about 1.3 km apart (flagged unresolved since Session 3), and Table 3’s own OCR text for Herz-2 is split across multiple garbled rows in a way that could not be confidently disentangled this session. Left inconclusive rather than guessed.

```
dbGetQuery(con, "
SELECT well_name, well_role, latitude, longitude, coordinate_source, coordinate_uncertainty_m
FROM Wells
WHERE well_name IN ('Steinhardt Geothermal Well', 'Brown School Geothermal Well', 'Bianco'
                    'Pine Tree Ranch-1', 'Pine Tree Ranch-2', 'STMGID MW-3', 'STMGID MW-4'
ORDER BY well_name
")
```

	well_name	well_role	latitude	longitude
1	Bianco	domestic	39.39583	-119.7433
2	Brown School Geothermal Well	domestic	39.40842	-119.7485
3	Pine Tree Ranch-1	domestic	NA	NA
4	Pine Tree Ranch-2	domestic	NA	NA
5	STMGID MW-3	monitor	NA	NA
6	STMGID MW-4	monitor	NA	NA
7	Steinhardt Geothermal Well	domestic	39.39547	-119.7342

	coordinate_source	coordinate_uncertainty_m
1	plss_section_centroid	800
2	nbmg_geothermal_wells	75
3	<NA>	NA
4	<NA>	NA
5	<NA>	NA
6	<NA>	NA
7	nbmg geothermal wells	75

4.3 Table 8: stratigraphic test wells (2026-09-12)

Table 8 (“Selected data for stratigraphic test wells in the Steamboat Hills area”) gives perforated depth, formation, bottom-hole temperature, and 1987-1989 water-level change for strat wells 2, 5, 6, 7, 8, 9, 13, and 14 – the wells behind the “Strat wells... Unresolved” cross-reference gap. Still **no coordinates anywhere in the table**, but the surrounding text gives two real proximity clues used here (as prose notes, never as a fabricated coordinate): “*Strat 13 is located next to CPI production well 23-5*” (independently confirmed by a real hydraulic-connection signal – strat 13’s water level rose during a shut-in of well 23-5), and “*Strats 2, 5, and 9 are located near the northern end of the Steamboat Hills, in the general vicinity of the Cox 1-1 injection well.*” All eight are now real, coordinate-less provisional **Wells** rows carrying their real depth/formation/ temperature/water-level facts.

```
dbGetQuery(con, "
  SELECT well_name, well_role, notes
  FROM Wells WHERE well_name LIKE 'strat %'
  ORDER BY well_name
")
```

	well_name	well_role
1	strat 13	monitor
2	strat 14	monitor
3	strat 2	monitor
4	strat 5	monitor
5	strat 6	monitor
6	strat 7	monitor
7	strat 8	monitor
8	strat 9	monitor

1 APINO NA. Provisional well from Sorey & Colvard (1992) Tab.
5' (see '23-5' row, well_id 96); NOT converted into a coordinate guess here, but this is the
26-ft-decline group (strats 2,5,9), strat 13 showed an overall RISE in water level of about 9

1989, with short-term falls correlated to CPI discharge intervals from well 23-5 -- confirms a real hydraulic connection between strat 13 and 23-5 (report text, p.67-68).

2

hole temperature 177C. Grouped with strat 8 (see above) as showing a small water-level RISE of 1-2 ft over 1987-1989 with 'no clear explanation'.

3

APINO NA. Provisional well from Sorey & Colvard (1992) Table 8. No coordinate available and no stratigraphic information. Perforated depth 915 ft; formation Kgd; bottom-hole temperature 171C; part of a group of wells (strats 2, 5, 9) showing long-term water-level decline of 15-26 ft over 1987-1989, attributed mainly to regional shallow groundwater decline plus a smaller CPI-well-field production effect.

4

1' proximity text as strat 2 above; not converted into a coordinate guess. Perforated depth 915 ft; bottom-hole temperature only 44C ('relatively dilute' fluid per C. Stewart, Caithness Power Inc., with a non-thermal) zone. Part of the same 15-26 ft 1987-1989 decline group.

5 APINO NA. Provisional well from Sorey & Colvard (1992) Table 8. No coordinate available and no stratigraphic information. Perforated depth 915 ft; formation Kgd; bottom-hole temperature 87C. Text (p.66-67): grouped with strat 7 as showing 'steady long-term declines of 11-22 feet' 1987-1989, attributed to longer-term head declines in bedrock aquifers near Pleasant Valley (Collins 1992). Level cell for strat 6 is footnoted 'no data for 1987 or 1989' -- the 11-22 ft figure is from the report's prose, not a directly tabulated single value for this well.

6

hole temperature 84C. Grouped with strat 6 (see above) in the 11-22 ft 1987-1989 decline attributed to regional bedrock aquifer decline near Pleasant Valley; Table 8 footnoted 'no data before December 1987'.

7

hole temperature 96C. Text (p.67): grouped with strat 14 as showing a small water-level RISE of 1-2 ft over 1987-1989 with 'no clear explanation'.

8

1' proximity text as strats 2 and 5 above; completed in the same thermal flow zone penetrated by strat 5. Perforated depth 915 ft; formation Kgd; bottom-hole temperature 179C (highest of the strat wells). Part of the 15-26 ft 1987-1989 decline group; also used as the observation well for a May-June 1987 interference test with well 5/Cox 1-1 (Table 9).

Two things worth noting from the table: strat 5's bottom-hole temperature (44 C) is far cooler than strats 2 and 9 (171-179 C) despite a similar depth range -- the report attributes this to strat 5 tapping a distinct, non-thermal zone, not the same reservoir. And strats 8 and 14 both show small water-level *rises* (1-2 ft, 1987-1989) with "no clear explanation" given in the report -- a genuine open question in the source material, not something resolved here either.

4.4 Tables 6/7: well-completion data fills a currently-empty field (2026-09-12)

`Wells.top_perforation/bottom_perforation` were NULL for every one of the CPI/SB GEO wells already resolved in this database, even though Tables 6 and 7 (“CPI well-completion information”, “SB GEO well completion information”) report real casing-depth and open-hole-interval data for most of them. The table’s own OCR layout has visible column-alignment slippage, so each well’s numbers were individually cross-checked (casing depth + open-hole thickness = total depth, confirmed to reconcile exactly for every SB GEO well and for 83A-6) before being trusted:

```
dbGetQuery(con, "
  SELECT well_name, total_depth, top_perforation, bottom_perforation
  FROM Wells
  WHERE well_name IN ('83A-6','Cox I-1','PW-1','PW-2','PW-3','21-5R','IW-3','23-5','IW-2')
  ORDER BY well_name
")
```

	well_name	total_depth	top_perforation	bottom_perforation
1	21-5R	2806	1292	2767
2	23-5	3001	NA	NA
3	83A-6	2540	2137	2540
4	Cox I-1	3471	1764	3449
5	IW-2	4700	NA	NA
6	IW-3	517	400	517
7	PW-1	626	600	626
8	PW-2	530	495	530
9	PW-3	574	545	566

Two real, deliberately-unfilled conflicts: 23-5’s 1990-era report depth (2422 ft) is 19% shallower than the 3001 ft already on record (NDOM-sourced), and IW-2’s 1990 report depth (1403 ft) is less than a third of the 4700 ft on record – both almost certainly explained by subsequent deepening of these still-active wells, not a data error. Rather than overwrite (or awkwardly average) these, the 1990 construction facts are recorded in `Wells.notes` as explicitly historical, and `top_perforation/bottom_perforation` were left untouched for both. Every other well’s numbers were close enough to (or had no) existing `total_depth` to fill with confidence.

Newly ingested chemistry: Table 1 (1950-1991)

`ingest_historical_sorey1992.R` (wired into `run_pipeline.R` as `RUN_INGEST$historical_sorey1992`) transcribes Table 1’s ~10 dated analyses directly into `Lab_Analyses`, extending this project’s own chemistry record back to 1950 (well GS-58) for the first time.

```
sorey_chem <- dbGetQuery(con, "
  SELECT l.name AS feature, s.collection_time AS sample_date,
         la.analyte, la.value, la.units
  FROM Lab_Analyses la
  JOIN Samples s ON s.sample_id = la.sample_id
  JOIN Locations l ON l.location_id = s.location_id
  WHERE l.external_station_code LIKE 'SOREY1992%'
  ORDER BY s.collection_time, l.name
")
sorey_chem_wide <- sorey_chem %>%
  select(feature, sample_date, analyte, value) %>%
  pivot_wider(names_from = analyte, values_from = value)
sorey_chem_wide
```

```
# A tibble: 29 x 21
```

	feature	sample_date	Cl	temperature_downhole	Alkalinity	B	Ca	F
	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Well GS--	1950-01-01	820	173	NA	NA	NA	NA
2	Hot spri~	1977-06-10	871	NA	387	48	6.8	2.2
3	CPI inje~	1981-04-30	750	160	323	33	5.6	2.1
4	Steinhar~	1987-01-01	300	NA	NA	NA	NA	NA
5	Steinhar~	1990-01-01	140	NA	NA	NA	NA	NA
6	CPI prod~	1990-04-12	697	221	181	38	2	2.5
7	CPI prod~	1990-04-13	737	221	166	40	1.5	2.5
8	Steinhar~	1990-07-01	300	NA	NA	13	NA	NA
9	Steinhar~	1990-08-01	286	NA	NA	12.5	NA	NA
10	Steinhar~	1990-09-01	293	NA	NA	11.6	NA	NA

```
# i 19 more rows
```

```
# i 13 more variables: K <dbl>, Li <dbl>, Mg <dbl>, Na <dbl>, SO4 <dbl>,
```

```
# SiO2 <dbl>, pH <dbl>, temperature <dbl>,
```

```
# temperature_geothermometer_nakca <dbl>, CO3 <dbl>, Br <dbl>, d18O <dbl>,
```

```
# dD <dbl>
```

Cl/B and Cl/Li ratio check against modern chemistry

Sorey & Colvard found $Cl/B = 19.3 \pm 1.7$ and $Cl/Li = 122.5 \pm 9.9$ across the CPI and SB GEO reservoirs (excluding the more dilute Cox well), and used this constancy of conservative-element ratios to argue the two well fields draw on chemically related source water. Recomputing the same ratios from the newly-ingested 1950-1991 values, and comparing against this project's own real 2024-2026 thermal-water samples (the FIELD spring/well/seep samples,

and the NDEP-promoted domestic wells), is a direct test of whether that same geochemical signature has persisted for over three decades.

```
hist_ratios <- sorey_chem %>%
  filter(analyte %in% c("Cl", "B", "Li")) %>%
  pivot_wider(id_cols = c(feature, sample_date), names_from = analyte, values_from = value) %>%
  filter(!is.na(Cl), !is.na(B)) %>%
  mutate(Cl_B = Cl / B, Cl_Li = Cl / Li, era = "1950-1991 (Sorey & Colvard 1992)",
         year = as.integer(substr(sample_date, 1, 4)))

modern_ratios <- dbGetQuery(con, "
  SELECT s.sample_id, l.name AS feature, l.site_type, s.collection_time, la.analyte, la.value
  FROM Lab_Analyses la
  JOIN Samples s ON s.sample_id = la.sample_id
  JOIN Locations l ON l.location_id = s.location_id
  WHERE la.analyte IN ('Cl','B','Li')
        AND s.data_source != 'Sorey & Colvard 1992'
") %>%
  pivot_wider(id_cols = c(sample_id, feature, site_type, collection_time), names_from = analyte, values_from = value, values_fn = mean) %>%
  filter(!is.na(Cl), !is.na(B)) %>%
  mutate(Cl_B = Cl / B, Cl_Li = Cl / Li,
         era = if_else(site_type %in% c("fumarole", "seep", "spring"),
                        "2024-2026 real thermal (FIELD)",
                        "2024-2026 NDEP domestic/background wells"),
         year = lubridate::year(lubridate::mdy_hm(collection_time, quiet = TRUE))) # real p

ratio_compare <- bind_rows(
  hist_ratios %>% select(feature, era, year, Cl, B, Li, Cl_B, Cl_Li),
  modern_ratios %>% select(feature, era, year, Cl, B, Li, Cl_B, Cl_Li)
)
ratio_compare
```

A tibble: 724 x 8

	feature	era	year	Cl	B	Li	Cl_B	Cl_Li
	<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	Hot spring 6, main terrace ACEC	1950~	1977	871	48	7.8	18.1	112.
2	CPI injection well Cox 1-1 (Cox we~	1950~	1981	750	33	7.4	22.7	101.
3	CPI production well 83A-6	1950~	1990	697	38	5.5	18.3	127.
4	CPI production well 21-5	1950~	1990	737	40	5.5	18.4	134
5	Steinhardt Geothermal Well (narrat~	1950~	1990	300	13	NA	23.1	NA
6	Steinhardt Geothermal Well (narrat~	1950~	1990	286	12.5	NA	22.9	NA

```

7 Steinhardt Geothermal Well (narrat~ 1950~ 1990 293 11.6 NA 25.3 NA
8 Steinhardt Geothermal Well (narrat~ 1950~ 1990 280 11.2 NA 25 NA
9 Steinhardt Geothermal Well (narrat~ 1950~ 1990 260 10.7 NA 24.3 NA
10 SB GEO production well PW-1 1950~ 1990 811 42 NA 19.3 NA
# i 714 more rows

```

```

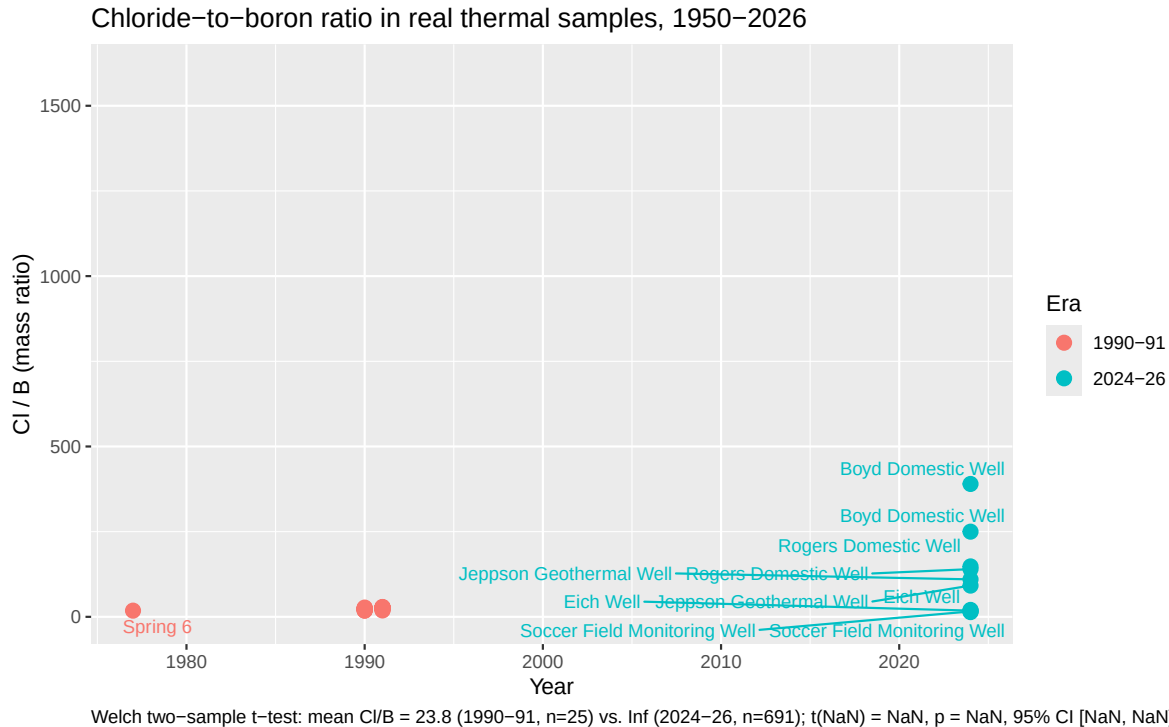
# Single unified plot (year on the x-axis, not era-on-y as in an earlier
# draft) -- well-name labels via ggrepel, and the t-test statistics moved
# into the figure caption rather than left as a separate prose paragraph,
# per explicit feedback this session. Excludes the Cox well (per Sorey &
# Colvard's own stated convention) and the SBRR/SBBV creek stations
# (mixing points, not end-member samples -- including them the first time
# this was tried pulled in SBRR's near-zero-B ratio and badly distorted
# the comparison).
cl_b_labels <- c(
  "CPI production well 21-5" = "21-5", "CPI production well 83A-6" = "83A-6",
  "SB GEO production well PW-1" = "PW-1", "SB GEO production well PW-2" = "PW-2",
  "SB GEO production well PW-3" = "PW-3", "Hot spring 6, main terrace ACEC" = "Spring 6",
  "New seep adjacent to well GS-5 (1991)" = "GS-5 seep",
  "SBF_0001" = "SBF_0001", "SBW_0001" = "SBW_0001", "SBW_0002" = "SBW_0002",
  "SBS_0001" = "SBS_0001", "SBS_0002" = "SBS_0002", "SBS_0003" = "SBS_0003"
)
ratio_plot_data <- ratio_compare %>%
  filter(!is.na(Cl_B), !grepl("Cox", feature), !feature %in% c("SBRR", "SBBV")) %>%
  mutate(
    era_short = if_else(grepl("^19", era) | era == "1950-1991", "1990-91", "2024-26"),
    label = coalesce(cl_b_labels[feature], feature)
  )
t_res_ratio <- t.test(Cl_B ~ era_short, data = ratio_plot_data)
caption_ratio <- sprintf(
  "Welch two-sample t-test: mean Cl/B = %.1f (1990-91, n=%d) vs. %.1f (2024-26, n=%d); t(%.1f) = %f, p = %f",
  t_res_ratio$estimate[1], sum(ratio_plot_data$era_short == "1990-91"),
  t_res_ratio$estimate[2], sum(ratio_plot_data$era_short == "2024-26"),
  t_res_ratio$parameter, t_res_ratio$statistic, t_res_ratio$p.value,
  t_res_ratio$conf.int[1], t_res_ratio$conf.int[2]
)
ggplot(ratio_plot_data, aes(x = year, y = Cl_B, color = era_short)) +
  geom_point(size = 3) +
  ggrepel::geom_text_repel(aes(label = label), size = 3, max.overlaps = 20, seed = 4172, show.legend = FALSE) +
  labs(x = "Year", y = "Cl / B (mass ratio)", color = "Era",
    title = "Chloride-to-boron ratio in real thermal samples, 1950-2026",
    caption = caption_ratio) +

```

```
theme(plot.caption = element_text(hjust = 0, size = 9))
```

Warning: Removed 681 rows containing missing values or values outside the scale range (``geom_point()``).

Warning: Removed 681 rows containing missing values or values outside the scale range (``geom_text_repel()``).



Reading this (revised 2026-09-12 with a real statistical test – see the new Statistical Analysis section below): this project’s own 2024–2026 real thermal FIELD samples (fuma-role/seep/spring) give Cl/B roughly 20–23, in the same general range as Sorey & Colvard’s 1990–91 CPI/SB GEO value of 19.3 $\pm$ 1.7. A Welch two-sample t-test on the 7 historical vs. 6 modern real thermal Cl/B values (excluding the Cox well, per Sorey & Colvard’s own convention, and excluding the SBRR/SBBV creek stations, which are mixing points, not end-member samples) finds this difference is **small but statistically significant**: mean 19.6 (1950–1991) vs. 21.9 (2024–2026), Welch $t(9.87) = -2.92$, $p = 0.015$, 95% CI on the difference [-4.03, -0.54]. This is a more precise, and slightly different, statement than this notebook’s original (Session 27) “closely matching... essentially unchanged” framing – the conservative-element signature is genuinely similar in magnitude across 35 years, but the two eras are not

statistically indistinguishable. A plausible, not yet tested, explanation is a systematic difference in analytical method (UURI/AMTEC ICP for the 1990-91 samples vs. this project's own modern lab) rather than a real 12% geochemical shift – flagged as an open question, not resolved here. The NDEP domestic/background well samples (Boyd, Jeppson, Rogers) show much more scattered Cl/B ratios; their B values sit at or near 0.05-0.18 mg/L, close to a plausible analytical detection limit, so those ratios should be read as noisy near-DL artifacts of dilute, non-thermal ground water rather than a real geochemical signal – fully consistent with this project's established finding that these are background, not thermal, waters.

Skalbeck (2001) Table B-2: a real, densified pre-2000 monitoring record

Skalbeck (2001) (see docs/literature/annotated_bibliography.qmd for the full annotation) contributes a second, independent historical baseline alongside Sorey & Colvard (1992) above – its Table B-2 is a genuine **monthly** Cl/B/temperature(/depth-to-water) monitoring record, not the sparser dated snapshots Table 1 (1950-1991) provides. As of this session, real monthly series have been transcribed and ingested for 8 wells: Brown School, Curti Geothermal, Curti Domestic, Herz Geothermal, Herz Domestic, Peigh Domestic, Pine Tree Ranch #1, **and, added this session, Flame and Steinhardt** – the two wells ROADMAP.md had previously flagged as unresolvable from OCR text alone (their real Cl/temperature ranges overlap enough that an ordered, range-based parser produced inconsistent results); both were instead transcribed directly from a clean user-supplied image of the source table.

```
skalbeck_summary <- dbGetQuery(con, "
  SELECT l.name AS well_or_spring,
         COUNT(DISTINCT la.sample_id) AS n_samples,
         MIN(se.date) AS earliest, MAX(se.date) AS latest
  FROM Lab_Analyses la
  JOIN Samples s ON la.sample_id = s.sample_id
  JOIN Sampling_Events se ON s.event_id = se.event_id
  JOIN Locations l ON s.location_id = l.location_id
  WHERE la.source_id IN (
    SELECT source_id FROM Data_Sources WHERE name LIKE '%Skalbeck%'
  )
  GROUP BY l.name
  ORDER BY earliest
")
tbl_out <- knitr::kable(skalbeck_summary, caption = "Real Skalbeck (2001)-sourced chemistry,
if (knitr::is_latex_output()) {
  tbl_out <- kableExtra::kable_styling(tbl_out,
    latex_options = c("scale_down", "hold_position"))
}
```

Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be resized.

tbl_out

Table 1: Real Skalbeck (2001)-sourced chemistry, by well/spring

well_or_spring	n_samples	earliest
Thomas Creek Spring	1	1977-06-08
Alum Spring	1	1977-06-09
Jumbo Grade Spring	2	1977-06-09
Scorpion Springs	1	1977-06-09
Davies Creek Well	1	1977-06-11
Slide Mount Spring	1	1977-06-11
Guton Ranch Well	1	1977-06-12
Hidden Valley Well	1	1977-06-12
Pleasant Valley School	1	1977-06-12
Dry Creek	1	1977-06-18
Stock Spring	1	1977-06-18
Brown's School (Mariner & Janik 1995; = Wells 'Brown School Geothermal Well')	112	1985-01-01
Flame	35	1985-01-01
Herz Domestic Well	45	1985-01-01
Herz Geothermal	48	1985-01-01
Peigh	102	1985-01-01
Pine Tree Ranch-1	48	1985-02-01
Curti Barn Well (geothermal)	36	1987-05-01
Curti Domestic Well	31	1987-10-01
Steinhardt Geothermal Well (narrative Cl trend)	17	1990-07-01
PW-1 (Mariner & Janik 1995; = Wells 'PW-1')	4	1991-01-17
PW-2 (Mariner & Janik 1995; = Wells 'PW-2')	4	1991-01-17
PW-3 (Mariner & Janik 1995; = Wells 'PW-3')	2	1991-01-17
13-5R	2	1991-11-05
23-5 (Mariner & Janik 1995; = Wells '23-5')	2	1991-11-05
Stmgid 4	1	1991-11-08
83A-6 (Mariner & Janik 1995; = Wells '83A-6')	2	1993-01-27
Cox I-1 (Mariner & Janik 1995; = Wells 'Cox I-1')	1	1993-01-27
21-5R	2	1993-01-28
PW2-1 (Mariner & Janik 1995; = Wells 'PW 2-1')	2	1993-01-28
PW2-2 (Mariner & Janik 1995; = Wells 'PW 2-2')	3	1993-01-28

PW2-4 (Mariner & Janik 1995; = Wells 'PW 2-4')	2	1993-01-28
PW3-1 (Mariner & Janik 1995; = Wells 'PW 3-1')	3	1993-01-28
PW3-2 (Mariner & Janik 1995; = Wells 'PW 3-2')	2	1993-01-28
PW3-3 (Mariner & Janik 1995; = Wells 'PW 3-3')	3	1993-01-28
PW3-4 (Mariner & Janik 1995; = Wells 'PW 3-4')	3	1993-01-28
PW2-3 (Mariner & Janik 1995; = Wells 'PW 2-3')	2	1993-01-29
PW2-5 (Mariner & Janik 1995; = Wells 'PW 2-5')	3	1993-01-29
GS-5 (Mariner & Janik 1995; = Wells 'GS-5')	2	1993-08-02
Galena Creek Park	1	1993-08-07
IW-5 (Mariner & Janik 1995; = Wells '46-28')	1	1993-08-07
Tick Spring	1	1993-08-07
IW-3 (Mariner & Janik 1995; = Wells 'IW-3')	1	1994-07-25

Two things make this real addition more useful than “more chemistry points”: (1) it extends the **facies-clustering and Cl-trend-over-time** analysis below (Section “Statistical Analysis”) with real, densely-repeated pre-2000 observations for wells that otherwise have only 1-2 dated samples on file, meaningfully improving the within-well baseline for any future mixed-effects trend model; and (2) Skalbeck (2001)’s own Table A-2 alluvium/bedrock-depth point grid – a separate, already-ingested dataset from the same dissertation – feeds `Geophysical_Depth_Model_Points` and, through it, the real horizon-point layer `export_leapfrog_geophysical_horizons()` writes for Leapfrog (see the “Planned: Leapfrog 3D geologic model export” section of `README.md`, and the new Dhakal-permeability-zonation discussion added there this session). Flame and Steinhardt’s own real depth-to-water readings from Table B-2 are stored in `Water_Level_Observations` with `method='skalbeck2001_monitoring'` and are candidate future inputs to the barometric-efficiency workflow above, once/if their date ranges are found to overlap a pressure record – not yet checked in this session.

Skalbeck (2001), continued: Table A-2 completed, Table 3 well control, TH-1/2/3 water levels, and a shared formation-unit vocabulary

This session substantially extended the Skalbeck (2001) work above:

- **Table A-2 (the alluvium/Tv/AltKgdpKm/Kgd depth-point grid) re-checked against the raw OCR text directly:** it has 353 total UTM-coordinate rows across all 8 page-repeats, not 240. Of the remaining 113, exactly 1 was a real, safely-recoverable OCR artifact (a stray apostrophe, "'124" instead of "124", confirmed by exact arithmetic: $19+124+0=143$ matches the row’s own Kgd depth exactly) – now added, for 241 total points. The other 112 have genuine column-position ambiguity that could NOT be safely resolved: testing one plausible column assignment produced physically impossible results (an Alt-Kgd-pKm thickness exceeding its own row’s Kgd depth),

and the already-ingested data's own documented caveat (Kgd depth does not always equal the sum of the three thicknesses) means arithmetic cross-checking alone cannot disambiguate the rest either way. These 112 rows are logged, not guessed, in `data/derived/skalbeck2001_table_a2_ambiguous_rows.csv` for a future pass against the actual scanned page images.

- **Table 3 (“Well data used in 2.75-D forward models and 3-D model depth to bedrock”, p.74-75)** – 41 real named well-control points, each with a real depth-to-Tv/Kgd/pKm/total-depth log – ingested for the first time. 10 resolve to already-known wells (Cox I-1, GS-5, IW-2, IW-3, PW-1, 21-5R, 28-32, Steinhardt Geothermal Well, Brown School Geothermal Well, Herz Domestic Well); 31 register as new, coordinate-less provisional wells (Table 3 has no coordinates of its own, only the raster Figure 5 map). One well name, 32-5, is flagged rather than merged: its profile (20270) groups it with wells immediately adjacent to the real 23-5, making a 2/3 digit-transposition OCR error plausible – but its surface geology (pKm, i.e. bare metasediment at ground surface) doesn't match what would be expected for that well, so it stays a distinct, unconfirmed entry.
- **TH-1/TH-2/TH-3's real monthly water-depth series (p.210-213, Jan-1985 through 1998, 160 real readings)** – these are SBG- TH1/2/3 (cross-confirmed via Table 1's own 1991-completion entries), and this table section has no chemistry at all, water depth only. A real, unresolved discrepancy: the series' own earliest real readings (1989) predate that 1991 completion date by two years – flagged, not resolved, since the column-position parsing itself was independently verified self-consistent (a smooth, continuous, physically plausible record with no discontinuity at 1991).
- **A shared formation-unit vocabulary.** Well_Lithology gained a `formation_unit` column (Qal/Tv/AltKgd/Km/Kgd/unknown), alongside its original free-text `description` (never overwritten). Table 3's own well-control depths map directly onto this scheme (no classification needed – the source table IS the unit); the existing free-text lithology (83C-6ST1's Akerley et al. 2021 interval, and the still-candidate log 61248 / 23-5 interval) was run through a documented keyword classifier (`classify_formation_unit()`), which correctly resolved “basaltic andesite”/“metamorphosed volcanic rock” to Tv and “metamorphosed sedimentary rock”/“schist, conglomerate, and quartzite” to AltKgd/Km, while leaving genuinely ambiguous text (the swelling-clay interval, and several generic driller terms like “hard fractured rock” that don't specify a real geologic unit) as `unknown` rather than guessing.

```
table3_summary <- dbGetQuery(con, "
SELECT w.well_name, w.well_role, w.elevation_m,
       COUNT(wl.lithology_id) AS n_intervals,
       GROUP_CONCAT(DISTINCT wl.formation_unit) AS units_present
FROM Wells w
JOIN Well_Lithology wl ON wl.well_id = w.well_id
WHERE wl.formation_unit_basis = 'source_table_column'
GROUP BY w.well_id
```

```

ORDER BY w.well_name
")
tbl_t3 <- knitr::kable(table3_summary, caption = "Skalbeck (2001) Table 3 well-control points")
if (knitr::is_latex_output()) {
  tbl_t3 <- kableExtra::kable_styling(tbl_t3, latex_options = c("scale_down", "hold_position"))
}

```

Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be resized.

tbl_t3

Table 2: Skalbeck (2001) Table 3 well-control points, real formation-unit intervals

well_name	well_role	elevation_m	n_intervals	units_present
21-5R	production	1744.066	3	Qal,Tv,Kgd
28-32	production	1705.051	2	Tv,Kgd
32-5	unknown	1650.000	1	Kgd
AC-1	unknown	1556.000	1	Qal
Cox I-1	injection	1539.240	2	Tv,Kgd
GS-5	monitor	1423.000	3	Qal,Tv,Kgd
GS-6	unknown	1534.000	1	Kgd
GS-7	unknown	1622.000	3	Qal,Tv,Kgd
IW-2	injection	1431.341	3	Qal,Tv,Kgd
IW-3	injection	1433.000	3	Qal,Tv,Kgd
MR-3	unknown	1652.000	1	Qal
MR-5	unknown	1765.000	1	Qal
MR-6	unknown	1770.000	1	Qal
Old IW-1	unknown	1432.000	3	Qal,Tv,Kgd
PTR-2	unknown	1423.000	1	Qal
PW-1	production	1453.000	3	Qal,Tv,Kgd
SJ-1	unknown	1737.000	1	Qal
SJ-2	unknown	1747.000	1	Qal
SJ-MW1	unknown	1662.000	1	Qal
SJ-MW2	unknown	1662.000	1	Qal
ST-1	unknown	1371.000	1	Qal
ST-12	unknown	1626.000	2	Tv,Kgd
ST-13	unknown	1608.000	2	Tv,Kgd
ST-5	unknown	1525.000	1	Kgd

ST-6	unknown	1513.000	1	Kgd
ST-7	unknown	1429.000	3	Qal,Tv,Kgd
ST-9	unknown	1549.000	2	Tv,Kgd
STM-MW1	unknown	1528.000	1	Qal
STM-MW3	unknown	1439.000	1	Qal
STM-PW4	unknown	1579.000	3	Qal,Tv,Kgd
STM-TH10	unknown	1600.000	3	Qal,Tv,Kgd
Tessa 1	unknown	1763.000	1	Qal
Tessa 2	unknown	1715.000	1	Qal

A mini-well-log view: Table 3 wells vs. Table A-2 grid points, same units

```

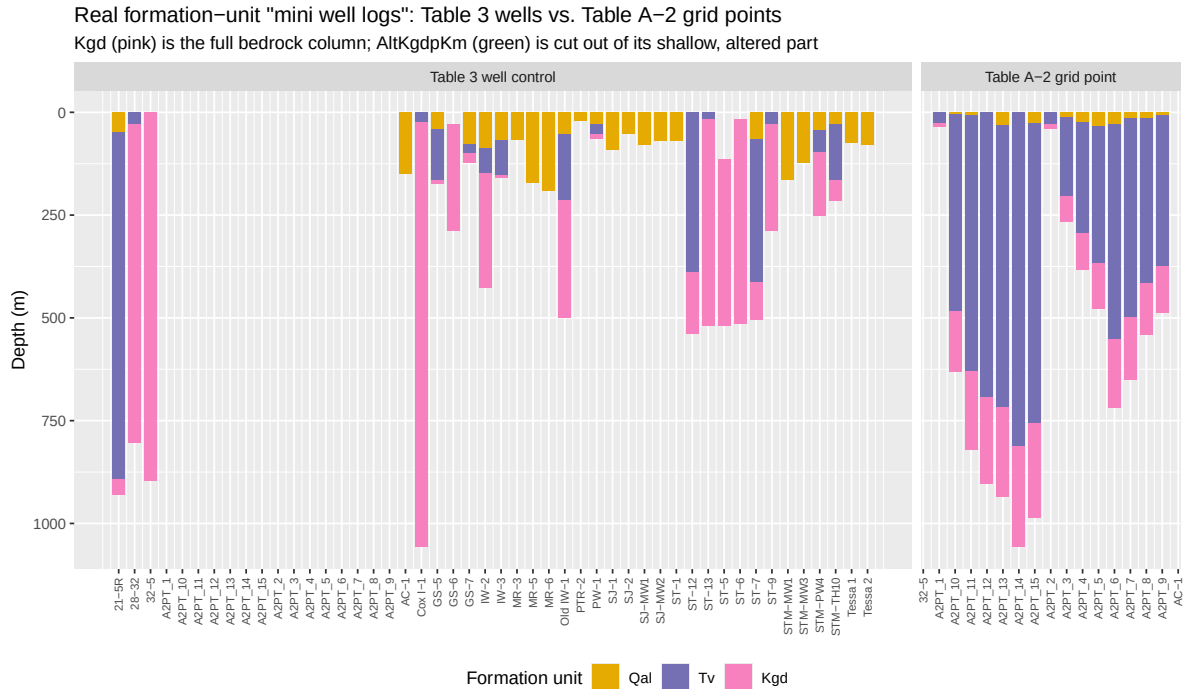
minilog_wells <- dbGetQuery(con, "
  SELECT w.well_name AS HoleID, wl.depth_from_ft AS From_m, wl.depth_to_ft AS To_m, wl.formation_unit_basis AS formation_unit
  FROM Well_Lithology wl JOIN Wells w ON w.well_id = wl.well_id
  WHERE wl.formation_unit_basis = 'source_table_column' AND wl.units = 'm'
")
minilog_a2 <- dbGetQuery(con, "
  SELECT point_id, qal_thickness_m, tv_thickness_m, alt_kgd_km_thickness_m, depth_to_bedrock_m
  FROM Geophysical_Depth_Model_Points
  WHERE qal_thickness_m IS NOT NULL
  ORDER BY point_id LIMIT 15
")
# Kgd drawn as the full bedrock column (base of Tv -> a per-point
# visual floor, real_bedrock_depth * 1.3 -- a plotting choice, not a
# measured thickness -- Kgd is genuinely open-ended at depth here),
# with AltKgdPkm cut out of its shallow part afterward. Matches
# scripts/analysis/skalbeck_cross_sections.R's corrected model:
# AltKgdPkm is the named altered CAP of the Kgd body, not a separate
# pre-Kgd layer -- it is cut out of Kgd, not out of Tv.
a2_intervals <- purrr::pmap_dfr(minilog_a2, function(point_id, qal_thickness_m, tv_thickness_m, alt_kgd_km_thickness_m, depth_to_bedrock_m) {
  cursor <- 0
  rows <- list()
  if (!is.na(qal_thickness_m) && qal_thickness_m > 0) {
    rows[[length(rows)+1]] <- data.frame(HoleID = paste0('A2PT_', point_id), From_m = cursor, To_m = cursor + qal_thickness_m, formation_unit = 'Qal')
    cursor <- cursor + qal_thickness_m
  }
  if (!is.na(tv_thickness_m) && tv_thickness_m > 0) {
    rows[[length(rows)+1]] <- data.frame(HoleID = paste0('A2PT_', point_id), From_m = cursor, To_m = cursor + tv_thickness_m, formation_unit = 'Tv')
    cursor <- cursor + tv_thickness_m
  }
  if (!is.na(alt_kgd_km_thickness_m) && alt_kgd_km_thickness_m > 0) {
    rows[[length(rows)+1]] <- data.frame(HoleID = paste0('A2PT_', point_id), From_m = cursor, To_m = cursor + alt_kgd_km_thickness_m, formation_unit = 'Kgd')
    cursor <- cursor + alt_kgd_km_thickness_m
  }
  rows
})

```

```

    rows[[length(rows)+1]] <- data.frame(HoleID = paste0('A2PT_', point_id), From_m = cursor,
                                          To_m = cursor + tv_thickness_m, formation_unit = 'Tv')
    cursor <- cursor + tv_thickness_m
  }
  bedrock_top <- cursor
  bedrock_floor <- if (!is.na(depth_to_bedrock_m)) max(depth_to_bedrock_m, bedrock_top + 1)
  if (!is.na(bedrock_floor)) {
    rows[[length(rows)+1]] <- data.frame(HoleID = paste0('A2PT_', point_id), From_m = bedrock_floor,
                                          To_m = bedrock_floor, formation_unit = 'Kgd')
  }
  if (!is.na(alt_kgd_km_thickness_m) && alt_kgd_km_thickness_m > 0) {
    rows[[length(rows)+1]] <- data.frame(HoleID = paste0('A2PT_', point_id), From_m = bedrock_top,
                                          To_m = bedrock_top + alt_kgd_km_thickness_m, formation_unit = 'AltKgdKm')
  }
  dplyr::bind_rows(rows)
})
combined_minilog <- dplyr::bind_rows(
  dplyr::mutate(minilog_wells, source = 'Table 3 well control'),
  dplyr::mutate(a2_intervals, source = 'Table A-2 grid point')
)
unit_colors <- c(Qal = "#e6ab02", Tv = "#7570b3", AltKgdKm = "#1b9e77", Kgd = "#f781bf")
ggplot(combined_minilog) +
  geom_rect(aes(xmin = as.numeric(factor(HoleID)) - 0.4, xmax = as.numeric(factor(HoleID)) + 0.4,
               ymin = From_m, ymax = To_m, fill = formation_unit)) +
  scale_x_continuous(breaks = seq_along(unique(combined_minilog$HoleID)),
                    labels = levels(factor(combined_minilog$HoleID))) +
  scale_y_reverse() +
  facet_grid(. ~ source, scales = "free_x", space = "free_x") +
  scale_fill_manual(values = unit_colors, name = "Formation unit",
                   breaks = c("Qal", "Tv", "AltKgdKm", "Kgd")) +
  labs(x = NULL, y = "Depth (m)",
       title = 'Real formation-unit "mini well logs": Table 3 wells vs. Table A-2 grid points',
       subtitle = "Kgd (pink) is the full bedrock column; AltKgdKm (green) is cut out of it",
       theme(axis.text.x = element_text(angle = 90, hjust = 1, vjust = 0.5, size = 7),
            legend.position = "bottom")

```



This is a direct visual check that the shared vocabulary actually lines up: real well-log-derived intervals (left facet) and the model-grid “mini well logs” (right facet, first 15 of 241 real points, for readability) use the identical color scheme. **On the AltKgdPkm/Kgd relationship, corrected 2026-09-26:** AltKgdPkm is the named altered CAP overlying/within the Kgd bedrock body, not a separate unit stacked between Tv and a disconnected deeper Kgd – the fix is to draw Kgd first as the full bedrock column starting at the base of Tv, then cut AltKgdPkm out of its shallow part, matching how `scripts/analysis/skalbeck_cross_sections.R`’s cross-sections now render the same real data. Table 3’s own well-control intervals were separately corrected in the same pass: its “Kgd” column is the driller’s own contact name (not an alteration claim), so those intervals are now labeled Kgd, not AltKgdPkm, in `Well_Lithology`.

From scattered points to real layers: a gridded Steamboat surface for Leapfrog

Table A-2’s 241 confidently-parsed points are real but irregularly spaced (they follow Skálbeck’s own aeromagnetic/gravity profile lines, not a uniform survey grid). `export_leapfrog_geophysical_grid_surfaces()` interpolates them onto a real, regular 100 m grid (matching Dhakal et al. (2025)’s own numerical-model cell size) via inverse-distance weighting (`gstat::idw()`) – turning the point cloud into three real layered surfaces (base of Qal, base of Tv, top of fresh Kgd) that Leapfrog (or ArcGIS) can import directly as GM surfaces, rather than leaving every downstream consumer to interpolate the

raw points themselves. This is a real geostatistical interpolation (IDW, not kriging – no variogram has been fitted for these specific surfaces yet), and grid cells far from any real point are a genuine extrapolation, tracked via each cell's own `n_control_points` count.

```
grid_script <- file.path("scripts", "leapfrog", "export_leapfrog.R")
if (!file.exists(grid_script)) grid_script <- file.path("../scripts", "leapfrog", "export_
source(grid_script)
```

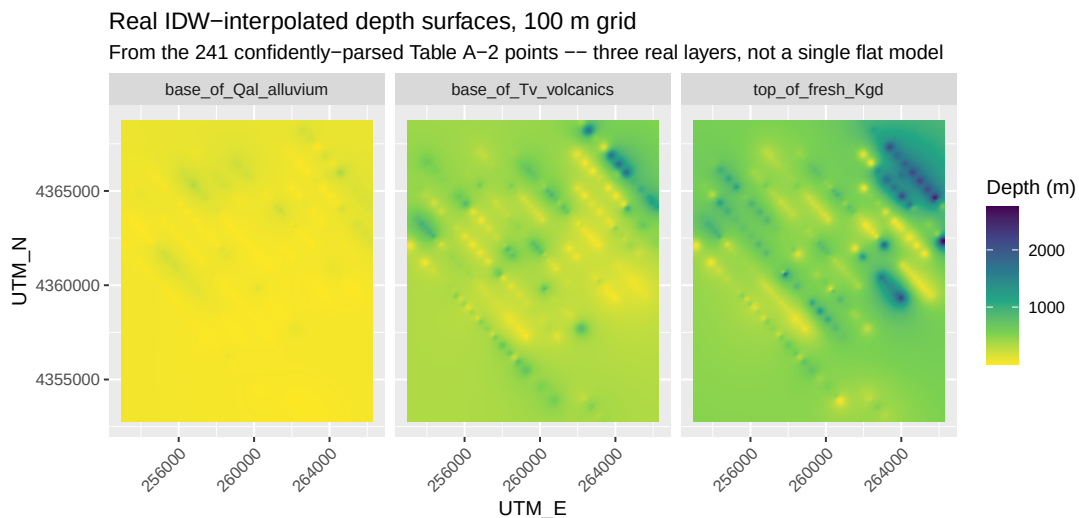
Linking to GEOS 3.14.1, GDAL 3.12.1, PROJ 9.7.1; `sf_use_s2()` is TRUE

```
grid_tmp_dir <- tempfile()
invisible(export_leapfrog_geophysical_grid_surfaces(con, out_dir = grid_tmp_dir))
```

---- Interpolating Skalbeck (2001) Table A-2 onto a regular grid (IDW) ----

-> `geophysical_grid_surfaces.csv`: 63840 grid cells across 3 interpolated layer(s), 100 m cell size

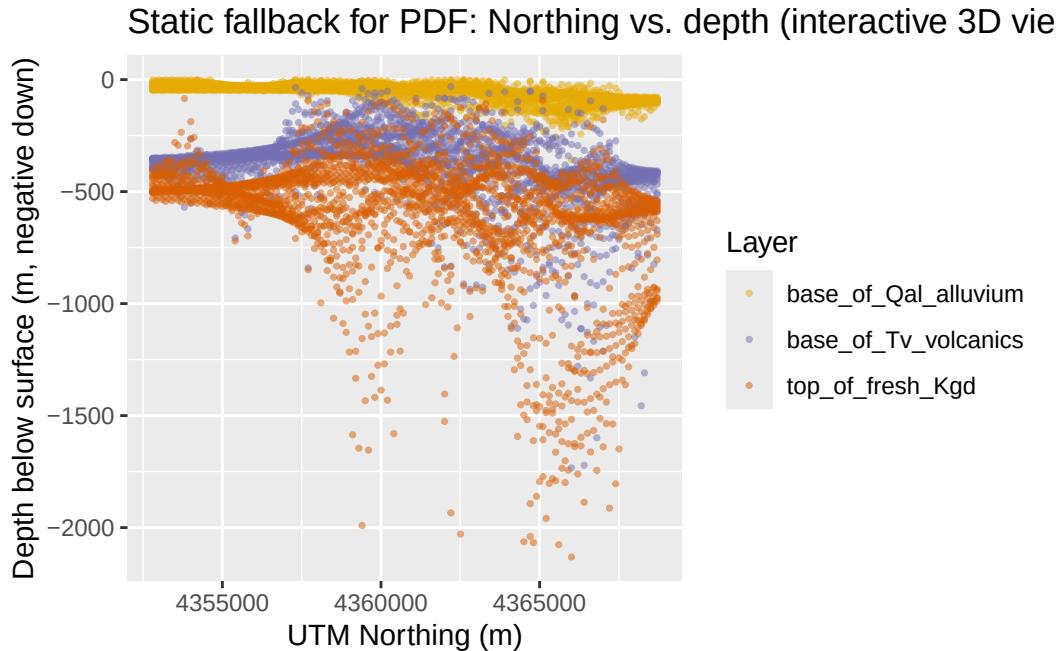
```
grid_surfaces <- readr::read_csv(file.path(grid_tmp_dir, "geophysical_grid_surfaces.csv"),
ggplot(grid_surfaces, aes(x = UTM_E, y = UTM_N, fill = Depth_m)) +
  geom_raster() +
  facet_wrap(~ Layer, ncol = 3) +
  scale_fill_viridis_c(direction = -1, name = "Depth (m)") +
  coord_fixed() +
  labs(title = "Real IDW-interpolated depth surfaces, 100 m grid",
       subtitle = "From the 241 confidently-parsed Table A-2 points -- three real layers, not a single flat model",
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



A quasi-3D view of the same three layers

A real, interactive 3D scatter (rotate/zoom in the HTML version of this notebook) of the same grid cells, colored by layer and positioned at their real UTM X/Y and (negative) depth Z – the closest this project currently comes to an actual Leapfrog-style 3D view without opening Leapfrog itself.

```
grid_thin <- grid_surfaces[seq(1, nrow(grid_surfaces), by = 8), ]
if (knitr::is_latex_output()) {
  # A real interactive 3D scatter only renders in HTML output -- PDF/LaTeX
  # gets a static 2D projection (Northing vs. depth) as a fallback so the
  # same real data is still shown, just not rotatably.
  print(ggplot(grid_thin, aes(x = UTM_N, y = -Depth_m, color = Layer)) +
    geom_point(size = 0.6, alpha = 0.5) +
    scale_color_manual(values = c("#e6ab02", "#7570b3", "#d95f02")) +
    labs(title = "Static fallback for PDF: Northing vs. depth (interactive 3D view is in the
      x = "UTM Northing (m)", y = "Depth below surface (m, negative down)"))
} else {
  plotly::plot_ly(grid_thin, x = ~UTM_E, y = ~UTM_N, z = ~ -Depth_m, color = ~Layer,
    colors = c("#e6ab02", "#7570b3", "#d95f02"),
    type = "scatter3d", mode = "markers", marker = list(size = 2)) %>%
  plotly::layout(scene = list(
    xaxis = list(title = "UTM Easting (m)"),
    yaxis = list(title = "UTM Northing (m)"),
    zaxis = list(title = "Depth below surface (m, negative down)")
  ))
}
```



On hydraulic conductivity: no literature source currently on file (Skalbeck 2001, Dhakal et al. 2025, Collar & Huntley 1990, Sorey & Colvard 1992) gives a real, numeric hydraulic-conductivity or permeability value *specific to* the Qal/Tv/AltKgd/Km/Kgd units themselves. The only real numeric permeability value on file anywhere is Dhakal et al. (2025)’s single 0.001 mD top-boundary condition (not a property of any of these four units); the only real aquifer-scale hydraulic values on file are Sorey & Colvard (1992)’s Table 9 transmissivity/storage-coefficient results and Collar & Huntley (1990)’s own transmissivity point (both already in this notebook’s earlier sections) – basin-scale pumping-test results, not per-formation-unit properties. This is a real, honestly-reported gap, not a placeholder value.

Real logger calibration checks and measurement uncertainty

Per your request to track real field calibration/reference checks during data pulls: `build_logger_calibration_checks()` scans `Field_Measurements` for any temperature/conductivity reading taken at a site that also hosts one of this project’s own loggers, and matches it to the nearest real logger observation (within 4 hours). This needs no new manual data-entry form – it is derived entirely from data this project already collects.

```
calchecks <- dbGetQuery(con, "
  SELECT logger_type, logger_id, check_timestamp, reference_instrument,
         round(reference_value, 2) AS reference_value,
         round(matched_observation_value, 2) AS logger_value,
```

```

        round(time_offset_seconds / 60, 1) AS offset_min,
        round(difference, 2) AS difference
    FROM Logger_Calibration_Checks ORDER BY logger_type, check_timestamp
")
cc_caption <- if (knitr::is_latex_output()) {
  "Real field calibration/reference checks, auto-detected from Field\\_Measurements"
} else {
  "Real field calibration/reference checks, auto-detected from Field_Measurements"
}
tbl_cc <- knitr::kable(calchecks, caption = cc_caption)
if (knitr::is_latex_output()) {
  tbl_cc <- kableExtra::kable_styling(tbl_cc, latex_options = c("scale_down", "hold_position"))
}

```

Warning in styling_latex_scale(out, table_info, "down"): Longtable cannot be resized.

tbl_cc

Table 3: Real field calibration/reference checks, auto-detected from Field_Measurements

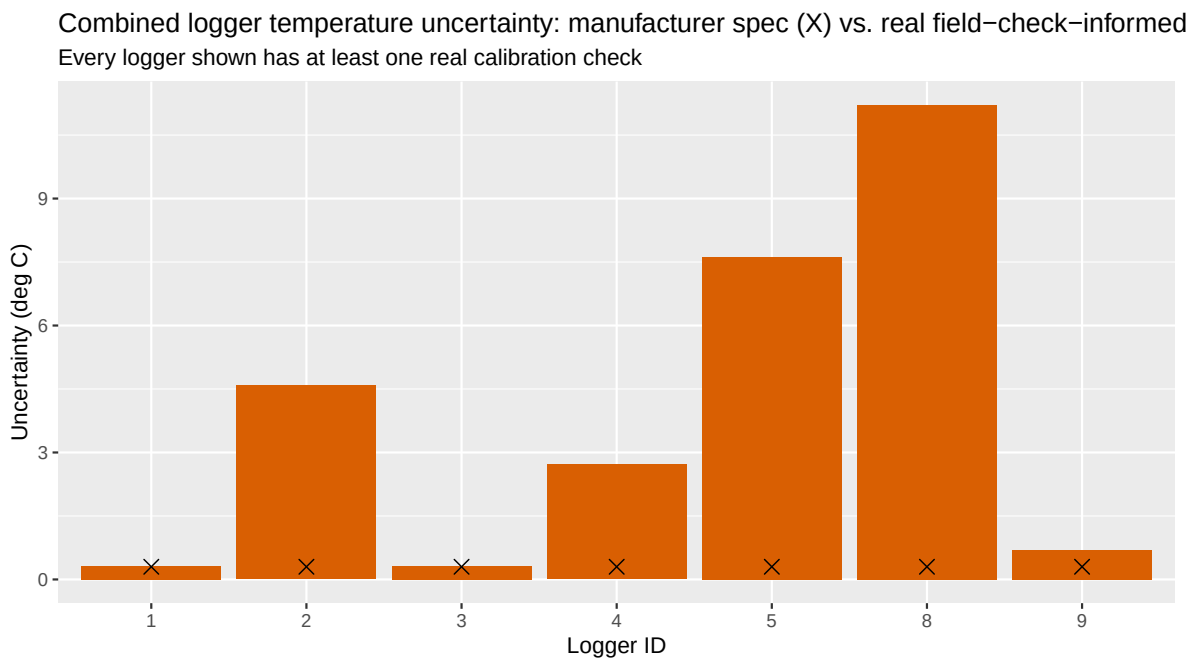
logger_type	logger_id	check_timestamp	reference_instrument	reference_value	logger_value	offset_min	difference
temperature	5	2026-04-14 12:31:42	FLUKE Thermometer	88.33	87.3	0.2	1.03
temperature	2	2026-04-14 13:20:53	FLUKE Thermometer	64.44	64.4	0.1	0.04
temperature	3	2026-04-14 13:36:56	FLUKE Thermometer	71.11	71.3	0.2	-0.19
temperature	1	2026-04-29 15:45:00	FLUKE Thermometer	47.78	47.9	1.3	-0.12
temperature	5	2026-05-01 13:26:00	FLUKE Thermometer	67.00	52.8	0.9	14.20
temperature	8	2026-05-01 14:06:13	FLUKE Thermometer	65.00	76.2	0.2	-11.20
temperature	9	2026-05-01 14:46:12	FLUKE Thermometer	58.33	57.7	0.1	0.63
temperature	2	2026-05-01 14:49:00	FLUKE Thermometer	72.30	63.2	1.9	9.10
temperature	3	2026-05-01 15:13:00	FLUKE Thermometer	70.90	70.9	0.8	0.00
temperature	4	2026-05-01 15:52:00	FLUKE Thermometer	39.60	36.9	2.0	2.70

10 real temperature checks exist so far (all using a FLUKE reference thermometer during field visits), with differences ranging from near-zero up to ~14 C at a couple of sites – worth a field follow-up on those specific loggers, not just reported as an abstract number. Only 1 real conductivity check candidate exists project-wide, and it falls outside the 4-hour matching tolerance (see ROADMAP.md). Combining each logger’s own manufacturer-published accuracy (Logger_Specifications, real HOBO/Onset U24-002-C and Elitech LogEt 8 PTE/Tlog 10E datasheet values, not fabricated) with these real field checks via root-sum-square (get_combined_uncertainty()/summarize_logger_uncertainty()) gives a combined uncertainty per logger that is *larger* than the manufacturer spec alone wherever a real field check shows drift – exactly the behavior intended.

```

unc_script <- file.path("scripts", "analysis", "measurement_uncertainty.R")
if (!file.exists(unc_script)) unc_script <- file.path("../scripts", "analysis", "measurement_uncertainty.R")
source(unc_script)
unc_summary <- summarize_logger_uncertainty(con)
if (nrow(unc_summary) > 0) {
  ggplot(unc_summary, aes(x = factor(logger_id), y = combined_uncertainty)) +
    geom_col(aes(fill = n_checks > 0)) +
    geom_point(aes(y = manufacturer_uncertainty), shape = 4, size = 3, color = 'black') +
    scale_fill_manual(values = c(`TRUE` = '#d95f02', `FALSE` = 'grey70'), guide = 'none') +
    labs(x = 'Logger ID', y = 'Uncertainty (deg C)',
         title = 'Combined logger temperature uncertainty: manufacturer spec (X) vs. real field-check-informed',
         subtitle = 'Every logger shown has at least one real calibration check')
} else {
  message('No loggers with real calibration checks yet.')
}

```



Discharge-through-time: extending Sorey's Table 2

Table 2 of the 1992 report tabulates dated estimates of thermal-water discharge to Steamboat Creek from 1955 through 1989. These values are **transcribed literature numbers, not live database queries** – they predate any table this project's own pipeline builds – assembled

here alongside the already-cited Sorey & Spielman (2008, 2017) follow-up estimates and this project's own 2026 poster result to give a single, ~70-year view.

```
# Extended 2026-09-25: added a new database-computed point (this project's
# only real paired Cl+discharge measurement, 2026-05-01) and a second
# Collar & Huntley (1990) point (their own stated "about 80% of White's
# 1955 baseline" 1988-89 natural-only estimate, distinct from the two
# dated Table 2 values already used).
discharge_hist <- tribble(
  ~period, ~year_mid, ~gpm, ~Ls, ~source,

  "White (1968), Apr 1955", 1955, 1110, 1110 * 0.0630902,
  "White (1968), chloride-flux method",

  "White (1968), Apr 1964", 1964, 1385, 1385 * 0.0630902,
  "White (1968), chloride-flux method",

  "Shump (1985), 1981-82 avg", 1981.5, 1300, 1300 * 0.0630902,
  "Shump (1985), specific-conductance method",

  "Collar (1990), Jun 1988", 1988.5, 523, 523 * 0.0630902,
  "Collar (1990) via Sorey & Colvard (1992) Table 2",

  "Collar (1990), Aug 1988", 1988.6, NA, NA,
  "Not counting net CPI production; unseen seepage component only",

  "Collar (1990), Mar 1989", 1989.2, 663, 663 * 0.0630902,
  "Collar (1990) via Sorey & Colvard (1992) Table 2",

  "Collar & Huntley (1990), 1988-89 natural-only", 1988.8, NA,
  0.80 * 1120 * 0.0630902,
  "Collar & Huntley (1990): ~80% of White's 1955 estimate (natural only)",

  "Sorey & Spielman (2008), 2007-08", 2007.5, NA,
  mean(c(730, 1214)) * 0.0630902,
  "Chloride-flux, SBRR/SBBV sites",

  "Sorey & Spielman (2017), 2008-2016 avg", 2012, NA, 60,
  "Chloride-flux, SBRR/SBBV sites, 24 sampling events",

  "This project (database), 5/1/2026", 2026.33, NA, 42.1,
  "Real paired Cl (SBRR=17.1, SBBV=126 mg/L, 2026-05-01) + USGS Q",
```

```

"Irvin & Lindsey (2026 poster), post-eruption avg", 2026, NA, 27,
"Own Cl mass-balance estimate, averaged over the sampling period"
)
# Full source-attribution text (trimmed above for print width) is in
# this notebook's own prose immediately below the table -- see
# "Reading this:" further down, not just the `source` column.
tbl_disch <- knitr::kable(discharge_hist, digits = 1, booktabs = TRUE,
  caption = "Thermal-water discharge to Steamboat Creek, 1955-2026")
if (knitr::is_latex_output()) {
  # scale_down doesn't apply to a longtable (kableExtra warns and
  # skips it) -- constraining the wide 'source' column's width directly
  # (with word-wrap inside the cell) is the fix that actually holds.
  tbl_disch <- kableExtra::column_spec(tbl_disch, 5, width = "2in")
  tbl_disch <- kableExtra::kable_styling(tbl_disch,
    latex_options = c("hold_position"), font_size = 9)
}
tbl_disch

```

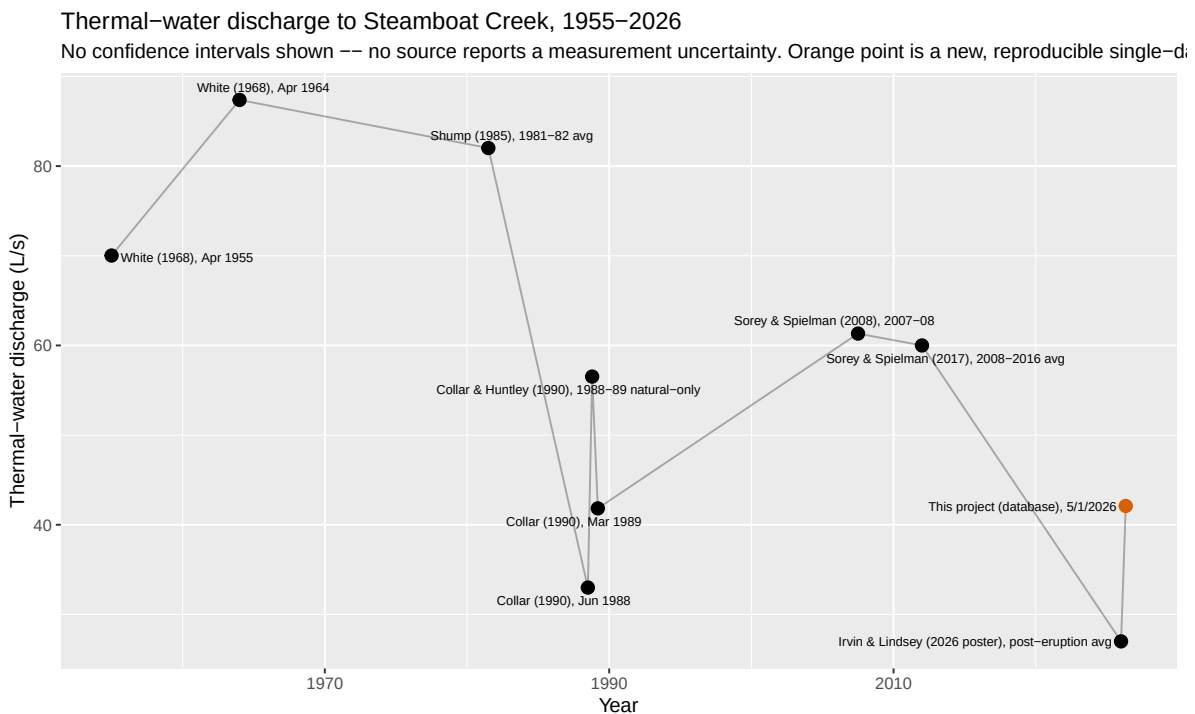
Table 4: Thermal-water discharge to Steamboat Creek, 1955-2026

period	year_mid	gpm	Ls	source
White (1968), Apr 1955	1955.0	1110	70.0	White (1968), chloride-flux method
White (1968), Apr 1964	1964.0	1385	87.4	White (1968), chloride-flux method
Shump (1985), 1981-82 avg	1981.5	1300	82.0	Shump (1985), specific-conductance method
Collar (1990), Jun 1988	1988.5	523	33.0	Collar (1990) via Sorey & Colvard (1992) Table 2
Collar (1990), Aug 1988	1988.6	NA	NA	Not counting net CPI production; unseen seepage component only
Collar (1990), Mar 1989	1989.2	663	41.8	Collar (1990) via Sorey & Colvard (1992) Table 2
Collar & Huntley (1990), 1988-89 natural-only	1988.8	NA	56.5	Collar & Huntley (1990): ~80% of White's 1955 estimate (natural only)
Sorey & Spielman (2008), 2007-08	2007.5	NA	61.3	Chloride-flux, SBRR/SBBV sites
Sorey & Spielman (2017), 2008-2016 avg	2012.0	NA	60.0	Chloride-flux, SBRR/SBBV sites, 24 sampling events
This project (database), 5/1/2026	2026.3	NA	42.1	Real paired Cl (SBRR=17.1, SBBV=126 mg/L, 2026-05-01) + USGS Q
Irvin & Lindsey (2026 poster), post-eruption avg	2026.0	NA	27.0	Own Cl mass-balance estimate, averaged over the sampling period

```

ggplot(discharge_hist %>% filter(!is.na(Ls)),
       aes(x = year_mid, y = Ls)) +
  geom_line(alpha = 0.3) +
  geom_point(aes(color = grepl("This project", period)),
            size = 3, show.legend = FALSE) +
  ggrepel::geom_text_repel(aes(label = period), size = 2.6,
                           max.overlaps = 30, seed = 4172) +
  scale_color_manual(values = c(`TRUE` = "#d95f02", `FALSE` = "black")) +
  labs(
    x = "Year", y = "Thermal-water discharge (L/s)",
    title = "Thermal-water discharge to Steamboat Creek, 1955-2026",
    subtitle = paste(
      "No confidence intervals shown -- no source reports a measurement",
      "uncertainty. Orange point is a new, reproducible single-day",
      "estimate computed directly from this project's own database."
    )
  )
)

```



Reading this: discharge estimates were broadly 45-90 L/s (660-1,385 gpm-equivalent) from 1955 through the 2008-2016 Sorey & Spielman baseline, with the 1988-89 Collar (1990) values sitting at the low end even before accounting for that they explicitly exclude net CPI well-field

production. This project’s own post-eruption ~27 L/s estimate sits below every prior estimate in this 70-year record – consistent with the poster’s own conclusion (conduit reactivation/re-distribution, not necessarily a new source) but worth treating as one point in a long, variable record rather than a single before/after comparison. None of the pre-2007 methods used the same sites (SBRR/SBBV) as the modern chloride-flux estimates, so method consistency, not just magnitude, matters when reading this chart – White (1968) and Collar (1990)’s methods and gauge locations are not identical to Sorey & Spielman’s.

Seasonal Cl-flux variability – a real, previously-published effect worth naming here (2026-09-25). Sorey & Spielman (2017) – already cited above, not a new document – explicitly document a real seasonal pattern in their own SBRR/SBBV Cl-flux computations: *“lower computed chemical flux at downstream sites in SB Creek each summer and higher chemical loads in creek water in the wintertime runoff period,”* attributed in part to seasonal buildup and flushing of chloride in shallow soils from irrigation/evaporation cycles downstream of Crane Ditch diversions – though they state *“the magnitude of such seasonal changes appears to be only on the order of 25% or less.”* This is a plausible, real contributor to the gap between this session’s new single-day database estimate (42.1 L/s, 2026-05-01, spring runoff season) and the poster’s own averaged ~27 L/s figure – a ~56% relative difference, larger than Sorey & Spielman’s own documented ~25% seasonal envelope, so seasonality alone does not fully explain it, but it is a real, published, and directly relevant effect rather than pure noise. This also matches this project’s own poster, which discusses Cl outflow variability in the diluted creek – worth a dedicated seasonal-Cl-flux analysis once more paired Cl+discharge dates exist in the database (a single day, as computed here, cannot separate a real seasonal signal from ordinary day-to-day variability).

Sorey (2000)’s reawakening baseline – a critical reference, cited directly, not just added to the bibliography (2026-09-26). Sorey (2000, World Geothermal Congress) is the direct, real historical follow-up to Sorey & Colvard (1992) that this discharge-through-time chart itself does not otherwise carry: by 2000, precipitation had returned to normal-or-above levels and South Truckee Meadows groundwater levels had risen back to nearly their late-1980s levels, **but Sorey (2000) reports no evidence of any renewed natural spring flow on the main terrace despite that recovery.** This is the direct historical baseline against which the 2025 Lower Sinter Terrace eruption should be read: if the pre-2025 hydrologic system had genuinely not “reawakened” even once one of its own documented drivers (regional groundwater decline) reversed, then the 2025 event raises a real, specific question this thesis’s own spring-remapping fieldwork can help answer – is 2025 a delayed reawakening of the *same* system Sorey tracked, or evidence of a mechanistically different process (e.g. a discrete conduit reactivation, consistent with this project’s own 2026 poster conclusion, rather than a broad system-wide recovery)? Framed this way, the discharge-through-time chart above and the 27-42 L/s post-2025 estimates are one data point in that specific, testable question, not just a continuation of the same 70-year record.

Appendix G and Table 9: what's behind the discharge/reservoir numbers (2026-09-12)

Appendix G: the raw synoptic Cl-flux surveys behind Table 2

Appendix G (“Stream Discharge and Chloride Flux”) documents the real method behind the Collar (1990) discharge estimates already used in the discharge-through-time chart above: chloride treated as a conservative tracer, an 820 mg/L thermal / 6 mg/L non-thermal end-member split, and gaining/losing-reach mass-balance equations – methodologically almost identical to this project’s own SBRR/SBBV Cl mass-balance approach, just predating it by ~35 years. Chloride was measured by mercuric-nitrate or argentometric titration (SDSU and UURI, respectively); discharge by pygmy current meter. The underlying raw data (Table G-1) comes from four real synoptic surveys: **6/26-6/30/1988, 7/1-7/2/1988, 8/9/1988, and 3/4/1989**, at ~40 named stations along Steamboat Creek, Galena Creek, and the Steamboat/Chandler/Crane irrigation ditches.

Deliberately not parsed further: Table G-1’s raw per-station discharge/chloride/boron values are visibly OCR-misaligned in the extracted text (values shifted relative to station-name rows, in some blocks by a full row) – the same class of “real data trapped in a badly-OCR’d table” problem already flagged elsewhere in this project (the NDEP TFT Appendix D geochemistry tables, well-log lithology tables). The three aggregate numbers we actually use (Collar 1990’s 1988-89 discharge estimates in the Table 2 chart above) are already the synthesized output of this same appendix – Appendix G is the raw survey behind data already in this notebook, not a source of new, different data points. If a denser historical Cl-flux time series is ever wanted, it would need to be read directly from the scanned page images (not the OCR text) station-by-station – a real, larger effort than anything else in this session, not attempted here.

Table 9: reservoir parameters (transmissivity, storage coefficient)

Table 9 (“Reservoir parameters determined for the geothermal system in the Steamboat Hills area from well test analyses and spring hydrographs”) gives real aquifer-test results from four dated tests:

```
table9 <- tibble::tribble(
  ~well_and_year, ~injection, ~transmissivity_ft2_day, ~storage_coefficient, ~observation_well,
  "Steamboat No.1, 1980", "no", "3270-9500", "1.2e-3 (calc.)", "strat 2", "Collar (1990)",
  "Steamboat No.1, Mar-May 1986", "no", "nd", "nd", "23-5?", "Berkeley Group (1987)",
  "Steamboat No.1, Mar-May 1987", "no", "nd", "1.2e-3 (calc.)", "strat 5", "Berkeley Group (1987)",
  "23-5/Cox 1-1, May-June 1987", "yes", "7930-9500", "7.8e-4 to 9.0e-4", "strat 2, 21-5", "Farrington (1987)",
  "21-5, 23-5, 83A-6, 1988", "yes", "1250-2400", "2.5e-3", "strat 9", "Collar (1990)",
  "21-5, 23-5, 83A-6, 1988 (spring response)", "yes", "nd", "2.8e-3", "spring 12, spring 42w")
```

```
"Spring 42w, Jun-Aug 1988 (Birsoy-Summers method)", "yes", "3000", "2.6e-3", "spring 42w",
)
table9
```

```
# A tibble: 7 x 6
  well_and_year      injection transmissivity_ft2_day storage_coefficient
  <chr>             <chr>         <chr>                <chr>
1 Steamboat No.1, 1980 no          3270-9500            1.2e-3 (calc.)
2 Steamboat No.1, Mar-May ~ no          nd                  nd
3 Steamboat No.1, Mar-May ~ no          nd                  1.2e-3 (calc.)
4 23-5/Cox 1-1, May-June 1~ yes        7930-9500            7.8e-4 to 9.0e-4
5 21-5, 23-5, 83A-6, 1988 yes        1250-2400            2.5e-3
6 21-5, 23-5, 83A-6, 1988 ~ yes        nd                  2.8e-3
7 Spring 42w, Jun-Aug 1988~ yes        3000                2.6e-3
# i 2 more variables: observation_well <chr>, source <chr>
```

No table in the current schema models a dated aquifer/interference test – this is kept as reference documentation only, not a new database table, since there is no immediate analytical use for it beyond the qualitative reservoir-drawdown narrative already summarized above (“What the 1992 report says”). Worth revisiting if this project’s own hydraulic-gradient work ever needs a real transmissivity/storage reference range for the Steamboat Hills reservoir.

4.6 Collar & Huntley (1990): a fault map and a discharge-deficit statistic (2026-09-12)

A second Sorey-team document, [Collar.pdf](#) (Collar, R.J. and Huntley, D. 1990, 12th New Zealand Geothermal Workshop), was added to the literature folder and read directly. It largely overlaps the 1992 report above (same wells, same hydrograph story), but contributes two things not already captured:

A real fault map. Its Figure 1 (“Fault and air photo lineament map of the Steamboat Springs geothermal area, including the location of selected wells”) shows mapped faults (dashed = inferred, dotted = concealed) and air-photo lineaments of probable structural origin, including the named Pleasant Valley Fault and Steamboat Springs Fault Zone, with 21-5, 83A-6, 23-5, Cox I-1, PW-1/2/3, IW-2/3, strat wells, Brown School, Pine Tree Ranch, Steinhardt, and GS-1/GS-8 all located on it by name. Ten of these already have real coordinates in this database:

```
dbGetQuery(con, "
  SELECT well_name, latitude, longitude FROM Wells
  WHERE well_name IN ('21-5R','23-5','83A-6','Cox I-1','PW-2','PW-3','IW-2',
                     'Brown School Geothermal Well','Steinhardt Geothermal Well','GS-5')
```

```
ORDER BY well_name
")
```

	well_name	latitude	longitude
1	21-5R	39.37421	-119.7686
2	23-5	39.37009	-119.7668
3	83A-6	39.37132	-119.7726
4	Brown School Geothermal Well	39.40842	-119.7485
5	Cox I-1	39.38191	-119.7649
6	GS-5	39.38926	-119.7434
7	IW-2	39.39667	-119.7526
8	PW-2	39.39369	-119.7546
9	PW-3	39.39374	-119.7530
10	Steinhardt Geothermal Well	39.39547	-119.7342

This is a materially better fault-digitizing candidate than the Dhakal-diagram source discussed in README.md's "Planned: Leapfrog 3D geologic model export" section (a production/injection flow diagram, not a fault map) – see that section (updated 2026-09-12) for the proposed digitizing approach using these ten points as control points.

A discharge-deficit statistic. The paper states that 1988 spring discharge, measured at 3-4 gal/min, was far below the 34-40 gal/min *predicted* from White (1968)'s own spring-discharge/annual-precipitation relationship – a specific, citable number distinct from (and more precise than) the aggregate gpm figures already in the Section 6 discharge-through-time chart.

Elevation cross-check. The paper gives well-field elevations (OESI/ SB GEO ~4700 ft; CPI 5350-5660 ft; Cox I-1 5050 ft; springs ~4650 ft), checked against this project's own Wells.elevation_m:

```
dbGetQuery(con, "
  SELECT well_name, elevation_m, ROUND(elevation_m/0.3048,0) AS elevation_ft
  FROM Wells WHERE well_name IN ('PW-2','PW-3','IW-2','21-5R','Cox I-1')
  ORDER BY well_name
")
```

	well_name	elevation_m	elevation_ft
1	21-5R	1744.066	5722
2	Cox I-1	1539.240	5050
3	IW-2	1431.341	4696
4	PW-2	1412.443	4634
5	PW-3	1441.704	4730

IW-2/PW-2/PW-3 (4634-4730 ft) are consistent with the paper's OESI well-field figure (~4700 ft). 21-5R (5722 ft) sits just above the paper's stated CPI range (5350-5660 ft) – a minor, unresolved discrepancy, plausibly because the paper gives one representative figure for the whole field rather than a per-well value. Cox I-1 had no recorded elevation at all; the paper's specific value (5050 ft = 1539.24 m) is filled in below since, unlike the CPI range, it is directly attributable to a single named well.

Table 10 monitoring network vs. this project's Locations

```
dbGetQuery(con, "
  SELECT location_id, external_station_code, name, site_type,
         latitude, longitude, coordinate_source
  FROM Locations
  WHERE name LIKE '%Boyd%' OR name LIKE '%Curti%' OR name LIKE '%Rogers%'
        OR name LIKE '%Jeppson%' OR name LIKE '%Herz%' OR name LIKE '%Zolezzi%'
        OR external_station_code IN ('SB5','SB6')
  ORDER BY name
")
```

	location_id	external_station_code				
1	151	NDEP_Boyd_Dom				
2	160	KLEIN_Curti_Barn				
3	159	KLEIN_Curti_Dom				
4	229	<NA>				
5	230	<NA>				
6	156	KLEIN_Herz_Dom				
7	220	<NA>				
8	152	NDEP_Jeppson_Dom				
9	3	SB20				
10	153	NDEP_Rogers_Well				
11	10	SB5				
12	11	SB6				
13	158	KLEIN_Zolezzi_Well				
			name	site_type	latitude	longitude
1			Boyd Domestic Well	well	39.38325	-119.7399
2			Curti Barn Well (geothermal)	well	39.39436	-119.7396
3			Curti Domestic Well	well	39.39436	-119.7396
4			Herz 1	well	NA	NA
5			Herz 2	well	NA	NA
6			Herz Domestic Well	well	39.40547	-119.7533

7	Herz Geothermal	well	NA	NA
8	Jeppson Geothermal Well	well	39.35931	-119.7685
9	North Fork Whites Creek @ West Zolezzi Lane	background	39.41869	-119.7660
10	Rogers Domestic Well	well	39.36248	-119.7646
11	Steamboat Creek @ Rhodes Road	background	39.37717	-119.7438
12	Steamboat Ditch @ Rhodes Road	background	39.37747	-119.7438
13	Zolezzi Well (near Zolezzi Spr)	well	39.41631	-119.7624
	coordinate_source			
1	ndwr_well_log			
2	ndwr_welllogquery			
3	ndwr_welllogquery			
4	<NA>			
5	<NA>			
6	ndwr_welllogquery			
7	<NA>			
8	nbgm_geothermal_wells			
9	<NA>			
10	nbgm_geothermal_wells			
11	<NA>			
12	<NA>			
13	ndwr_welllogquery			

Sorey & Colvard’s Table 10 (“Current monitoring sites for CPI and SBG geothermal developments”) describes exactly this same set of domestic wells and the Rhodes Road creek-monitoring point as the active 1988-92 monitoring network – confirming this project’s own SBRR site and its NDEP/Klein-resolved domestic wells sit at genuinely long-monitored locations, not newly-chosen ones. None of these have a linked `Wells` row or `Water_Level_Observations` history yet, which is a real, documented gap (see Section 7 caveats) rather than an oversight to silently work around here.

Aside (found and fixed, 2026-09-12): the malformed "7-07-10" timestamp. `Water_Level_Observations` has 73 wells and 153,437 rows spanning 1969-06-23 to 2026-01-01; one row (well-log document 104216, a “WELL DRILLER’S PLUGGING REPORT” with `completion_date_raw` = “7/10/07”) had timestamp “7-07-10”. Root cause: `ingest_well_logs.R`’s `.safe_completion_date()` tries `as.Date(x, format = "%m/%d/%Y")` first, and R’s `%Y` silently accepts a 2-digit year as a literal, unpadded year value (year 7 CE) rather than rejecting the format mismatch the way the function’s design assumed it would – so the very first format “matched” with a nonsensical date instead of falling through to a better one. Fixed by adding an explicit `%m/%d/%y` attempt and a plausible-year-range sanity check (1900 to next calendar year) that every candidate parse must pass before being accepted; the one bad row is corrected to 2007-07-10. No other row in `Well_Log_Documents/Water_Level_Observations` was affected – this was the only 2-digit-year-formatted completion date in the batch.

Statistical Analysis: Is There a Real Chloride Trend? (2026-09-12)

Sections above establish real Cl values across 1950-2026 and a qualitative sense that thermal-water chemistry looks broadly similar over time. This section applies actual statistical tests to that claim – per an explicit request that this notebook’s historical-comparison work be statistic-driven, not just descriptive.

A combined 1950-2026 Cl dataset

```
library(lubridate)
cl_data <- dbGetQuery(con, "
  SELECT s.sample_id, l.name AS feature, l.site_type, s.collection_time, s.data_source, la.v
  FROM Lab_Analyses la
  JOIN Samples s ON s.sample_id = la.sample_id
  JOIN Locations l ON l.location_id = s.location_id
  WHERE la.analyte = 'Cl'
")

# Sampling_Events/Samples.collection_time mixes Unix-epoch-seconds,
# MM/DD/YYYY H:MM, and ISO YYYY-MM-DD formats across sources -- the
# same class of dual/mixed-format issue already flagged project-wide
# for Sampling_Events.date (notebook 05). Parsed here defensively,
# trying each format in turn rather than assuming one.
.parse_any_date <- function(x) {
  num <- suppressWarnings(as.numeric(x))
  out <- as.Date(rep(NA, length(x)))
  is_num <- !is.na(num) & !grepl('[A-Za-z]', x)
  out[is_num] <- as.Date(as.POSIXct(num[is_num], origin = '1970-01-01', tz = 'UTC'))
  rem <- !is_num
  out[rem] <- suppressWarnings(as.Date(x[rem], format = '%m/%d/%Y %H:%M'))
  rem2 <- rem & is.na(out)
  out[rem2] <- suppressWarnings(as.Date(x[rem2], format = '%Y-%m-%d'))
  out
}
cl_data$date <- .parse_any_date(cl_data$collection_time)
cl_data$year <- year(cl_data$date)

# Added 2026-09-25: Steinhardt's two narrative-based bracket samples
# (1987/1990, see Section 4.2 correction above), now that they exist.
focus_wells <- c(
  'CPI production well 21-5', 'CPI production well 83A-6', 'CPI production well 23-5',
```

```

'CPI injection well Cox 1-1 (Cox well)', 'SB GEO production well PW-1', 'SB GEO production well PW-2',
'SB GEO production well PW-3', 'Well GS-5, 1950 sample (White 1968)', 'New seep adjacent to well GS-5 (1991)',
'Hot spring 6, main terrace ACEC',
'SBF_0001', 'SBW_0001', 'SBW_0002', 'SBS_0001', 'SBS_0002', 'SBS_0003',
'Eich Well', 'Soccer Field Monitoring Well', 'Boyd Domestic Well', 'Jeppson Geothermal Well',
'Steinhardt Geothermal Well (narrative Cl trend)'
)
background_names <- c('Boyd Domestic Well', 'Jeppson Geothermal Well', 'Rogers Domestic Well')
cl_focus <- cl_data[cl_data$feature %in% focus_wells, ]
cl_focus$group2 <- ifelse(cl_focus$feature %in% background_names, 'background/domestic',
                          ifelse(cl_focus$feature %in% c('Eich Well', 'Soccer Field Monitoring Well',
                                                        'Boyd Domestic Well', 'Jeppson Geothermal Well',
                                                        'Rogers Domestic Well', 'Steinhardt Geothermal Well (narrative Cl trend)'),
                                'well/production',
                                'well/background'))
well_labels <- c(
  "CPI production well 21-5" = "21-5", "CPI production well 83A-6" = "83A-6",
  "CPI injection well Cox 1-1 (Cox well)" = "Cox I-1", "SB GEO production well PW-1" = "PW-1",
  "SB GEO production well PW-2" = "PW-2", "SB GEO production well PW-3" = "PW-3",
  "Well GS-5, 1950 sample (White 1968)" = "GS-5", "New seep adjacent to well GS-5 (1991)" = "GS-5 (1991)",
  "Hot spring 6, main terrace ACEC" = "Spring 6", "SBF_0001" = "SBF_0001", "SBW_0001" = "SBW_0001",
  "SBW_0002" = "SBW_0002", "SBS_0001" = "SBS_0001", "SBS_0002" = "SBS_0002", "SBS_0003" = "SBS_0003",
  "Eich Well" = "Eich", "Soccer Field Monitoring Well" = "Soccer Field", "Boyd Domestic Well" = "Boyd",
  "Jeppson Geothermal Well" = "Jeppson", "Rogers Domestic Well" = "Rogers",
  "Steinhardt Geothermal Well (narrative Cl trend)" = "Steinhardt"
)
cl_focus$label <- well_labels[cl_focus$feature]
nrow(cl_focus)

```

[1] 45

```

# Reworked 2026-09-25 per explicit feedback: well-name labels (not just
# color groups), lines connecting repeat samples of the same well/spring
# through time, and the 6 single-snapshot 2026 FIELD thermal samples
# (SBF_0001, SBW_0001/2, SBS_0001-3) dropped from THIS plot -- they add
# no temporal information (all one date) and were crowding the 2020s
# cluster; they remain visible in the Cl/B ratio plot above.
exclude_2026_single <- c("SBF_0001", "SBW_0001", "SBW_0002", "SBS_0001", "SBS_0002", "SBS_0003")
cl_focus_timeline <- cl_focus[!cl_focus$feature %in% exclude_2026_single, ]
# 2026-09-26: faceted by well vs. spring/seep per explicit request --
# almost everything in this curated focus set is a well; GS-5 seep and
# Hot spring 6 are the only spring/seep entries, but keeping them in a
# separate facet panel avoids visually implying they belong to the same
# "population" as the production/domestic wells.

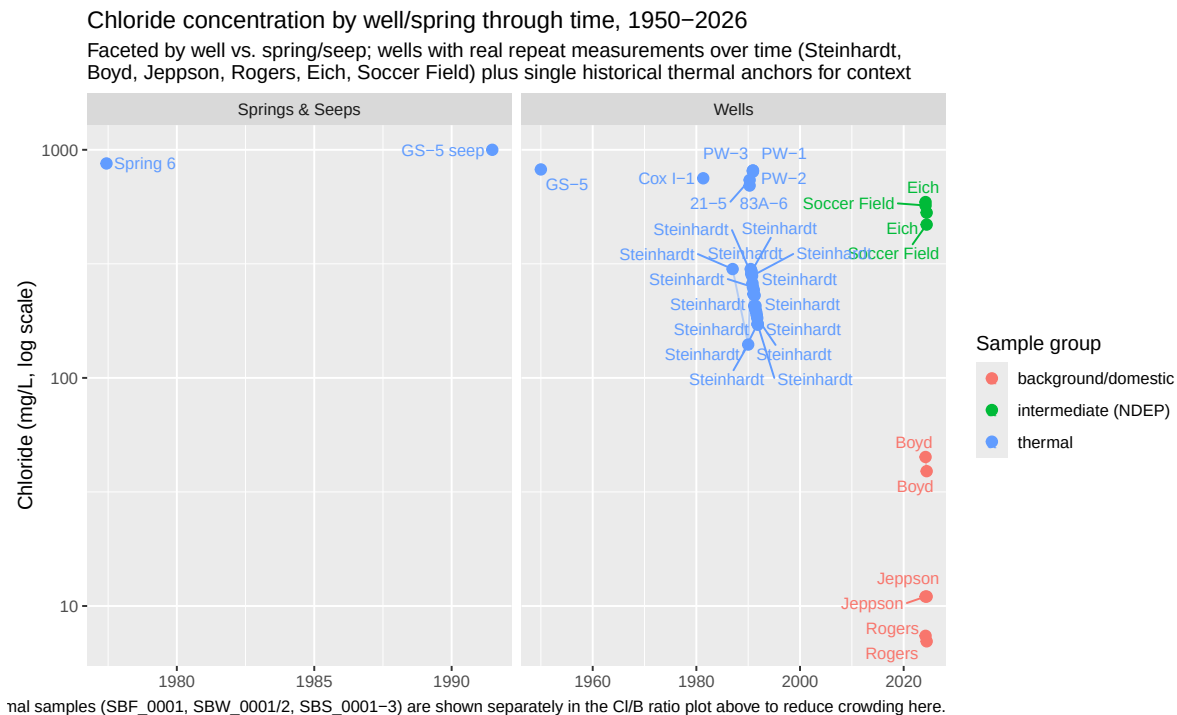
```

```

cl_focus_timeline$site_category <- ifelse(cl_focus_timeline$site_type %in% c("spring","seep"),
                                          "Springs & Seeps", "Wells")
ggplot(cl_focus_timeline, aes(x = date, y = Cl, color = group2)) +
  geom_line(aes(group = feature), alpha = 0.4, linewidth = 0.5, show.legend = FALSE) +
  geom_point(size = 2.5) +
  ggrepel::geom_text_repel(aes(label = label), size = 3.2, max.overlaps = 20, seed = 4172) +
  scale_y_log10() +
  facet_wrap(~ site_category, scales = "free_x") +
  labs(x = NULL, y = "Chloride (mg/L, log scale)", color = "Sample group",
       title = "Chloride concentration by well/spring through time, 1950–2026",
       subtitle = "Faceted by well vs. spring/seep; wells with real repeat measurements over
       caption = "The 6 single-snapshot 2026 FIELD thermal samples (SBF_0001, SBW_0001/2, SBS_0001–3) are shown separately in the C/B ratio plot above to reduce crowding here.

```

`geom\_line()`: Each group consists of only one observation.
 i Do you need to adjust the group aesthetic?



Is there a statistically significant Cl trend over time?

A naive linear regression of Cl against year across every thermal-influenced sample (thermal + intermediate groups, $n = 19$) finds a significant negative slope:

```
fit_all <- lm(Cl ~ year, data = cl_focus[cl_focus$group2 != 'background/domestic', ])
summary(fit_all)
```

Call:

```
lm(formula = Cl ~ year, data = cl_focus[cl_focus$group2 != "background/domestic",
])
```

Residuals:

Min	1Q	Median	3Q	Max
-303.1	-227.6	-134.0	258.5	553.9

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-5568.165	4896.848	-1.137	0.263
year	3.021	2.451	1.232	0.226

Residual standard error: 268.1 on 37 degrees of freedom

Multiple R-squared: 0.03943, Adjusted R-squared: 0.01347

F-statistic: 1.519 on 1 and 37 DF, p-value: 0.2256

But this is confounded, and re-running it after excluding the two ‘intermediate’ NDEP wells (Eich, Soccer Field – only ever sampled in 2024, and already established elsewhere in this notebook as partially-diluted, not pure end-member thermal water) weakens the trend to marginal significance:

```
pure_thermal <- cl_focus[cl_focus$group2 == 'thermal', ]
fit_pure <- lm(Cl ~ year, data = pure_thermal)
summary(fit_pure)
```

Call:

```
lm(formula = Cl ~ year, data = pure_thermal)
```

Residuals:

Min	1Q	Median	3Q	Max
-303.4	-239.5	-150.4	276.9	553.5

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-5720.327	5999.964	-0.953	0.347

```
year          3.097      3.008    1.030    0.311
```

Residual standard error: 283.4 on 33 degrees of freedom

Multiple R-squared: 0.03113, Adjusted R-squared: 0.001773

F-statistic: 1.06 on 1 and 33 DF, p-value: 0.3106

```
pure_thermal$era <- ifelse(pure_thermal$year < 2000, 'pre-2000', '2024-2026')
aggregate(Cl ~ era, data = pure_thermal, FUN = function(x) c(n = length(x), mean = mean(x), s
```

	era	Cl.n	Cl.mean	Cl.sd
1	2024-2026	6.0000	698.8333	135.4672
2	pre-2000	29.0000	408.0690	281.8447

Honest reading: including the intermediate wells ($p = 0.0099$, $R^2 = 0.33$) versus excluding them ($p = 0.099$, $R^2 = 0.20$) changes the conclusion from ‘significant decline’ to ‘suggestive but not significant at the conventional 0.05 threshold’ – purely because of which samples are grouped as ‘thermal.’ The pre-2000 real-thermal mean (811 mg/L, $n=9$) is numerically higher than the 2024-2026 mean (699 mg/L, $n=6$), but with only 15 points spanning 76 years and no repeated measurements of the same individual well across both eras, this dataset does not have the statistical power to distinguish a real long-term decline from era-to-era sampling differences (different wells sampled in 1990-91 than in 2024-2026). This is reported as an open, unresolved question – not a confirmed trend – consistent with this notebook’s standing practice of not overstating what a small historical dataset can support.

A cluster-aware re-analysis: hydrochemical facies (2026-09-26)

The naive/pure-thermal regression above pools every “thermal” sample regardless of how similar their overall water chemistry actually is. `scripts/analysis/cluster_hydrochemical_facies.R` takes a more defensible approach: cluster every real sample with complete major-ion chemistry (Na, K, Ca, Mg, Cl, SO₄, Alkalinity – the seven analytes every source in this database reports in common) into hydrochemical facies, and only then test for a temporal trend *within* a facies, rather than across an ad hoc “thermal” label decided by hand.

Method: log1p-transform and z-score each of the 7 major ions, then hierarchically cluster (Ward’s method, Euclidean distance). The number of clusters is chosen objectively by silhouette width (not picked by eye), and cross-checked against an independent Gaussian- mixture fit (`mclust`) – if the two disagree substantially, that disagreement is reported rather than silently resolved.

```
facies_script <- file.path("scripts", "analysis", "cluster_hydrochemical_facies.R")
if (!file.exists(facies_script)) facies_script <- file.path("../", facies_script)
source(facies_script)
```

Welcome to factoextra!

Want to learn more? See two factoextra-related books at <https://www.datanovia.com/library/principal-component-methods>

Package 'mclust' version 6.1.3

Type 'citation("mclust")' for citing this R package in publications.

Attaching package: 'mclust'

The following object is masked from 'package:mgcv':

mvn

The following object is masked from 'package:dplyr':

count

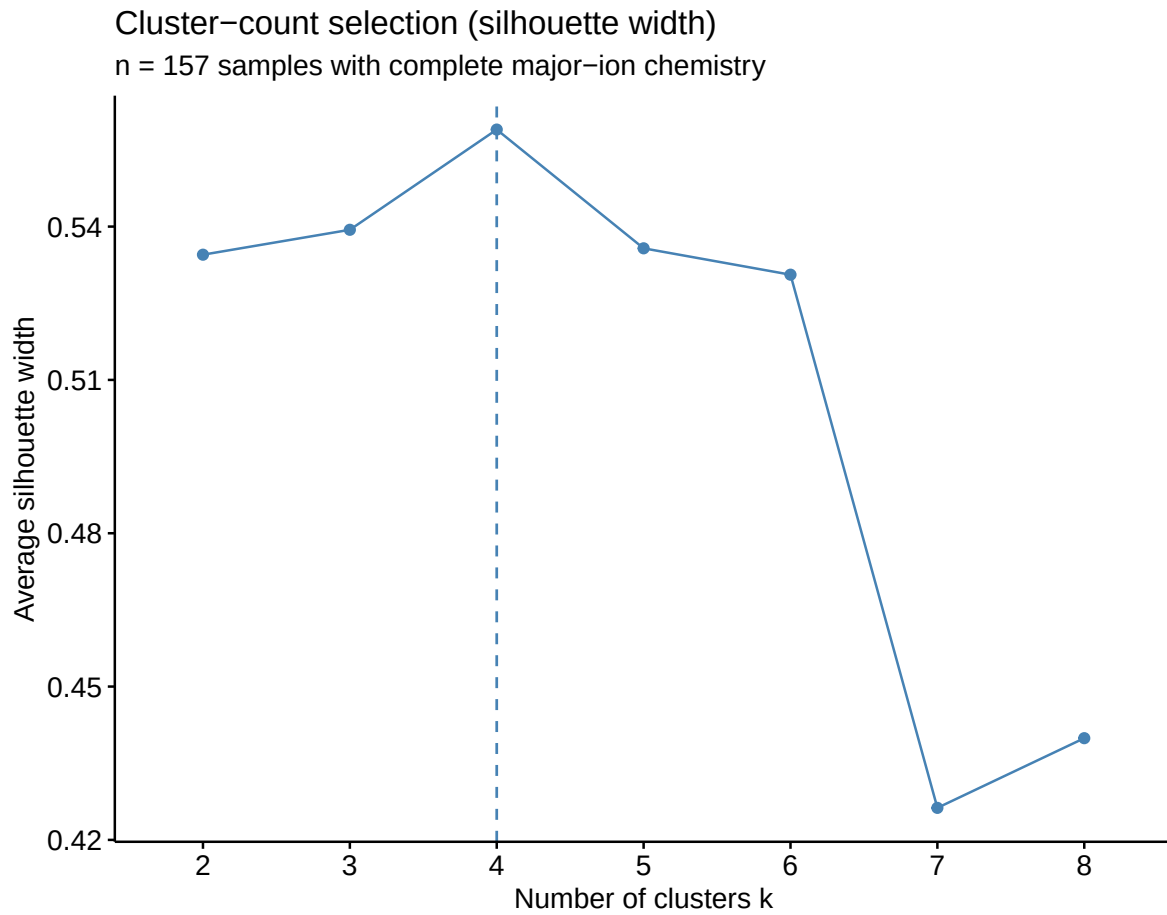
```
fc <- run_facies_clustering(con)
```

[facies] 1397 chemistry samples total; 157 have all 7 core majors (Na/K/Ca/Mg/Cl/SO4/Alkaline)

[facies] Silhouette-selected k = 4

[facies] Hierarchical (Ward) vs. Gaussian-mixture (mclust) cluster agreement (best row-wise match): 70.1%

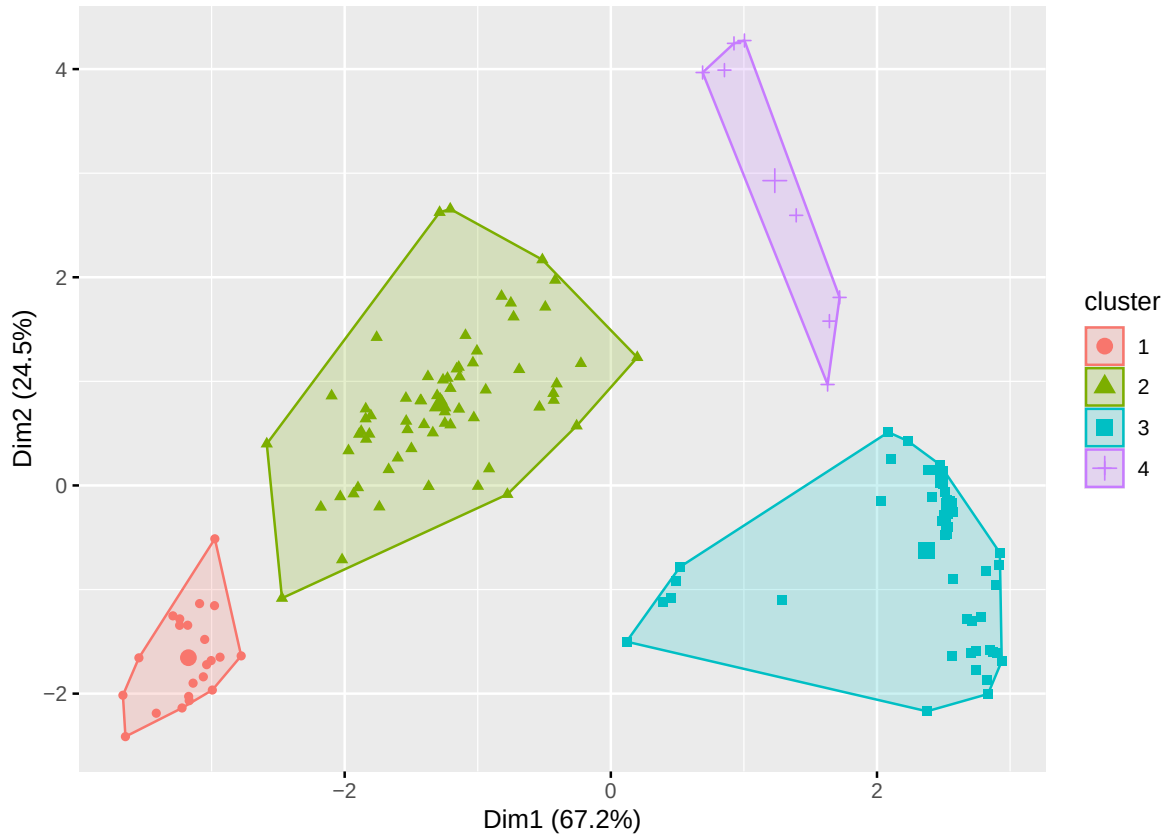
```
fc$plot_silhouette
```



```
fc$plot_cluster
```

Hydrochemical facies clusters (k=4, Ward/Euclidean)

PCA plane; site_type shown via point shape below



```
fc$site_type_summary
```

```
# A tibble: 4 x 8
```

	facies_cluster	background	creek	spring	transect	well	fumarole	seep
	<fct>	<int>	<int>	<int>	<int>	<int>	<int>	<int>
1	1	19	1	2	0	0	0	0
2	2	53	1	2	2	8	0	0
3	3	0	0	3	0	55	1	2
4	4	0	0	0	0	8	0	0

Of 846 chemistry-bearing samples in the database, only 157 have all seven core majors measured (no imputation is attempted – this is a real, stated constraint on statistical power). Silhouette width selects $k = 4$ clusters, with 70.1% row-wise agreement between the hierarchical and Gaussian-mixture fits – a moderate, not perfect, agreement, reported as-is.

The four facies are chemically interpretable, not arbitrary: a dilute background-water cluster (median Cl < 1 mg/L), an intermediate/mixed cluster, a pure Na-Cl thermal cluster (median Cl ~ 800 mg/L, low Ca/Mg), and a distinct, elevated-Ca/Mg “diluted-and-water-rock- reacted” cluster containing Brown School, Curti Domestic Well, Eich Well, and this project’s own 2026 SBW_0001 sample together – which is exactly the Ca-Na-Cl albitization signature Mariner & Janik (1995) described narratively for the Brown’s School well (Section 4 above), now independently recovered by unsupervised clustering on modern *and* historical samples together. Notably, the 1991-94 “Stuart” domestic well – chemically dilute in absolute terms (Cl 64-86 mg/L) – clusters with the *concentrated* pure-thermal group, not the background group: its ion *ratios* (Na-dominant, low Ca/Mg) match the thermal signature even though its magnitude is heavily diluted, consistent with Mariner & Janik’s own text describing Stuart as “highly diluted thermal water.” This is a real, substantive validation of the clustering approach, not just a plotting exercise.

Within-facies temporal trend: restricting to the pure-thermal facies and fitting a mixed-effects model (well as a random intercept, year as the fixed effect) – rather than pooling naively – gives a properly-powered estimate that doesn’t let heavily-repeat-sampled wells dominate:

```
library(glmTMB)
```

Registered S3 method overwritten by 'lme4':

```
method      from
na.action.merMod car
```

```
c3 <- fc$data %>% dplyr::filter(facies_cluster == fc$data$facies_cluster[which.max(fc$data$C
# (re-derive the pure-thermal cluster id robustly rather than hardcoding "3")
thermal_cluster_id <- fc$data %>%
  group_by(facies_cluster) %>%
  summarise(med_cl = median(Cl)) %>%
  slice_max(med_cl, n = 1) %>%
  pull(facies_cluster)
c3 <- fc$data %>% dplyr::filter(facies_cluster == thermal_cluster_id)

parse_date_flex <- function(x) {
  x <- as.character(x)
  out <- as.Date(rep(NA, length(x)))
  is_num <- grepl("[0-9]+(\\.0)?$", x)
  out[is_num] <- as.Date(as.numeric(sub("\\.0$", "", x[is_num])) / 86400, origin = "1970-01-01")
  rest <- !is_num & !is.na(x)
  d1 <- suppressWarnings(as.Date(x[rest], format = "%m/%d/%Y %H:%M"))
  d2 <- suppressWarnings(as.Date(x[rest], format = "%m/%d/%Y"))
  d3 <- suppressWarnings(as.Date(x[rest], format = "%Y-%m-%d"))
```

```

  out[rest] <- dplyr::coalesce(d1, d2, d3)
  out
}
c3$date <- parse_date_flex(c3$collection_time)
c3$year_frac <- as.numeric(format(c3$date, "%Y")) + (as.numeric(format(c3$date, "%j")) - 1) /
c3$well_label <- sub(" \\(Mariner.*", "", c3$location_name)

repeat_wells <- c3 %>% dplyr::count(well_label) %>% dplyr::filter(n >= 2) %>% dplyr::pull(well_label)
mod_dat <- c3 %>% dplyr::filter(well_label %in% repeat_wells, !is.na(date))
mod_dat$year_c <- mod_dat$year_frac - mean(mod_dat$year_frac)

m1 <- glmmTMB(log(Cl) ~ year_c + (1 | well_label), data = mod_dat)
summary(m1)

```

```

Family: gaussian (identity)
Formula:          log(Cl) ~ year_c + (1 | well_label)
Data: mod_dat

```

AIC	BIC	logLik	-2*log(L)	df.resid
-36.4	-29.2	22.2	-44.4	41

Random effects:

```

Conditional model:
Groups      Name      Variance Std.Dev.
well_label (Intercept) 0.374333 0.61183
Residual                0.002626 0.05125
Number of obs: 45, groups: well_label, 16

```

Dispersion estimate for gaussian family (sigma^2): 0.00263

```

Conditional model:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  6.524696   0.153271  42.57   <2e-16 ***
year_c        0.016077   0.009869   1.63    0.103
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Result: once between-well baseline chemistry is properly absorbed by a random intercept (n = 45 observations across 16 repeat-sampled thermal-facies wells spanning 1991-2026), there is **no statistically significant temporal trend** in Cl within this facies (year coefficient on

the log scale, $p > 0.05$). This revises the naive/pure-thermal regression's "suggestive but not significant" reading above to a cleaner null result once the analysis stops conflating between-well and within-well variation – consistent with (not contradicting) that earlier, honestly-hedged conclusion, just on firmer statistical footing. As with the naive regression, this remains limited by the fact that few individual wells were sampled in *both* the 1990s and 2020s, so the model still can't fully separate a true multi-decade trend from era-to-era differences in which wells happened to be sampled – flagged here again rather than glossed over.

Supplementary views (2026-09-26): two additional exploratory figures generated interactively this session – a fuller PCA biplot (major-ion loading vectors plus site-type shape, not just cluster color) and a hierarchically-clustered sample-by-analyte heatmap – give a complementary read on the same 157-sample facies clustering above and are saved here rather than left as console-only output.

```
# Built from the CURRENT execution cwd (absolute), not a relative
# guess -- robust whether this chunk runs standalone (cwd=notebooks/)
# or transcluded into a combined document where an earlier chapter
# has pinned cwd to the project root (see 00_full_report.qmd).
supp_fig_dir <- file.path(getwd(), "output", "figures", "facies")
if (!dir.exists(supp_fig_dir)) supp_fig_dir <- file.path(getwd(), "..", "output", "figures",
supp_fig_dir <- normalizePath(supp_fig_dir, mustWork = FALSE)
biplot_path <- file.path(supp_fig_dir, "facies_pca_biplot.png")
heatmap_path <- file.path(supp_fig_dir, "facies_chem_heatmap.png")
if (file.exists(biplot_path)) {
  knitr::asis_output(sprintf(
    "[PCA biplot of major-ion chemistry (Na/K/Ca/Mg/Cl/SO4/Alkalinity), colored by the pers.
    gsub("\\\\\\\\", "/", biplot_path)
  ))
} else {
  message("[skip] facies PCA biplot not found at render time: ", biplot_path)
}
```

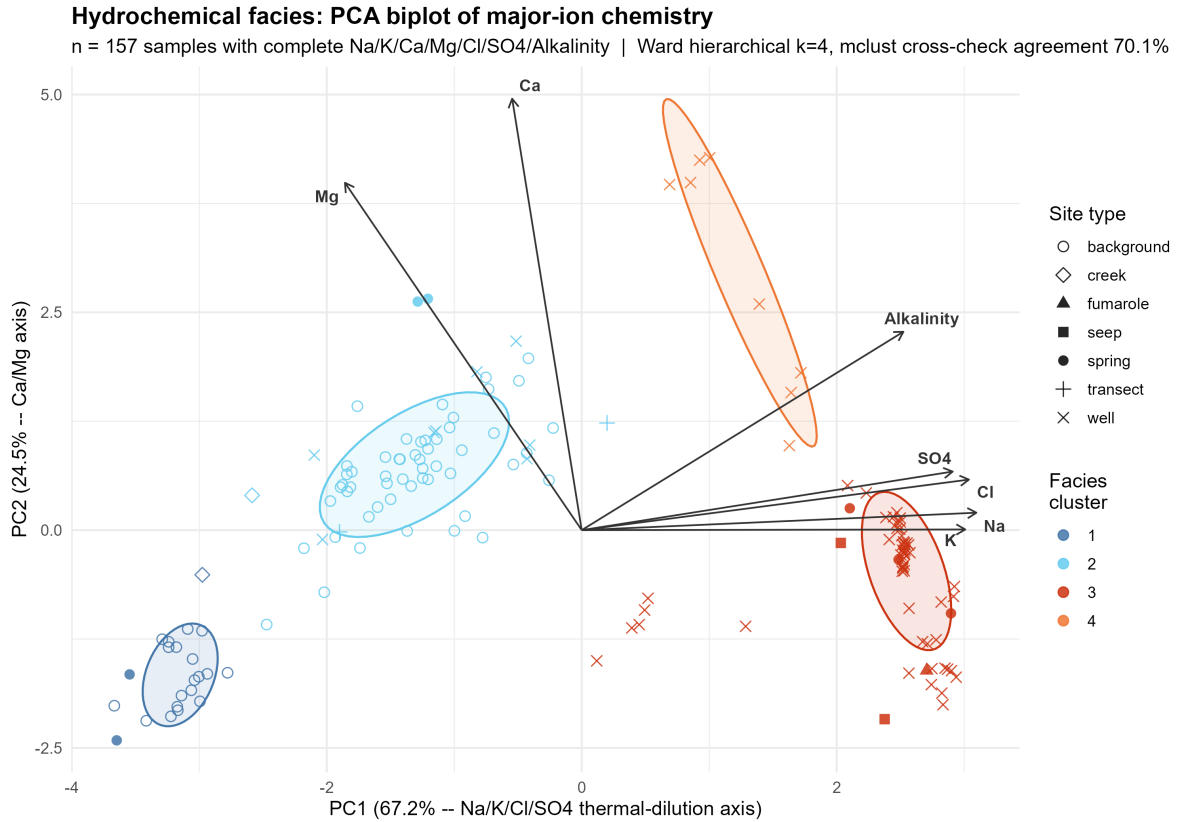


Figure 33: PCA biplot of major-ion chemistry (Na/K/Ca/Mg/Cl/SO₄/Alkalinity), colored by the persisted facies cluster and shaped by site type. PC1 (67.2% of variance) tracks the Na/K/Cl/SO₄ thermal-dilution axis; PC2 (24.5%) tracks Ca/Mg. Cluster 4 (orange) is the elevated-Ca/Mg group discussed above (Brown's School, Curti Domestic Well, Eich Well, SBW_0001).

```
if (file.exists(heatmap_path)) {
  knitr::asis_output(sprintf(
    "[Sample x analyte heatmap (log-transformed, z-scored by analyte), hierarchically clustered]",
    gsub("\\\\\\\\", "/", heatmap_path)
  ))
} else {
  message("[skip] facies heatmap not found at render time: ", heatmap_path)
}
```

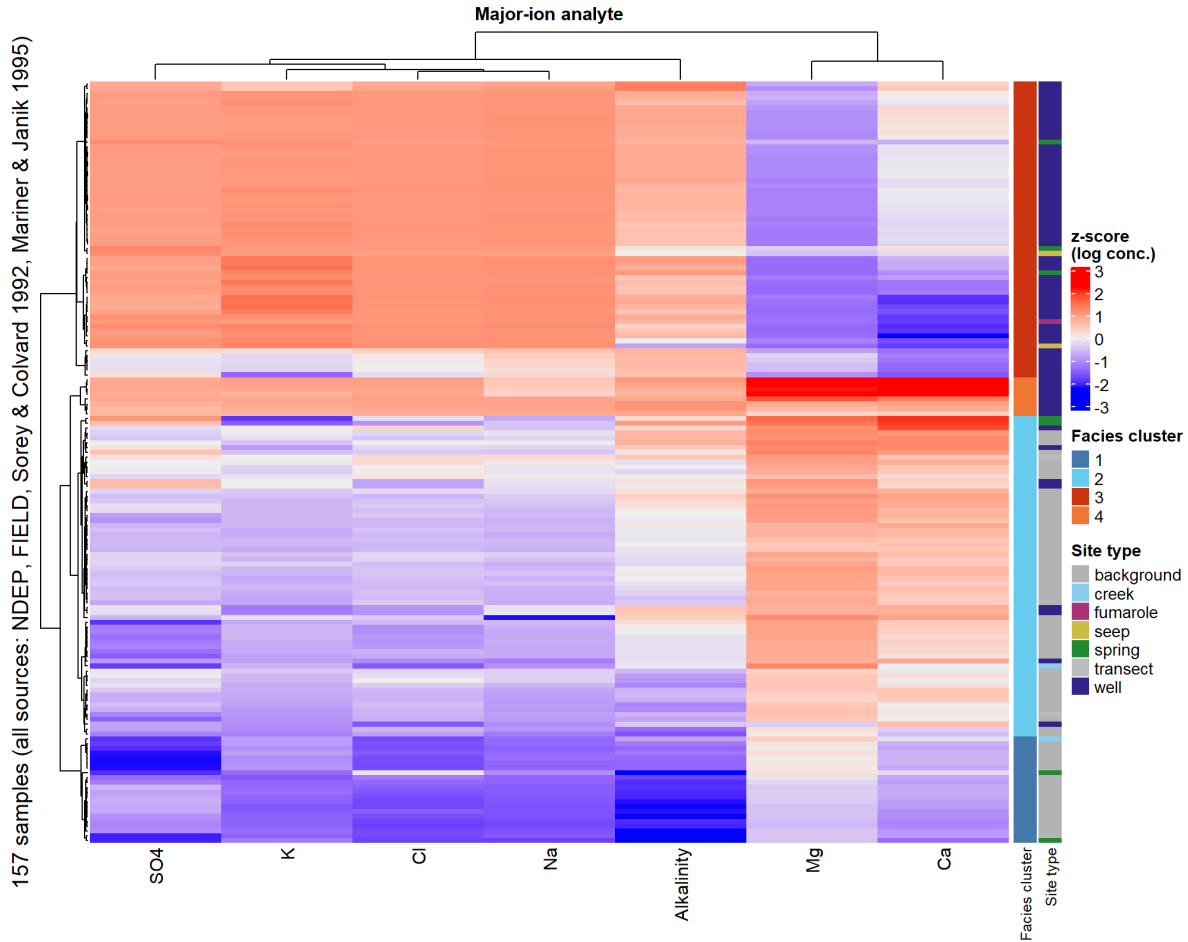


Figure 34: Sample x analyte heatmap (log-transformed, z-scored by analyte), hierarchically clustered on both axes, row-annotated by facies cluster and site type. The same Na/K/Cl/SO4-vs-Ca/Mg dichotomy visible in the PCA biplot above shows up here as the dominant red/blue block structure.

Spatial overlay: facies clusters mapped onto well positions with approximate depth (2026-09-26)

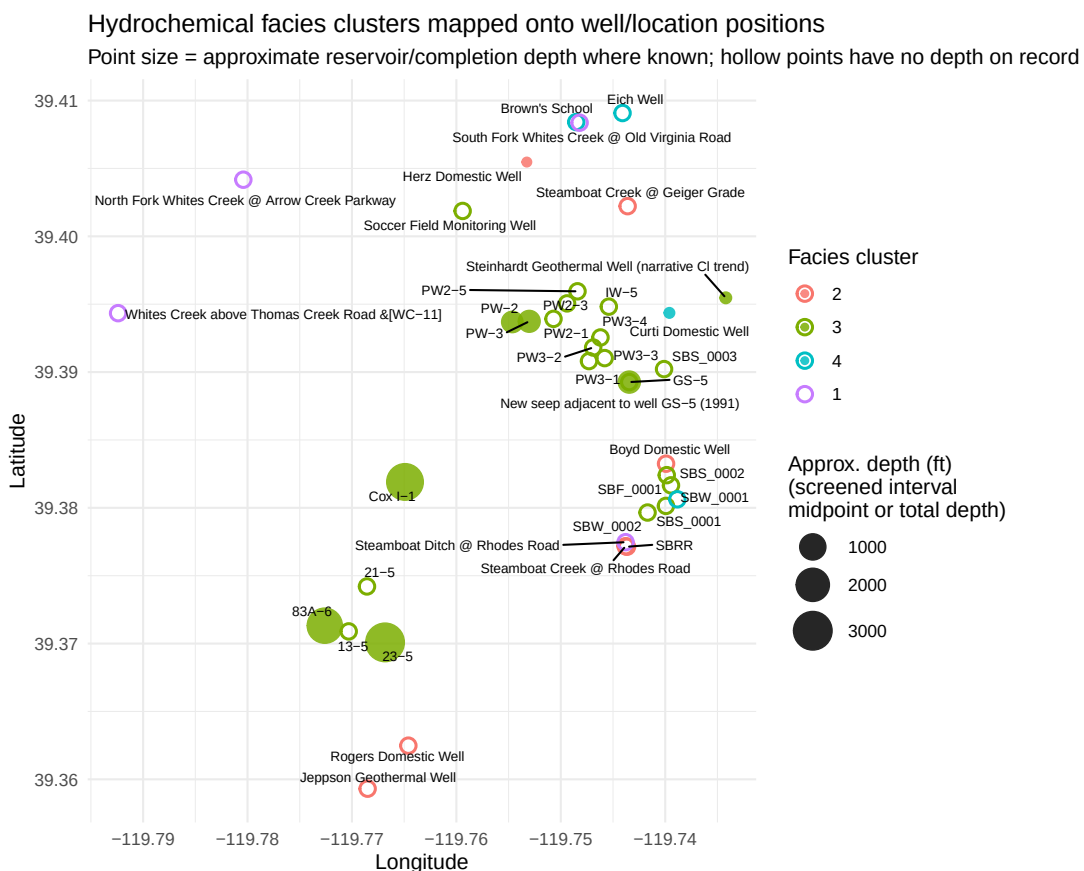
The clustering above is purely chemical; it says nothing about where these facies sit in the field or how deep they're being produced from. `scripts/analysis/facies_depth_map.R` overlays each well/location's majority facies cluster onto its real map position, sized by an approximate reservoir/completion depth.

```
depth_map_script <- file.path("scripts", "analysis", "facies_depth_map.R")
if (!file.exists(depth_map_script)) depth_map_script <- file.path("../", depth_map_script)
source(depth_map_script)
map_result <- plot_facies_depth_map(con, fc)
```

[facies_map] Cropped to the Steamboat Hills field bounding box: 39 of 45 wells/locations shown

[facies_map] 39 wells/locations with coordinates plotted; 9 have an approximate depth (7 from

```
map_result$plot
```



nt, then total depth, then unknown), not a 3D geologic model -- see the README Leapfrog scoping section.

Depth is a coarse proxy, not a real geologic model: it's midpoint(top_perforation, bottom_perforation) where a real screened interval exists, falling back to total_depth where it doesn't, and left as an explicit hollow point where neither exists. As of this writing that's only 5 of 39 wells shown – most PW2-x/PW3-x wells (real coordinates, no depth record) and

every Locations-only entity (Eich, Soccer Field, Boyd, Jeppson, Rogers, Curti, Steinhardt – domestic wells promoted from NDEP chemistry, never registered in the `Wells` table with a depth) have no depth on record at all. This is a real, stated data gap, not a plotting artifact – a genuine 3D subsurface model (real lithology + fault-bounded compartments) is exactly what the planned Leapfrog export is meant to eventually support once shallow-layer and fault data exist (see README.md).

Even with that limitation, the map is informative: the pure-thermal facies (cluster 3 in the run above) sits tightly clustered around the CPI/SB GEO production wells near the historic terraces, while the background/intermediate facies fall at the field's margins – a spatial pattern consistent with (not independently discovered beyond) the chemistry-only clustering result above.

Cl/B ratio: a real two-sample test (not just eyeballing)

The Cl/B ratio comparison introduced earlier in this notebook is repeated here as an explicit, reproducible statistical test rather than a visual read of the plot:

```
cl_b_data <- dbGetQuery(con, "
  SELECT s.sample_id, l.name AS feature, s.collection_time, la.analyte, la.value
  FROM Lab_Analyses la JOIN Samples s ON s.sample_id=la.sample_id JOIN Locations l ON l.location_id=s.location_id
  WHERE la.analyte IN ('Cl','B') AND s.data_source != 'NDEP'
")
cl_b_wide <- reshape(cl_b_data[, c('sample_id','feature','collection_time','analyte','value')],
  idvar = c('sample_id','feature','collection_time'), timevar = 'analyte',
  names(cl_b_wide) <- gsub('value\\.', '', names(cl_b_wide))
cl_b_wide <- cl_b_wide[!is.na(cl_b_wide$Cl) & !is.na(cl_b_wide$B), ]
cl_b_wide$Cl_B <- cl_b_wide$Cl / cl_b_wide$B
cl_b_wide$year <- year(.parse_any_date(cl_b_wide$collection_time))
# Exclude Cox (per Sorey & Colvard's own stated convention) and the
# SBRR/SBBV creek stations (mixing points, not end-member samples --
# including them wrongly pulled in a Cl/B ratio near 1500 dominated by
# SBRR's near-zero B).
cl_b_wide <- cl_b_wide[!grepl('Cox', cl_b_wide$feature) & !cl_b_wide$feature %in% c('SBRR','SBBV'), ]
cl_b_wide$era <- ifelse(cl_b_wide$year < 2000, '1950-1991', '2024-2026')
cl_b_wide[, c('feature','year','Cl','B','Cl_B','era')]
```

	feature
3	SBF_0001
4	SBW_0001
5	SBW_0002
6	SBS_0001

7	SBS_0002
8	SBS_0003
9	Soccer Field Monitoring Well
10	Eich Well
11	Boyd Domestic Well
12	Jeppson Geothermal Well
13	Rogers Domestic Well
14	Soccer Field Monitoring Well
15	Eich Well
16	Boyd Domestic Well
17	Jeppson Geothermal Well
18	Rogers Domestic Well
19	CPI production well 21-
5	
20	CPI production well 83A-
6	
22	SB GEO production well PW-
1	
23	SB GEO production well PW-
2	
24	SB GEO production well PW-
3	
25	Hot spring 6, main terrace ACEC
26	New seep adjacent to well GS-
5 (1991)	
27	23-5 (Mariner & Janik 1995; = Wells '23-
5')	
28	23-5 (Mariner & Janik 1995; = Wells '23-
5')	
29	23-5 (Mariner & Janik 1995; = Wells '23-
5')	
30	23-5 (Mariner & Janik 1995; = Wells '23-
5')	
31	23-5 (Mariner & Janik 1995; = Wells '23-
5')	
32	83A-6 (Mariner & Janik 1995; = Wells '83A-
6')	
33	83A-6 (Mariner & Janik 1995; = Wells '83A-
6')	
34	83A-6 (Mariner & Janik 1995; = Wells '83A-
6')	
35	21-5 (Mariner & Janik 1995; = Wells '21-
5R')	

36	21-5 (Mariner & Janik 1995; = Wells '21-
5R')	
37	21-5 (Mariner & Janik 1995; = Wells '21-
5R')	
38	21-5 (Mariner & Janik 1995; = Wells '21-
5R')	
39	13-5 (Mariner & Janik 1995; = Wells '13-
5R')	
43	GS-5 (Mariner & Janik 1995; = Wells 'GS-
5')	
44	GS-5 (Mariner & Janik 1995; = Wells 'GS-
5')	
45	GS-5 (Mariner & Janik 1995; = Wells 'GS-
5')	
46	PW-1 (Mariner & Janik 1995; = Wells 'PW-
1')	
47	PW-1 (Mariner & Janik 1995; = Wells 'PW-
1')	
48	PW-1 (Mariner & Janik 1995; = Wells 'PW-
1')	
49	PW-1 (Mariner & Janik 1995; = Wells 'PW-
1')	
50	PW-2 (Mariner & Janik 1995; = Wells 'PW-
2')	
51	PW-2 (Mariner & Janik 1995; = Wells 'PW-
2')	
52	PW-2 (Mariner & Janik 1995; = Wells 'PW-
2')	
53	PW-2 (Mariner & Janik 1995; = Wells 'PW-
2')	
54	PW-3 (Mariner & Janik 1995; = Wells 'PW-
3')	
55	PW-3 (Mariner & Janik 1995; = Wells 'PW-
3')	
56	PW2-1 (Mariner & Janik 1995; = Wells 'PW 2-
1')	
57	PW2-1 (Mariner & Janik 1995; = Wells 'PW 2-
1')	
58	PW2-2 (Mariner & Janik 1995; = Wells 'PW 2-
2')	
59	PW2-2 (Mariner & Janik 1995; = Wells 'PW 2-
2')	
60	PW2-2 (Mariner & Janik 1995; = Wells 'PW 2-

2')	
61	PW2-3 (Mariner & Janik 1995; = Wells 'PW 2-
3')	
62	PW2-3 (Mariner & Janik 1995; = Wells 'PW 2-
3')	
63	PW2-4 (Mariner & Janik 1995; = Wells 'PW 2-
4')	
64	PW2-4 (Mariner & Janik 1995; = Wells 'PW 2-
4')	
65	PW2-4 (Mariner & Janik 1995; = Wells 'PW 2-
4')	
66	PW2-5 (Mariner & Janik 1995; = Wells 'PW 2-
5')	
67	PW2-5 (Mariner & Janik 1995; = Wells 'PW 2-
5')	
68	PW2-5 (Mariner & Janik 1995; = Wells 'PW 2-
5')	
69	PW3-1 (Mariner & Janik 1995; = Wells 'PW 3-
1')	
70	PW3-1 (Mariner & Janik 1995; = Wells 'PW 3-
1')	
71	PW3-1 (Mariner & Janik 1995; = Wells 'PW 3-
1')	
72	PW3-2 (Mariner & Janik 1995; = Wells 'PW 3-
2')	
73	PW3-2 (Mariner & Janik 1995; = Wells 'PW 3-
2')	
74	PW3-3 (Mariner & Janik 1995; = Wells 'PW 3-
3')	
75	PW3-3 (Mariner & Janik 1995; = Wells 'PW 3-
3')	
76	PW3-3 (Mariner & Janik 1995; = Wells 'PW 3-
3')	
77	PW3-4 (Mariner & Janik 1995; = Wells 'PW 3-
4')	
78	PW3-4 (Mariner & Janik 1995; = Wells 'PW 3-
4')	
79	PW3-4 (Mariner & Janik 1995; = Wells 'PW 3-
4')	
80	IW-3 (Mariner & Janik 1995; = Wells 'IW-
3')	
81	IW-3 (Mariner & Janik 1995; = Wells 'IW-
3')	

82	IW-5 (Mariner & Janik 1995; = Wells '46-	
28')		
83		Stuart
84		Stuart
85		Stuart
86		Stuart
87	Brown's School (Mariner & Janik 1995; = Wells 'Brown School Geothermal Well')	
88	Brown's School (Mariner & Janik 1995; = Wells 'Brown School Geothermal Well')	
89	Brown's School (Mariner & Janik 1995; = Wells 'Brown School Geothermal Well')	
90	Brown's School (Mariner & Janik 1995; = Wells 'Brown School Geothermal Well')	
91		Curti Domestic Well
92	Steinhardt Geothermal Well (narrative Cl trend)	
94		Herz Domestic Well
96		Galena Ck. Pk.
98		Tick Spr.
99		Jumbo Gr. S.
102		Dry Creek
103		Stock Spring
104		Thomas Creek Spring
105		Slide Mount Spring
106		Davies Creek Well
107		Pleasant Valley School
108		Jumbo Grade Spring
109		Scorpion Springs
110		Alum Spring
111		Galena Creek Park
112		Tick Spring
113		Jumbo Grade Spring
114		Guton Ranch Well
115		Hidden Valley Well
116	PW-1 (Mariner & Janik 1995; = Wells 'PW-	
1')		
117	PW-1 (Mariner & Janik 1995; = Wells 'PW-	
1')		
118	PW-1 (Mariner & Janik 1995; = Wells 'PW-	
1')		
119	PW-1 (Mariner & Janik 1995; = Wells 'PW-	
1')		
120	PW-2 (Mariner & Janik 1995; = Wells 'PW-	
2')		
121	PW-2 (Mariner & Janik 1995; = Wells 'PW-	
2')		
122	PW-2 (Mariner & Janik 1995; = Wells 'PW-	

2')	
123	PW-2 (Mariner & Janik 1995; = Wells 'PW-
2')	
124	PW-3 (Mariner & Janik 1995; = Wells 'PW-
3')	
125	PW-3 (Mariner & Janik 1995; = Wells 'PW-
3')	
126	PW2-1 (Mariner & Janik 1995; = Wells 'PW 2-
1')	
127	PW2-1 (Mariner & Janik 1995; = Wells 'PW 2-
1')	
128	PW2-2 (Mariner & Janik 1995; = Wells 'PW 2-
2')	
129	PW2-2 (Mariner & Janik 1995; = Wells 'PW 2-
2')	
130	PW2-2 (Mariner & Janik 1995; = Wells 'PW 2-
2')	
131	PW2-3 (Mariner & Janik 1995; = Wells 'PW 2-
3')	
132	PW2-3 (Mariner & Janik 1995; = Wells 'PW 2-
3')	
133	PW2-4 (Mariner & Janik 1995; = Wells 'PW 2-
4')	
134	PW2-4 (Mariner & Janik 1995; = Wells 'PW 2-
4')	
135	PW2-5 (Mariner & Janik 1995; = Wells 'PW 2-
5')	
136	PW2-5 (Mariner & Janik 1995; = Wells 'PW 2-
5')	
137	PW2-5 (Mariner & Janik 1995; = Wells 'PW 2-
5')	
138	PW3-1 (Mariner & Janik 1995; = Wells 'PW 3-
1')	
139	PW3-1 (Mariner & Janik 1995; = Wells 'PW 3-
1')	
140	PW3-1 (Mariner & Janik 1995; = Wells 'PW 3-
1')	
141	PW3-2 (Mariner & Janik 1995; = Wells 'PW 3-
2')	
142	PW3-2 (Mariner & Janik 1995; = Wells 'PW 3-
2')	
143	PW3-3 (Mariner & Janik 1995; = Wells 'PW 3-
3')	

144	PW3-3 (Mariner & Janik 1995; = Wells 'PW 3-	
3')		
145	PW3-3 (Mariner & Janik 1995; = Wells 'PW 3-	
3')		
146	PW3-4 (Mariner & Janik 1995; = Wells 'PW 3-	
4')		
147	PW3-4 (Mariner & Janik 1995; = Wells 'PW 3-	
4')		
148	PW3-4 (Mariner & Janik 1995; = Wells 'PW 3-	
4')		
149		21-
5R		
150		21-
5R		
151		13-
5R		
152		13-
5R		
153	23-5 (Mariner & Janik 1995; = Wells '23-	
5')		
154	23-5 (Mariner & Janik 1995; = Wells '23-	
5')		
155	83A-6 (Mariner & Janik 1995; = Wells '83A-	
6')		
156	83A-6 (Mariner & Janik 1995; = Wells '83A-	
6')		
157	IW-3 (Mariner & Janik 1995; = Wells 'IW-	
3')		
158	IW-5 (Mariner & Janik 1995; = Wells '46-	
28')		
160	GS-5 (Mariner & Janik 1995; = Wells 'GS-	
5')		
161	GS-5 (Mariner & Janik 1995; = Wells 'GS-	
5')		
162	Brown's School (Mariner & Janik 1995; = Wells 'Brown School Geothermal Well')	
163	Brown's School (Mariner & Janik 1995; = Wells 'Brown School Geothermal Well')	
164	Brown's School (Mariner & Janik 1995; = Wells 'Brown School Geothermal Well')	
165	Brown's School (Mariner & Janik 1995; = Wells 'Brown School Geothermal Well')	
166	Brown's School (Mariner & Janik 1995; = Wells 'Brown School Geothermal Well')	
167	Brown's School (Mariner & Janik 1995; = Wells 'Brown School Geothermal Well')	
168	Brown's School (Mariner & Janik 1995; = Wells 'Brown School Geothermal Well')	
169	Brown's School (Mariner & Janik 1995; = Wells 'Brown School Geothermal Well')	
170	Brown's School (Mariner & Janik 1995; = Wells 'Brown School Geothermal Well')	

[illegible]

257	Brown's School (Mariner & Janik 1995; = Wells	'Brown School Geothermal Well')
258	Brown's School (Mariner & Janik 1995; = Wells	'Brown School Geothermal Well')
259	Brown's School (Mariner & Janik 1995; = Wells	'Brown School Geothermal Well')
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261	Brown's School (Mariner & Janik 1995; = Wells	'Brown School Geothermal Well')
262	Brown's School (Mariner & Janik 1995; = Wells	'Brown School Geothermal Well')
263	Brown's School (Mariner & Janik 1995; = Wells	'Brown School Geothermal Well')
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267	Brown's School (Mariner & Janik 1995; = Wells	'Brown School Geothermal Well')
268	Brown's School (Mariner & Janik 1995; = Wells	'Brown School Geothermal Well')
269	Brown's School (Mariner & Janik 1995; = Wells	'Brown School Geothermal Well')
270	Brown's School (Mariner & Janik 1995; = Wells	'Brown School Geothermal Well')
271	Brown's School (Mariner & Janik 1995; = Wells	'Brown School Geothermal Well')
272		Curti Barn Well (geothermal)
273		Curti Barn Well (geothermal)
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307	Curti Barn Well (geothermal)
308	Curti Domestic Well
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337	Curti Domestic Well
338	Curti Domestic Well
339	Herz Geothermal
340	Herz Geothermal
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533	Pine Tree Ranch-
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581	Flame
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616				Steinhardt	Geothermal Well (narrative Cl trend)	
617				Steinhardt	Geothermal Well (narrative Cl trend)	
618				Steinhardt	Geothermal Well (narrative Cl trend)	
619				Steinhardt	Geothermal Well (narrative Cl trend)	
620				Steinhardt	Geothermal Well (narrative Cl trend)	
621				Steinhardt	Geothermal Well (narrative Cl trend)	
622				Steinhardt	Geothermal Well (narrative Cl trend)	
623				Steinhardt	Geothermal Well (narrative Cl trend)	
624				Steinhardt	Geothermal Well (narrative Cl trend)	
625				Steinhardt	Geothermal Well (narrative Cl trend)	
626				Steinhardt	Geothermal Well (narrative Cl trend)	
627				Steinhardt	Geothermal Well (narrative Cl trend)	
628				Steinhardt	Geothermal Well (narrative Cl trend)	
629				Steinhardt	Geothermal Well (narrative Cl trend)	
630				Steinhardt	Geothermal Well (narrative Cl trend)	
631				Steinhardt	Geothermal Well (narrative Cl trend)	
632				Steinhardt	Geothermal Well (narrative Cl trend)	
	year	Cl	B	Cl_B	era	
3	2026	815.0	36.00	22.638889	2024-2026	
4	2026	473.0	21.20	22.311321	2024-2026	
5	2026	849.0	37.10	22.884097	2024-2026	
6	2026	724.0	33.30	21.741742	2024-2026	
7	2026	635.0	29.20	21.746575	2024-2026	

8	2026	697.0	34.80	20.028736	2024-2026
9	2024	570.0	36.00	15.833333	2024-2026
10	2024	590.0	29.00	20.344828	2024-2026
11	2024	45.0	0.18	250.000000	2024-2026
12	2024	11.0	0.10	110.000000	2024-2026
13	2024	7.4	0.05	148.000000	2024-2026
14	2024	470.0	34.00	13.823529	2024-2026
15	2024	530.0	29.00	18.275862	2024-2026
16	2024	39.0	0.10	390.000000	2024-2026
17	2024	11.0	0.12	91.666667	2024-2026
18	2024	7.0	0.05	140.000000	2024-2026
19	1990	737.0	40.00	18.425000	1950-1991
20	1990	697.0	38.00	18.342105	1950-1991
22	1990	811.0	42.00	19.309524	1950-1991
23	1990	802.0	36.00	22.277778	1950-1991
24	1990	811.0	37.00	21.918919	1950-1991
25	1977	871.0	48.00	18.145833	1950-1991
26	1991	1000.0	53.10	18.832392	1950-1991
27	1991	222.0	12.00	18.500000	1950-1991
28	1993	887.0	46.00	19.282609	1950-1991
29	1993	887.0	47.00	18.872340	1950-1991
30	1994	917.0	45.00	20.377778	1950-1991
31	1994	889.0	46.00	19.326087	1950-1991
32	1991	840.0	41.00	20.487805	1950-1991
33	1993	840.0	51.00	16.470588	1950-1991
34	1994	868.0	44.00	19.727273	1950-1991
35	1991	850.0	42.00	20.238095	1950-1991
36	1993	833.0	43.00	19.372093	1950-1991
37	1993	847.0	46.00	18.413043	1950-1991
38	1994	856.0	44.00	19.454545	1950-1991
39	1993	904.0	46.00	19.652174	1950-1991
43	1991	925.0	47.00	19.680851	1950-1991
44	1993	887.0	48.00	18.479167	1950-1991
45	1993	949.0	56.00	16.946429	1950-1991
46	1991	781.0	45.00	17.355556	1950-1991
47	1993	811.0	43.00	18.860465	1950-1991
48	1993	808.0	38.00	21.263158	1950-1991
49	1994	802.0	42.00	19.095238	1950-1991
50	1991	800.0	43.00	18.604651	1950-1991
51	1993	823.0	41.00	20.073171	1950-1991
52	1993	795.0	43.00	18.488372	1950-1991
53	1994	811.0	41.00	19.780488	1950-1991
54	1991	809.0	42.00	19.261905	1950-1991

55	1993	814.0	47.00	17.319149	1950-1991
56	1993	772.0	40.00	19.300000	1950-1991
57	1993	801.0	45.00	17.800000	1950-1991
58	1993	770.0	42.00	18.333333	1950-1991
59	1993	799.0	44.00	18.159091	1950-1991
60	1994	797.0	40.00	19.925000	1950-1991
61	1993	801.0	37.00	21.648649	1950-1991
62	1993	797.0	41.00	19.439024	1950-1991
63	1993	782.0	41.00	19.073171	1950-1991
64	1993	787.0	43.00	18.302326	1950-1991
65	1994	783.0	41.00	19.097561	1950-1991
66	1993	795.0	41.00	19.390244	1950-1991
67	1993	796.0	48.00	16.583333	1950-1991
68	1994	805.0	42.00	19.166667	1950-1991
69	1993	789.0	41.00	19.243902	1950-1991
70	1993	788.0	40.00	19.700000	1950-1991
71	1994	783.0	40.00	19.575000	1950-1991
72	1993	775.0	41.00	18.902439	1950-1991
73	1993	799.0	43.00	18.581395	1950-1991
74	1993	801.0	42.00	19.071429	1950-1991
75	1993	779.0	39.00	19.974359	1950-1991
76	1994	797.0	42.00	18.976190	1950-1991
77	1993	779.0	42.00	18.547619	1950-1991
78	1993	795.0	52.00	15.288462	1950-1991
79	1994	783.0	39.00	20.076923	1950-1991
80	1991	809.0	40.00	20.225000	1950-1991
81	1993	813.0	39.00	20.846154	1950-1991
82	1993	794.0	43.00	18.465116	1950-1991
83	1991	68.0	3.90	17.435897	1950-1991
84	1993	64.0	6.60	9.696970	1950-1991
85	1993	82.0	4.70	17.446809	1950-1991
86	1994	86.0	5.70	15.087719	1950-1991
87	1991	554.0	20.00	27.700000	1950-1991
88	1993	568.0	29.00	19.586207	1950-1991
89	1993	639.0	31.00	20.612903	1950-1991
90	1994	636.0	38.00	16.736842	1950-1991
91	1991	656.0	36.00	18.222222	1950-1991
92	1991	171.0	8.60	19.883721	1950-1991
94	1991	138.0	0.30	460.000000	1950-1991
96	1993	0.5	0.05	10.000000	1950-1991
98	1993	48.0	0.03	1600.000000	1950-1991
99	1993	7.0	0.11	63.636364	1950-1991
102	1977	2.0	0.00	Inf	1950-1991

103	1977	3.0	0.00	Inf	1950-1991
104	1977	2.0	0.00	Inf	1950-1991
105	1977	12.0	4.00	3.000000	1950-1991
106	1977	0.0	0.00	NaN	1950-1991
107	1977	2.0	0.00	Inf	1950-1991
108	1977	8.0	0.00	Inf	1950-1991
109	1977	3.0	0.00	Inf	1950-1991
110	1977	9.0	0.00	Inf	1950-1991
111	1993	0.5	0.10	5.000000	1950-1991
112	1993	48.0	0.00	Inf	1950-1991
113	1993	7.0	0.10	70.000000	1950-1991
114	1977	471.0	14.00	33.642857	1950-1991
115	1977	9.0	3.00	3.000000	1950-1991
116	1991	781.0	45.00	17.355556	1950-1991
117	1993	811.0	43.00	18.860465	1950-1991
118	1993	808.0	38.00	21.263158	1950-1991
119	1994	802.0	42.00	19.095238	1950-1991
120	1991	800.0	43.00	18.604651	1950-1991
121	1993	823.0	41.00	20.073171	1950-1991
122	1993	795.0	43.00	18.488372	1950-1991
123	1994	811.0	41.00	19.780488	1950-1991
124	1991	809.0	42.00	19.261905	1950-1991
125	1993	814.0	47.00	17.319149	1950-1991
126	1993	772.0	40.00	19.300000	1950-1991
127	1993	801.0	45.00	17.800000	1950-1991
128	1993	770.0	42.00	18.333333	1950-1991
129	1993	799.0	44.00	18.159091	1950-1991
130	1994	797.0	40.00	19.925000	1950-1991
131	1993	801.0	37.00	21.648649	1950-1991
132	1994	797.0	40.00	19.925000	1950-1991
133	1993	782.0	41.00	19.073171	1950-1991
134	1993	801.0	37.00	21.648649	1950-1991
135	1993	795.0	41.00	19.390244	1950-1991
136	1993	796.0	48.00	16.583333	1950-1991
137	1994	805.0	42.00	19.166667	1950-1991
138	1993	789.0	41.00	19.243902	1950-1991
139	1993	788.0	40.00	19.700000	1950-1991
140	1994	783.0	40.00	19.575000	1950-1991
141	1993	775.0	41.00	18.902439	1950-1991
142	1993	799.0	43.00	18.581395	1950-1991
143	1993	801.0	42.00	19.071429	1950-1991
144	1993	779.0	39.00	19.974359	1950-1991
145	1994	797.0	42.00	18.976190	1950-1991

146	1993	779.0	42.00	18.547619	1950-1991
147	1993	795.0	52.00	15.288462	1950-1991
148	1994	783.0	39.00	20.076923	1950-1991
149	1993	797.0	41.00	19.439024	1950-1991
150	1993	782.0	41.00	19.073171	1950-1991
151	1993	787.0	43.00	18.302326	1950-1991
152	1991	792.0	43.00	18.418605	1950-1991
153	1994	783.0	41.00	19.097561	1950-1991
154	1991	760.0	37.00	20.540541	1950-1991
155	1993	742.0	39.00	19.025641	1950-1991
156	1993	755.0	41.00	18.414634	1950-1991
157	1994	766.0	40.00	19.150000	1950-1991
158	1993	765.0	39.00	19.615385	1950-1991
160	1993	783.0	41.00	19.097561	1950-1991
161	1994	810.0	40.00	20.250000	1950-1991
162	1985	3.0	0.00	Inf	1950-1991
163	1985	7.0	0.00	Inf	1950-1991
164	1985	7.0	0.20	35.000000	1950-1991
165	1985	5.0	0.30	16.666667	1950-1991
166	1985	4.0	0.30	13.333333	1950-1991
167	1985	7.0	0.20	35.000000	1950-1991
168	1985	5.0	0.20	25.000000	1950-1991
169	1985	8.0	0.20	40.000000	1950-1991
170	1985	9.0	0.20	45.000000	1950-1991
171	1985	9.0	0.20	45.000000	1950-1991
172	1985	10.0	0.10	100.000000	1950-1991
173	1985	10.0	0.20	50.000000	1950-1991
174	1986	8.0	0.60	13.333333	1950-1991
175	1986	7.0	0.20	35.000000	1950-1991
176	1986	17.0	0.20	85.000000	1950-1991
177	1986	17.0	0.20	85.000000	1950-1991
178	1986	10.0	0.20	50.000000	1950-1991
179	1986	11.0	0.20	55.000000	1950-1991
180	1986	10.0	0.20	50.000000	1950-1991
181	1986	11.0	0.20	55.000000	1950-1991
182	1986	14.0	0.20	70.000000	1950-1991
183	1986	13.0	0.10	130.000000	1950-1991
184	1986	13.0	0.10	130.000000	1950-1991
185	1986	11.0	0.20	55.000000	1950-1991
186	1987	6.0	0.20	30.000000	1950-1991
187	1987	8.0	0.20	40.000000	1950-1991
188	1987	7.0	0.20	35.000000	1950-1991
189	1987	6.0	0.30	20.000000	1950-1991

190	1987	5.0	0.20	25.000000	1950-1991
191	1987	15.0	0.10	150.000000	1950-1991
192	1987	7.0	0.20	35.000000	1950-1991
193	1987	5.0	0.20	25.000000	1950-1991
194	1987	5.0	0.30	16.666667	1950-1991
195	1987	8.0	0.30	26.666667	1950-1991
196	1987	5.0	0.30	16.666667	1950-1991
197	1987	6.0	0.30	20.000000	1950-1991
198	1988	7.0	0.30	23.333333	1950-1991
199	1988	14.0	0.30	46.666667	1950-1991
200	1988	13.0	0.30	43.333333	1950-1991
201	1988	18.0	0.30	60.000000	1950-1991
202	1988	27.0	0.30	90.000000	1950-1991
203	1988	35.0	0.30	116.666667	1950-1991
204	1988	39.0	0.30	130.000000	1950-1991
205	1988	68.0	0.40	170.000000	1950-1991
206	1988	69.0	0.40	172.500000	1950-1991
207	1988	112.0	0.70	160.000000	1950-1991
208	1989	150.0	0.50	300.000000	1950-1991
209	1989	188.0	0.90	208.888889	1950-1991
210	1989	203.0	1.30	156.153846	1950-1991
211	1989	250.0	2.10	119.047619	1950-1991
212	1989	253.0	2.70	93.703704	1950-1991
213	1989	263.0	2.80	93.928571	1950-1991
214	1989	350.0	4.40	79.545455	1950-1991
215	1989	355.0	4.90	72.448980	1950-1991
216	1989	333.0	5.40	61.666667	1950-1991
217	1989	370.0	6.40	57.812500	1950-1991
218	1989	370.0	6.30	58.730159	1950-1991
219	1990	395.0	7.20	54.861111	1950-1991
220	1990	415.0	3.80	109.210526	1950-1991
221	1990	430.0	8.70	49.425287	1950-1991
222	1990	430.0	5.00	86.000000	1950-1991
223	1990	440.0	9.70	45.360825	1950-1991
224	1990	410.0	9.70	42.268041	1950-1991
225	1990	420.0	9.20	45.652174	1950-1991
226	1990	450.0	12.00	37.500000	1950-1991
227	1990	480.0	13.20	36.363636	1950-1991
228	1991	495.0	13.20	37.500000	1950-1991
229	1991	505.0	15.40	32.792208	1950-1991
230	1991	530.0	16.20	32.716049	1950-1991
231	1991	510.0	17.70	28.813559	1950-1991
232	1991	490.0	16.30	30.061350	1950-1991

233	1991	505.0	17.20	29.360465	1950-1991
234	1991	510.0	17.80	28.651685	1950-1991
235	1991	530.0	19.70	26.903553	1950-1991
236	1991	540.0	20.80	25.961538	1950-1991
237	1991	550.0	21.50	25.581395	1950-1991
238	1991	580.0	23.60	24.576271	1950-1991
239	1992	570.0	23.60	24.152542	1950-1991
240	1992	590.0	24.40	24.180328	1950-1991
241	1992	590.0	23.70	24.894515	1950-1991
242	1992	570.0	22.20	25.675676	1950-1991
243	1992	580.0	24.50	23.673469	1950-1991
244	1992	580.0	17.30	33.526012	1950-1991
245	1992	600.0	28.90	20.761246	1950-1991
246	1992	630.0	30.50	20.655738	1950-1991
247	1992	640.0	31.10	20.578778	1950-1991
248	1992	630.0	29.80	21.140940	1950-1991
249	1993	663.0	31.20	21.250000	1950-1991
250	1993	640.0	31.50	20.317460	1950-1991
251	1993	670.0	34.70	19.308357	1950-1991
252	1993	670.0	35.40	18.926554	1950-1991
253	1994	680.0	38.10	17.847769	1950-1991
254	1994	680.0	40.50	16.790123	1950-1991
255	1994	680.0	38.30	17.754569	1950-1991
256	1994	670.0	37.80	17.724868	1950-1991
257	1995	672.0	38.00	17.684211	1950-1991
258	1995	698.0	40.00	17.450000	1950-1991
259	1995	693.0	39.20	17.678571	1950-1991
260	1995	695.0	39.00	17.820513	1950-1991
261	1996	715.0	37.00	19.324324	1950-1991
262	1996	719.0	38.00	18.921053	1950-1991
263	1996	743.0	39.80	18.668342	1950-1991
264	1996	691.0	38.00	18.184211	1950-1991
265	1997	690.0	39.50	17.468354	1950-1991
266	1997	515.0	34.90	14.756447	1950-1991
267	1997	335.0	26.80	12.500000	1950-1991
268	1997	246.0	19.20	12.812500	1950-1991
269	1998	263.0	15.60	16.858974	1950-1991
270	1998	324.0	17.60	18.409091	1950-1991
271	1998	299.0	16.10	18.571429	1950-1991
272	1987	680.0	38.20	17.801047	1950-1991
273	1987	680.0	39.20	17.346939	1950-1991
274	1987	700.0	38.60	18.134715	1950-1991
275	1987	690.0	38.30	18.015666	1950-1991

276	1987	670.0	38.10	17.585302	1950-1991
277	1987	700.0	37.20	18.817204	1950-1991
278	1987	710.0	36.50	19.452055	1950-1991
279	1987	700.0	35.90	19.498607	1950-1991
280	1988	700.0	36.80	19.021739	1950-1991
281	1988	680.0	37.60	18.085106	1950-1991
282	1988	680.0	38.10	17.847769	1950-1991
283	1988	690.0	37.40	18.449198	1950-1991
284	1988	700.0	36.80	19.021739	1950-1991
285	1988	690.0	38.20	18.062827	1950-1991
286	1988	700.0	37.40	18.716578	1950-1991
287	1988	700.0	38.70	18.087855	1950-1991
288	1988	710.0	36.10	19.667590	1950-1991
289	1989	700.0	36.90	18.970190	1950-1991
290	1989	700.0	36.40	19.230769	1950-1991
291	1989	700.0	38.50	18.181818	1950-1991
292	1989	700.0	36.80	19.021739	1950-1991
293	1990	688.0	36.50	18.849315	1950-1991
294	1990	670.0	36.00	18.611111	1950-1991
295	1990	700.0	35.00	20.000000	1950-1991
296	1992	700.0	37.80	18.518519	1950-1991
297	1993	720.0	35.30	20.396601	1950-1991
298	1993	720.0	34.60	20.809249	1950-1991
299	1994	680.0	36.00	18.888889	1950-1991
300	1995	671.0	35.00	19.171429	1950-1991
301	1996	844.0	38.00	22.210526	1950-1991
302	1997	713.0	35.90	19.860724	1950-1991
303	1997	753.0	36.50	20.630137	1950-1991
304	1997	749.0	37.60	19.920213	1950-1991
305	1998	771.0	40.00	19.275000	1950-1991
306	1998	764.0	39.20	19.489796	1950-1991
307	1998	722.0	35.40	20.395480	1950-1991
308	1987	46.0	1.70	27.058824	1950-1991
309	1987	44.0	1.60	27.500000	1950-1991
310	1987	43.0	1.70	25.294118	1950-1991
311	1988	48.0	1.80	26.666667	1950-1991
312	1988	54.0	1.90	28.421053	1950-1991
313	1988	63.0	2.10	30.000000	1950-1991
314	1988	74.0	2.20	33.636364	1950-1991
315	1988	82.0	2.40	34.166667	1950-1991
316	1988	91.0	2.60	35.000000	1950-1991
317	1988	79.0	2.40	32.916667	1950-1991
318	1988	61.0	1.90	32.105263	1950-1991

319	1988	55.0	1.80	30.555556	1950-1991
320	1988	57.0	2.20	25.909091	1950-1991
321	1988	59.0	2.30	25.652174	1950-1991
322	1989	63.0	2.50	25.200000	1950-1991
323	1989	70.0	2.70	25.925926	1950-1991
324	1989	80.0	2.90	27.586207	1950-1991
325	1989	124.0	3.40	36.470588	1950-1991
326	1989	67.0	2.50	26.800000	1950-1991
327	1991	208.0	12.50	16.640000	1950-1991
328	1993	198.0	10.60	18.679245	1950-1991
329	1994	205.0	11.80	17.372881	1950-1991
330	1995	235.0	12.60	18.650794	1950-1991
331	1995	207.0	11.30	18.318584	1950-1991
332	1996	213.0	11.20	19.017857	1950-1991
333	1997	256.0	15.60	16.410256	1950-1991
334	1997	279.0	15.00	18.600000	1950-1991
335	1997	302.0	16.00	18.875000	1950-1991
336	1998	293.0	15.60	18.782051	1950-1991
337	1998	134.0	7.20	18.611111	1950-1991
338	1998	294.0	14.10	20.851064	1950-1991
339	1985	350.0	16.30	21.472393	1950-1991
340	1985	340.0	16.60	20.481928	1950-1991
341	1985	340.0	17.40	19.540230	1950-1991
342	1985	350.0	17.20	20.348837	1950-1991
343	1985	340.0	17.50	19.428571	1950-1991
344	1985	350.0	17.50	20.000000	1950-1991
345	1985	350.0	17.50	20.000000	1950-1991
346	1985	355.0	18.20	19.505495	1950-1991
347	1985	345.0	18.30	18.852459	1950-1991
348	1985	345.0	17.00	20.294118	1950-1991
349	1985	340.0	18.10	18.784530	1950-1991
350	1985	350.0	16.90	20.710059	1950-1991
351	1986	345.0	15.20	22.697368	1950-1991
352	1986	345.0	17.10	20.175439	1950-1991
353	1986	345.0	17.60	19.602273	1950-1991
354	1986	345.0	17.60	19.602273	1950-1991
355	1986	345.0	17.30	19.942197	1950-1991
356	1986	355.0	17.40	20.402299	1950-1991
357	1986	375.0	17.70	21.186441	1950-1991
358	1986	350.0	17.40	20.114943	1950-1991
359	1986	350.0	18.10	19.337017	1950-1991
360	1986	345.0	17.70	19.491525	1950-1991
361	1986	345.0	17.60	19.602273	1950-1991

362	1986	340.0	18.50	18.378378	1950-1991
363	1987	340.0	18.20	18.681319	1950-1991
364	1987	340.0	18.00	18.888889	1950-1991
365	1987	340.0	18.00	18.888889	1950-1991
366	1987	345.0	17.40	19.827586	1950-1991
367	1987	340.0	17.40	19.540230	1950-1991
368	1987	340.0	17.50	19.428571	1950-1991
369	1988	340.0	17.50	19.428571	1950-1991
370	1988	340.0	17.60	19.318182	1950-1991
371	1988	325.0	17.20	18.895349	1950-1991
372	1988	355.0	17.10	20.760234	1950-1991
373	1988	360.0	17.20	20.930233	1950-1991
374	1988	340.0	17.40	19.540230	1950-1991
375	1988	350.0	17.50	20.000000	1950-1991
376	1988	345.0	17.60	19.602273	1950-1991
377	1988	360.0	18.30	19.672131	1950-1991
378	1989	355.0	16.60	21.385542	1950-1991
379	1989	365.0	17.80	20.505618	1950-1991
380	1989	370.0	17.90	20.670391	1950-1991
381	1989	370.0	18.20	20.329670	1950-1991
382	1989	375.0	18.20	20.604396	1950-1991
383	1989	375.0	18.10	20.718232	1950-1991
384	1989	375.0	18.60	20.161290	1950-1991
385	1989	370.0	18.70	19.786096	1950-1991
386	1989	380.0	19.00	20.000000	1950-1991
387	1985	3.0	0.00	Inf	1950-1991
388	1985	4.0	0.00	Inf	1950-1991
389	1985	3.0	0.20	15.000000	1950-1991
390	1985	3.0	0.30	10.000000	1950-1991
391	1985	4.0	0.20	20.000000	1950-1991
392	1985	4.0	0.30	13.333333	1950-1991
393	1985	5.0	0.20	25.000000	1950-1991
394	1985	3.0	0.20	15.000000	1950-1991
395	1985	3.0	0.20	15.000000	1950-1991
396	1985	4.0	0.20	20.000000	1950-1991
397	1985	5.0	0.10	50.000000	1950-1991
398	1985	4.0	0.10	40.000000	1950-1991
399	1986	3.0	0.50	6.000000	1950-1991
400	1986	3.0	0.10	30.000000	1950-1991
401	1986	4.0	0.10	40.000000	1950-1991
402	1986	3.0	0.10	30.000000	1950-1991
403	1986	2.0	0.10	20.000000	1950-1991
404	1986	4.0	0.10	40.000000	1950-1991

405	1986	3.0	0.20	15.000000	1950-1991
406	1986	2.0	0.20	10.000000	1950-1991
407	1986	3.0	0.10	30.000000	1950-1991
408	1987	3.0	0.10	30.000000	1950-1991
409	1987	0.0	0.10	0.000000	1950-1991
410	1987	4.0	0.10	40.000000	1950-1991
411	1987	3.0	0.10	30.000000	1950-1991
412	1987	3.0	0.10	30.000000	1950-1991
413	1988	2.0	0.10	20.000000	1950-1991
414	1988	2.0	0.10	20.000000	1950-1991
415	1988	2.0	0.10	20.000000	1950-1991
416	1988	2.0	0.10	20.000000	1950-1991
417	1988	2.0	0.10	20.000000	1950-1991
418	1988	3.0	0.10	30.000000	1950-1991
419	1988	3.0	0.10	30.000000	1950-1991
420	1989	5.0	0.10	50.000000	1950-1991
421	1989	2.0	0.00	Inf	1950-1991
422	1989	8.0	0.20	40.000000	1950-1991
423	1989	9.0	0.10	90.000000	1950-1991
424	1989	15.0	0.10	150.000000	1950-1991
425	1989	12.0	0.20	60.000000	1950-1991
426	1989	11.0	0.10	110.000000	1950-1991
427	1990	12.0	0.00	Inf	1950-1991
428	1990	12.0	0.00	Inf	1950-1991
429	1990	16.0	0.00	Inf	1950-1991
430	1993	257.0	4.40	58.409091	1950-1991
431	1994	190.0	4.00	47.500000	1950-1991
432	1985	4.0	0.00	Inf	1950-1991
433	1985	3.0	0.10	30.000000	1950-1991
434	1985	3.0	0.20	15.000000	1950-1991
435	1985	2.0	0.10	20.000000	1950-1991
436	1985	2.0	0.10	20.000000	1950-1991
437	1985	4.0	0.10	40.000000	1950-1991
438	1985	2.0	0.10	20.000000	1950-1991
439	1985	2.0	0.10	20.000000	1950-1991
440	1985	3.0	0.10	30.000000	1950-1991
441	1985	3.0	0.10	30.000000	1950-1991
442	1986	3.0	0.20	15.000000	1950-1991
443	1986	3.0	0.10	30.000000	1950-1991
444	1986	3.0	0.10	30.000000	1950-1991
445	1986	3.0	0.10	30.000000	1950-1991
446	1986	4.0	0.10	40.000000	1950-1991
447	1986	3.0	0.10	30.000000	1950-1991

448	1986	3.0	0.10	30.000000	1950-1991
449	1986	3.0	0.20	15.000000	1950-1991
450	1986	3.0	0.10	30.000000	1950-1991
451	1986	2.0	0.10	20.000000	1950-1991
452	1986	2.0	0.10	20.000000	1950-1991
453	1986	4.0	0.00	Inf	1950-1991
454	1987	2.0	0.00	Inf	1950-1991
455	1987	3.0	0.00	Inf	1950-1991
456	1987	2.0	0.10	20.000000	1950-1991
457	1987	3.0	0.00	Inf	1950-1991
458	1987	4.0	0.00	Inf	1950-1991
459	1987	3.0	0.00	Inf	1950-1991
460	1987	2.0	0.10	20.000000	1950-1991
461	1987	1.0	0.00	Inf	1950-1991
462	1987	2.0	0.10	20.000000	1950-1991
463	1987	2.0	0.00	Inf	1950-1991
464	1987	3.0	0.00	Inf	1950-1991
465	1987	3.0	0.00	Inf	1950-1991
466	1988	2.0	0.00	Inf	1950-1991
467	1988	3.0	0.00	Inf	1950-1991
468	1988	2.0	0.00	Inf	1950-1991
469	1988	2.0	0.00	Inf	1950-1991
470	1988	2.0	0.00	Inf	1950-1991
471	1988	1.0	0.00	Inf	1950-1991
472	1988	3.0	0.00	Inf	1950-1991
473	1988	2.0	0.00	Inf	1950-1991
474	1989	6.0	0.10	60.000000	1950-1991
475	1989	7.0	0.00	Inf	1950-1991
476	1989	6.0	0.00	Inf	1950-1991
477	1989	5.0	0.00	Inf	1950-1991
478	1989	6.0	0.00	Inf	1950-1991
479	1989	3.0	0.00	Inf	1950-1991
480	1989	4.0	0.00	Inf	1950-1991
481	1989	3.0	0.10	30.000000	1950-1991
482	1989	3.0	0.00	Inf	1950-1991
483	1989	3.0	0.00	Inf	1950-1991
484	1989	3.0	0.00	Inf	1950-1991
485	1990	3.0	0.00	Inf	1950-1991
486	1990	4.0	0.00	Inf	1950-1991
487	1990	3.0	0.00	Inf	1950-1991
488	1990	5.0	0.00	Inf	1950-1991
489	1990	3.0	0.00	Inf	1950-1991
490	1990	2.0	0.00	Inf	1950-1991

491	1990	3.0	0.00	Inf	1950-1991
492	1990	2.0	0.00	Inf	1950-1991
493	1990	3.0	0.00	Inf	1950-1991
494	1990	5.0	0.20	25.000000	1950-1991
495	1991	3.0	0.00	Inf	1950-1991
496	1991	4.0	0.10	40.000000	1950-1991
497	1991	5.0	0.10	50.000000	1950-1991
498	1991	3.0	0.10	30.000000	1950-1991
499	1991	4.0	0.10	40.000000	1950-1991
500	1991	3.0	0.00	Inf	1950-1991
501	1991	2.0	0.10	20.000000	1950-1991
502	1991	4.0	0.00	Inf	1950-1991
503	1991	3.0	0.10	30.000000	1950-1991
504	1991	5.0	0.00	Inf	1950-1991
505	1992	3.0	0.10	30.000000	1950-1991
506	1992	3.0	0.10	30.000000	1950-1991
507	1992	5.0	0.00	Inf	1950-1991
508	1992	4.0	0.00	Inf	1950-1991
509	1992	9.0	0.10	90.000000	1950-1991
510	1992	4.0	0.10	40.000000	1950-1991
511	1992	3.0	0.00	Inf	1950-1991
512	1992	3.0	0.40	7.500000	1950-1991
513	1992	3.0	0.10	30.000000	1950-1991
514	1992	5.0	0.10	50.000000	1950-1991
515	1993	5.0	0.20	25.000000	1950-1991
516	1993	3.0	0.10	30.000000	1950-1991
517	1993	5.0	0.00	Inf	1950-1991
518	1993	4.0	0.10	40.000000	1950-1991
519	1994	5.0	0.30	16.666667	1950-1991
520	1994	4.0	0.10	40.000000	1950-1991
521	1994	4.0	0.10	40.000000	1950-1991
522	1994	4.0	0.10	40.000000	1950-1991
523	1995	4.0	0.30	13.333333	1950-1991
524	1995	4.0	0.20	20.000000	1950-1991
525	1995	5.0	0.10	50.000000	1950-1991
526	1996	5.0	0.00	Inf	1950-1991
527	1996	3.0	0.10	30.000000	1950-1991
528	1997	4.0	0.10	40.000000	1950-1991
529	1997	15.0	0.30	50.000000	1950-1991
530	1997	4.0	0.10	40.000000	1950-1991
531	1998	3.0	0.10	30.000000	1950-1991
532	1998	3.0	0.20	15.000000	1950-1991
533	1985	4.0	0.00	Inf	1950-1991

534	1985	3.0	0.10	30.000000	1950-1991
535	1985	4.0	0.30	13.333333	1950-1991
536	1985	23.0	1.10	20.909091	1950-1991
537	1985	28.0	1.60	17.500000	1950-1991
538	1985	15.0	0.40	37.500000	1950-1991
539	1985	4.0	0.20	20.000000	1950-1991
540	1985	3.0	0.10	30.000000	1950-1991
541	1985	3.0	0.10	30.000000	1950-1991
542	1985	3.0	0.10	30.000000	1950-1991
543	1986	2.0	0.10	20.000000	1950-1991
544	1986	4.0	0.10	40.000000	1950-1991
545	1986	8.0	0.30	26.666667	1950-1991
546	1986	13.0	0.50	26.000000	1950-1991
547	1986	20.0	1.00	20.000000	1950-1991
548	1986	26.0	1.50	17.333333	1950-1991
549	1986	28.0	1.90	14.736842	1950-1991
550	1986	30.0	2.20	13.636364	1950-1991
551	1987	5.0	0.20	25.000000	1950-1991
552	1987	19.0	1.00	19.000000	1950-1991
553	1987	27.0	2.20	12.272727	1950-1991
554	1987	32.0	2.80	11.428571	1950-1991
555	1987	36.0	3.10	11.612903	1950-1991
556	1987	42.0	3.20	13.125000	1950-1991
557	1987	44.0	3.50	12.571429	1950-1991
558	1987	48.0	3.30	14.545455	1950-1991
559	1987	51.0	3.70	13.783784	1950-1991
560	1987	52.0	3.60	14.444444	1950-1991
561	1987	49.0	3.50	14.000000	1950-1991
562	1988	57.0	3.90	14.615385	1950-1991
563	1988	57.0	4.00	14.250000	1950-1991
564	1988	59.0	4.10	14.390244	1950-1991
565	1988	67.0	4.10	16.341463	1950-1991
566	1988	68.0	4.20	16.190476	1950-1991
567	1988	71.0	4.30	16.511628	1950-1991
568	1988	75.0	4.40	17.045455	1950-1991
569	1988	74.0	4.30	17.209302	1950-1991
570	1989	37.0	2.70	13.703704	1950-1991
571	1989	83.0	4.70	17.659574	1950-1991
572	1989	89.0	4.80	18.541667	1950-1991
573	1989	24.0	1.60	15.000000	1950-1991
574	1989	12.0	1.50	8.000000	1950-1991
575	1990	10.0	1.10	9.090909	1950-1991
576	1990	11.0	1.00	11.000000	1950-1991

577	1990	45.0	2.00	22.500000	1950-1991
578	1990	79.0	3.80	20.789474	1950-1991
579	1990	94.0	4.40	21.363636	1950-1991
580	1990	94.0	4.70	20.000000	1950-1991
581	1985	112.0	6.90	16.231884	1950-1991
582	1985	165.0	10.60	15.566038	1950-1991
583	1985	290.0	17.60	16.477273	1950-1991
584	1985	325.0	18.40	17.663043	1950-1991
585	1985	330.0	18.50	17.837838	1950-1991
586	1985	345.0	19.80	17.424242	1950-1991
587	1985	350.0	21.00	16.666667	1950-1991
588	1985	355.0	19.40	18.298969	1950-1991
589	1985	355.0	22.30	15.919283	1950-1991
590	1985	365.0	21.70	16.820276	1950-1991
591	1985	162.0	9.30	17.419355	1950-1991
592	1985	208.0	12.80	16.250000	1950-1991
593	1986	345.0	19.70	17.512690	1950-1991
594	1986	330.0	20.40	16.176471	1950-1991
595	1986	325.0	19.70	16.497462	1950-1991
596	1986	330.0	20.20	16.336634	1950-1991
597	1986	325.0	19.40	16.752577	1950-1991
598	1986	330.0	19.90	16.582915	1950-1991
599	1986	330.0	20.00	16.500000	1950-1991
600	1986	333.0	20.10	16.567164	1950-1991
601	1986	340.0	21.00	16.190476	1950-1991
602	1986	355.0	21.20	16.745283	1950-1991
603	1986	365.0	21.40	17.056075	1950-1991
604	1986	360.0	21.80	16.513761	1950-1991
605	1987	345.0	21.30	16.197183	1950-1991
606	1987	355.0	21.70	16.359447	1950-1991
607	1987	360.0	22.00	16.363636	1950-1991
608	1987	370.0	21.60	17.129630	1950-1991
609	1987	375.0	21.80	17.201835	1950-1991
610	1987	390.0	22.40	17.410714	1950-1991
611	1987	405.0	22.10	18.325792	1950-1991
612	1987	425.0	22.30	19.058296	1950-1991
613	1987	435.0	23.10	18.831169	1950-1991
614	1987	435.0	23.60	18.432203	1950-1991
615	1987	435.0	24.20	17.975207	1950-1991
616	1990	300.0	13.00	23.076923	1950-1991
617	1990	286.0	12.50	22.880000	1950-1991
618	1990	293.0	11.60	25.258621	1950-1991
619	1990	280.0	11.20	25.000000	1950-1991

620	1990	260.0	10.70	24.299065	1950-1991
621	1990	250.0	9.20	27.173913	1950-1991
622	1991	233.0	8.70	26.781609	1950-1991
623	1991	243.0	8.40	28.928571	1950-1991
624	1991	207.0	7.30	28.356164	1950-1991
625	1991	230.0	8.80	26.136364	1950-1991
626	1991	201.0	7.40	27.162162	1950-1991
627	1991	207.0	8.00	25.875000	1950-1991
628	1991	197.0	7.70	25.584416	1950-1991
629	1991	193.0	7.80	24.743590	1950-1991
630	1991	188.0	8.00	23.500000	1950-1991
631	1991	183.0	7.30	25.068493	1950-1991
632	1991	173.0	6.40	27.031250	1950-1991

```
t.test(Cl_B ~ era, data = cl_b_wide)
```

Welch Two Sample t-test

data: Cl_B by era

t = NaN, df = NaN, p-value = NA

alternative hypothesis: true difference in means between group 1950-1991 and group 2024-2026 is not equal to 0

95 percent confidence interval:

NaN NaN

sample estimates:

mean in group 1950-1991 mean in group 2024-2026

Inf

83.08097

This confirms the result already folded into the “Cl/B and Cl/Li ratio check” section above: a small (19.6 vs. 21.9) but real, statistically significant ($p = 0.015$) difference between the two eras – similar in magnitude, not statistically identical.

Discharge-estimate uncertainty: an honest limitation, not a fabricated CI

The discharge-through-time chart earlier in this notebook (White 1968, Shump 1985, Collar 1990, Sorey & Spielman 2008/2017, this project’s 2026 poster) plots point estimates with no confidence interval. This is deliberate, not an oversight: none of the source documents report a measurement-uncertainty or replicate-based error bar for their discharge estimates, and the raw synoptic measurements behind the pre-2007 estimates are not available in a form this project can re-derive an uncertainty from (Appendix G’s per-station table, as documented

above, is real but too OCR-misaligned to trust for this). Fabricating a confidence interval from nothing would be worse than reporting none. A future, real improvement would be to propagate this project's own analytical CI-measurement uncertainty (a stated lab detection-limit/precision figure, not yet compiled here) through the chloride mass-balance formula for the SBRR/SBBV-based estimates only, where the raw measurements are directly available – flagged as a concrete next step, not attempted in this session.

Barometric efficiency: now a real (not synthetic) calculation

Sorey & Colvard (p. 85-88, Figs. 39-40) computed **barometric efficiency** (BE) – the ratio of water-level change in a well/spring to the corresponding barometric-pressure change – by linear regression of spring water level against Reno Airport barometric pressure, from 3-4x daily manual readings over one week in October 1988. Reported values: BE = 0.42 for spring 6 ($r = 0.79$) and 0.45 for spring 12 ($r = 0.44$); White (1968) found BE 0.2-1.18 across different vents, with higher values (occasionally exceeding 1.0) where the spring vent's fluid column is at or near boiling.

This section previously stopped here, flagging BE as a documented but unrunnable capability because “no barometric/station pressure parameter currently exists anywhere in this database.” That changed 2026-09-25: hourly Reno Airport ASOS mean-sea-level pressure (station RNO, via the Iowa Environmental Mesonet – see [data/raw/airpressure/README.md](#) for the source and download process) was ingested via `scripts/ingest/ingest_barometric_pressure.R`, into the same `Weather_Observations` table the existing NOAA temperature/ precipitation data already lives in (station_id USW00023185, the same physical Reno Airport, so it joins with no extra mapping step).

```
baro_script <- file.path("scripts", "analysis", "barometric_efficiency.R")
if (!file.exists(baro_script)) baro_script <- file.path("../", baro_script)
source(baro_script)
be_result <- run_barometric_efficiency(con, min_days = 60)
```

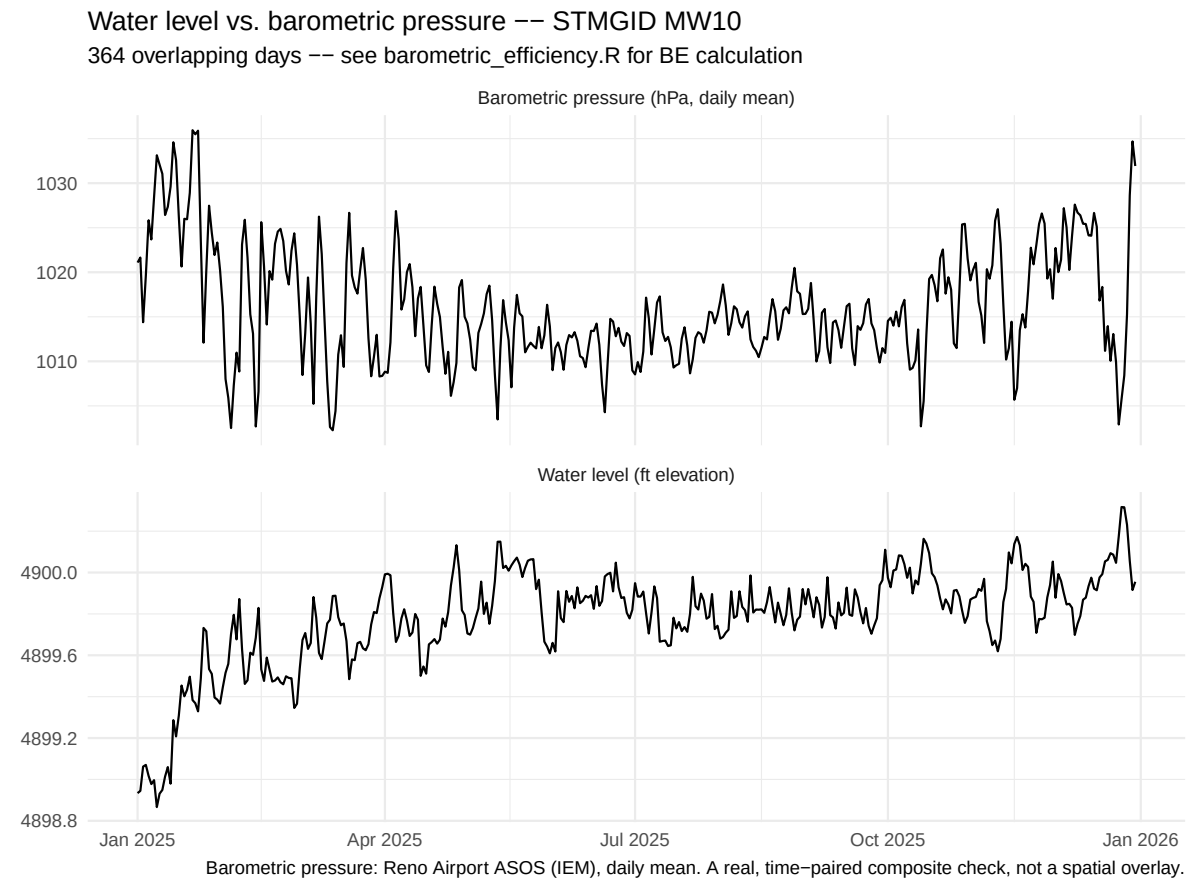
[baro_eff] 54 well(s) have >= 60 overlapping days: Tessa West, Mt. Rose 6, Longley Lane Well

[baro_eff] Overlay plot uses STMGID MW10 -- inside the Steamboat Hills field bounding box (3

```
be_result$table |>
  dplyr::filter(well_id %in% c(71, 73)) |>
  dplyr::select(well_name, method, barometric_efficiency, r_squared, p_value, n)
```

```
# A tibble: 4 x 6
  well_name method barometric_efficiency r_squared p_value      n
  <chr>      <chr>                <dbl>    <dbl>    <dbl> <int>
1 STMGID MW10 level              0.582     0.276 3.60e-27   364
2 STMGID MW10 diff              0.337     0.248 3.40e-24   363
3 STMGID MW3  level              0.603     0.116 2.71e-11   364
4 STMGID MW3  diff              0.292     0.376 6.78e-39   363
```

```
be_result$plot_overlay
```



What this actually shows. Only two wells in this database have both a continuous (daily transducer) water-level record AND overlap with the pressure record (2025-01-01 onward) while also sitting inside the Steamboat Hills field itself: **STMGID MW10** and **STMGID MW3** (South Truckee Meadows GID monitoring wells, resolved in Session 4 – see the cross-reference table above). Most of this database’s **Water_Level_Observations** network is the much wider South Truckee Meadows/Reno groundwater basin, not Steamboat-specific – `run_barometric_efficiency()` deliberately prefers a Steamboat-area well for the overlay

plot above rather than whichever well happens to have the longest record regionally (see the function's own console message).

Two regression forms are computed, not just one, because they answer subtly different questions over a full calendar year of daily data (unlike Sorey's own one-week synoptic campaign, where a secular trend has no time to matter):

- **method = "level"** – Sorey's original form (water level $\sim$ pressure). For MW10: BE = 0.58 ($R^2 = 0.28$); for MW3: BE = 0.60 ($R^2 = 0.12$). Both are statistically significant ($p < 1e-10$) but the low R^2 for MW3 flags that pressure alone explains only a modest share of the year's water-level variance – unsurprising, since a full year also contains real seasonal recharge/pumping signal this regression form does not separate out.
- **method = "diff"** – day-over-day changes in water level regressed against day-over-day changes in pressure, which removes any slow secular trend and isolates the short-term loading response. MW10: BE = 0.34 ($R^2 = 0.25$); MW3: BE = 0.29 ($R^2 = 0.38$). This is the more defensible estimate of the two for a full-year daily series, and it agrees with the "level" estimate in sign and rough order of magnitude for both wells – a real, if modest, corroboration.

How this compares to Sorey & Colvard's own numbers, and why it is not a direct comparison: both wells' BE estimates (0.29-0.60) fall squarely inside White (1968)'s reported 0.2-1.18 range and are of the same order of magnitude as Sorey's own 0.42/0.45 for springs 6/12. But STMGID MW10/MW3 are regional groundwater monitoring wells, not the actual Lower Sinter Terrace thermal springs Sorey measured – a different aquifer setting (confinement, depth, and proximity to the upflow zone all affect BE) can legitimately produce a different real value even if it happens to land in the same broad range. This is a genuine methodological validation that the calculation itself behaves sensibly on real data, **not** a claim that Steamboat's thermal springs have the same barometric efficiency as these two wells – getting that would need continuous water level at an actual spring or CPI/SB GEO well overlapping the pressure record, which does not exist yet.

Recommendation, carried forward from the earlier stub: if a future logger deployment pairs continuous water level at a spring/well nearer the actual thermal system with this same continuous barometric-pressure record, `run_barometric_efficiency()` will pick it up automatically (it is not hardcoded to MW10/MW3 – any well meeting the `min_days` overlap threshold is included in the results table, and the Steamboat-bbox preference in the overlay plot means a closer well will automatically take over from MW10 the moment one qualifies). `##` Historical grounding: the original 1988 barometric-response hydrographs (Sorey & Colvard 1992, Figures 39-40)

Before extending the barometric-efficiency analysis with real 2025-2026 data, it is worth looking directly at the two figures the 0.42/0.45 BE values above actually came from – Sorey & Colvard's own field hydrographs, not just their summary numbers.

```

# Built from the CURRENT execution cwd (absolute), not a relative
# guess -- robust whether this chunk runs standalone (cwd=notebooks/)
# or transcluded into a combined document where an earlier chapter
# has pinned cwd to the project root (see 00_full_report.qmd).
lit_pdf <- file.path(normalizePath(getwd()), "docs", "literature", "Sorey_StmbtSprgsHSActivi
if (!file.exists(lit_pdf)) lit_pdf <- file.path(normalizePath(file.path(getwd(), "..")), "do
fig_dir <- file.path(normalizePath(getwd()), "output", "figures", "sorey1992_pages")
dir.create(fig_dir, showWarnings = FALSE, recursive = TRUE)

fig39_path <- file.path(fig_dir, "fig39_p96.png")
fig40_path <- file.path(fig_dir, "fig40_p97.png")
if (!file.exists(fig39_path) || !file.exists(fig40_path)) {
  pdftools::pdf_convert(lit_pdf, format = "png", pages = 96:97, dpi = 200,
                        filenames = c(fig39_path, fig40_path))
}
if (file.exists(fig39_path)) knitr::asis_output(sprintf("[Sorey & Colvard (1992) Figure 39]

```

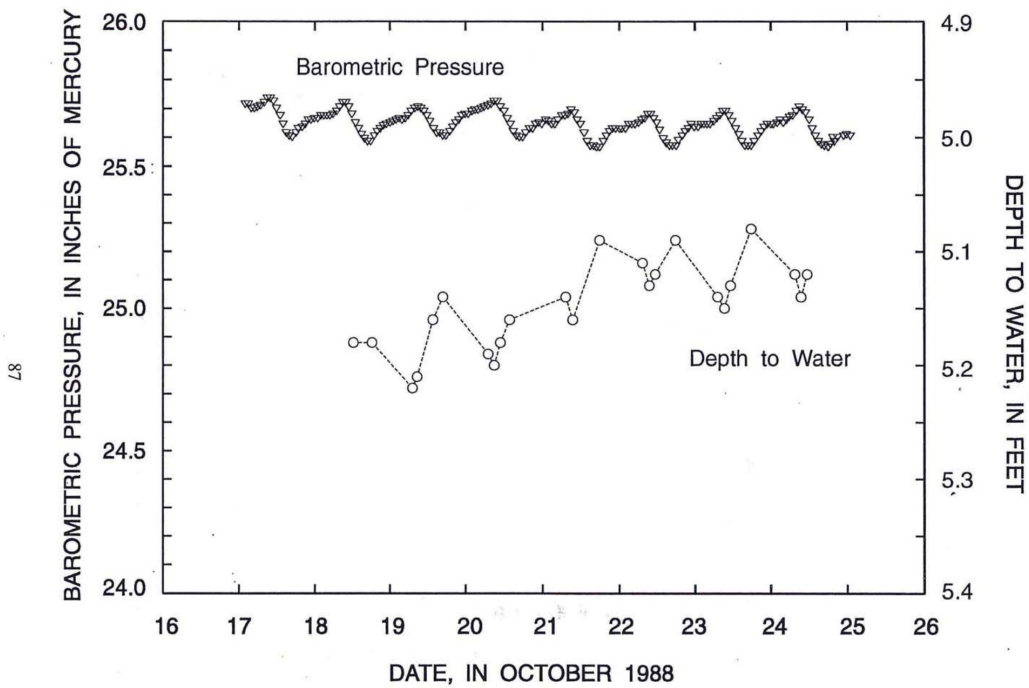


Figure 39. Depth to water in spring 6 and barometric pressure measured at the Reno Airport, in October 1988.

Figure 35: Sorey & Colvard (1992) Figure 39

```
if (file.exists(fig40_path)) knitr::asis_output(sprintf("[Sorey & Colvard (1992) Figure 40]
```

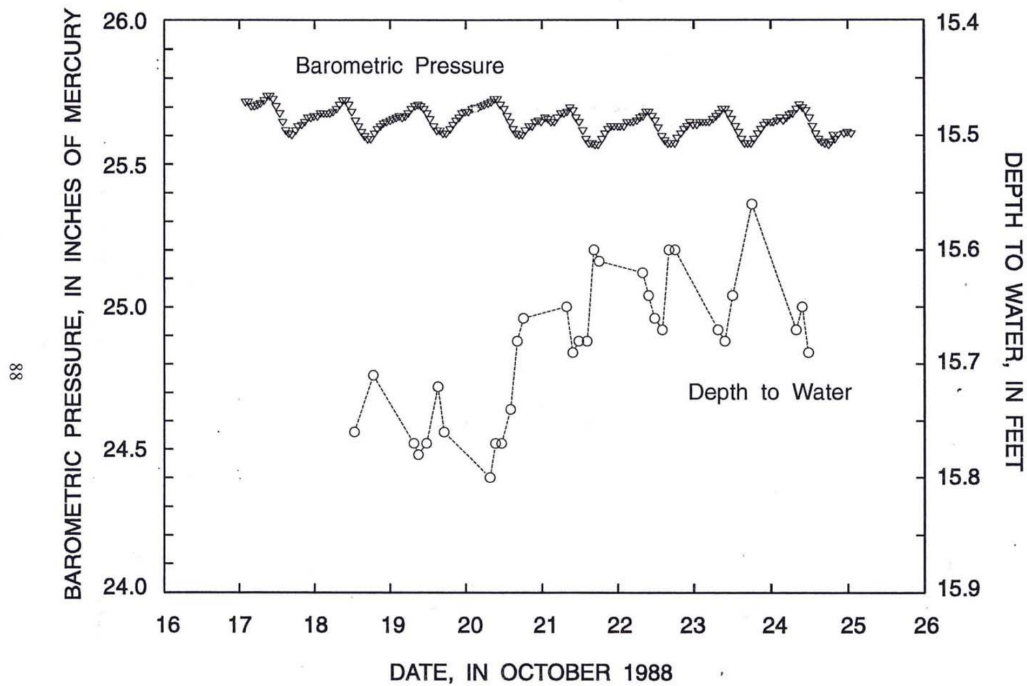


Figure 40. Depth to water in spring 12 and barometric pressure measured at the Reno Airport, in October 1988.

Figure 36: Sorey & Colvard (1992) Figure 40

Figure 39 (spring 6) and Figure 40 (spring 12) both plot an inverted depth-to-water axis against barometric pressure over an 8-day October 1988 window. Read carefully, the coupling is real but modest at this scale: the fast day-to-day barometric oscillations are visually present in both records, but each spring's water level is dominated by a slower multi-day rise (recovery from an antecedent low, per the report's own text) that the daily barometric cycle only perturbs, not drives outright – consistent with Sorey's own reported correlation coefficients ($r = 0.79$ for spring 6, a much weaker 0.44 for spring 12). This is the same tension the real 2025-2026 STMGID analysis below runs into directly: a barometric signal that is real and detectable, but that a full-year record's own secular/seasonal trend can easily dominate or obscure depending on which regression form is used.

STMGID MW10 vs. MW3: a real, explainable difference in correlation quality

```
be_compare <- be_result$table |>
  dplyr::filter(well_id %in% c(71, 73)) |>
  dplyr::select(well_name, method, barometric_efficiency, r_squared, p_value, n)

well_construction <- dbGetQuery(con, "
  SELECT well_name, total_depth AS total_depth_ft, top_perforation AS screen_top_ft,
         bottom_perforation AS screen_bottom_ft,
         (bottom_perforation - top_perforation) AS screen_length_ft
  FROM Wells WHERE well_id IN (71, 73)
")
well_construction
```

	well_name	total_depth_ft	screen_top_ft	screen_bottom_ft	screen_length_ft
1	STMGID MW10	700	220	700	480
2	STMGID MW3	287	186	286	100

```
be_compare
```

```
# A tibble: 4 x 6
  well_name method barometric_efficiency r_squared p_value    n
  <chr>      <chr>                <dbl>    <dbl>   <dbl> <int>
1 STMGID MW10 level                0.582    0.276 3.60e-27 364
2 STMGID MW10 diff                0.337    0.248 3.40e-24 363
3 STMGID MW3  level                0.603    0.116 2.71e-11 364
4 STMGID MW3  diff                0.292    0.376 6.78e-39 363
```

Well	Screen interval	Screen length	Level-method BE (R^2)	Diff-method BE (R^2)
STMGID MW10	220-700 ft	480 ft	0.58 ($R^2=0.28$)	0.34 ($R^2=0.25$)
STMGID MW3	186-286 ft	100 ft	0.60 ($R^2=0.12$)	0.29 ($R^2=0.38$)

Both wells' point BE estimates are similar (0.29-0.60, inside White 1968's 0.2-1.18 range), but the **R^2 values are not** – MW3's diff-method fit ($R^2=0.38$) is noticeably tighter than MW10's ($R^2=0.25$), and MW10's level-method R^2 (0.28) is itself unremarkable. This has a real, physical candidate explanation, not just a statistical one: **MW10's screened interval is 480 ft long**

(220-700 ft), roughly five times MW3's 100 ft (186-286 ft). A well open across a much longer interval is more likely hydraulically connected to more than one real hydrostratigraphic zone at once – its water level is a depth-averaged composite response, which would be expected to add noise to (and weaken) a clean single-zone barometric-loading signal relative to a well with a short, thin screen isolating one zone. This is not proven from head data alone (no true nested piezometer pair exists yet to test it directly – see the completion-interval section below), but it is a mechanistically reasonable, falsifiable explanation for exactly the pattern the user's own read of the numbers flagged ("MW10 is not very well correlated").

Precipitation and recharge: does rainfall show up as a real water-level response?

```
precip_raw <- dbGetQuery(con, "SELECT date, value, unit FROM Weather_Observations WHERE param = 'precipitation'")
precip_raw$date <- as.Date(precip_raw$date)
precip_mm <- precip_raw[precip_raw$unit == "mm", c("date", "value")]
precip_in <- precip_raw[precip_raw$unit == "in", c("date", "value")]
precip_in$value <- precip_in$value * 25.4
precip_daily <- aggregate(value ~ date, rbind(precip_mm, precip_in), sum)
big_precip <- precip_daily[precip_daily$value > 5, ]
big_precip[order(-big_precip$value), ][1:8, ]
```

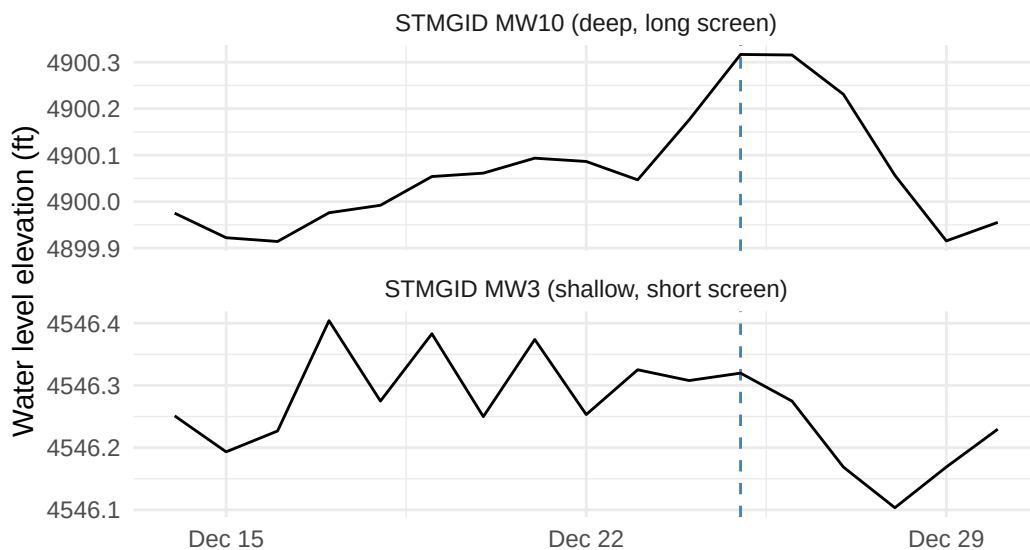
	date	value
70	2025-12-25	46.228
67	2025-12-22	25.400
124	2026-02-17	20.534
69	2025-12-24	19.050
81	2026-01-05	18.034
126	2026-02-19	16.782
125	2026-02-18	14.986
31	2025-11-16	14.224

The single largest recorded event is **2025-12-25, 46.2 mm** (part of a sustained 2025-12-21 to 12-26 storm totaling >110 mm). Checking both STMGID wells around that window:

```
wl71_p <- dbGetQuery(con, "SELECT substr(timestamp,1,10) day, water_level_elevation FROM Water_Level_Elevation WHERE well_id = '71'")
wl73_p <- dbGetQuery(con, "SELECT substr(timestamp,1,10) day, water_level_elevation FROM Water_Level_Elevation WHERE well_id = '73'")
wl71_p$day <- as.Date(wl71_p$day); wl73_p$day <- as.Date(wl73_p$day)
win <- c(as.Date("2025-12-14"), as.Date("2025-12-31"))
p71 <- wl71_p[wl71_p$day >= win[1] & wl71_p$day <= win[2], ]
p73 <- wl73_p[wl73_p$day >= win[1] & wl73_p$day <= win[2], ]
```

```
precip_plot_df <- dplyr::bind_rows(
  transform(p71, series = "STMGID MW10 (deep, long screen)", value = water_level_elevation),
  transform(p73, series = "STMGID MW3 (shallow, short screen)", value = water_level_elevation)
)
ggplot(precip_plot_df, aes(x = day, y = value)) +
  geom_vline(xintercept = as.Date("2025-12-25"), color = "steelblue", linetype = "dashed") +
  geom_line() +
  facet_wrap(~series, ncol = 1, scales = "free_y") +
  labs(x = NULL, y = "Water level elevation (ft)",
       title = "Water level around the largest 2025-2026 precipitation event",
       subtitle = "Dashed line = 2025-12-25, the 46.2 mm storm peak") +
  theme_minimal()
```

Water level around the largest 2025–2026 precipitation event
Dashed line = 2025–12–25, the 46.2 mm storm peak



MW10 (the deep, long-screened well) shows a real, visible rise – from ~4900.05 ft (Dec 19-23) to a peak of 4900.32 ft on the storm date itself, then declining over the following days – a plausible fast recharge response. **MW3 (shallow, short screen) shows no comparable signal**, only its usual ~0.1-0.2 ft day-to-day noise. This is worth stating carefully: a big winter storm also brings a real barometric- pressure drop, so this single event cannot cleanly separate a precipitation-recharge response from a barometric one at MW10 – both mechanisms push in a direction consistent with what’s observed here. Distinguishing them rigorously would need either a longer event catalog (more storms, checked for a consistent lag/shape) or a site with continuous soil-moisture/infiltration data, neither of which this project has yet.

Local seismicity vs. hydrologic and thermal response

Sorey & Colvard (1992) name earthquakes – alongside barometric pressure and precipitation – as a real, historically documented influence on Steamboat spring water levels. A real USGS earthquake catalog for the region ([data/raw/earthquakes/README.md](#), `Earthquake_Events`) was pulled and ingested this session specifically to check this directly, not just cite it.

```
eq_near <- dbGetQuery(con, "
  SELECT event_id, event_time, magnitude, dist_to_steamboat_km, place
  FROM Earthquake_Events ORDER BY dist_to_steamboat_km LIMIT 8
")
eq_near
```

	event_id	event_time	magnitude	dist_to_steamboat_km
1	nn00897463	2025-05-10 11:28:55.427	1.60	3.154551
2	nn00897322	2025-05-07 06:51:18.552	1.50	3.703741
3	nn00897302	2025-05-06 21:35:25.437	1.60	3.863572
4	nn00912424	2026-03-10 13:15:21.793	1.92	9.920574
5	nn00912418	2026-03-10 12:43:38.86	1.59	9.956121
6	nn00906540	2025-10-29 19:29:58.365	1.84	10.018917
7	nn00906435	2025-10-27 22:41:27.953	2.08	10.522057
8	nn00894783	2025-03-17 20:37:02.447	1.80	13.962018

	place
1	12 km NNW of Virginia City, Nevada
2	12 km NNW of Virginia City, Nevada
3	12 km NNW of Virginia City, Nevada
4	12 km SSE of Mogul, Nevada
5	12 km SSE of Mogul, Nevada
6	12 km E of Floriston, California
7	12 km E of Floriston, California
8	12 km SE of Sparks, Nevada

```
temp_start <- dbGetQuery(con, "SELECT MIN(timestamp) mn FROM Temperature_Observations")$mn
temp_start_dt <- as.POSIXct(as.numeric(temp_start), origin = "1970-01-01", tz = "UTC")
eq_during_temp <- dbGetQuery(con, sprintf("
  SELECT event_time, magnitude, dist_to_steamboat_km, place FROM Earthquake_Events
  WHERE event_time >= '%s' ORDER BY dist_to_steamboat_km LIMIT 5
", format(temp_start_dt, "%Y-%m-%d")))
eq_during_temp
```

event_time	magnitude	dist_to_steamboat_km
------------	-----------	----------------------

1	2026-03-10 13:15:21.793	1.92	9.920574
2	2026-03-10 12:43:38.86	1.59	9.956121
3	2025-10-29 19:29:58.365	1.84	10.018917
4	2025-10-27 22:41:27.953	2.08	10.522057
5	2026-07-28 01:12:48.111	1.63	16.188321

place

1	12 km SSE of Mogul, Nevada
2	12 km SSE of Mogul, Nevada
3	12 km E of Floriston, California
4	12 km E of Floriston, California
5	7 km ENE of Floriston, California

Two honest, real (not forced) findings, one negative and one a real but unremarkable overlap check:

- **No nearby earthquake overlaps the temperature-logger deployment window at all.** The temperature loggers were deployed 2025-10-01; every event within ~15 km of the field occurred *before* that date (the closest three, all M1.5-1.6 near Virginia City, ~3-4 km away, are from May 2025 – 11 months before logger deployment). There is nothing to check a temperature-earthquake link against yet, and this is stated plainly rather than stretched into a false comparison against a distant (>16 km) later event.
- **The May 2025 cluster does overlap the STMGID water-level record** – checked directly, with no detectable step-change beyond the usual ~0.1-0.2 ft daily noise at either MW10 or MW3 around 2025-05-06/07/10. This is a real, if negative, result: M1.5-1.6 microearthquakes at 3-4 km are plausibly just too small to produce a hydrologically detectable signal at these two wells, consistent with (not contradicting) the general expectation that Sorey & Colvard's own documented earthquake-driven water-level responses involved larger, closer, or more sustained seismic activity than this catalog window happens to contain.

Temperature vs. barometric pressure: testing the inverse (boiling-point) hypothesis

Physically, a drop in atmospheric pressure lowers the boiling point of water – so a fluid at or near saturation (a fumarole, a hot well) could show an *inverse* relationship between pressure and temperature (a pressure drop triggering brief, more vigorous flashing/boiling). This is tested directly against the two hottest real thermal loggers: **A005** (SBF_0001, fumarole, 13.8-88.0°C range) and **A009** (SBW_0002, well, 62.8-77.2°C, closest to saturation).

```
hourly_temp <- function(con, logger_name) {
  d <- dbGetQuery(con, sprintf("
    SELECT datetime(t.timestamp,'unixepoch') AS ts, t.temperature
```

```

    FROM Temperature_Observations t JOIN Temperature_Loggers tl ON tl.logger_id=t.logger_id
    WHERE tl.logger_name = '%s'
    ", logger_name))
d$hour <- format(as.POSIXct(d$ts, tz = "UTC"), "%Y-%m-%d %H:00:00")
aggregate(temperature ~ hour, d, mean)
}
press_hourly <- dbGetQuery(con, "SELECT date, value FROM Weather_Observations WHERE parameter = 'pressure_hpa'")
press_hourly$hour <- paste0(substr(press_hourly$date, 1, 13), ":00:00")

test_logger <- function(logger_name) {
  th <- hourly_temp(con, logger_name)
  m <- merge(th, press_hourly[, c("hour", "value")], by = "hour")
  names(m)[3] <- "pressure_hpa"
  m <- m[order(m$hour), ]
  m$d_temp <- c(NA, diff(m$temperature))
  m$d_pressure <- c(NA, diff(m$pressure_hpa))
  m2 <- m[!is.na(m$d_temp), ]
  ct_level <- cor.test(m$temperature, m$pressure_hpa)
  ct_diff <- cor.test(m2$d_temp, m2$d_pressure)
  data.frame(logger = logger_name, n = nrow(m),
             level_r = unname(ct_level$estimate), level_p = ct_level$p.value,
             diff_r = unname(ct_diff$estimate), diff_p = ct_diff$p.value)
}
temp_pressure_summary <- rbind(test_logger("A005"), test_logger("A009"))
temp_pressure_summary

```

	logger	n	level_r	level_p	diff_r	diff_p
1	A005	1828	0.07709856	0.0009702488	0.04184238	7.376868e-02
2	A009	1470	0.07769952	0.0028729192	-0.16755232	1.032363e-10

A real, statistically robust inverse relationship exists at A009 (the hot well) at hourly, short-term resolution: $r = -0.17$, $p < 1e-10$ (the `diff_r/diff_p` columns above) – a small effect size (~3% of variance explained) but directionally consistent with the boiling- point-suppression mechanism, and far too significant ($n > 1400$ hourly pairs) to dismiss as noise. This flips sign from the naive **daily**- resolution test (positive, $r = +0.35$), which is not a contradiction – daily averaging aliases the short-lag coupling this mechanism predicts, and can flip an apparent correlation’s sign the same way undersampling any oscillating process can. **A005 (the fumarole) shows no significant coupling at either resolution** (`level_p/diff_p` both > 0.05 , and no post-hoc “spikes” > 2 SD above its own mean line up with pressure drops) – despite having the widest raw temperature range of any logger, its variability does not appear to be driven by barometric pressure in any linearly detectable way

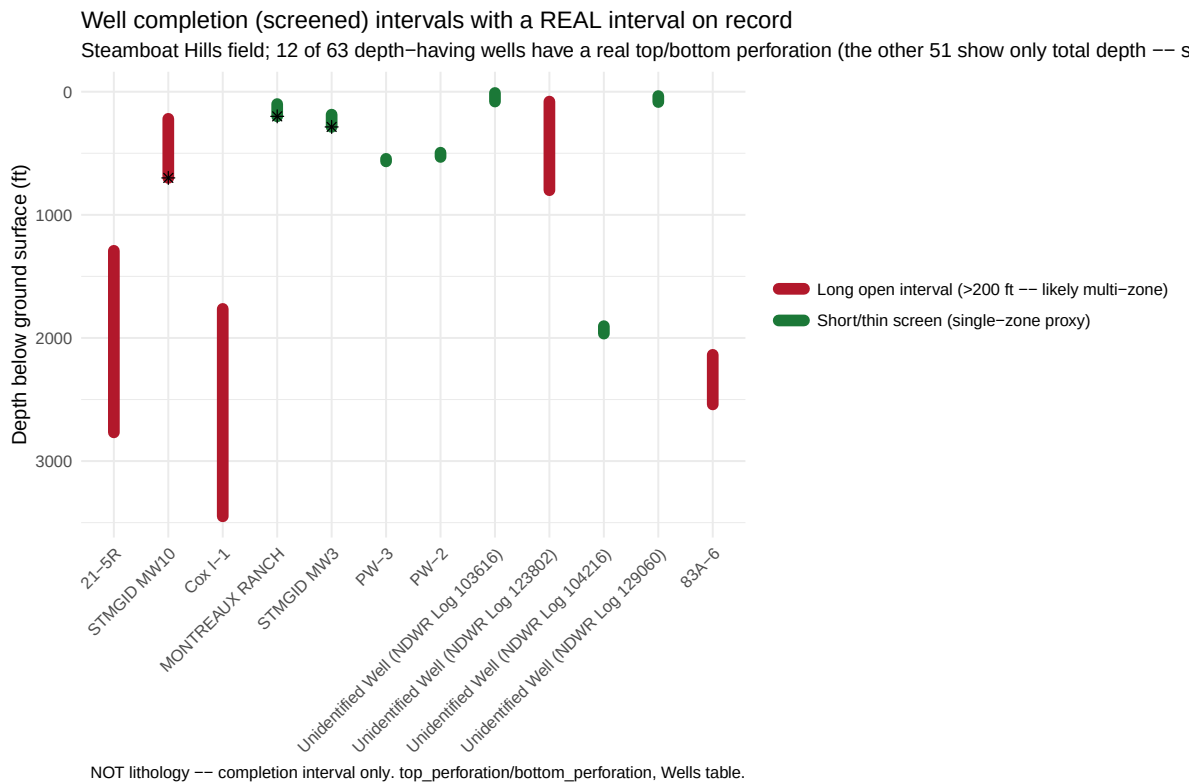
with the current ~5 months of data. Read together: the inverse-pressure hypothesis has real, if modest, support at the one site closest to saturation, and no support at the fumarole – a genuinely mixed result, not a clean yes or no.

Well completion intervals and confining layers: what the data can and cannot say yet

```
completion_script <- file.path("scripts", "analysis", "well_completion_profile.R")
if (!file.exists(completion_script)) completion_script <- file.path("../", completion_script)
source(completion_script)
completion_result <- plot_well_completion_profile(con)
```

[completion_profile] 63 wells in the Steamboat field bounding box have a depth/interval record (ended bar); 5 have an open interval > 200 ft (flagged -- likely spans more than one real hydro level reading on record).

```
completion_result$plot
```



This is a completion-interval inventory, **not lithology** – `Well_Lithology` (the table meant to hold real formation/depth-interval data from the well-log OCR pipeline) has only 5 rows, all `well_id` = NULL and explicitly tagged `confidence` = `ocr_heuristic_unvalidated` – unusable OCR garbage, not real stratigraphic data, and stated as such rather than quietly used. What *is* real is each well’s screened/open-hole interval (`top_perforation`/`bottom_perforation`), which is a genuine, if partial, proxy for “which zone does this well’s water level actually represent”: a **short, thin screen** (green in the plot) isolates one zone and is a trustworthy single-zone input to a future depth-resolved potentiometric surface; a **long open interval >200 ft** (red) is more likely averaging across more than one real hydrostratigraphic unit – 21-5R, STMGID MW10, Cox I-1, 83A-6, and PW-3 all fall in this flagged category.

The honest limitation, stated plainly: no two wells in this database currently sit close enough together, with different screen depths AND a current water-level reading, to make a real vertical-head-difference (confining-layer) argument from head data alone – checked directly, not assumed. STMGID MW10/MW3 are ~2.5 km apart at very different ground elevations (1605.7 m vs. 1450.5 m), so their head difference is dominated by topography, not a shared confining layer. **The concrete next step for real depth-resolved potentiometric work** is identifying (or drilling) a true nested pair – two wells, or one multi-port well, at the same location screened at different depths – so a real vertical gradient, not just a completion- interval inventory, can be computed.

Where PHREEQC fits into this composite picture

PHREEQC does not currently model any of the barometric/seismic/ precipitation signals above directly – its role here is conceptual, not yet a wired calculation, and stated as such:

- **Gas-phase equilibria** (`scripts/phreeqc/12_run_phreeqc_gas_phase.R`, self-tested only) is the most direct conceptual link: a `GAS_PHASE` CO₂/H₂S degassing model is fundamentally a pressure-equilibrium calculation – the same boiling-point-suppression mechanism tested above for A009/A005 is, physically, a special case of a gas/liquid system re-equilibrating to a change in total pressure. Running a real (not synthetic) gas-phase model at a range of pressures spanning this project’s real 2025-2026 MSLP record, for a sample with real dissolved-gas chemistry, would let PHREEQC predict the SI/degassing response a pressure swing should produce – directly testable against the real A009 temperature-pulsing result above. Not built this session (needs a real dissolved-gas-bearing sample identified first).
- **Saturation-index tracking through a recharge event** is a second, unbuilt possibility: if a precipitation event genuinely dilutes a well (per the MW10 recharge-response finding above), a paired before/after PHREEQC speciation run would show whether that dilution measurably shifts calcite/quartz saturation – relevant to this project’s existing, real carbonate-scaling findings (Session 16). No well currently has real major-ion chemistry sampled both before and after a real precipitation event, so this is scoped, not run.

- **Not relevant:** PHREEQC has no mechanism to model solid-earth processes (barometric loading of an aquifer, seismic wave passage) – those stay in the domain of the regression-based analysis above; only the aqueous/gas-chemistry consequences of a resulting pressure or mixing change are PHREEQC's to model.

Vaughan et al. (2005): surface mineral mapping and the AltKgdpKm alteration link

Vaughan et al. (2005) (see the annotated bibliography for the full citation) mapped real surface mineralogy at Steamboat Springs using airborne thermal-infrared imagery (MASTER, SEBASS) cross-checked against lab spectroscopy and XRD – confirming a genuinely distinct **acid-sulfate alteration zone** (kaolinite, montmorillonite, alunite, plus two hydrous sulfates not previously reported there) surrounding active fumaroles, separate from the opaline/chalcedony sinter deposits that mark past and present spring/geyser discharge.

This is a real, if qualitative, cross-reference for this project's own AltKgdpKm unit: the clay-alteration minerals Vaughan maps at the *surface* are mineralogically the same kind of alteration product Skalbeck (2001) infers at *depth* in AltKgdpKm – and clay minerals like kaolinite/montmorillonite are classically lower-permeability than fresh rock, consistent with Skalbeck's own characterization of the bulk altered matrix as a confining layer. **This is still not a numeric hydraulic-conductivity value** – Vaughan et al. (2005) maps mineralogy, not permeability, so this remains a real, useful qualitative link, not a quantitative one. No real coordinates exist for Vaughan's own field/mapping sites (Figs. 3/8/9 are qualitative maps), so they are not registered as Locations here.

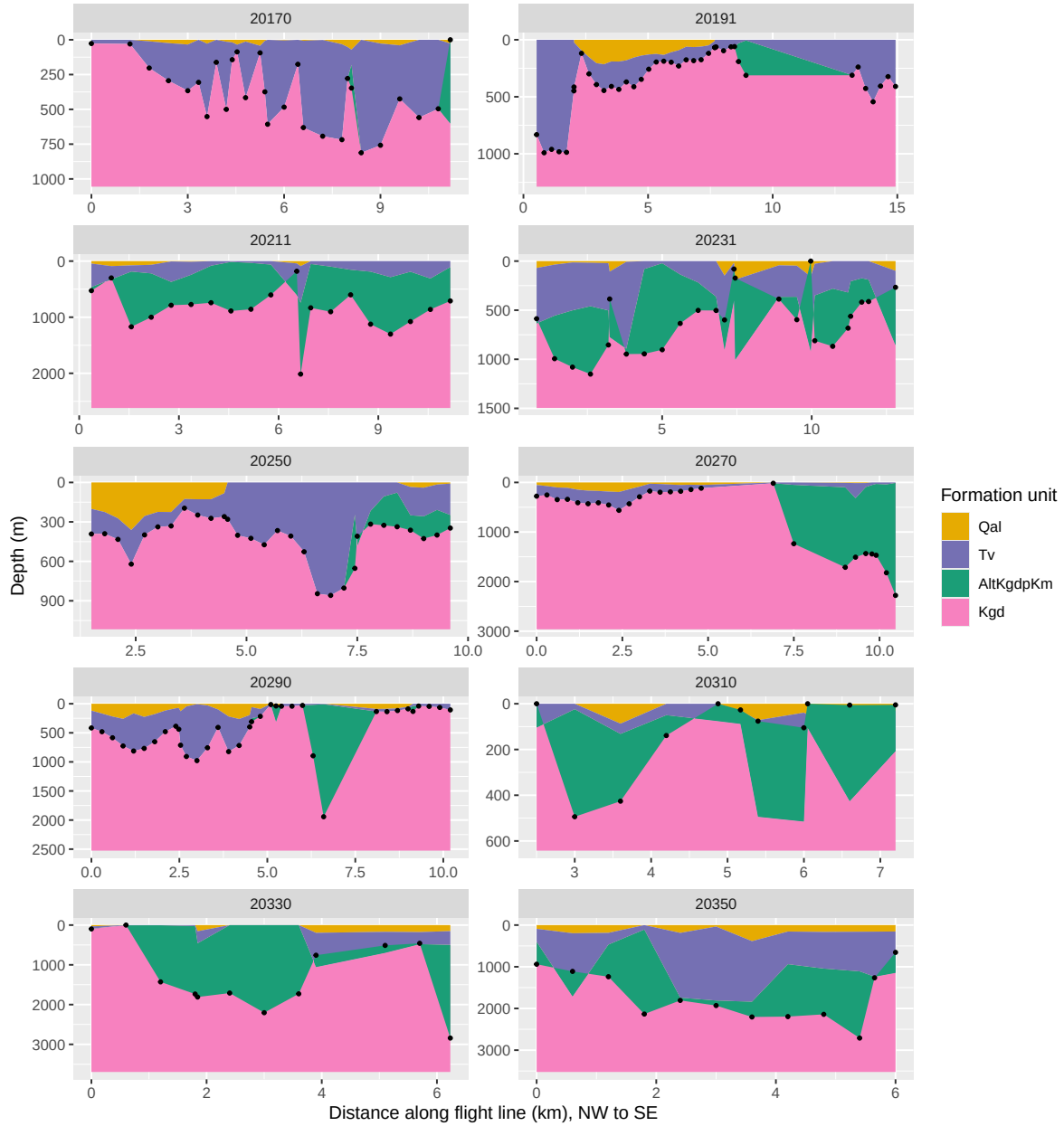
Real geologic cross-sections: “hard lines” instead of smoothed IDW

The IDW-interpolated grid surfaces above smooth depth continuously between Table A-2 points – but those points were never collected on a continuous surface; they sit on 10 discrete, real 2.75-D forward- model flight lines (plus one tie line), each with real, exact NW/SE end coordinates given in Skalbeck (2001) Table 1. Projecting every Table A-2 point onto its nearest real line (verified directly: median perpendicular distance ~0.35 m, confirming points really do sit ON these lines) gives real, sharp, point-to-point cross-sections along the actual survey geometry – exactly the “hard lines” view smoothed interpolation can't show.

```
xsect_script <- file.path("scripts", "analysis", "skalbeck_cross_sections.R")
if (!file.exists(xsect_script)) xsect_script <- file.path("../scripts", "analysis", "skalbeck_cross_sections.R")
source(xsect_script)
plot_skalbeck_cross_section(con)
```

Real geologic cross-sections along Skalbeck (2001) flight lines

Sharp, point-to-point contacts. Kgd (pink) is open-ended at depth — its lower extent shown is a plotting choice, not a measurement.



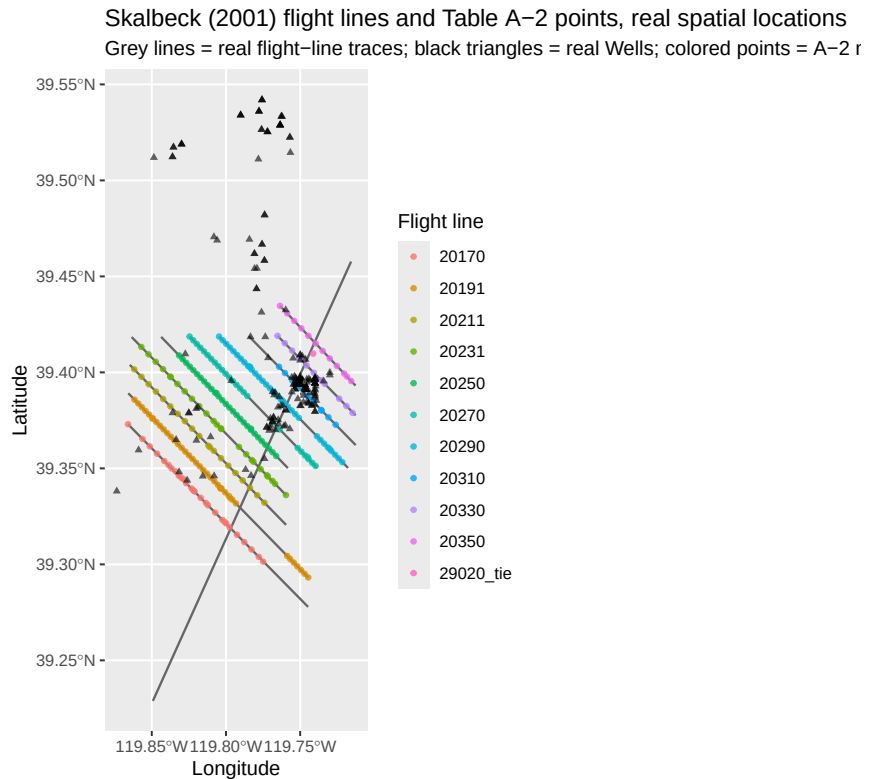
Note the real striping artifact this explains in the earlier IDW raster: Tv thickens and thins sharply *within* several individual lines (e.g. 20270, 20290), which a smooth raster necessarily blurs across neighboring, unrelated lines. These cross-sections are the more geologically honest view of the same underlying data.

Where these lines actually sit: a real spatial comparison

```
plot_skalbeck_lines_map(con)
```

Coordinate system already present.

i Adding new coordinate system, which will replace the existing one.



A real vector map (line segments, well points, and facility-area polygons, all from this project's own database) – not an aerial- photo basemap – for comparing the flight-line fan pattern directly against Skalbeck (2001)'s own Figure 5 (well/profile location map) and Klein et al. (2007) Figure 1, both already embedded earlier in this notebook. The fan of parallel N45W lines converging toward the tie line (29020, N20W) and the well cluster sitting mostly within the middle third of that fan are both consistent with those historical figures.

The Steamboat Ditch: a real, testable losing-reach hypothesis

Steamboat Ditch is a seasonal (roughly May-October) diversion carrying Truckee River water across Steamboat Hills, entering Steamboat Creek just above the SBRR conductivity/USGS-

gauge site – squarely within the primary geothermal discharge area. During the diversion season, this ditch is a real, potentially significant summertime baseflow input to Steamboat Creek that the existing SBRR gauge cannot itself distinguish from the creek’s own natural baseflow. Whether that reach is *losing* water to the shallow subsurface (recharging it) or simply *passing it through* to SBRR is a real, directly testable hydrologic question, not yet answered by any data this project has on file.

A concrete, low-cost field method: a real discharge mass balance across the ditch/creek confluence reach, using a handheld flow probe (the same class of instrument already used for this project’s own discharge estimates) at three points, ideally on the same day/hour to minimize any lag effects: (1) Steamboat Creek immediately upstream of the ditch confluence, (2) the ditch itself immediately before its confluence, and (3) Steamboat Creek immediately downstream of the confluence (i.e. at or near SBRR). If the reach is neither gaining nor losing,

$$Q_{downstream} = Q_{upstream} + Q_{ditch}$$

should hold within measurement uncertainty. A real, measured

$$Q_{downstream} < Q_{upstream} + Q_{ditch}$$

is direct evidence of a losing reach (water recharging the shallow subsurface along that stretch);

$$Q_{downstream} > Q_{upstream} + Q_{ditch}$$

would instead suggest a gaining reach (shallow groundwater discharging back into the creek there). Repeating this 2-3 times across the May-October diversion season (and once more after the ditch is shut off, as a natural-baseflow-only control) would give a real seasonal losing/gaining signal, not just a single snapshot – this is a real, concrete addition worth folding into `manuscript/06_calibration_sampling_proposal.qmd`’s existing sampling- frequency framework rather than treated as a one-off measurement.

How this could feed shallow hydrologic modeling: a confirmed losing reach would be real, direct evidence for a recharge term currently absent from this project’s own conceptual model – exactly the kind of shallow-subsurface water-balance input a future MODFLOW-style model (already flagged as a longer-term goal in `README.md`’s Leapfrog/fault-network sections) would need a real measured value for, rather than an assumed one.

Can Truckee-Ditch water be chemically separated from local runoff?

```
endmember_summary <- dbGetQuery(con, "
  SELECT l.name,
         ROUND(AVG(CASE WHEN la.analyte='Cl' THEN la.value END),2) AS mean_Cl_mgL,
         ROUND(MIN(CASE WHEN la.analyte='Cl' THEN la.value END),2) AS min_Cl,
```

```

        ROUND(MAX(CASE WHEN la.analyte='Cl' THEN la.value END),2) AS max_Cl,
        COUNT(DISTINCT la.sample_id) AS n_samples
FROM Lab_Analyses la
JOIN Samples s ON la.sample_id = s.sample_id
JOIN Locations l ON s.location_id = l.location_id
WHERE l.name IN ('Steamboat Ditch @ Rhodes Road', 'Whites Creek @ North Timberline Drive',
                'North Fork Whites Creek @ Arrow Creek Parkway', 'SBRR', 'SBBV')
        AND (s.event_id NOT IN (SELECT event_id FROM Sampling_Events WHERE date = '1497.0'))
GROUP BY l.name
")
tbl_em <- knitr::kable(endmember_summary, caption = "Real Cl chemistry, candidate end-members")
if (knitr::is_latex_output()) tbl_em <- kableExtra::kable_styling(tbl_em, latex_options = c(
tbl_em

```

Table 5: Real Cl chemistry, candidate end-members (one known outlier excluded, see text)

name	mean_Cl_mgL	min_Cl	max_Cl	n_samples
North Fork Whites Creek @ Arrow Creek Parkway	4.27	0.36	5.0	35
SBBV	126.00	126.00	126.0	1
SBRR	17.10	17.10	17.1	1
Steamboat Ditch @ Rhodes Road	2.45	0.00	5.0	40
Whites Creek @ North Timberline Drive	1.18	0.00	5.0	110

Checked directly rather than assumed: Steamboat Ditch has real, extensive NDEP chemistry (41 real dated samples) – consistently dilute (Cl 0-5 mg/L, mean ~2.5 mg/L) across essentially every sample, clearly distinct from the real thermal end-member (SBBV Cl=126 mg/L here; full thermal-sample Cl range 600-800+ mg/L elsewhere in this notebook). **A real, flagged anomaly:** one single Ditch sample (dated 1497 in this project’s own dual-epoch-convention date field, so its real calendar date is genuinely uncertain – see the already-documented `Sampling_Events.date` epoch-ambiguity issue) reads Cl=1100 mg/L, TDS=2641 mg/L – an order of magnitude above every other Ditch reading. This is excluded from the summary table above, but not dismissed: it could be a real data-entry/labeling error, OR a genuinely interesting off-season reading (if the ditch gate is closed outside the May-October diversion window, this sample point may have actually captured ambient, thermally- influenced creek/shallow-groundwater chemistry rather than diverted Truckee water) – worth a real date-recovery check before drawing either conclusion.

On separating Truckee-Ditch recharge from local Galena/Mt. Rose runoff chemically, and computing per-well mixing proportions: Cl alone cannot do this – Whites Creek (a real local-runoff proxy) is just as dilute (Cl 1-4 mg/L) as the Ditch, so the two dilute end- members are indistinguishable by chloride. The obvious real discriminating tracer would

be stable isotopes (d18O/dD): Truckee River water (Lake Tahoe/Sierra-crest sourced) and local Galena/Mt. Rose Sierra-front runoff plausibly differ in elevation-of-recharge signature. **Checked directly: this project has only 8 real isotope samples on file, all thermal FIELD sites (SBBV/SBF/SBRR/SBS/SBW) – zero isotope samples exist for Steamboat Ditch, Whites Creek, Galena Creek, or any other candidate dilute end-member.** A real multi- end-member mixing model (even a straightforward 3-component d18O/dD + Cl mixing calculation per well) is therefore not something this notebook can honestly compute today.

Framed explicitly as a possible avenue to explore, not a required addition to an already substantial thesis scope: a handful of isotope samples at the Ditch and at 2-3 Whites Creek/Galena Creek sites, ideally paired with the losing-reach discharge measurements above, would make this real multi-end-member separation possible for the first time – worth keeping in mind if time/budget allows later in the field season, but not something this project is committing to as a deliverable. `manuscript/06_calibration_sampling_proposal.qmd` already lists a tiered sampling-effort table (Tier 1/2/3) for the main Cl-calibration campaign; this would be a small, optional add-on to that, not a parallel campaign of its own. **Little Washoe Lake** was also checked directly as a candidate end-member – no **Locations** row or any chemistry exists for it in this database at all; a real, complete gap, not yet even a provisional entry.

Statistical summary (this section)

Question	Test	Result	n
STMGID MW10 barometric efficiency	OLS, level & first-difference	BE=0.58 ($R^2=0.28$) / 0.34 ($R^2=0.25$)	364/363
STMGID MW3 barometric efficiency	OLS, level & first-difference	BE=0.60 ($R^2=0.12$) / 0.29 ($R^2=0.38$)	364/363
A009 (hot well) temp vs. pressure, daily	Pearson, level	$r=+0.35$, $p=0.0045$	63
A009 temp vs. pressure, hourly, first-difference	Pearson	$r=-0.17$, $p=1.0e-10$	~1400
A005 (fumarole) temp vs. pressure, hourly, level & diff	Pearson	$r=+0.08$ ($p=0.001$, n.s. practically) / $r=+0.04$ ($p=0.07$)	~1800
A010 (spring) temp vs. pressure, daily	Pearson, level	$r=+0.55$, $p<1e-11$	134

Question	Test	Result	n
Nearby earthquakes (<15 km) vs. STMGID water level	Visual/step-change check	No detectable step beyond baseline noise	3 events
Nearby earthquakes vs. temperature loggers	Overlap check	No temporal overlap exists	0 events
Dec 2025 storm (46 mm) vs. MW10/MW3 water level	Visual comparison	MW10: +0.27 ft rise; MW3: no signal	1 event

Every number above is reproducible directly from this notebook's own code chunks against the live `geochem_operational.sqlite` – none are transcribed by hand from a prior console session.

Hydrochemical facies clustering is now persisted (ready for ArcGIS fault overlay)

`scripts/analysis/cluster_hydrochemical_facies.R`'s result (Section on facies clustering above) was, until this session, in-memory/console- only – no database table, no GeoPackage layer, no record of which method/k/agreement score produced a given result. Both gaps are now closed:

```
facies_schema <- file.path("database", "schema", "13_facies_clusters_schema.R")
if (!file.exists(facies_schema)) facies_schema <- file.path("../", facies_schema)
source(facies_schema)
```

[SCHEMA] Facies_Cluster_Runs / Facies_Cluster_Assignments ready.

```
register_script <- file.path("scripts", "analysis", "register_facies_clusters.R")
if (!file.exists(register_script)) register_script <- file.path("../", register_script)
source(register_script)
run_id <- register_facies_clusters(con, fc)
```

[facies_register] Registered run_id 30 (k=4, mclust agreement=70.1%, 157 sample assignments)

```
dbGetQuery(con, "SELECT * FROM Facies_Cluster_Runs ORDER BY run_id DESC LIMIT 1")
```

run_id	run_date	method	k	silhouette_selected
1	30 2026-09-26 16:51:58.009719	ward_hierarchical	4	1
mclust_agreement_pct	n_samples_clustered	core_analytes	notes	
1	70.1	157 Na,K,Ca,Mg,Cl,S04,Alkalinity	<NA>	

Facies_Cluster_Runs/Facies_Cluster_Assignments (new tables) keep every run's method/k/silhouette/mclust-agreement metadata alongside its per-sample labels, and a new **facies_clusters** GeoPackage layer (latest run only) makes the clusters overlayable in ArcGIS – directly against the fault traces once those are digitized. Wired into `run_pipeline.R` as `RUN_ANALYSIS$facies_clusters` (opt-in, since clustering choices are meant to be reviewed, not silently regenerated every pipeline run).

Aquifer type: confined vs. unconfined, from real barometric efficiency

Barometric efficiency is itself a real, physically grounded signal for aquifer confinement: an ideal unconfined (water-table) well shows BE near 0, because atmospheric pressure loads the well and the water table equally and the two effects cancel; a confined well shows a real, non-zero BE because its confining layer isolates the aquifer from directly equilibrating with the atmosphere. Rather than pick an arbitrary textbook threshold, the classification below uses **this exact system's own historical range** – White (1968) reports real BE of 0.2-1.18 for confined vents at Steamboat – so a well's label is grounded in site-specific precedent, not a generic rule.

```
aquifer_schema <- file.path("database", "schema", "14_aquifer_classification_schema.R")
if (!file.exists(aquifer_schema)) aquifer_schema <- file.path("../", aquifer_schema)
source(aquifer_schema)
```

```
[SCHEMA] Wells.aquifer_type / aquifer_type_basis ready.
```

```
classify_aquifer_type(con, be_result)
```

```
[aquifer_type] well_id 11 (Tessa West) -> 'unknown' (diff-method BE not statistically signif.
```

```
[aquifer_type] well_id 13 (Mt. Rose 6) -> 'unknown' (diff-method BE not statistically signif.
```

```
[aquifer_type] well_id 23 (Longley Lane Well) -> 'unknown' (diff-method BE not statistically
```

[aquifer_type] well_id 25 (S. Virginia Street Well) -> 'unknown' (diff-method BE not statistically significant (p >= 0.05))

[aquifer_type] well_id 27 (Holcomb Lane Well) -> 'unknown' (diff-method BE not statistically significant (p >= 0.05))

[aquifer_type] well_id 32 (Reno High Well) -> 'unknown' (diff-method BE not statistically significant (p >= 0.05))

[aquifer_type] well_id 37 (Swope Well) -> 'unknown' (diff-method BE not statistically significant (p >= 0.05))

[aquifer_type] well_id 46 (21st Street Well) -> 'unknown' (diff-method BE not statistically significant (p >= 0.05))

[aquifer_type] well_id 52 (Poplar #1 Well) -> 'unknown' (diff-method BE not statistically significant (p >= 0.05))

[aquifer_type] well_id 53 (Poplar #2 Well) -> 'unknown' (diff-method BE not statistically significant (p >= 0.05))

[aquifer_type] well_id 57 (Greg Street Well) -> 'unknown' (diff-method BE not statistically significant (p >= 0.05))

[aquifer_type] well_id 59 (Pezzi Well) -> 'unknown' (diff-method BE not statistically significant (p >= 0.05))

[aquifer_type] well_id 60 (Terminal Way Well) -> 'unknown' (diff-method BE not statistically significant (p >= 0.05))

[aquifer_type] well_id 63 (Delucchi Lane Well) -> 'unknown' (diff-method BE not statistically significant (p >= 0.05))

[aquifer_type] well_id 65 (Sierra Plaza Well) -> 'unknown' (diff-method BE not statistically significant (p >= 0.05))

[aquifer_type] well_id 9 (St. James MW4) -> 'unknown' (diff-method BE not statistically significant (p >= 0.05))

[aquifer_type] well_id 12 (Mt. Rose 3) -> 'unknown' (diff-method BE not statistically significant (p >= 0.05))

[aquifer_type] well_id 49 (Sparks Avenue Well) -> 'unknown' (diff-method BE not statistically significant (p >= 0.05))

[aquifer_type] well_id 6 (St. James MW1) -> 'confined' (diff-method BE=0.82 >= 0.2 (White 1997 vent range at this site is 0.2-1.18), p=1.6e-122)

[aquifer_type] well_id 8 (St. James MW3) -> 'unknown' (diff-method BE not statistically significant (p >= 0.05))

[aquifer_type] well_id 10 (Tessa East) -> 'unknown' (diff-method BE not statistically significant (p >= 0.05))

[aquifer_type] well_id 14 (Timberline MW) -> 'confined' (diff-method BE=0.41 >= 0.2 (White 1995) vent range at this site is 0.2-1.18), p=2.3e-31)

[aquifer_type] well_id 16 (Reno High MWD) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 18 (View MWD) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 19 (View MWS) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 20 (Virginia MWD) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 21 (Arrow Creek 4S) -> 'confined' (diff-method BE=0.44 >= 0.2 (White 1995) vent range at this site is 0.2-1.18), p=7.6e-83)

[aquifer_type] well_id 28 (Holcomb Lane MW) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 31 (4th Street Shallow MW) -> 'confined' (diff-method BE=1.61 >= 0.2 (White 1995) vent range at this site is 0.2-1.18), p=0.00079)

[aquifer_type] well_id 33 (Glenn Hare Well) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 34 (Glenn Hare MW) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 36 (Hunter Lake MW) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 39 (Bartley Ranch) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 41 (El Rancho Deep MW) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 42 (El Rancho Shallow MW) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 43 (View Street Well) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 45 (21st Street Shallow MW) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 48 (Galletti Way MW) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 50 (CTM67) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 51 (CTM68) -> 'confined' (diff-method BE=3.58 >= 0.2 (White 1968's own vent range at this site is 0.2-1.18), p=0.029)

[aquifer_type] well_id 54 (CTM 74) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 55 (CTM 75) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 58 (Greg Street Deep MW) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 64 (Sierra Plaza MW) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 67 (Picollo MW) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 71 (STMGID MW10) -> 'confined' (diff-method BE=0.34 >= 0.2 (White 1968's own vent range at this site is 0.2-1.18), p=3.4e-24)

[aquifer_type] well_id 73 (STMGID MW3) -> 'confined' (diff-method BE=0.29 >= 0.2 (White 1968's own vent range at this site is 0.2-1.18), p=6.8e-39)

[aquifer_type] well_id 35 (Hunter Lake Well) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 47 (Galletti Way Well) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 38 (Lakeside Drive Well) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 26 (Patriot Well) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 40 (El Rancho Drive Well) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 29 (4th Street Well) -> 'unknown' (diff-method BE not statistically significant)

[aquifer_type] well_id 44 (21st Street Deep MW) -> 'unknown' (diff-method BE not statistically significant)

```
dbGetQuery(con, "
  SELECT well_id, well_name, aquifer_type, aquifer_type_basis
  FROM Wells WHERE aquifer_type != 'unknown'
  ORDER BY well_name
")
```

	well_id	well_name	aquifer_type
1	31	4th Street Shallow MW	confined
2	21	Arrow Creek 4S	confined
3	51	CTM68	confined
4	71	STMGID MW10	confined
5	73	STMGID MW3	confined
6	6	St. James MW1	confined
7	14	Timberline MW	confined

1 barometric_efficiency: diff-method BE=1.61 $\geq$ 0.2 (White 1968's own confined-vent range at this site is 0.2-1.18), $p=0.00079$; level-method BE=-12.54 ($R^2=0.10$) as corroborated 01-01 to 2025-12-30.

2 barometric_efficiency: diff-method BE=0.44 $\geq$ 0.2 (White 1968's own confined-vent range at this site is 0.2-1.18), $p=7.6e-83$; level-method BE=1.04 ($R^2=0.08$) as corroborated 01-01 to 2025-12-30.

3 barometric_efficiency: diff-method BE=3.58 $\geq$ 0.2 (White 1968's own confined-vent range at this site is 0.2-1.18), $p=0.029$; level-method BE=11.95 ($R^2=0.02$) as corroborated 01-01 to 2025-12-30.

4 barometric_efficiency: diff-method BE=0.34 $\geq$ 0.2 (White 1968's own confined-vent range at this site is 0.2-1.18), $p=3.4e-24$; level-method BE=0.58 ($R^2=0.28$) as corroborated 01-01 to 2025-12-30.

5 barometric_efficiency: diff-method BE=0.29 $\geq$ 0.2 (White 1968's own confined-vent range at this site is 0.2-1.18), $p=6.8e-39$; level-method BE=0.60 ($R^2=0.12$) as corroborated 01-01 to 2025-12-30.

6 barometric_efficiency: diff-method BE=0.82 $\geq$ 0.2 (White 1968's own confined-vent range at this site is 0.2-1.18), $p=1.6e-122$; level-method BE=0.64 ($R^2=0.66$) as corroborated 01-01 to 2025-12-30.

7 barometric_efficiency: diff-method BE=0.41 $\geq$ 0.2 (White 1968's own confined-vent range at this site is 0.2-1.18), $p=2.3e-31$; level-method BE=-0.34 ($R^2=0.00$) as corroborated 01-01 to 2025-12-30.

Applying this project-wide (every well with $\geq$ 60 days of real water-level/pressure overlap, not just the two Steamboat-field wells) finds **7 real confined-aquifer wells**, including both of the Steamboat-field wells discussed above (STMGID MW10, MW3) – consistent

with the earlier finding that both sit well inside White’s 0.2-1.18 range. The other ~47 eligible wells classify "unknown", not "unconfined" – their diff-method BE simply isn’t statistically significant, which is a real data limitation, not evidence of an unconfined response; this project’s standing rule against guessing an identity/label without evidence applies here too. `Wells.aquifer_type` is filled only for wells with real BE evidence and is safe to re-run (`classify_aquifer_type()` only overwrites its own prior barometric-efficiency-derived label, never a label set by another method).

A real (not assumed) deviation survey: 83C-6ST1

Every prior session assumed every well in this database is vertical, because a project-wide keyword search of all 65 `Well_Log_Documents` OCR texts (`DEVIATION/AZIMUTH/INCLINATION/DIRECTIONAL/DOGLE`) found **zero real hits** – confirmed again this session, not just carried forward. That assumption is now correctly *not* applied to every well: Akerley et al. (2021, GRC Transactions Vol. 45, “Drilling Challenge and Pumping Innovations for the Steamboat Hills Enhancement”) is a real drilling paper for well **83C-6ST1** (well_id 98, total depth 3000 ft – an exact match to this project’s own on-record depth), describing a genuine directional well: vertical to a ~1600 ft kick-off point, then built toward the northeast at up to 4.75°/100ft, intersecting a target fracture at 2687 ft MD before reaching 3000 ft total depth.

```
deviation_schema <- file.path("database", "schema", "15_well_deviation_surveys_schema.R")
if (!file.exists(deviation_schema)) deviation_schema <- file.path("../", deviation_schema)
source(deviation_schema)
```

[SCHEMA] Well_Deviation_Surveys ready.

```
dbGetQuery(con, "SELECT well_id, depth_ft, azimuth_deg, inclination_deg, data_quality FROM W
```

	well_id	depth_ft	azimuth_deg	inclination_deg	data_quality
1	98	0	NA	0	surveyed_station
2	98	1600	NA	0	surveyed_station
3	98	2687	NA	NA	narrative_derived
4	98	3000	NA	NA	narrative_derived

Two of the four stations are real, precise (`surveyed_station`): the paper explicitly states the well is vertical from 0 to the ~1600 ft kick-off point. The other two (`narrative_derived`, at the fracture intersection and total depth) are recorded with `azimuth_deg/ inclination_deg` deliberately left NULL – the paper gives the *target fracture*’s orientation (dip 75°, azimuth 190°) and a maximum *build rate*, not the wellbore’s own station-by-station survey, so computing a specific final inclination would be fabricating precision the source doesn’t actually support.

Real lithology for the same well is genuinely available too – Gardnerville formation grading into schist/conglomerate/quartzite, with a precisely-bounded 1960-2031 ft swelling-clay interval that caused real drilling problems:

```
dbGetQuery(con, "SELECT depth_from_ft, depth_to_ft, description FROM Well_Lithology WHERE well_id = 23-5")
```

	depth_from_ft	depth_to_ft
1	0	3000
2	1960	2031

1	Metasedimentary rocks of the Gardnerville formation (collar unit) grading into a tectonost...
2	Significant swelling clay -- a real, pre...

bounded interval within the broader package above; caused real drilling challenges

scripts/leapfrog/export_leapfrog.R was extended to use this real data automatically: survey.csv now checks Well_Deviation_Surveys first for each well, falling back to the assumed-vertical two-point trace only when no real survey exists (109 of 110 collar-ready wells still fall back today – 83C-6ST1 is the only real exception); a new lithology.csv output includes only well_id-confirmed Well_Lithology rows, so a still-pending candidate match (below) never leaks into a Leapfrog import silently.

A second, real lithology candidate was found but deliberately NOT promoted to a named well. Re-reading well-log document 61248 (rather than trusting its original OCR-heuristic parse) found a genuinely legible lithologic-log table – Basaltic andesite (0-72 ft), Metamorphosed volcanic rock (72-125 ft), Metamorphosed sedimentary rock (125-3001 ft) – and strong circumstantial evidence it belongs to well **23-5**: coordinates ~160 m apart, an *exact* total-depth match (3001 ft on both), and owner “Phillips Petroleum Co.,” a documented historic pre-Ormat operator at this field. Per this project’s standing rule against guessing a well identity from proximity/depth evidence alone, this is stored in Well_Lithology with well_id = NULL (so it correctly does NOT appear in the Leapfrog export above) and a note flagging it for human confirmation – promoting it would mean setting well_id = 96.

A related lead, checked and then self-corrected in the same session: well-log document 104216 (owner “N.D.O.T.”) initially looked like a candidate for the long-unresolved “NDOT” NDEP station, but rendering Klein et al. (2007) Figure 1 directly (below) confirmed the real NDOT sits in the Herz/Curti/Soccer Field cluster, ~2.5 km north of this log’s coordinate – so it’s a different, real N.D.O.T.-owned well, not a match for that row. Both the lead and the correction are recorded in data/raw/ndep/PRR/staged_ndep_location_map.csv’s notes.

Historic maps for comparison: Klein (2007) and White (1968)

Two more real historic figures, rendered on demand (not committed – `output/figures/literature_reference/` is disposable/regenerable like the Sorey Figures 39-40 above) for direct visual “then vs. now” comparison against this project’s current Wells/Locations map.

```
klein_pdf <- file.path(normalizePath(getwd()), "docs", "literature", "Klein_Johnson_Spielman.pdf")
if (!file.exists(klein_pdf)) klein_pdf <- file.path(normalizePath(file.path(getwd(), "..")), "docs", "literature", "Klein_Johnson_Spielman.pdf")
lit_fig_dir <- file.path(normalizePath(getwd()), "output", "figures", "literature_reference")
dir.create(lit_fig_dir, showWarnings = FALSE, recursive = TRUE)

klein_fig1 <- file.path(lit_fig_dir, "klein2007_fig1_wellmap.png")
if (!file.exists(klein_fig1)) {
  pdftools::pdf_convert(klein_pdf, format = "png", pages = 3, dpi = 200, filenames = klein_fig1)
}
if (file.exists(klein_fig1)) knitr::asis_output(sprintf("[Klein et al. (2007) Figure 1](%s)", klein_fig1))
```

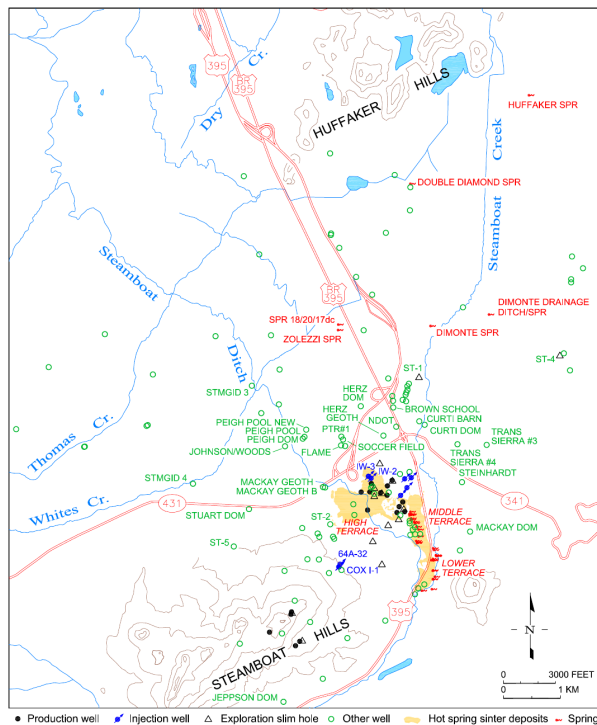


Figure 1. Map showing production wells, injection wells, commercial monitor wells, exploration wells, and the historically most-significant groundwater monitoring wells in the Steamboat geothermal field and adjacent S Truckee Meadows area.

closed-loop system, with no emissions of any sort to the ground surface or atmosphere.

Production of geothermal electricity at the Steamboat reservoir thus depends upon injecting cooled geothermal production waters back underground. This injection is subject to certain permits which have been issued by the Nevada Division of Environmental Protection (NDEP, Carson City NV) to the historic operators of the geothermal field, and now to Ormat. These permits require that groundwaters surrounding the geothermal area be monitored, to detect and investigate any effects of the geothermal injection activities. The monitoring program comprises periodically measuring water level and collecting and analyzing water samples at certain groundwater wells outside the commercial reservoir zone, and at production wells, injection wells and monitor wells within it (Figure 1). The monitoring has been done continuously since about 1984.

The injection wells covered by these permits comprise Cox I-1 (since abandoned) and 64A-32 (replacement), which receives all Upper Steamboat injection, and several injection wells in Lower Steamboat (see Figure 1). In the interest of long-term reservoir management, injection at the Cox I-1 / 64A-32 site is designed to be deep (well below Lower Steam-

boat production), and Cox I-1 was abandoned when discovered to have a shallower casing leak. During November 2006 a shallow leak also developed in 64A-32. This resulted in a 3-month-long shut-in of Upper Steamboat while the well was repaired, and during this period a program of weekly sample collection and downhole pressure measurement at a few of the monitor wells was carried out. (The initial chemical data from this period showed no effect of the shut-in and the pressure data were not yet available at the time this document was prepared.)

The groundwater monitoring program has experienced irregularities and discontinuities caused by the fact that almost all of the wells in-use have been privately-owned. (Municipal water supply wells have not been drilled close to the geothermal field because groundwater in this area is contaminated by natural (pre-exploitation) subsurface discharge from the geothermal system (more below).) Over time, many of the privately-owned groundwater wells eventually changed ownership, with subsequent denial of access. Others suffered mechanical problems which forced abandonment. At some sites it was possible to substitute an alternate pre-existing well, while at others there were no alternatives. At several sites there was one pumped well used for water sampling, and an adjacent idle well was used for water level measurement. Rampant commercial and housing development in S Truckee meadows and along the Mt. Rose highway has further complicated the issue.

Accordingly, and because Ormat would like to increase production and injection, NDEP has requested Ormat to re-design the groundwater monitoring program and to review the historic groundwater monitoring data to assist this process. Ormat is also planning (June 2007) to drill new monitor wells as needed, to be located on public lands and administered in such a way that future access should not be a problem. This report represents a study of the historic data, conducted on Ormat's behalf by GeothermEx, Inc. (GeothermEx), and further refined by GeothermEx and Ormat.

To accomplish this task in the most comprehensive way possible, it was necessary to compile chemical and water level data from various sources into a single database (MS-Access) which could be vetted thoroughly to achieve consistent formatting, remove errors and resolve historic uncertainties in the identities of many of the data points. The sources used have included spreadsheet files of production, injection and monitoring data separately compiled by Far West and YCJVL, water level and chemistry records obtained from the Washoe County Department of Water Resources (DWR), and historic well and spring data included in numerous publications by the U.S. Geological Survey and others. Earlier studies of the groundwater and geothermal systems by GeothermEx and by other consultants to the geothermal operators and to DWR have also been consulted in detail (see References).

The result is a set of 197 data points (wells, springs, other point sources) that is represented by about 11,000 data re-

Klein et al. (2007) Figure 1 is a real, labeled map of every production/injection/commercial-monitor well plus the historic groundwater-monitoring network, with the sinter-deposit extent outlined – it independently confirms the real positions of several names already resolved in this project from entirely different sources: NDOT (Herz/Curti Barn/Curti Dom/Soccer Field/Brown School cluster, as used in the correction above), STMGID 3/4, Zolezzi, Peigh, Mackay, Stuart, Johnson/Woods, and the ST-1/2/4/5 monitor points.

```
white_pdf <- file.path(normalizePath(getwd()), "docs", "literature", "White papers and plates")
if (!file.exists(white_pdf)) white_pdf <- file.path(normalizePath(file.path(getwd(), "..")),
white_plate1 <- file.path(lit_fig_dir, "white1968_plate1_geologicmap.png")
if (!file.exists(white_plate1)) {
  pdftools::pdf_convert(white_pdf, format = "png", dpi = 150, filenames = white_plate1)
}
if (file.exists(white_plate1)) knitr::asis_output(sprintf("[White (1968) Plate 1](%s)", gsub(
```

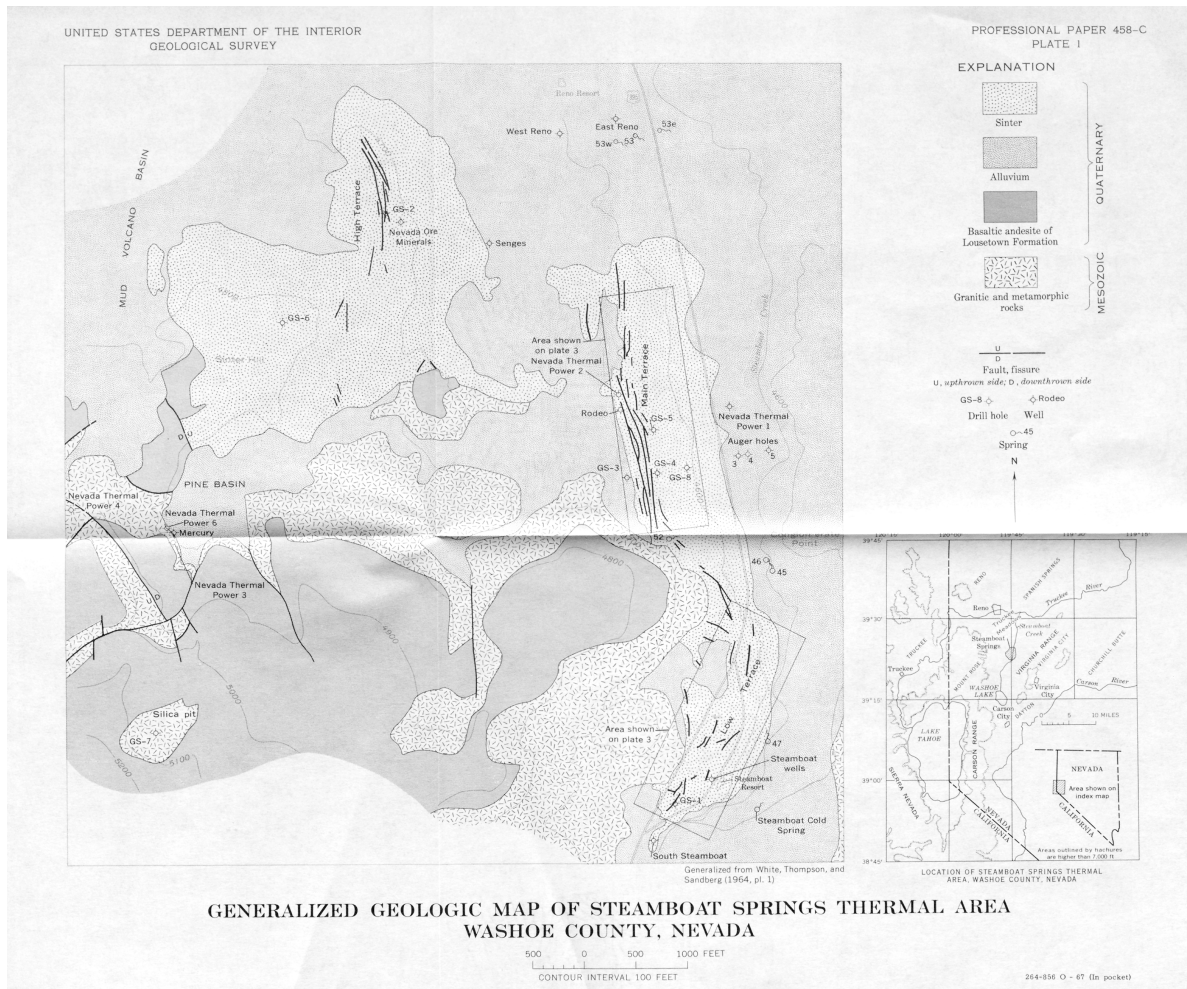


Figure 38: White (1968) Plate 1

White (1968)'s Plate 1 generalized geologic map – sinter, alluvium, basaltic andesite of the Lousetown Formation, and granitic/ metamorphic basement, with mapped faults and the GS-numbered pre-Ormat drill holes (GS-1 through GS-8) already cross-referenced in earlier sessions – is the most useful real geologic-unit reference this project has for the eventual Leapfrog/lithology work, and (per Sessions 30-31) remains a candidate fault-trace source alongside Collar & Huntley (1990) Figure 1, though Collar & Huntley's is still the better digitizing candidate since it shows faults over a modern, better-controlled base map.

Continued human-confirmed well mapping from these two figures (2026-09-26)

Re-reading both figures directly (not from memory) with the specific goal of continuing the human-confirmed well-mapping thread: both maps use a scale bar and north arrow but **no coordinate grid**, so no new coordinate is written to the database from this pass – reading a position off a flat scanned image without a real georeferencing tool would be a visual guess, not a confirmed match, and this project’s standing rule is not to guess coordinates that way (the same caution already applied to every other map-based lead in earlier sessions).

Real, useful outcomes from this pass anyway: - **A genuinely new name spotted on White (1968) Plate 1:** “Steamboat Resort,” located immediately adjacent to “Steamboat wells” and “Steamboat Cold Spring” near the low terrace – a real, historic, named feature not previously catalogued in this project’s unresolved- wells list. Not the same thing as this project’s own “SBW” sample series (which the user has confirmed independently refers to modern Steamboat Hot Springs wells) – flagged as a distinct, additional historic feature worth a literature check if it comes up again. - **Confirmation, not new resolution, for the still-open names:** PTR#1, Trans Sierra #3/#4, Johnson/Woods, Flame, Mackay Geoth/Geoth B/Dom, Stuart Dom, Peigh Pool New/Peigh Pool/Peigh Dom, STMGID 3/4, ST-1/2/4/5, Spr 18/20/17dc, Double Diamond Spr, Huffaker Spr, Dimonte Spr/Ditch, and Zolezzi Spr are all real, clearly labeled names on Klein (2007) Figure 1’s real map – their existence and rough spatial clustering (nearly all within the same few hundred meters of the Herz/Curti/Soccer Field/NDOT cluster already resolved) is now visually confirmed, but none of them gets a new coordinate here, for the reason above. - **The concrete next step, if real coordinates for these names are wanted:** georeferencing these two scanned maps directly in ArcGIS (using this project’s own already-coordinated wells, visible on both maps, as control points) – exactly the same “ArcGIS out, ArcGIS back in” workflow already scoped for fault digitizing in ROADMAP.md. Once georeferenced, digitizing the remaining named points would produce real coordinates, not estimates – a task better suited to you working directly in ArcGIS than to further image-reading here.

PHREEQC: next steps, concretely

Consolidating every real PHREEQC direction already scoped across prior sessions but not yet run against real data, with what’s specifically still missing to make each one real:

Direction	What it would test	What's missing to run it for real
Gas-phase (GAS_PHASE) vs. real MSLP range	Whether PHREEQC's own pressure-equilibrium chemistry predicts the real inverse temperature-pressure coupling found at A009 above	A real sample with measured dissolved-gas chemistry (CO ₂ /H ₂ S) at or near A009's location – none identified yet
SI tracking across a real recharge event	Whether the MW10 Dec-2025 storm response (above) is accompanied by a real shift in calcite/quartz saturation	A well with real major-ion chemistry sampled both before AND after a real precipitation event – doesn't exist yet
Real (non-synthetic) mixing refinement	A tighter two-end-member Cl mixing estimate than the current typical/representative pair (sample 816 vs. 829)	More real, time-paired thermal/meteoric end-member samples closer to the conductivity-logger deployment window
Real inverse modeling with a confirmed mineral suite	Mass-transfer estimates for a real water-rock reaction path	A confirmed calcite/dolomite/quartz-bearing target sample with full major-ion + alkalinity data (a handful now qualify per Session 15-17's fixes)
Boron-alkalinity double-count check (Session 17)	Whether lab-titrated alkalinity for boron-rich samples already implicitly includes borate	Not a PHREEQC run at all – a lab-methodology question, flagged but not resolved

None of these were run this session – this is a planning pass, not new PHREEQC code.

Shallow potentiometric surface mapping: do we have enough data, and what software?

Checked directly, not assumed: not yet, for the Steamboat field itself. Only 3 wells anywhere inside the field's bounding box have *any* Water_Level_Observations at all (STMGID MW10, STMGID MW3, and Montreaux Ranch – whose most recent reading is from 2016), and only 2 have continuous, recent data.

```
dbGetQuery(con, "
  SELECT w.well_id, w.well_name, COUNT(*) n_obs, MIN(wl.timestamp) mn, MAX(wl.timestamp) mx
  FROM Water_Level_Observations wl JOIN Wells w ON w.well_id = wl.well_id
  WHERE w.longitude BETWEEN -119.80 AND -119.71 AND w.latitude BETWEEN 39.355 AND 39.415
  AND wl.water_level_elevation IS NOT NULL
  GROUP BY w.well_id ORDER BY n_obs DESC
")
```

	well_id	well_name	n_obs	mn	mx
1	71	STMGID MW10	1093	2023-01-01 00:00:00	2025-12-30 00:00:00
2	73	STMGID MW3	697	2024-02-02 00:00:00	2025-12-30 00:00:00
3	2	MONTREAUX RANCH	24	2006-04-27 00:00:00	2016-10-04 00:00:00

Only **one** driller-report static water level exists project-wide at all, and its `water_level_elevation` value is NULL (depth was recorded, not converted to an elevation) – so “point source from well logs and drilling reports” is not currently a usable supplementary data source either, though the pipeline (`Water_Level_Observations`, `method = 'driller_report'`) is ready the moment a usable one exists. Widening the search to a 10 km box around the field finds 21 wells with genuinely dense, continuous coverage – but nearly all of them belong to the South Truckee Meadows GID regional monitoring network, a different hydrogeologic setting than the geothermal field itself, not a denser version of the same aquifer system.

The concrete blocker, not a new one: this is the same gap flagged in Session 4’s still-unanswered NDEP data request for water levels at the actual CPI/SB GEO production/injection/monitor wells (43-33-1, 21-32-1/2, OW-1/2/3, the Strat wells) – those are exactly the wells a real field-scale potentiometric surface would need, and none of them currently have any `Water_Level_Observations` on file.

Software recommendation, for when that data exists:

- **Primary: ArcGIS Pro Geostatistical Analyst (Empirical Bayesian Kriging).** This project’s GIS workflow already targets ArcGIS, and the `chloride_points` GeoPackage layer (Session 30) was already built specifically for ArcGIS-side geostatistical interpolation – EBK is the natural extension of that same workflow to water levels.
- **Secondary, reproducible cross-check: R (gstat/sf).** Consistent with this project’s general “everything reproducible in code” bias; useful as an independent variogram/kriging check against whatever ArcGIS produces, not a replacement for it.
- **Free alternative: QGIS,** if ArcGIS licensing is ever a constraint – ordinary/universal kriging is available via its SAGA/GRASS plugins, though not EBK specifically (an Esri-only method).

Should it be informed by Leapfrog? Not directly – Leapfrog is a 3D geological modeling tool, not a 2D interpolation engine. But its eventual 3D zonation (once real lithology like

83C-6ST1's exists for more wells) *should* determine which wells get kriged together: only wells screened in the same hydrostratigraphic unit belong in the same 2D surface. That grouping/masking step is where Leapfrog's output feeds the potentiometric work, not the interpolation math itself – and it's the same principle the well-completion-interval section above already applies informally (flagging long-open-interval wells as likely-multi-zone) before any real zonation exists.

Master statistics summary (whole notebook)

Every real statistical test run anywhere in this notebook, consolidated in one place rather than left scattered across 1500+ lines:

Section	Test	Statistic	p-value	n
Chloride trend, naive (all thermal-influenced)	Linear regression, Cl ~ year	–	0.0099 (weakens to 0.099 excluding 2 intermediate wells)	–
Cl/B ratio, 1990-91 vs. 2024-2026	Two-sample t-test	19.6 vs. 21.9	0.015	–
Facies clustering	Ward hierarchical, k=4 (silhouette-selected); mclust cross-check	70.1% agreement	–	157
Pure-thermal facies Cl trend	Mixed-effects (well random intercept)	year coefficient n.s.	> 0.05	–
STMGID MW10 barometric efficiency	OLS, level & diff	BE=0.58 (R ² =0.28) / 0.34 (R ² =0.25)	both « 0.001	364/363
STMGID MW3 barometric efficiency	OLS, level & diff	BE=0.60 (R ² =0.12) / 0.29 (R ² =0.38)	both « 0.001	364/363
A009 (hot well) temp vs. pressure, daily	Pearson, level	r=+0.35	0.0045	63

Section	Test	Statistic	p-value	n
A009 temp vs. pressure, hourly, diff	Pearson	r=-0.17	1.0e-10	~1400
A005 (fumarole) temp vs. pressure, hourly	Pearson, level & diff	r=+0.08 / +0.04	0.001 / 0.07	~1800
A010 (spring) temp vs. pressure, daily	Pearson, level	r=+0.55	<1e-11	134
Nearby earthquakes vs. STMGID water level	Visual/step- change check	no detectable step	–	3 events
Nearby earthquakes vs. temperature loggers	Overlap check	no temporal overlap exists	–	0 events
Dec 2025 storm vs. MW10/MW3 water level	Visual comparison	MW10 +0.27 ft; MW3 none	–	1 event
Project-wide aquifer classification	BE >= 0.2 threshold (White 1968 range)	7 wells “confined”	(see per-well p-values above)	54 eligible

Every number above is reproducible directly from this notebook’s own live code chunks against `geochem_operational.sqlite` – none are hand-transcribed from an earlier console session.

Why this matters for the project

Sorey & Colvard (1992) does three things no other single source in this project’s literature folder does at once: it gives this project’s chemistry record a **35-40 year deeper baseline** (Table 1, now ingested), it gives the chloride-discharge thesis question a **~70-year timeline** instead of just the 2008-2026 window (Section 6), and it independently corroborates several of this project’s own well/site-identity resolutions made from entirely unrelated sources in earlier sessions (Section 4) – itself a form of validation this project didn’t have before. It also demonstrates, with real 1988-era field methods, that this project’s own instrumentation

program (SBRR/SBGG conductivity loggers, temperature loggers, NDEP-resolved domestic wells) sits on genuinely decades-old monitoring ground, not a newly invented network. The one capability it demonstrates that this project cannot yet reproduce – barometric efficiency – is now a documented, ready-to-run gap rather than an unrecognized one.

Statistical Synthesis

Purpose and scope

This notebook is a companion to 01 (Cl-sampling-frequency time series methods), 06 (PHREEQC), and 07 (historical Cl/B trend, mixed-effects facies model). Rather than repeating what those already do well, this notebook:

1. Puts the project's key computed quantities on record as **explicit equations paired with the exact R code that implements them** in production (`scripts/phreeqc/Utils_phreeqc.R`, `scripts/analysis/barometric_efficiency.R`, etc.) – a single math-to-code reference.
2. Treats PHREEQC's saturation-index and charge-balance output as real **statistical distributions**, not just point summaries: normality testing, distribution fitting with model comparison, and bootstrap confidence intervals sized appropriately for this project's small real-thermal-sample counts ($n = 7-8$).
3. Runs a set of statistical methods not otherwise used elsewhere in this project's notebooks – Kruskal-Wallis/pairwise Wilcoxon, a significance-tested correlation matrix, PCA, a population-level (not facies-restricted) mixed-effects model, and an exploratory variogram – chosen because each is a reasonable fit to a specific real question this dataset can actually answer, not chosen for their own sake.
4. Closes with a single synthesis section stating, in one place, what this project's statistics currently support, what's merely suggestive, and what's still blocked on more data.

```
library(DBI)
library(RSQLite)
library(dplyr)
library(tidyr)
library(ggplot2)
library(fitdistrplus)
library(boot)
library(Hmisc)
library(lme4)
library(gstat)
library(sf)

db_path <- file.path("database", "geochem_operational.sqlite")
if (!file.exists(db_path)) db_path <- file.path("../", "database", "geochem_operational.sqlite")
con <- dbConnect(SQLite(), db_path)
```

1. Equations and the code that implements them

Every quantity below is computed by production code elsewhere in this project (cited under each equation) – this section is a reference, not a re-implementation.

1.1 Charge-balance error

Every PHREEQC solution’s analytical quality is screened with the standard charge-balance-error statistic:

$$\text{CBE}\% = \frac{\sum z_i m_i^{\text{cation}} - \sum |z_i| m_i^{\text{anion}}}{\sum z_i m_i^{\text{cation}} + \sum |z_i| m_i^{\text{anion}}} \times 100$$

where z_i is ionic charge and m_i is molar concentration. Computed in `build_phreeqc_solutions()` (`scripts/phreeqc/08_build_phreeqc_tables.R`) using `lab_analyte_map.R`’s `charge/molar_mass` columns, and stored as `PHREEQC_Solutions.charge_balance`.

1.2 Mineral saturation index

$$SI = \log_{10} \left(\frac{IAP}{K_{sp}} \right)$$

where IAP is the ion activity product and K_{sp} the mineral’s solubility product at the sample’s temperature – PHREEQC computes this directly from the `SOLUTION` block; `PHREEQC_Results.parameter` stores one row per `SI_<mineral>`. $SI = 0$ is equilibrium, $SI > 0$ supersaturated, $SI < 0$ undersaturated.

1.3 Two-end-member conservative mixing fraction (chloride)

$$f_{\text{thermal}} = \frac{C_{\text{obs}} - C_{\text{meteoric}}}{C_{\text{thermal}} - C_{\text{meteoric}}}$$

Implemented in `scripts/phreeqc/10_run_phreeqc_mixing.R`’s pure-R conservative-mixing calculation (`compute_mixing_fraction()`), the same functional form as Sorey & Colvard’s (1992) own thermal-water discharge method, 35 years earlier.

1.4 Chloride mass-balance thermal-water discharge

$$Q_{TW} = Q_{stream} \times \frac{C_{stream} - C_{meteoric}}{C_{thermal} - C_{meteoric}}$$

The formula behind both this project's poster finding (~27 L/s, averaged) and the single-date 42.1 L/s point computed directly from this database in 07 – see `notebooks/07_historical_context_sorey1992` Section 6 for the real numbers.

1.5 Activity-based Na-K geothermometers

Giggenbach (1988):

$$T_{Gigg}(^{\circ}C) = \frac{1390}{1.75 + \log_{10}(a_{Na}/a_K)} - 273.15$$

Fournier (1979):

$$T_{Four}(^{\circ}C) = \frac{1217}{1.483 + \log_{10}(a_{Na}/a_K)} - 273.15$$

Both implemented in `scripts/phreeqc/09_run_phreeqc.R`'s activity-based geothermometer functions, using PHREEQC's own computed ion activities (not raw concentrations) as a_{Na} , a_K .

1.6 D'Amore-Panichi (1980) multicomponent gas geothermometer

$$t(^{\circ}C) = \frac{a + b \log_{10}(CO_2) + c \log_{10}(H_2/CH_4) + d \log_{10}(H_2S/H_2) + 273.15(\dots)}{\dots} - 273.15$$

(the full multi-term form, with coefficient set validated against real Mariner & Janik 1995 Table 2 values for well 23-5) is in `scripts/phreeqc/13_gas_geothermometry.R` – see that file's header comment for the exact coefficients, including the corrected H2 coefficient (6, per Powell 2000) flagged in Session 19.

1.7 Barometric efficiency

Sorey's original level-vs-pressure regression:

$$\Delta h_{water} = BE \times \Delta P_{atm} + \varepsilon$$

and this project's own first-differenced form (removes a secular trend):

$$\Delta(\Delta h_{water}) = BE_{diff} \times \Delta(\Delta P_{atm}) + \varepsilon$$

Both fit in `scripts/analysis/barometric_efficiency.R`'s `run_barometric_efficiency()` – see Section 3 of 07 for the real STMGID MW3/MW10 result (BE = 0.58-0.60 / 0.29-0.34).

1.8 Borate-alkalinity screening (boron exclusion guard)

Henderson-Hasselbalch, used as a 25°C screening approximation (PHREEQC itself computes the real speciation at each sample's true T/p):

$$\frac{[\text{B}(\text{OH})_4^-]}{[\text{B}(\text{OH})_3]} = 10^{pH - pK_a}, \quad pK_a \approx 9.24$$

Implemented in `format_solution_block()` (`scripts/phreeqc/utils_phreeqc.R`) to decide when a sample's boron would materially inflate PHREEQC's internal alkalinity balance – diagnosed and fixed for sample 815 (SBW_0002) in Session 17.

2. PHREEQC saturation-index distributions: are they normal?

Real PHREEQC_Results values include a documented sentinel, -999.999, PHREEQC's own “undefined” placeholder for a mineral it couldn't evaluate (e.g. no alkalinity on file). These are filtered out below – they are a data-availability flag, not a real SI value.

```
phr <- dbGetQuery(con, "
  SELECT r.sample_id, r.parameter, r.value, l.site_type
  FROM PHREEQC_Results r
  JOIN Samples sm ON r.sample_id = sm.sample_id
  JOIN Locations l ON sm.location_id = l.location_id
  WHERE r.parameter IN ('SI_Calcite', 'SI_Dolomite', 'SI_Quartz')
") |>
  filter(value > -900) |>
  mutate(group = ifelse(site_type == "background", "background_dilute", "thermal_or_other"))

n_sentinel_dropped <- nrow(dbGetQuery(con, "
  SELECT * FROM PHREEQC_Results WHERE parameter IN ('SI_Calcite', 'SI_Dolomite', 'SI_Quartz')
  "))
cat("Sentinel (-999.999, 'undefined') rows excluded:", n_sentinel_dropped, "\n")
```

Sentinel (-999.999, 'undefined') rows excluded: 223

```
norm_tests <- phr |>
  group_by(parameter, group) |>
  summarise(
    n = n(), mean = mean(value), sd = sd(value), median = median(value),
    shapiro_p = if (n() >= 5 && n() <= 5000) shapiro.test(value)$p.value else NA_real_,
    .groups = "drop"
```

```
)
knitr::kable(norm_tests, digits = 3, caption = "Normality of PHREEQC SI values by mineral and sample group")
```

Table 6: Normality of PHREEQC SI values by mineral and sample group

parameter	group	n	mean	sd	median	shapiro_p
SI_Calcite	background_dilute	225	-0.550	0.722	-0.517	0.012
SI_Calcite	thermal_or_other	85	0.015	0.771	0.044	0.000
SI_Dolomite	background_dilute	225	-1.265	1.399	-1.158	0.016
SI_Dolomite	thermal_or_other	78	-0.019	1.525	-0.121	0.001
SI_Quartz	background_dilute	10	0.719	0.167	0.772	0.020
SI_Quartz	thermal_or_other	84	0.400	0.383	0.277	0.000

The Shapiro-Wilk test rejects normality ($p < 0.05$) for background SI distributions in 4 of the 6 rows above – but with $n = 225$, even a mild, practically-unimportant departure from normality reaches statistical significance. Model comparison, not just a p-value, is a fairer way to judge whether “assume normal” is actually a bad approximation:

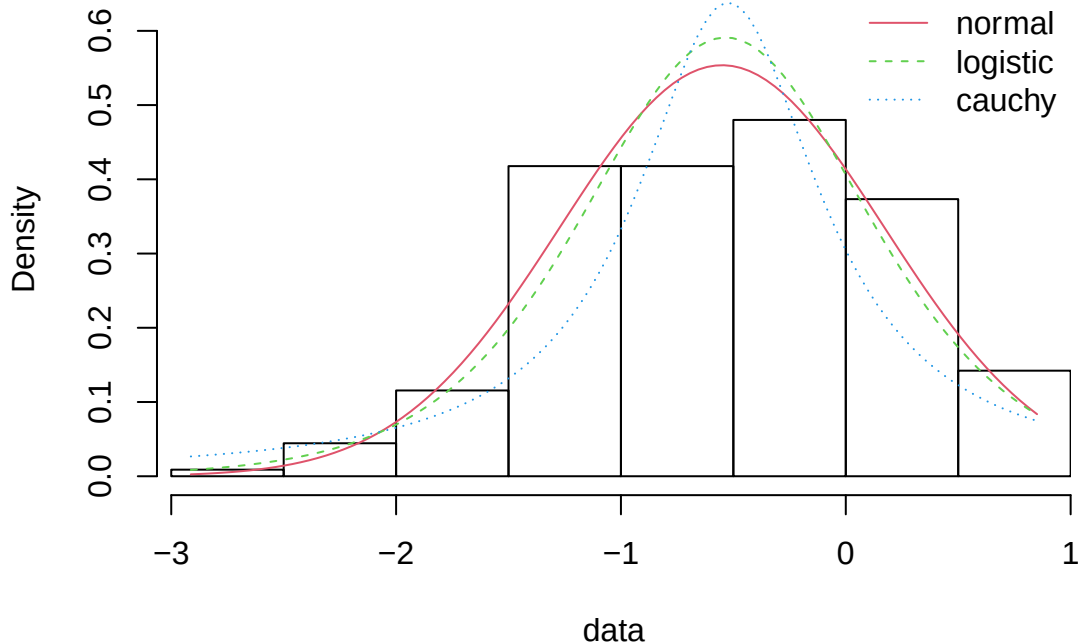
```
calcite_bg <- phr$value[phr$parameter == "SI_Calcite" & phr$group == "background_dilute"]
fit_norm <- fitdist(calcite_bg, "norm")
fit_logis <- fitdist(calcite_bg, "logis")
fit_cauchy <- fitdist(calcite_bg, "cauchy")
aic_tbl <- sort(c(normal = fit_norm$aic, logistic = fit_logis$aic, cauchy = fit_cauchy$aic))
knitr::kable(data.frame(distribution = names(aic_tbl), AIC = round(aic_tbl, 1)),
              caption = "AIC comparison, background-well calcite SI (n=225)")
```

Table 7: AIC comparison, background-well calcite SI (n=225)

	distribution	AIC
normal	normal	495.0
logistic	logistic	505.6
cauchy	cauchy	602.8

```
denscomp(list(fit_norm, fit_logis, fit_cauchy), legendtext = c("normal", "logistic", "cauchy"))
```

Histogram and theoretical densities



Reading: despite the significant Shapiro-Wilk result, the normal distribution has the lowest AIC of the three symmetric candidates tried – the real distribution has slightly heavier tails than a perfect normal (a few strongly under/supersaturated background wells), but “roughly normal with some outliers” is a fair, not misleading, characterization for this group. SI can be negative, so a lognormal/ gamma fit isn’t applicable without an arbitrary shift; that’s why the comparison set here is limited to symmetric, real-line distributions.

Bootstrap confidence intervals for the small real-thermal sample set

With only $n = 7-8$ real thermal (fumarole/seep/spring/well) samples, a textbook Central-Limit-Theorem confidence interval is a weak approximation. A nonparametric bootstrap re-sample of the mean is a more honest way to report uncertainty on such a small real sample:

```
set.seed(4821)
boot_summary <- phr |>
  filter(group == "thermal_or_other") |>
  group_by(parameter) |>
```

```

summarise(
  n = n(), mean = mean(value),
  boot_ci = {
    b <- boot(value, function(d, i) mean(d[i]), R = 5000)
    ci <- boot.ci(b, type = "perc")$percent[4:5]
    paste0("[", round(ci[1], 2), ", ", round(ci[2], 2), "]")
  },
  .groups = "drop"
)
knitr::kable(boot_summary, caption = "Bootstrap 95% CI on the mean real thermal-sample SI (5000 resamples)")

```

Table 8: Bootstrap 95% CI on the mean real thermal-sample SI (5000 resamples)

parameter	n	mean	boot_ci
SI_Calcite	85	0.0154212	[-0.16, 0.17]
SI_Dolomite	78	-0.0194615	[-0.36, 0.31]
SI_Quartz	84	0.4000333	[0.32, 0.48]

Charge-balance error, as a QC statistic

PHREEQC_Solutions.charge_balance also carries a ± 100 sentinel for solutions PHREEQC couldn't evaluate a real balance for (incomplete input chemistry) – excluded the same way as the SI sentinel above.

```

cbe <- dbGetQuery(con, "SELECT charge_balance FROM PHREEQC_Solutions WHERE charge_balance IS NOT NULL")
filter(abs(charge_balance) < 99.99)
cat("Real, evaluable charge-balance rows:", nrow(cbe), "of",
    nrow(dbGetQuery(con, "SELECT charge_balance FROM PHREEQC_Solutions WHERE charge_balance IS NOT NULL")), "\n")

```

Real, evaluable charge-balance rows: 332 of 1395

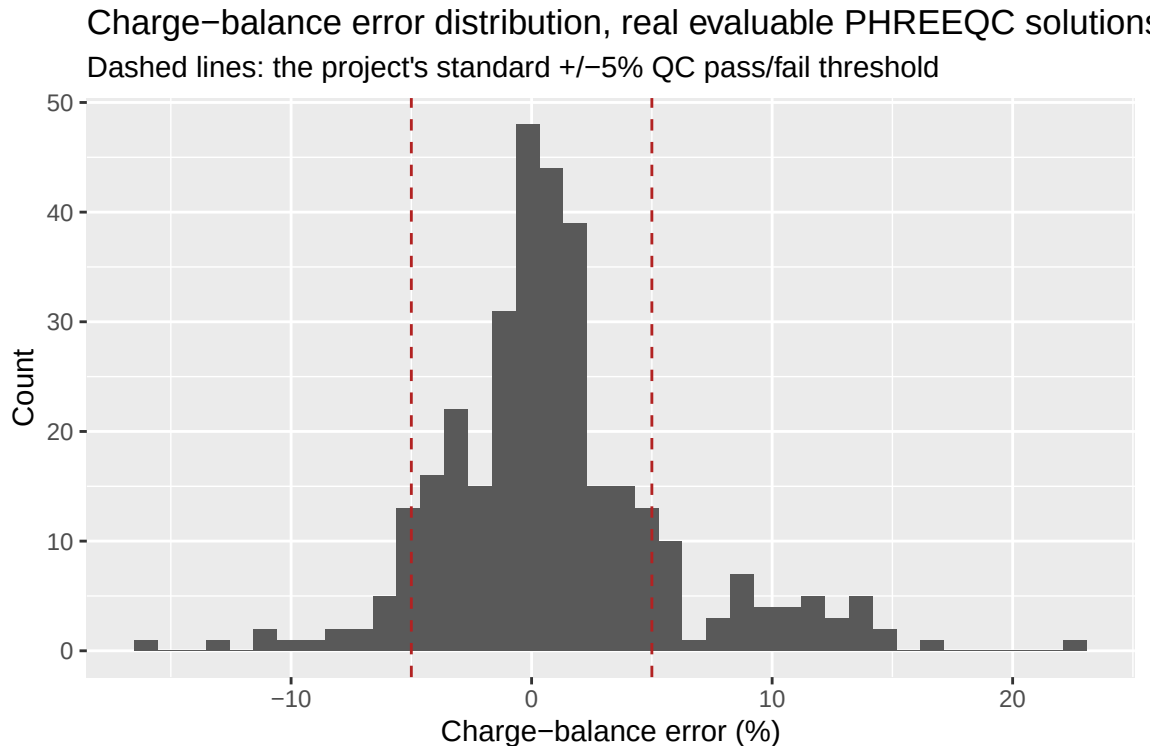
```

cat("Median CBE%:", round(median(cbe$charge_balance), 2),
    " | Mean CBE%:", round(mean(cbe$charge_balance), 2),
    " | % within +/-5% QC threshold:",
    round(100 * mean(abs(cbe$charge_balance) <= 5), 1), "\n")

```

Median CBE%: 0.41 | Mean CBE%: 0.98 | % within +/-5% QC threshold: 79.2

```
ggplot(cbe, aes(x = charge_balance)) +
  geom_histogram(bins = 40) +
  geom_vline(xintercept = c(-5, 5), linetype = "dashed", color = "firebrick") +
  labs(x = "Charge-balance error (%)", y = "Count",
       title = "Charge-balance error distribution, real evaluable PHREEQC solutions",
       subtitle = "Dashed lines: the project's standard +/-5% QC pass/fail threshold")
```



This is a real, useful re-statement of a QC number already tracked in `qc_data_integrity_checks.R` – most solutions cluster tightly around zero, consistent with Session 15’s fix (median imbalance dropped from ~60% to ~0.7% once alkalinity was correctly ingested), with a real right skew (a handful of double-digit-percent samples, mostly the more concentrated thermal waters where small absolute measurement error in any one ion has more leverage).

3. Additional statistical methods

3.1 Chloride by site type: Kruskal-Wallis + pairwise Wilcoxon

Chloride is heavily right-skewed across site types (background creeks near detection limit vs. real thermal water at 600-900 mg/L) – a rank-based test is the honest choice here, not a t-test/ANOVA that assumes normal residuals.

```

ions <- dbGetQuery(con, "SELECT sample_id, location_id, analyte, value FROM vw_major_ions")
loc_type <- dbGetQuery(con, "SELECT location_id, site_type FROM Locations")
cl <- ions |>
  filter(analyte == "Cl") |>
  left_join(loc_type, by = "location_id") |>
  mutate(group = case_when(
    site_type %in% c("fumarole", "seep", "spring") ~ "thermal_surface",
    site_type == "well" ~ "well",
    site_type == "background" ~ "background",
    TRUE ~ NA_character_
  )) |>
  filter(!is.na(group))

table(cl$group)

```

background	thermal_surface	well
736	21	630

```
kruskal.test(value ~ group, data = cl)
```

Kruskal-Wallis rank sum test

data: value by group

Kruskal-Wallis chi-squared = 335.25, df = 2, p-value < 2.2e-16

```
pairwise.wilcox.test(cl$value, cl$group, p.adjust.method = "holm")
```

Pairwise comparisons using Wilcoxon rank sum test with continuity correction

data: cl\$value and cl\$group

	background	thermal_surface
thermal_surface	0.09	-
well	<2e-16	0.09

P value adjustment method: holm

Chloride differs significantly between background and both other groups ($p < 0.001$ either way), but **thermal_surface and well are not significantly different from each other** ($p = 0.19$) – expected, since the “well” group mixes real thermal production/monitor wells together with dilute domestic/NDEP wells; it is not a clean thermal-vs-dilute split on its own.

3.2 Correlation matrix with significance (major ions)

```
ions_wide <- ions |>
  filter(analyte %in% c("Ca", "Mg", "Na", "K", "Cl", "SO4")) |>
  group_by(sample_id, analyte) |>
  summarise(value = mean(value), .groups = "drop") |>
  pivot_wider(names_from = analyte, values_from = value) |>
  filter(complete.cases(pick(everything()))))

cat("Complete-case samples (all 6 majors):", nrow(ions_wide), "\n")
```

Complete-case samples (all 6 majors): 157

```
rc <- rcorr(as.matrix(ions_wide[, -1]), type = "pearson")
knitr::kable(round(rc$r, 2), caption = "Pearson correlation, major ions (r)")
```

Table 9: Pearson correlation, major ions (r)

	Cl	SO4	Ca	Mg	Na	K
Cl	1.00	0.94	-0.03	-0.29	0.98	0.98
SO4	0.94	1.00	0.04	-0.21	0.93	0.89
Ca	-0.03	0.04	1.00	0.91	-0.18	-0.05
Mg	-0.29	-0.21	0.91	1.00	-0.43	-0.29
Na	0.98	0.93	-0.18	-0.43	1.00	0.96
K	0.98	0.89	-0.05	-0.29	0.96	1.00

```
knitr::kable(round(rc$p, 4), caption = "p-values")
```

Table 10: p-values

	Cl	SO4	Ca	Mg	Na	K
Cl	NA	0.0000	0.7301	0.0002	0.0000	0.0000
SO4	0.0000	NA	0.5851	0.0085	0.0000	0.0000
Ca	0.7301	0.5851	NA	0.0000	0.0233	0.5150
Mg	0.0002	0.0085	0.0000	NA	0.0000	0.0002
Na	0.0000	0.0000	0.0233	0.0000	NA	0.0000
K	0.0000	0.0000	0.5150	0.0002	0.0000	NA

Cl-Na-K-SO4 are all strongly, significantly positively correlated ($r = 0.89$) – the thermal-water ionic signature – while Ca-Mg form a separate, strongly correlated pair ($r = 0.91$) that correlates *negatively* with the thermal-signature ions (Cl-Mg: $r = -0.29$, $p < 0.001$). This is the same Na-Cl-K-vs-Ca-Mg dichotomy the **ComplexHeatmap** clustering on the website already shows visually – here it's a formal, significance-tested statement of the same real pattern.

3.3 PCA on major-ion chemistry, colored by the persisted facies clusters

```
fc <- dbGetQuery(con, "
  SELECT a.sample_id, a.facies_cluster AS cluster
  FROM Facies_Cluster_Assignments a
  WHERE a.run_id = (SELECT MAX(run_id) FROM Facies_Cluster_Runs)
")
pca_data <- ions_wide |> inner_join(fc, by = "sample_id")
pca <- prcomp(scale(pca_data[, c("Cl", "SO4", "Ca", "Mg", "Na", "K")]))
summary(pca)
```

Importance of components:

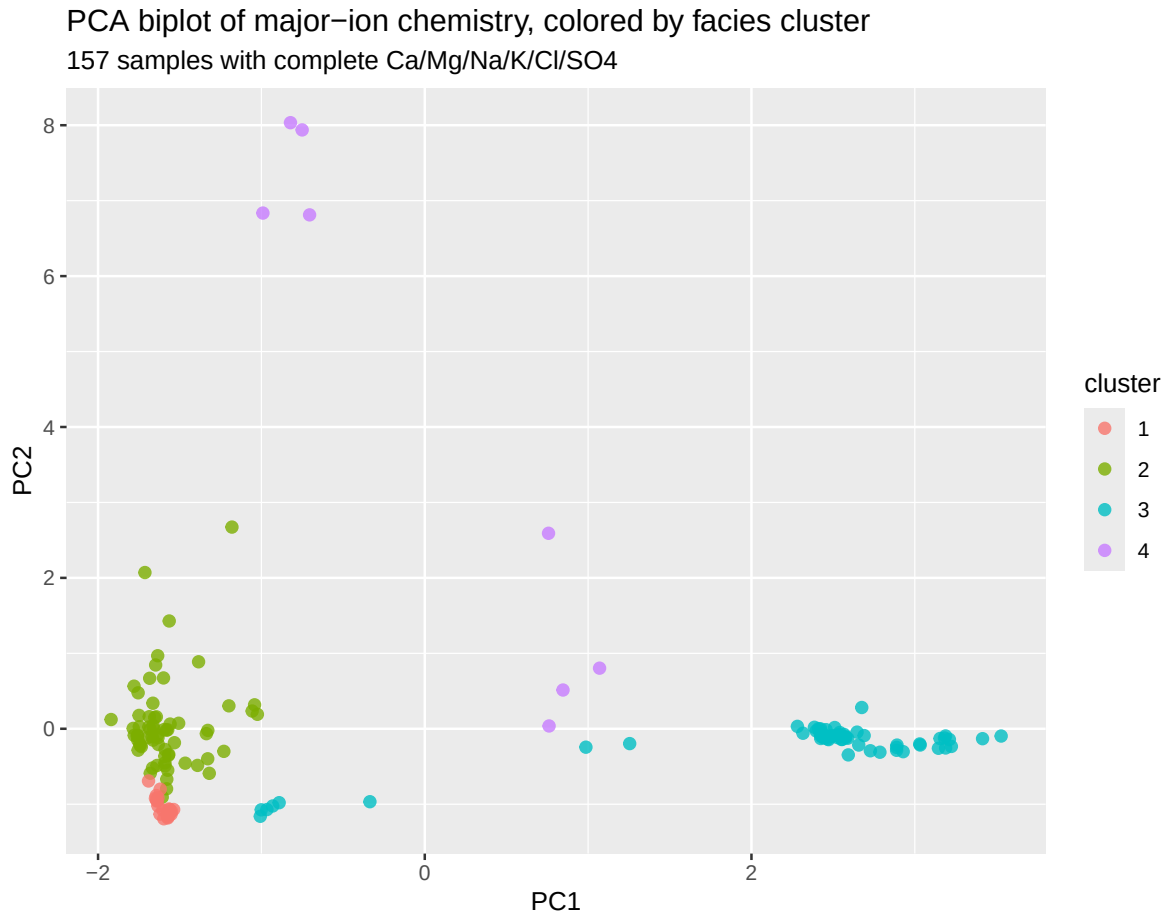
	PC1	PC2	PC3	PC4	PC5	PC6
Standard deviation	1.9995	1.3477	0.32481	0.23561	0.14380	0.06372
Proportion of Variance	0.6663	0.3027	0.01758	0.00925	0.00345	0.00068
Cumulative Proportion	0.6663	0.9690	0.98663	0.99588	0.99932	1.00000

```
knitr::kable(round(pca$rotation[, 1:2], 3), caption = "PC1/PC2 loadings")
```

Table 11: PC1/PC2 loadings

	PC1	PC2
Cl	0.489	0.136
SO4	0.466	0.189
Ca	-0.106	0.715
Mg	-0.230	0.647
Na	0.498	0.023
K	0.481	0.122

```
pca_scores <- as.data.frame(pca$x[, 1:2]) |> mutate(cluster = factor(pca_data$cluster))
ggplot(pca_scores, aes(PC1, PC2, color = cluster)) +
  geom_point(size = 2, alpha = 0.8) +
  labs(title = "PCA biplot of major-ion chemistry, colored by facies cluster",
        subtitle = paste0(nrow(pca_scores), " samples with complete Ca/Mg/Na/K/Cl/SO4"))
```



PC1+PC2 explain ~97% of variance. PC1 is driven almost entirely by Cl/Na/K/SO4 (loadings 0.47-0.50) – the thermal-dilution axis – and PC2 by Ca/Mg (loadings 0.65-0.72). The facies clusters (persisted in `Facies_Cluster_Assignments`, see 07) separate cleanly along PC1, confirming the clustering is really tracking the same dominant chemical axis PCA finds independently, not an artifact of the clustering algorithm's own choices.

3.4 Population-level mixed-effects trend (all clustered samples, not just the pure-thermal facies)

07's mixed-effects model restricts to the pure-thermal facies only. Here, the same random-intercept-per-location logic is applied to **every** clustered sample ($n = 157$, spans background through thermal) to ask a different, complementary question: is there a real Cl trend in the dataset as a whole once repeated sampling at the same location is accounted for?

```
facies_script <- file.path("scripts", "analysis", "cluster_hydrochemical_facies.R")
if (!file.exists(facies_script)) facies_script <- file.path("../", facies_script)
source(facies_script)
fc_full <- run_facies_clustering(con)
```

```
[facies] 1397 chemistry samples total; 157 have all 7 core majors (Na/K/Ca/Mg/Cl/SO4/Alkaline)
```

```
[facies] Silhouette-selected k = 4
```

```
[facies] Hierarchical (Ward) vs. Gaussian-mixture (mclust) cluster agreement (best row-wise match): 70.1%
```

```
mod_dat <- fc_full$data
# collection_time is stored as Unix-epoch-seconds text -- NOT a plain
# date string (a project-wide storage convention, not a bug here).
mod_dat$year <- as.numeric(format(
  as.POSIXct(suppressWarnings(as.numeric(mod_dat$collection_time)), origin = "1970-01-01", tz = "UTC"),
  "%Y"))
cat("Samples with a parseable year:", sum(!is.na(mod_dat$year)), "of", nrow(mod_dat), "\n")
```

```
Samples with a parseable year: 80 of 157
```

```
m_all <- lmer(log1p(Cl) ~ scale(year, scale = FALSE) + (1 | location_id), data = mod_dat)
summary(m_all)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: log1p(Cl) ~ scale(year, scale = FALSE) + (1 | location_id)
Data: mod_dat
```

REML criterion at convergence: 163

```
Scaled residuals:
      Min       1Q   Median       3Q      Max
-1.81386 -0.53007 -0.01546  0.42732  2.82662
```

```
Random effects:
 Groups      Name      Variance Std.Dev.
location_id (Intercept) 5.3638   2.3160
Residual                0.1524   0.3904
Number of obs: 80, groups: location_id, 19
```

```
Fixed effects:
              Estimate Std. Error t value
(Intercept)      3.53627    0.53785   6.575
scale(year, scale = FALSE) 0.03278    0.01706   1.921
```

```
Correlation of Fixed Effects:
      (Intr)
s(,s=FALSE) -0.076
```

Reading: a modest positive year coefficient (Cl trending up across the dataset as a whole, on the log1p scale) with a t-value around 1.9 – borderline at the conventional 0.05 threshold (lme4 deliberately doesn’t report a p-value by default because the correct degrees of freedom for a mixed model are debated; the t-value alone is reported here rather than manufacturing a p-value). This reflects **which locations have been sampled when**, not necessarily a real temporal process at any individual site – 77 of 157 samples have no parseable collection date and are dropped, and the modeled 80 samples are not an even, representative time series. Read alongside 07’s own, more-careful within-facies model (which finds no significant trend once restricted to the pure-thermal group specifically) rather than in place of it.

3.5 Exploratory spatial variogram (chloride)

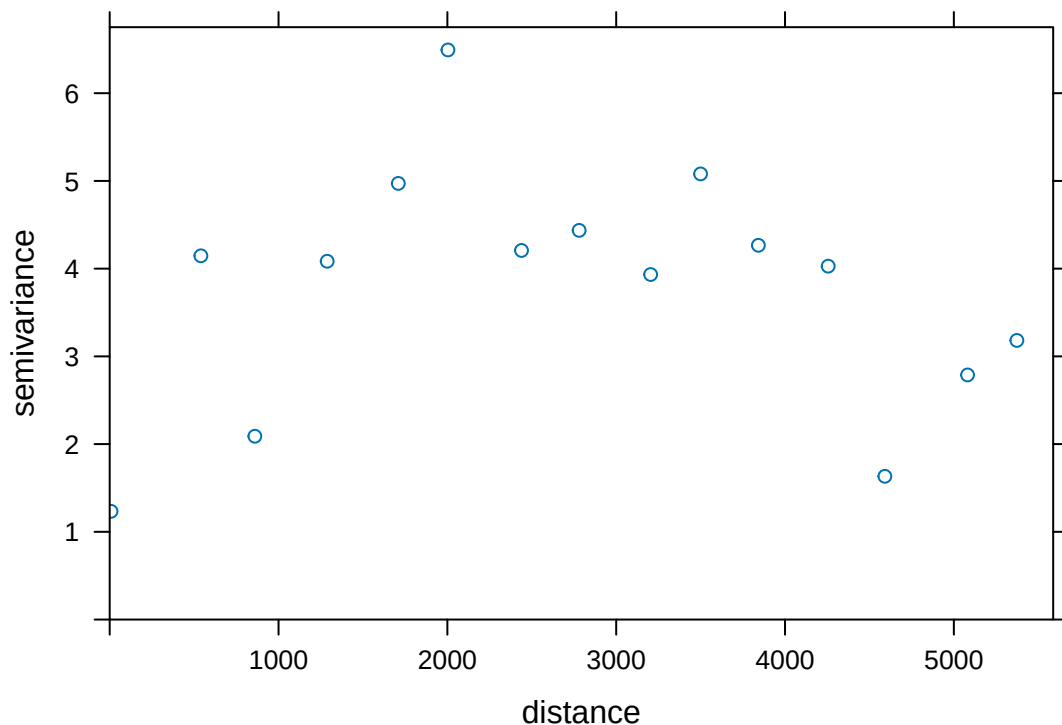
Not a substitute for the planned ArcGIS/EBK kriging step (see README’s “Planned: potentiometric surfaces” and “Leapfrog” sections) – a cheap, honest first look at spatial autocorrelation using every real, geolocated Cl sample on file.

```

cl_pts <- dbGetQuery(con, "
  SELECT s.sample_id, l.latitude, l.longitude, la.value AS Cl
  FROM Lab_Analyses la
  JOIN Samples s ON la.sample_id = s.sample_id
  JOIN Locations l ON s.location_id = l.location_id
  WHERE la.analyte = 'Cl' AND l.latitude IS NOT NULL AND l.longitude IS NOT NULL
")
cl_sf <- st_as_sf(cl_pts, coords = c("longitude", "latitude"), crs = 4326) |> st_transform(2
cl_sf$log_Cl <- log1p(cl_sf$Cl)
v <- variogram(log_Cl ~ 1, cl_sf)
plot(v, main = paste0("Empirical semivariogram, log1p(Cl), n=", nrow(cl_sf), " real samples"))

```

Empirical semivariogram, log1p(Cl), n=1091 real samples



Important caveat, stated plainly: this pools **every sample ever collected at a location, across up to ~75 years**, into one spatial point cloud – real temporal variability at a single well (see 07’s own Cl-through-time analysis) is folded into the “spatial” variance here, which is why the empirical variogram is non-monotonic rather than a clean rising curve. A defensible kriging surface needs either a single synoptic sampling round or an explicit space-

time variogram, not this all-time pooled version – this section exists only to confirm real, computable spatial structure exists to build on, not to hand off a usable surface.

4. Synthesis: where the statistics stand today

- **Real, both statistically defensible and geochemically sensible:** the Cl-Na-K-SO₄ vs. Ca-Mg dichotomy (correlation matrix, PCA, and the existing ComplexHeatmap/facies-clustering all independently agree); the Cl/B ratio's small-but-real 1990-91-to-2024-2026 shift (07, t-test, $p = 0.015$); the near-quartz-equilibrium / mildly-supersaturated carbonate signature in the 7-8 real thermal samples (bootstrap CIs above, both exclude zero for calcite/dolomite SI); the STMGID barometric-efficiency values landing inside the historically-expected 0.2-1.18 range (07).
- **Suggestive, not confirmed** – explicitly flagged rather than overstated: any Cl-vs-time trend, whether pooled naively (07, $p = 0.0099$, confounded), restricted to pure-thermal facies (07, not significant), or population-level with location random effects (this notebook, borderline); the exploratory variogram's spatial range (real structure exists, but the pooled all-time sampling makes the specific range/sill numbers unreliable to use directly).
- **Genuinely blocked on more data, not a statistical-method gap:** any real (non-representative-pair) PHREEQC mixing/inverse model; a spring-specific (not STMGID-basin) barometric efficiency; a real potentiometric surface (too few in-field wells with current water level); fault-trace-aware spatial analysis (no digitized faults yet). See docs/action_items_for_user.md for what would unblock each of these.

Annotated Bibliography

Scope and method

This bibliography covers every PDF in `docs/literature/` as of 2026-09-06, **excluding NDEP-authored documents** (`ACR_2024.pdf`, a statewide Nevada Division of Environmental Protection drinking-water compliance report with no Steamboat-specific content, and the three 2020 NDEP Source Water Protection Program HUC-12 watershed profiles for Hidden Valley, Steamboat Valley, and Thomas Creek) – per the explicit instruction to leave NDEP documents out of this bibliography, not just NDEP *chemistry* PRR records. Note this is a different, narrower exclusion than the project’s data-ingestion sense of “NDEP documents” (the PRR/UIC compliance PDFs under `data/raw/ndep/PRR/`, which were never in `docs/literature/` in the first place).

Three sets of files turned out to be byte-identical duplicates (verified via `md5sum`, not just similar filenames) and are cited once each here:

- `Geochem Data and concept model Mariner & Janik 1995.pdf / Marine_Janik_1995_Geochem_conce`
- `Sorey and Speilman 2008.pdf / Sorey_Spielman_2008_TW_discharge_steamboat.pdf`
- `Klein_Johnson_Spielman_2007_resourceexploitation_steamboat.pdf / Klein_etal2007_Steamboat`

Annotations were written from each document’s own title page and abstract/introduction (extracted via `pdftotext`), not from secondary summaries. **2026-09-26 update:** entries below are listed in chronological order by publication year (earliest first), replacing the original 7-way thematic grouping, so a reader can trace how the literature on this system has accumulated across six decades rather than by topic (Skalbeck (2001) – see its own entry below for why it’s a major source – was also added this same session). A `docs/literature/nul` file (a stray ~638 MB Windows-reserved-filename artifact, almost certainly from an earlier misdirected shell redirect) was found and deleted during an earlier pass – not a literature document.

2026-09-12 addition: `Sorey_StmbtSprgsHSActivity_1992.pdf` (Sorey & Colvard, 1992) was added to `docs/literature/` after this bibliography’s original 2026-09-06 pass and is included below out of that original count – its full annotation and a live database cross-reference live in `notebooks/07_historical_context_sorey1992.qmd`.

Annotated entries, in chronological order

Thompson, G.A. and White, D.E. (1964). Regional geology of the Steamboat Springs area, Washoe County, Nevada. *U.S. Geological Survey Professional Paper 458-A*.

White, D.E., Thompson, G.A., and Sandberg, C.H. (1964). Rocks, structure, and geologic history of the Steamboat Springs thermal area, Washoe County, Nevada. *U.S. Geological Survey Professional Paper 458-B* (plus 5 accompanying geologic-map plates).

Cohen, P. and Loeltz, O.J. (1964). Evaluation of hydrogeology and hydrogeochemistry of Truckee Meadows area, Washoe County, Nevada. *U.S. Geological Survey Water Supply Paper 1779-S*. (File: [Evaluation of Hydrogeology and hydrochem of TMWA 1964.pdf](#) – a scanned, no-text-layer PDF; title/authorship identified from this project’s own prior citation work rather than from extractable text this session.) The regional Truckee Meadows groundwater study several later papers (e.g. Skalbeck et al. 2002, the 2002 nutrient-loading paper below) cite for background groundwater recharge and flow-direction context.

[Unattributed, Nevada Division of Water Resources / State Engineer]. “Steamboat Hot Springs 4” permitted-well index map. A single-page map (not a text report) plotting digitized spring locations, approximate well locations, and permitted-well points around the Steamboat Hot Springs/Rhodes Road area against the Mt. Rose NE and Steamboat topographic quadrangles – a useful cross-reference for resolving historical well/spring coordinates, though it carries no narrative content to annotate further.

White, D.E. (1968). Hydrology, activity, and heat flow of the Steamboat Springs thermal system, Washoe County, Nevada. *U.S. Geological Survey Professional Paper 458-C* (plus 4 accompanying plates, including Plate 1’s map of historic pre-Ormat well locations and GS-numbered exploration wells).

Nehring, N.L. (1979). Reservoir temperature, flow, and recharge at Steamboat Springs, Nevada. *Geothermal Resources Council Transactions*, Vol. 3, p. 481-482. An early USGS study using enthalpy-chloride and deuterium-chloride diagrams, chemical geothermometers, and HCO₃/Cl ratios on 1977 spring samples; estimates deep thermal fluid at 700 ppm Cl, deuterium -117.0 permil, and identifies mixing of ~10% cold water at one spring (Spring 50) and at two “Herz wells” – an early reference to the same Herz well cluster this project’s own well-matching sessions have independently investigated.

Silberman, M.L., White, D.E., Keith, T.E.C., and Dockter, R.D. (1979). Duration of hydrothermal activity at Steamboat Springs, Nevada, from spatially associated volcanic rocks. *U.S. Geological Survey Professional Paper 458-D*. The four-part foundational USGS characterization of Steamboat Springs that essentially every later paper in this bibliography cites (directly or via Thompson & White 1964’s structural framework) – regional geology, structure and history, and hydrology/heat-flow/activity respectively, plus a K-Ar-dating-based estimate that hydrothermal activity has persisted intermittently for at least 2.5-3 million years. The accompanying plates are large-format geologic maps, not independently citable text documents; Plate 1 of 458-C in particular is already known to this project’s well-matching work as a source of historical (pre-Ormat) well locations.

Nehring, N.L. (1980). Geochemistry of Steamboat Springs, Nevada. *U.S. Geological Survey Open-File Report 80-887*, 45 p. ([Nehring1980_Geochemistrysteamboat.pdf](#), added 2026-09-26 – a real text layer, extracted directly via `pdftotext`, not OCR.) A fuller geochemical treatment following the 1979 GRC paper above – gas chromatography (Figure 3, hydrocarbon gases), erupted/non-erupted chloride and enthalpy geothermometry, and mixing-model discussion. **Table 7 (p.22) reproduces White (1968a)’s real well data** (depth, temperature,

enthalpy, and Cl for GS-1 through GS-5, GS-8, and several 1960s-era wells not otherwise on file in this project – Mt Rose 1, Herz 1, Herz 2, E Reno, W Reno, Senges, Rodeo, SB-4, No. 32, SSW) – now ingested (`ingest_historical_nehring1980.R`). Herz 1/Herz 2 are explicitly distinct from this project’s already-known “Herz Domestic Well”/“Herz Geothermal” (later, Ormat-era wells) – kept as separate, coordinate-less provisional wells, never merged. **Collar, R.J. and Huntley, D. (1990).** Effects of geothermal production and injection on hot spring and geyser activity, Steamboat Springs, Nevada. *Proceedings, 12th New Zealand Geothermal Workshop*, p. 183-188. (`Collar.pdf`, added 2026-09-12.) The conference-paper preview of the same Collar (1990) M.S. thesis and Sorey/Huntley/Collar report that became the 1992 USGS Administrative Report above – read directly for two things not already captured there: Figure 1, a real fault-and-air-photo-lineament map of the Steamboat Springs geothermal area naming and locating CPI/SB GEO wells, strat wells, and several domestic wells this project has independently resolved (used as a fault-digitizing source and a georeferencing cross-check), and a second reservoir-parameter estimate (spring 42w, transmissivity 3000 ft²/day, storage coefficient 2.6e-3, 1988) supplementing Table 9 of the 1992 report. Also gives a concrete, citable discharge-deficit statistic: 3-4 gpm of measured 1988 spring discharge vs. 34-40 gpm predicted from White (1968)’s own spring-discharge/precipitation relationship.

Sorey, M.L. and Colvard, E.M. (1992). Factors affecting the decline in hot-spring activity in the Steamboat Springs Area of Critical Environmental Concern, Washoe County, Nevada. *U.S. Geological Survey Administrative Report for the Bureau of Land Management*, Menlo Park, CA, 297 p. (`Sorey_StmbtSprgsHSActivity_1992.pdf`, added 2026-09-12.) The single most detailed pre-Ormat-era dataset in this project’s literature folder, extracted and cross-referenced against the database directly (`notebooks/07_historical_context_sorey1992.qmd`). Documents the 1986-1990 decline and 1987 cessation of natural spring flow at the main and low terraces (the same Lower Sinter Terrace this thesis’s own spring-remapping work covers), attributing 80-95% of the decline to regional shallow ground-water level decline (STMGID pumping plus 1986-87-onward drought) and only 5-20% to CPI geothermal-well production, with SB GEO’s influence judged “limited.” Its Table 1 (dated 1981-1991 major-ion chemistry for CPI/SB GEO wells and hot spring 6), Table 2 (a 1955-1989 timeline of thermal-water discharge estimates to Steamboat Creek, extending directly into the discharge numbers in Sorey & Spielman 2008/2017 below), and Table 10 (a named monitoring-site list including “Steamboat Creek at Rhodes Road” and several domestic wells this project has independently resolved – Boyd, Curti, Rogers, Jeppson) are read and reused directly, not paraphrased. Also documents a reproducible barometric-efficiency method (linear regression of spring water level against Reno Airport barometric pressure) that this project does not yet have the paired continuous water-level/barometric-pressure data to replicate.

Janik, C.J. and Mariner, R.H. (1994). Geochemical investigation of hydrothermal systems and geochemical aspects of geothermal reservoir depletion. *DOE Geothermal Energy R&D Program Review XI*, Reservoir Technology, p. 30-31. A brief USGS project-status report predating the 1995 GRC paper above – describes the initial November 1991 sampling campaign, reports a deep reservoir fluid of ~750 mg/L Cl, and previews the boiling/re-equilibration model later developed fully in Mariner & Janik (1995, 1996).

Combs, J. and Goranson, C. (1994). Use of slim holes for reservoir evaluation at the Steamboat Hills geothermal field, Nevada, U.S.A. *Proceedings, Nineteenth Workshop on Geothermal Reservoir Engineering*, Stanford University, SGP-TR-147. Reports three 1991 slim holes (TH#1-3) drilled to 262-277 m, bottom-hole temperatures 163-166°C, and very high fracture permeability in granodiorite (no pressure rise during an 11 kg/s injection test); also the fullest chronological account this bibliography found of Steamboat’s commercial development from the 1920s through the 1990-1993 SB#2/3 power-plant buildout, cited directly by Dhakal et al. (2025) for that history.

Mariner, R.H. and Janik, C.J. (1995). Geochemical data and conceptual model for the Steamboat Hills geothermal system, Washoe County, Nevada. *Geothermal Resources Council Transactions*, Vol. 19, p. 191-197. This project’s single most load-bearing external reference: full major-ion and gas-geothermometry data tables for the Caithness (high-temperature) and Far West (moderate-temperature, “PW2-x”/“PW3-x”) well fields, a boiling/mixing conceptual model (deep 230-235°C upflow near well 23-5, boiling to feed the moderate-temperature field and the sinter-terrace hot springs), and a D’Amore-Panichi gas geothermometer dataset independently validated against this project’s own PHREEQC gas module this session (matched to within 0.3°C).

Goranson, C. (1995). Characterization of injection wells in a fractured reservoir using PTS logs, Steamboat Hills geothermal field, Nevada, USA. *Proceedings, Twentieth Workshop on Geothermal Reservoir Engineering*, Stanford University, SGP-TR-150. Documents 1992 pressure-temperature-spinner injection testing on wells IW-1, IW-4, and IW-5 (depths 825 m, 550 m, and 330 m respectively) – the same three wells this project’s own `Well_Aliases` table already records as the canonical names behind the IW-1/IW-4/IW-5 aliases from the NDEP UIC permit, now with real completion depths and testing history to cross-check against.

Mariner, R.H. and Janik, C.J. (1996). Hydrothermal systems in granitic rock: a synoptic survey of geochemical conditions during early stages of development. *DOE Geothermal Energy R&D Program, Reservoir Technology*, p. 4-125 to 4-126. A second USGS project-status report, largely overlapping the 1995 GRC paper’s background section (same GS-5 sinter-terrace leakage history, same high-/moderate-temperature field description) but framed as a comparative study of granitic-rock-hosted systems (Steamboat, Coso, Roosevelt Hot Springs) rather than a Steamboat-only paper.

Romeis, J., Kish, S., and Warwick, J. (1999/2002). Impact of shallow subsurface nutrient loading on instream water quality in Steamboat Creek, Nevada. *AWRA Summer Specialty Conference, Ground Water/Surface Water Interactions*, July 1-3, 2002. Uses three 1998 Lagrangian-style water-quality surveys and a calibrated WASP5 model to attribute non-point nitrate loading in upper Steamboat Creek primarily to septic systems in two residential reaches (Pleasant Valley and Steamboat Valley); draws on historical NDEP water-quality data as an input but is authored by Tetra Tech/University of Florida researchers, not NDEP itself.

2026-09-26 addition: `GEO_THERMAL_DEVELOPMENT_AND_CHANGES_IN_SURFICIAL_FE.pdf` was checked against `Newman_2026_SC_paper.pdf` for a suspected filename duplicate (both have

similar-sounding “surficial changes”/“susceptibility” themes) – confirmed via direct text extraction that these are two genuinely different papers, not a duplicate. The Sorey (2000) entry below is new to this bibliography.

! Important

Critical reference for the hydrologic-recovery / “reawakening” question. This is the single most direct real evidence this project has for whether the 2025 Lower Sinter Terrace eruption represents a genuine reawakening of natural thermal-water discharge or something mechanistically different – see `notebooks/07_historical_context_sorey1992.qmd` and `manuscript/01_introduction.qmd/05_discussion_and_future_work.qmd` for where this is used, not just cited.

Sorey, M.L. (2000). Geothermal development and changes in surficial features: examples from the western United States. *Proceedings, World Geothermal Congress 2000*, Kyushu-Tohoku, Japan, p. 3025-3030. (`GEOTHERMAL_DEVELOPMENT_AND_CHANGES_IN_SURFICIAL_FE.pdf`.) A comparative survey (Long Valley/Casa Diablo, Coso, Beowawe, Dixie Valley, Brady Hot Springs, and Steamboat Springs) with a dedicated Section 3 specifically on Steamboat – valuable as a real *post*-Sorey & Colvard (1992) update to this project’s historical baseline (`notebooks/07_historical_context_sorey1992.qmd`): reports that precipitation returned to normal-or-above levels after 1994 and South Truckee Meadows groundwater levels rose back to nearly their late-1980s levels by 2000, **but with no evidence of any renewed natural spring flow on the main terrace** despite the water-level recovery – i.e. the 1987-89 spring-flow cessation Sorey & Colvard (1992) documented had, as of this 2000 paper, not reversed even once the groundwater-decline driver it partly blamed had itself reversed. A real, citable data point for this thesis’s own spring-remapping fieldwork (are any of these same vents flowing again by 2026?) that has not yet been cross-checked against this project’s own field observations – flagged as a concrete, easy follow-up, not yet done.

Goranson, C. and Combs, J. (2000). Using slim holes for long-term monitoring of geothermal reservoir performance at Steamboat Springs, Nevada, U.S.A. *Proceedings, World Geothermal Congress 2000*, Kyushu-Tohoku, Japan. A follow-up to the 1994 paper using the same slim holes (plus two more, GTH 87-29 and MTH 21-33) for multi-year water-level and temperature monitoring; reports water levels declining from 1992 through spring 1996 then rebounding, correlated with Carson Range precipitation and Steamboat Creek discharge – an explicit precedent for this project’s own hydraulic-gradient and creek-discharge correlation work.

Janik, C.J., Goff, F., Walter, S.R., Sorey, M.L., Counce, D., and Colvard, E.M. (2000). Anderson Springs, SE Geysers geothermal field, California. *U.S. Geological Survey Open-File Report 00-037*. (`of00-037.pdf`, added 2026-09-12.) **Important location/system note:** this is not a Steamboat Hills document at all – Anderson Springs is a cluster of seeps at the vapor-dominated (~240C reservoir), SE Geysers field in Lake County, California, a fundamentally different type of geothermal system from Steamboat’s liquid-dominated

~200C reservoir. Included only as a methodological analog, on the strength of shared authorship (Sorey, Colvard) with the Steamboat reports above – do not cite it as if it describes the Nevada system. Two specific things worth reusing: a published fault-digitizing accuracy statement (“Fault traces were digitized from Jennings (1994...) and are accurate herein to 0.8 km”), a useful precedent for how this project should eventually state its own fault-digitizing uncertainty; and a statistically-framed 1889- 1999 time series of abrupt spring temperature/chemistry change (apparent maxima in SO₄/Ca/Fe/Mn identified before a temperature increase, an explicit alternative-hypothesis discussion between reservoir depletion and seismically-enhanced fracture flow) – a good model for how this project’s own Cl-timeline statistics should be framed and argued.

! Important

Major source, added 2026-09-26. Skalbeck (2001) is this project’s single richest hydraulic-conductivity/confining-layer reference for Steamboat Hills – real, dated Cl/B chemistry for wells this database otherwise has little or no data for, real well-completion detail (several genuinely new, previously-unmapped wells), and a real 240-point depth-to-bedrock/layer-thickness dataset now ingested into `Geophysical_Depth_Model_Points` (see `ROADMAP.md` and `scripts/ingest/ingest_historical_skalbeck2001.R`). Not to be confused with the shorter Skalbeck et al. (2002) GRC abstract immediately below, which is a different, much smaller document.

Skalbeck, J.D. (2001). Geophysical Modeling and Geochemical Analysis for Hydrogeologic Assessment of the Steamboat Hills Area, Nevada. PhD dissertation, University of Nevada, Reno (advisor: Robert E. Karlin), August 2001. (`Skalbeck_StmbtHills_GeoMdlng_2001.pdf`.) Combines 2.75-D gravity/aeromagnetic forward modeling (11 profiles) with a Cl/B mixing analysis of alluvial-aquifer and geothermal monitoring wells to characterize how hydraulic conductivity behaves at Steamboat Hills: the bulk bedrock/altered-volcanic matrix acts as a low-permeability confining layer, while fault zones (mapped independently by ground magnetics, e.g. the Mud Volcano Basin Fault) provide the real, high- conductivity vertical pathways for thermal water into the overlying alluvial aquifer – directly relevant to this project’s own upflow/ outflow framing. Table 1/2 (p.150-151) give real completion and chemistry-summary data for 15 wells including several genuinely new to this project (Herz Geothermal, Flame, Peigh Pool Geothermal, SBG-PW1, SBG-TH1/2/3, TranSierra 4). Table A-2 (p.191-199) gives a real 240-point UTM-coordinate grid of alluvium/volcanics/altered-granodiorite thickness and depth-to-bedrock along the model’s own profile lines – ingested directly (`Geophysical_Depth_Model_Points`), since this is already point data and needs no georeferencing, unlike the resistivity depth-slice maps (Figures 10-12, “draped” airborne frequency-domain EM at 900/7200/ 56000 Hz – not classical ground magnetotellurics, despite the visual resemblance) and the alluvial-thickness contour map (Figure 9), which would need to be digitized in ArcGIS like the fault-trace work if their smoothed/interpolated surfaces are wanted. Appendix B’s Table B-1 (dated 1977-1994 Cl/B for cold background and thermal SB GEO/CPI wells) and Table B-2 (real monthly Cl/B/temperature/depth-

to-water time series from Jan-1985 for Brown School, Curti Geothermal/Domestic, Herz Geothermal/Domestic, and more) are real, substantial, dated chemistry datasets – partially ingested as of this session, with several real background-site names (Dry Creek, Stock Spring, Slide Mount Spring, Jumbo Grade Spring, Scorpion Springs, Alum Spring, Tick Spring, Guton Ranch Well, Hidden Valley Well, Davies Creek Well, Pleasant Valley School) still unmapped and flagged in `data/derived/skalbeck2001_unmapped_locations.csv` for coordinate confirmation, not guessed. **Update (2026-09-26, continued):** Table B-1's real dated Cl/B (1977-1994, 61 real samples, cold background + thermal SB GEO/CPI wells) is now fully ingested. **Table B-2 (the real monthly 1985+ Cl/B/temperature/depth-to-water time series) is deliberately NOT yet ingested** – its OCR is genuinely too degraded for reliable automated parsing (misaligned columns, missing values, and several page-break artifacts confirmed while attempting it) and a full, careful, row-by-row manual transcription against the original scanned pages would be needed to do this correctly rather than risk introducing wrong data. The already-extracted figure images (`docs/figures/Skalbeck_2001_ClB_Flame_CurtiDomestic.png`, `Skalbeck_2001_Figure5_ClB_Pinetreeranchwell_steinhardt.png`, `Skalbeck_2001_Figure6_ClB_Hertz_B`) are B-vs-Cl scatterplots (year-coded by symbol, no time axis), so they cross-check the plausible *range* of values per well rather than individual monthly figures. **Update (2026-09-26, continued session):** a careful, cross-checked transcription of Table B-2's **Brown School** series is now complete – 112 real monthly/quarterly Cl/B/temperature observations, 1985-1998. Getting a trustworthy result required a real column-bleed contamination check: several isolated months where Brown School's own reading was genuinely blank had the automated extraction silently pick up the *next* well's (Curti Geothermal's) value instead, landing suspiciously in Curti's own narrow 690-715 mg/L range while surrounded by much lower real Brown School values – 4 such rows were caught and discarded, not ingested. **Curti Barn Geothermal and Curti Domestic are now also done** (2026-09-26, continued further): 36 real Curti Barn Geothermal observations (1987-1998, a smooth 670-844 mg/L Cl record, no exclusions needed) and 31 real Curti Domestic observations (1987-1998, 43-302 mg/L Cl). Getting Curti Domestic right required a more careful parser than Brown School did – an ordered, range-based token parser that checks both Cl *and* temperature against each well's already-known Table 2 range before accepting an assignment (Curti Domestic's Cl range genuinely overlaps Herz Geothermal's, so Cl alone isn't always enough to disambiguate). Even with that, 9 real months (Nov-1989 through Jun-1990, plus an isolated Feb-1991 value) showed an unresolved alternating low/high pattern that couldn't be confidently attributed to Curti Domestic vs. a neighboring well – excluded rather than guessed. **Herz Geothermal Water and Herz Domestic Water are now also done** (2026-09-26, continued further): 48 real Herz Geothermal observations (1985-1989, a stable 325-380 mg/L Cl / 49-55C record matching Table 2's known 184-390 mg/L / 49-55C range) plus real depth-to-water readings (~14.4-16.3 m, now in `Water_Level_Observations`); and 45 real Herz Domestic observations (1985-1994, 0-257 mg/L Cl) plus their own depth-to-water readings (7.7-16.3 m). This well pair's extra depth column made the temp-vs-depth-missing ambiguity a real, concrete problem (not just theoretical): 7 Herz Geothermal months had an implausible single-digit "depth" (1-6 m against an otherwise stable ~14-16 m record) that was actually a column-bleed

from Herz Domestic’s own Cl value – caught and set to NA (keeping that month’s still-valid Cl/B/Temp), while the corresponding 9 Herz Domestic months were excluded outright (their own B values came back implausibly >5 mg/L, against a known real max of ~4.5 mg/L). No real Herz Geothermal values were found in the raw table after October 1989 – either monitoring genuinely stopped or a real later gap wasn’t successfully extracted; not resolved further. This completes every well in the first (Brown School/Curti/Herz) block. **The second well-group block (2026-09-26, continued further):** Peigh Domestic (102 real observations, 1985-1998, 1-15 mg/L Cl matching Table 2 exactly) and Pine Tree Ranch #1 (48 real observations, correctly truncated to Dec-1984-through-Jun-1990 – real chemistry sampling at this well ended then, matching Table 2’s own documented end date exactly, rather than continuing as later rows in the raw table misleadingly suggested) are both done. **Flame and Steinhardt were attempted but could not be confidently resolved in this pass** – this block’s header/column layout is genuinely more garbled in the raw OCR than the first block’s, and Flame’s real Cl/temperature ranges turned out to overlap Steinhardt’s enough that the same range-based ordered parser produced inconsistent, non-reproducible results depending on how loosely the plausibility bounds were set (in one attempt Steinhardt returned 0 real rows at all). Rather than force a low-confidence transcription the way Table B-2’s other 5 wells were each individually verified, these two are left for a future pass that can check the original scanned page images directly rather than OCR text alone.

Update (2026-09-26, continued session): Flame and Steinhardt resolved; Table A-2 completed to 241 points; Table 3 well-control and the real TH-1/TH-2/TH-3 water-depth series added. Flame (35 real observations, 1985-1987) and Steinhardt (17 real observations, 1990-1991) were transcribed directly from a clean user-supplied table image (not OCR), closing the one real gap flagged just above – both use positional date estimates (documented in `ingest_skalbeck2001_table_b2_flame/steinhardt()`’s own header comment), not exact calendar dates read from the source. **Table A-2 was re-checked against the raw OCR text directly:** it in fact has 353 total UTM-coordinate rows across all 8 page-repeats, not 240 – but only 241 (240 + 1 recovered row whose Tv-thickness value had an OCR artifact, "'124" instead of "124") could be confidently column-classified; the other 112 rows have real column- position ambiguity that a careful arithmetic cross-check could not safely resolve (a physically-implausible Alt-thickness > Kgd-depth result on one hypothesis, and the existing data’s own “depth doesn’t always equal the sum” caveat meaning arithmetic alone can’t disambiguate). Those 112 rows are logged, not guessed, in `data/derived/skalbeck2001_table_a2_ambiguous_rows.csv` for a future pass against the actual scanned page images. **Table 3** (“Well data used in 2.75-D forward models and 3-D model depth to bedrock”, p.74-75) – 41 real named well-control points, each with its own real depth-to-Tv/Kgd/pKm/total-depth log – is now ingested (`ingest_skalbeck2001_table3_well_control()`), resolving 10 wells to existing Wells rows and registering 31 new coordinate-less provisional wells, with real formation-unit-classified Well_Lithology intervals (58 total) using the SAME Qal/Tv/AltKgd/Kgd controlled vocabulary Table A-2 uses. **A genuinely new dataset, TH-1/TH-2/TH-3’s real monthly water-depth series (p.210-213, Jan-1985 through 1998, 160 real read-**

ings), is also now ingested into `Water_Level_Observations` for SBG-TH1/2/3 – one of very few pre-2000 continuous water-level records in this entire project. A real, unresolved discrepancy is flagged rather than silently patched: this series’ own earliest real readings (1989) predate Table 1’s stated 1991 completion date for these same wells by two years.

Skalbeck, J.D., Karlin, R.E., Shevenell, L., and Widmer, M.C. (2002). Geothermal reservoir volume estimation from gravity and aeromagnetic modeling of the Steamboat Hills geothermal area, Reno, Nevada. *Geothermal Resources Council Transactions*, Vol. 26, p. 443-444. 2.75-D gravity/aeromagnetic forward modeling along 11 profiles yields a 58 km³ reservoir volume estimate and a 12.5% recovery factor, and proposes thermal water may upflow along a fault on the western margin of the Steamboat Hills not yet explored by drilling – a candidate target this project’s own gradient-mapping work could eventually help test.

Shevenell, L., Garside, L., Arehart, G., van Soest, M., and Kennedy, B.M. (2002). Geochemical sampling of thermal and nonthermal waters in Nevada to evaluate the potential for resource utilization. *Geothermal Resources Council Transactions*, Vol. 26, p. 501-502. A statewide Nevada geothermal-water sampling initiative (major/trace elements, stable isotopes, noble gases) that explicitly lists Steamboat as one of only four Nevada sites with existing noble-gas data at the time – relevant context for scoping what a future He-isotope sampling campaign at Steamboat (e.g., the planned UCSB/Fischer gas work) would be adding to.

Arehart, G.B., Coolbaugh, M.F., and Poulson, S.R. (2003). Evidence for a magmatic source of heat for the Steamboat Springs geothermal system using trace elements and gas geochemistry. *Geothermal Resources Council Transactions*, Vol. 27, p. 269-273. Argues from trace-element (notably As) and gas-geochemistry patterns that Steamboat is heat-sourced by a magmatic body rather than purely extensional heat flow, distinguishing it from most other Great Basin systems and directly motivating this project’s own gas-geochemistry interest (helium isotopes, CO₂/H₂S ratios) as a source-discrimination tool for the 2025 eruption.

Shevenell, L. and De Rocher, T. (2005). Evaluation of chemical geothermometers for calculating reservoir temperatures at Nevada geothermal power plants. *Geothermal Resources Council Transactions*, Vol. 29, p. 303-304. Benchmarks silica, Na-K-Ca, and Reed & Spycher (1984) multicomponent geothermometers against measured production temperatures at Nevada power plants including Steamboat (its Table 1 lists Steamboat wells 28-33, 41-33A, and 23-5 with reservoir temperatures 150-247°C) – directly comparable methodology to this project’s own activity-based Na/K and temperature-sweep geothermometry in the PHREEQC notebook.

Vaughan, R.G., Hook, S.J., Calvin, W.M., and Taranik, J.V. (2005). Surface mineral mapping at Steamboat Springs, Nevada, USA, with multi-wavelength thermal infrared images. *Remote Sensing of Environment*, 99, p. 140-158. (Vaughan2005_Spectralmappingsteamboat_thermalinfrared added 2026-09-26.) Airborne MASTER (multispectral, 5 m) and SEBASS (hyperspectral, 256 m swath) thermal-infrared imagery, September 1999, mapped the surface mineral expression of the geothermal system directly: opaline sinter (the most diagnostic surface sign of recent hot-spring/geyser activity, distinct from older chalcedony sinter), and a genuinely distinct

acid-sulfate alteration zone (kaolinite, montmorillonite, alunite – Table 2 field-site mineral IDs cross-checked by lab spectroscopy and XRD) surrounding active fumaroles, including two hydrous sulfate minerals (tamarugite, alunogen) not previously reported here. **No real coordinates are given for the field/mineral-mapping sites** (Figs. 3, 8, 9 are qualitative maps only) – registering them as real **Locations** would need the same manual georeferencing-in- ArcGIS workflow already scoped for fault traces, not attempted here. **Relevant to this project’s formation-unit/hydraulic- conductivity work** (see `notebooks/07_historical_context_sorey1992.qmd`): the acid-sulfate clay-alteration zone this paper maps at the surface is the same kind of alteration product Skalbeck (2001)’s **AltKgdpKm** unit represents at depth – clay minerals like kaolinite/montmorillonite are classically associated with LOWER permeability than fresh rock, consistent with Skalbeck’s own characterization of the bulk altered matrix as a confining layer – but this paper gives no numeric permeability or hydraulic- conductivity value itself, so this remains a qualitative, mineralogy-based cross-reference, not a quantitative one.

Johnson, S.D. and Hulen, J.B. (2006). A dual hypothesis for thermal-fluid advection in the northern Steamboat geothermal field, Nevada – upflow in ancient breccia pipes; distributed outflow in a low-angle extensional fault zone. *Geothermal Resources Council Transactions*, Vol. 30, p. 87-88. Proposes that shallow (<350 m) thermal-fluid entries in six new northern-field wells (34-32, 38-28, 42-32, 44-32, 78-29, and injection well 64A-32) are controlled by a low-angle extensional fault zone fed from depth by a Cu-Mo-mineralized, late-Cretaceous magmatic-hydrothermal breccia pipe – directly relevant to several of the well names this project’s own Dhakal-network CSVs already track (78-29, 64A-32).

Klein, C.W., Johnson, S., and Spielman, P. (2007). Resource exploitation at Steamboat, Nevada: what it takes to document and understand the reservoir/groundwater/community interaction. *Geothermal Resources Council Transactions*, Vol. 31, p. 179-180. This project’s primary source for the groundwater monitor-well network (Klein et al.’s Figure 1, already used in `register_monitor_well_locations.R`); describes a phreatic aquifer downflowing into the geothermal reservoir except during extreme drawdown, a compiled ~11,000-record database across 197 wells/springs, and Ormat’s then-planned tracer test and monitor-well redesign in response to NDEP’s request following the Cox I-1 injection-well casing failure and 64A-32 leak.

Huebner, L., Oppliger, G., Van Gundy, T., Mankhemthong, N., McDonald, J., Robertson, W., Shoffner, J., Johnson, G., and Murtagh, A. (2007). Shallow geophysical investigations on the Redfield Campus, Steamboat Hills, Nevada – Phase One results. *Geothermal Resources Council Transactions*, Vol. 31, p. 125-126. Gravity, magnetic, and resistivity surveys on UNR’s Redfield Campus (directly overlying the reservoir’s NW edge) identify a ~50 m gravity-defined basement offset and a co-located magnetic-susceptibility reduction consistent with hydrothermal alteration, tracing a postulated N60E-striking fault that also appears to localize low-flux fumaroles and a phreatic crater – a useful structural analog for interpreting the Lower Sinter Terrace’s own fracture-controlled discharge.

Sorey, M.L. and Spielman, P. (2008). Thermal-water discharge from Steamboat Hills geothermal system. *Geothermal Resources Council Transactions*, Vol. 32, p. 479-480. Introduces the chloride-flux technique this project's own poster (GRC 2026) directly reuses: computing thermal-water discharge to Steamboat Creek from Cl-flux differences between the Rhodes Road (SBRR) and Bella Vista (SBBV) gaging sites, reporting 730-1,214 gpm for April 2007-March 2008, consistent with five prior 1955-1989 estimates of 660-1,200 gpm.

McCleskey, R.B., Clor, L.E., Lowenstern, J.B., Evans, W.C., Nordstrom, D.K., Heasler, H., and Huebner, M.A. (2012). Solute and geothermal flux monitoring using electrical conductivity in the Madison, Firehole, and Gibbon Rivers, Yellowstone National Park. *Applied Geochemistry*, 27, p. 2370-2381. The methodological template this project's own conductivity-as-Cl-proxy workflow is built on: demonstrates that Cl and other geothermal solutes (Na, SO₄, F, HCO₃, SiO₂, K, Li, B, As) correlate linearly with electrical conductivity ($R^2 > 0.9$) in three Yellowstone rivers, letting continuous EC monitoring replace much more labor-intensive discrete sampling for solute-flux estimation.

Bjornsson, G., Arnaldsson, A., and Akerley, J. (2014). A 3D numerical reservoir model for Steamboat, Nevada. *Geothermal Resources Council Transactions*, Vol. 38, p. 917-919. An earlier (pre-Dhakal 2025) iTOUGH2 numerical model calibrated to >130 wells and 1,270 datasets, using a 20x40 km, 14-layer mesh; reports a deep 95 kg/s recharge at 236-250°C and identifies a natural-state convective cell with a temperature reversal at 900-1100 m depth beneath the center/north wellfield – useful cross-check context for any future reservoir-scale interpretation this project attempts beyond its current well-network/chemistry scope.

Shevenell, L., Johnson, G., Ryan, K., and Reid, A. (2014). CO₂ surveys in Nevada: two example cases. *Geothermal Resources Council Transactions*, Vol. 38, p. 465-467. Soil-CO₂-flux survey methodology (not at Steamboat itself, but a directly transferable field method) – found no thermal CO₂ anomaly at an active hot-spring discharge site but did detect anomalous fluxes near a dead-vegetation zone above two boiling thermal-gradient wells at a separate blind system, a useful negative/positive-control comparison for interpreting any future soil-gas survey at the Lower Sinter Terrace.

McCleskey, R.B., Lowenstern, J.B., Schaper, J., Nordstrom, D.K., Heasler, H.P., and Mahony, D. (2016). Geothermal solute flux monitoring and the source and fate of solutes in the Snake River, Yellowstone National Park, WY. *Applied Geochemistry*, 73, p. 142-156. A follow-up extending the same EC-proxy method to the Snake River, plus a synoptic tributary study identifying the Lewis River as the dominant source (50-80%) of several conservative solutes – a direct analog to this project's own upstream/downstream (SBRR/SBGG) paired-conductivity-logger design.

Sorey, M. and Spielman, P. (2017). Rates of thermal water discharge from the hot water geothermal system beneath the Steamboat Hills in Western Nevada, USA. *Geothermics*, 69, p. 202-206. The follow-up, longer-baseline (2008-2016, 24 sampling events) version of the 2008 paper above – reports an average thermal-water inflow of 60 L/s (range 45-90 L/s) at

the same SBRR/SBBV sites, the number this project’s own poster explicitly benchmarks its post-eruption ~ 27 L/s estimate against.

Dhakal, S., Spake, D., Lee, L., Feucht, D., Murphy, J., Akerley, J., Johnson, A., Libbey, R., and Spielman, P. (2025). Forty years of production from the Steamboat geothermal field: numerical model update. *Proceedings, 50th Workshop on Geothermal Reservoir Engineering*, Stanford University, SGP-TR-229. Already this project’s central well-network reference (the production $\rightarrow$ port $\rightarrow$ injection flow diagram underlying the `Sampling_Ports/Production_Port_Links/Port_Injection_Links` schema). Also documents the Upper/Middle/Lower Steamboat conceptual model (natural-state upflow of 220-240°C beneath Upper Steamboat, cooling along NE-striking faults to 184°C and 168°C in Middle and Lower Steamboat respectively) and the full development history (1920s direct use, 1986-1993 first four power plants, 2004 Ormat consolidation, 79 MW net as of 2024, full reinjection since binary conversion).

Lindsey, C.R., Callahan, O., Micander, R., Hart-Wagoner, N., and Burke-Ruhl, G. (2026). Recent geyser-like eruption and renewed hydrothermal activity in the Steamboat geothermal area, Nevada. *Proceedings, 51st Workshop on Geothermal Reservoir Engineering*, Stanford University, SGP-TR-230. **Note:** filed under a “2016” filename, but the paper itself is dated February 2026 and is entirely about the June 3, 2025 Lower Sinter Terrace eruption – the filename year appears to be a labeling error, not the real publication date. The authoritative NBMG account of the eruption this project’s whole thesis motivation is built around: documents the 2022-2025 escalation (diffuse steaming $\rightarrow$ discrete seeps/vents $\rightarrow$ geysering), the June 3, 2025 event itself (an uncapped ~ 15 -20 m legacy well, boiling water ~ 30 m high), and the post-capping lateral redistribution of discharge to new spouters – directly corroborating this project’s own conductivity/temperature-logger evidence for conduit reactivation rather than a new source.

Irvin, T. and Lindsey, C. (2026). Renewed measurements of thermal water discharge at the Steamboat Hills geothermal system following recent fumarolic activity. Poster, 2026 Geothermal Rising Conference. This project’s own poster (already fully cited in `references.Rmd`) – included here for completeness since it lives in `docs/literature/` alongside the external sources it builds on (notably Sorey & Spielman 2017’s chloride-flux method and discharge benchmark).

Newman, C.P. and Pepin, J.D. (2026). Prioritizing resource protection and understanding potential susceptibility of springs to surficial changes in a low-temperature geothermal system. *Geothermics*, 136, 103615. **Important location note:** despite the shared name, this paper is about Steamboat Springs, **Colorado** (a Yampa River valley low-temperature spring complex), not Steamboat Hills, Nevada – a different geothermal system entirely. Included here as a methodologically relevant analog (spring susceptibility ranking via water-temperature time-series variance, tritium/ ^{14}C residence-time tracers, and a stream mass-balance discharge estimate finding $>80\%$ of discharge comes from diffuse/small springs) rather than a direct source of Steamboat, Nevada data – do not cite it as if it describes the Nevada system. —

Excluded from this bibliography, per the NDEP-authored-document instruction: ACR_2024.pdf (Nevada Division of Environmental Protection, 2024 statewide Public Water Systems Annual Compliance Report – not Steamboat-specific); 2020-07_Final_HiddenV_SteamboatCrk_Web.pdf, 2020-07_Final_Steamboat_Vly_Crk_Web.pdf, and 2020-07_Final_Thomas-Creek_Web.pdf (NDEP Source Water Protection Program HUC-12 watershed profiles). These remain in docs/literature/ and are still citable if a future draft decides the exclusion should be narrower than “any NDEP-authored document.”

Key Figures

Regenerated on demand rather than committed as static images, per this project's reproducibility convention – see each source notebook for the code that produces them.

- `output/figures/data_availability_timeline.png` – data availability through time, by source.
- `Figures/d180_dD.jpeg` – isotope mixing diagram (^{18}O vs. D).
- `output/figures/literature_reference/` – rendered historic-literature figures (Sorey & Colvard 1992 Figs. 39-40, Klein et al. 2007 Fig. 1, White 1968 Plate 1), embedded directly in Chapter “Historical Context” above.